Neurons Through the Looking Glass: Constructing an Interacting Quantum Field in Two Dimensions

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Construction, conditional quantitative bounds and renormalized local squares.
Dated revision guides retain their historical status.
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Abstract

We construct an interacting quantum field from random fields called *process neurons*. We combine these building blocks and change their probability weights through a local quartic interaction. Two explicit architectures, built from random sign changes or randomly colored cells, reach the established massive, weakly coupled two-dimensional Wick-quartic field along one deterministic parameter sequence. This sequence is fixed before any measurement is chosen and works for both architectures.

The construction preserves *reflection positivity*, the positivity condition used to recover a quantum Hilbert space from Euclidean fields. At every finite stage the normalized interacting law has this property across the two central coordinate lines. We prove weak convergence of the field laws on $H^{-2}_{\mathrm{loc}}(\mathbb{R}^2)$, convergence of every power of the local seminorms, and convergence of all polynomial moments of any finite collection of smooth, compactly supported measurements chosen afterwards. The joint compact-test correlation distributions also converge at every order.

We establish the process-neuron, Gaussian-reference and fixed-volume Wick estimates needed for this construction. Attributed classical results supply the infinite-volume target and its reconstruction as a relativistic quantum field in one space and one time dimension; we verify their hypotheses at the points of use. Finite width counts whole random processes, each using countably many random variables globally. The common limiting sequence is existential, with no effective end-to-end parameter schedule or computational advantage established. The result gives a precise route from these two neural architectures to the specified interacting field.

Current guide: every dyadic accuracy

A process neuron is a random pattern on the plane. The construction adds independent patterns and changes their probability weights through a local quartic interaction. Its limiting law is the original interacting field. The new result makes a quantitative part of that construction explicit: after choosing an allowed finite collection of measurements, every dyadic accuracy 2^{-q}, $q \geq 3$, has terminating finite outer stages and a sufficient count of the original process neurons. The separately accepted UDC implementation produces and independently checks the symbolic prescription when its configured finite capacities suffice.

The field, target and two architectures are unchanged. The measurement family consists of an ordered one to four unweighted translates of the original product bump, with canonical dyadic centers in $[-8, 8]^2$ and denominator exponents at most sixteen. Repetitions, singular observation laws, and overlapping, touching or separated supports retain the full ambient dimension. A separate $1/5$ request remains available. The guarantee controls the joint bounded-Lipschitz law of these chosen measurements.

The proof follows the joint law of a partial field and the next replacement direction under the actual interaction weight. Coercivity controls field moments; density domination controls direction moments; a centered fourth-order comparison controls adjacent normalized laws. The argument keeps every Bell and observation-product term and then sums the replacements. Volume-dependent Nelson and physical coupling estimates select the later stages. All five arrows end at the same original pressure-identified field.

Write $a = 2^{-p}$, $\epsilon = 2^{-t}$ and $\eta = 2^{-d}$ for the regulator, stabilizer and per-arrow allocation. Appendix K gives the complete prescription, proofs and exact source qualifications. Its $N_{\mathrm{new}} = 2^{e_{\mathrm{new}}}$ is strictly below the older sufficient $2^{2^{w_{\mathrm{old}}}}$ at the same stages for every allowed q. The exponents depend on the selected volume, and the proof retains $\lceil t/2 \rceil$ at both parities. On the eleven older settings the stages and sharper width agree exactly with SVA/SVC.

The earlier reading guides and appendices are preserved as dated results with their named versions. Their eleven-setting domains and statements about then-pending software describe those versions. The current extension is UDA-STAGE-RELATIVE-v2, with the separately accepted UDC-dyadic-accuracy-v1 finite-resource implementation. The mathematical every-q theorem and the software's success or resource refusal have different scopes. Section K.8 specifies the current request, certificate, source roots, verifier and capacities. Acceptance of the two source packets is an internal exact-version decision; this manuscript integration has its own fresh review gate.

The finite-positive-stage strong-L^4 laws, every bounded-bilinear identity, physical common space, classical/source and NR1 qualifications remain literal premises. Strict AP6 supplies the numerical target route; the original fixed g_* remains unknown. Counts are astronomical sufficient whole-plane process-neuron counts, with countably many internal variables. Canonical least inner indices, a global finite-weight simulator, practical or necessary costs, full-field rates, broader models and priority remain separate. Public v1 is unchanged.

Current reading guide: 28 October 2026

A process neuron is a random pattern on the plane. Adding independent patterns and changing their probability weights through a local quartic interaction produces the laws studied here. Removing inner approximations and enlarging the interaction box reaches the original interacting field.

This revision improves the sufficient width when a supported finite collection of measurements and an accuracy are chosen first. The sharper relative estimate now follows the accuracy-selected outer stages, giving $N_{\mathrm{new}} = 2^{16\lceil t/2\rceil+20p+4d+544}$, strictly below the older sufficient $N_{\mathrm{old}} = 2^{2^{w_{\mathrm{old}}}}$ for every supported request. Here $a = 2^{-p}$ is the regulator, $\varepsilon = 2^{-t}$ the stabilizer, and $\eta = 2^{-d}$ the per-arrow allocation. Theorem J.2 proves the enclosure, ceiling and odd-t rounding argument. The four later errors, actual field and target are unchanged. Widths remain astronomical sufficient counts of whole processes, each with countably many internal spatial variables.

The domain is precise: ordered tuples of one to four unweighted canonical dyadic translates of the original product bump, centers in $[-8,8]^2$ with denominator exponent at most sixteen, and exactly the eleven tolerances $1/5, 2^{-3}, \ldots, 2^{-12}$. Repetitions, singular observation covariances, overlapping, touching and separated supports all keep their original ambient dimension. For the four distinct corners $(-8,-8),(-8,8),(8,-8),(8,8)$ at tolerance 2^{-12}, the sufficient new width is $2^{15668170148}$; the same-stage old width is $2^{2^{1958531694}}$. Their outer stages and error allocation coincide. The odd-t unshifted scalar example at $1/5$ has sufficient width $2^{13265754396}$; Section J.4 gives its separate stages.

A separately versioned symbolic backend implements this rule. Proposition J.3 gives the accepted SVC contract: independent arithmetic verification, source/code bindings, parser limits and explicit refusal/resource behavior. Proof, software identity, finite checks and internal model review have separate roles. Checksums establish bytes; classical and random-element premises retain their qualifications.

The key premise is the complete strong-L^4 reference-law interface at every selected stage: strong measurability, zero Bochner means, global sign symmetry and matching for every bounded bilinear form. Point-pair covariance alone does not supply that interface. The physical own-variance Wick and full neural/Gaussian normalizers have separate roles. Strict AP6, P1–P6/source/classical assumptions, NR1 sharper-author versus larger-blind-core scopes and the unknown original fixed g_* remain.

The qualitative construction in Theorem I.2 retains its inclusive cap $0 < u/m^2 \le 1/8000$ under its named premises and its all-later-test quantifiers. The present finite-observation schedules use the strict quantitative cap and do not compute that construction's canonical least inner indices. Theorem I.1 retains local H^{-2} moment convergence; the separate SEQ-HS result retains physical/target H^{-s} moment finiteness for every real $s > 1$.

The accepted 107-page baseline's mathematical blocks and bibliography are preserved. Its 27 and 26 October guides and fixed-1/5/old-ACC results are dated checkpoints; Appendix J gives the current result. The preserved SVA-era unlaunched-SVC status describes its mathematics-only checkpoint; the later accepted SVC decision supersedes that implementation status. Full-field rates, practical/necessary complexity, broader models and priority remain separate. Public v1 is unchanged.

This integration candidate awaits fresh source, proof and visual review. The reversible delta, block audit and local proof map accompany it; the coordinator owns acceptance.

Historical revision guide: 27 October 2026

A process neuron is a random pattern on the plane. Adding independent patterns gives a reference field; weighting their probabilities by a local quartic interaction turns it into an interacting law. We remove the four inner approximations, then enlarge the interaction box. Reflection positivity survives at every finite stage across the two central coordinate lines and permits the classical quantum interpretation of the limiting law. The detailed construction and classical attribution remain in the original paper below.

The accepted qualitative statement is Theorem I.2. Under the literal accepted original, source and classical premises in Appendix I, it extends the original construction to $0 < u/m^2 \leq 1/8000$, including equality. One unchanged deterministic sequence serves both original architectures and every compact measurement, polynomial, correlation test and permitted reflection list chosen afterwards. The local-moment proof is given in Theorem I.1. MOM formally concerns the original local H^{-2} seminorms; the separately derived SEQ-HS statement gives all-real-$s > 1$ physical/target moment finiteness, without asserting moment convergence at other s.

The quantitative question fixes measurements and an accuracy first. For a nonzero scalar bump at $m = 1$, $u = 10^{-10}$, $\ell = 1000$, a relative normalization argument reduces one sufficient width to $2^{549756142138}$. For ordered tuples of one to four supported dyadic bump translates, the separate vector result gives $2^{549756142144}$ at accuracy $1/5$. Both are immense counts of whole random processes. Each process retains countably many internal spatial variables. The exact scopes, outer stages, error allocations and proofs are in Proposition I.5 and Table 7.

The bounded accuracy search of Theorem I.6 uses the older full-normalizer vector backend. It supports exactly $1/5$ and $2^{-3}, \ldots, 2^{-12}$ for the stated typed family. It selects a sufficient tuple for each request. At this 27 October checkpoint, the sharper fixed-$1/5$ vector result remained separate. The later selected-stage combination is now given in Appendix J. All quantitative target uses retain the strict cap $0 < u/m^2 < 1/8000$ and literal AP6/P1–P6 premises. The originally fixed g_* remains numerically unknown.

Theorem I.7 proves that both routes reach the identical original Fröhlich–Simon law and compares them through their same physical box. It provides a sufficient physical-box label for one represented observation when certified source/norm data are supplied. It leaves the original four canonical least indices existential. The finite-observation-to-field reverse inference fails in general, as shown by an explicit smooth-field kernel example. The construction's global neural Schwartz extension, practical or necessary complexity, wider models/dimensions and historical priority remain separate questions.

The 91-page accepted private baseline is preserved mathematically and its complete bibliography is unchanged. The original g_* theorem and 26 October revision guide remain below as the historical route; the new appendix states precisely which later premises yield the inclusive qualitative cap. Full classical/source, NR1 author-versus-blind-core, reconstruction and earlier adverse-version qualifications remain visible. All accepted input results are internally reviewed at exact conditional scopes. This integrated source/PDF is a new private candidate for fresh source, proof and rendered-editorial review; its own acceptance is pending. The published v1 remains unchanged.

The following private revision guide is the retained 26 October guide. Its references lead to the earlier quantitative appendix. The current results begin in Appendix I; the public proof map identifies the mathematical locations in this edition. Original exact-version source bindings and the historical integration checks are retained in the private research archive.

Historical private revision guide: 26 October 2026

This private post-v1 candidate preserves the accepted construction's mathematical statements, proofs and equations. The abstract and main theorem continue to describe the original observable-independent sequence, with its original g_* assumption and noneffective selection. The new quantitative and reconstruction material is collected in Appendix H.

The subsequent chain proves local shell-response decay, converts it into a finite interaction-box error, and controls all five approximation errors in one fixed finite-observation metric. For a nonzero smooth product bump at $m = 1$, $u = 10^{-10}$ and $\ell = 1000$, it specifies an exact sufficient tuple for both process-neuron architectures with error at most $65/4096 < 1/5$. An earlier sufficient tuple and its slightly smaller certified error sum remain visible. The optimized tuple has smaller parameters at the same requested tolerance; its width remains astronomical.

The explicit numerical coupling uses a separately proved pressure-target route under its named classical and finite-interface premises. It supplies no numerical lower bound for the originally fixed g_*. The original universal sequence remains intact. Corrected REC2 applies attributed Fröhlich–Simon/Osterwalder–Schrader reconstruction to that pressure-identified target, keeping the precise half-plane observable domain.

The historical quantitative assertions have exact-version application maps in the private research archive. The public PROOF-MAP and companion index identify the mathematical locations and the limits of this public export. Earlier in-house reviews retain their exact scope. The 26 October candidate status is historical; version-specific review is summarized in the accompanying REVIEW document. The published v1 artifacts remain unchanged.

Finite process-neuron width counts whole random processes, each with internal whole-plane randomness. The certificate is a sufficient mathematical bound, with sampling cost and necessary complexity left separate. Classical target construction, clustering and quantum reconstruction retain attribution.

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1 Introduction

1.1 The question

Can simple random building blocks be assembled into an interacting quantum field? We study this question by starting with random patterns on a plane, combining them, and changing the probabilities of the resulting patterns to introduce an interaction. The goal is to prove that their probability laws approach a specified quantum field as the approximations are removed.

The building blocks are *process neurons*: whole-plane random sign fields. In one architecture a neuron is formed from two independent random sign processes, one for each coordinate. In the other, a random division of the plane into rectangular cells receives random signs. A width-N field is a normalized sum of N such neurons. Before the interaction is introduced the neurons are independent; reweighting their joint probability law introduces the interaction. The width counts these random processes. Each process uses countably many random variables globally, so finite width does not mean a finite list of parameters encoding the entire plane.

Our target is the massive, weakly coupled two-dimensional Wick-quartic field, usually denoted φ_2^4. Here "quartic" means that the interaction involves the fourth power of the field, with the subtractions needed to define its continuum limit. We first work in a Euclidean plane. Classical reconstruction then identifies the limiting model with a relativistic quantum field in one space and one time dimension. The main construction result concerns this particular model and weak-coupling regime.

1.2 Why the looking glass?

Reflection gives a way to compare measurements on opposite halves of the plane. *Reflection positivity* requires certain averages formed from a measurement and its reflected counterpart to be nonnegative. In the reconstruction of a quantum theory, this supplies the positivity needed for the Hilbert-space inner product. The precise condition concerns the full probability law, including its interactions.

We preserve this condition across the two central coordinate lines at every finite stage of the construction. The limiting field has full Euclidean symmetry and reflection positivity across every line, provided by the classical theory. The looking glass in the title points to this mathematical role of reflection.

1.3 From random patterns to a field

There are several approximations to control. We increase the number of neurons, remove a stabilizing term, enlarge an auxiliary region, and remove a short-distance regulator. We then enlarge the physical region in which the quartic interaction acts. These are ordered limiting steps, assembled into one common sequence for both architectures.

The sequence is chosen once, before selecting measurements. For example, one may later choose smooth weighted averages of the field over bounded regions: their joint polynomial moments converge along that same sequence. The field laws themselves also converge in a space of distributions, which allows the limiting field to be rougher than an ordinary function. This is an existence result for the full sequence; it does not give an effective rule for how many neurons or how large a region achieves a chosen accuracy.

The quantitative supplement in Appendix H addresses a specified finite observation under

separately named premises. It supplies a concrete sufficient example while retaining this universal sequence's existential meaning.

The target and the classical tools have a substantial history. Fröhlich and Simon constructed its infinite-volume states and proved their regularity [8]; Albeverio, Høegh-Krohn and Zegarlinski proved a weak-coupling uniqueness theorem that identifies them [2]; Guerra, Rosen and Simon developed the free-boundary lattice approximation and the correlation inequalities used below [10]; and Simon and Griffiths proved the finite-system inequality we need [23]. Relativistic reconstruction uses the Osterwalder–Schrader theorem [19, 20] in the measure-level formulation of [8, Thm. 1.0]. These results supply the classical inputs to our architecture-specific construction. Section 14 compares that construction with earlier neural-field work.

The next subsection states the result precisely. Readers seeking an overview can then follow the three layers in Section 1.5 and the contribution summary in Section 1.6; the definitions and proofs develop the details.

1.4 The result in brief

Fix $m > 0$, a length $\ell > 0$ and $u > 0$ with $0 < u/m^2 < g_*$ (Definition 3.2). For $b \in \{\mathrm{prod}, \mathrm{Mon}\}$ let $P^b_{N,\varepsilon,\Omega,a;\Lambda}$ be the normalized annealed neural law of Definition 2.10, a Borel probability on the Polish space $X = H^{-2}_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{R})$. Theorem 4.1 states that there is one deterministic sequence $(a_n, R_n, \varepsilon_n, N_n)_{n\geq 1}$, common to both architectures and depending only on m, ℓ, u, a fixed localization system and fixed parameter enumerations, such that with the axis-parallel physical squares $\Lambda_n = (-n\ell, n\ell)^2$ and auxiliary squares $\Omega_n = [-R_n, R_n]^2$ the laws $P^b_n = P^b_{N_n,\varepsilon_n,\Omega_n,a_n;\Lambda_n}$ satisfy:

(A) weak convergence on X to the X-valued version ν of the classical target, with uniformly bounded and convergent powers of every local seminorm;

(B) convergence of all polynomial moments of every finite tuple of real compactly supported smooth tests chosen after the sequence;

(C) existence and weak distributional convergence of the joint correlation distributions of every order on compactly supported joint tests;

(D) identification of the limits with restrictions of the classical tempered Schwinger distributions;

(E) reflection positivity of every finite stage across the two central coordinate lines, for bounded half-plane observables and for finite polynomial forms in compact joint tests, with convergence of these forms.

Theorem 4.2 records, as attributed classical results with project-derived verification steps, the Euclidean invariance, translation mixing and ergodicity, all-line reflection positivity, exponential regularity, tempered Schwinger distributions, non-Gaussianity and $1+1$-dimensional Wightman reconstruction of the target.

1.5 What is proved, what is attributed

The paper has three layers.

(i) *Classical inputs* (Section 3). Published theorems C1–C5 are imported with attribution, in formulations whose hypotheses are matched to the application at the point of use, and C6

lists standard tools. Their proofs are not audited here, and the original Osterwalder–Schrader and Streater–Wightman texts were not available (Appendix D).

(ii) *Foundations proved here.* The process neurons and their finite-width central reflection positivity (Section 6, Appendix A); the Gaussian reference fields, including the finite-region mass tilt and the common Gaussian space with Borel distribution-valued versions (Section 7, Appendix B); the fixed-volume Wick theory (Section 8, Appendix C); the field-space facts (Section 5); and the fixed-volume chain of normalized laws (Section 9).

(iii) *Construction and target.* The reduced continuum lemmas (Section 10): the continuum limit with its Wick identification, the whole-field volume limit with all compact moments, and the target properties. The common sequence, the local moments and the joint correlations (Sections 11–12). The assembly (Section 13).

The main theorem assumes C1–C6 only. An earlier conditional version of this construction assumed, besides these, nine project premises; Appendix E records for each of them whether it is now a theorem of this paper or no longer used, and where the replacing argument is.

1.6 Contribution and organization

Our contribution is an architecture-specific realization and synthesis. In each of the two architectures the width-N field $F_N = N^{-1/2}\sum_{i=1}^{N} g_i$ is a normalized sum of N independent copies of a whole-plane process neuron with $|g| = 1/2$, $\mathbb{E}g = 0$ and the endpoint covariance $e^{-\mu|z|_1}/4$, $|z|_1 = |z_1| + |z_2|$, $\mu = 1/a$. On these fields we place normalized annealed Gibbs laws: the joint law of all neuron randomness is reweighted by $e^{-V(F_N)}$ and normalized, where for a centered axis-parallel physical square Λ inside an auxiliary square $\Omega = [-R, R]^2$,

$$V(F_N) = u\int_\Lambda (F_N^4 - 6c_a F_N^2 + 3c_a^2)\,dz + \varepsilon\int_{\Omega\setminus\Lambda} F_N^4\,dz - \frac{\kappa_a^2}{2}\int_\Omega F_N^2\,dz, \qquad \kappa_a^2 = a^{-2} - m^2.$$

The fixed-volume limits, the growing-volume target and the joint correlation statements above are proved for both architectures along one sequence. The infinite-volume target, its Osterwalder–Schrader properties and its reconstruction are classical results used with attribution. The reflection-positivity mechanism, the density-deformation mechanism and the endpoint covariance kernel all have prior instances (Section 14). We make no historical-priority claim.

Section 2 fixes notation and the order of the limiting stages. Section 3 states the classical inputs and fixes the admissible pair. Section 4 states the results. Sections 5–8 state the foundations, with proofs in Appendices A–C. Sections 9–12 prove the construction and Section 13 assembles it. Sections 14 and 15 discuss prior work and limitations.

2 Setting, notation and conventions

This section fixes the objects that appear in the main theorem. It introduces the model parameters and the three covariances, lists the order in which the approximations are removed (Section 2.2), and defines the field spaces in which convergence is measured, the two neuron architectures, the interaction potential and the normalized laws. A first reading can concentrate on Definitions 2.1, 2.6, 2.8 and 2.10 together with the list of stages; the remaining conventions can be consulted where they are cited.

2.1 Parameters, geometry and covariances

Points of the plane are $z = (z_1, z_2)$; $|z|$ is the Euclidean norm, $|z|_1 = |z_1| + |z_2|$ and $|z|_\infty = \max(|z_1|, |z_2|)$. In the reconstruction (Theorem 4.2) Euclidean time is z_2, following [8, p. 495]. $|A|$ is Lebesgue measure, $\mathcal{L}(\mathbb{R}^2)$ the Lebesgue σ-algebra and $\mathcal{B}(\cdot)$ a Borel σ-algebra. $\mathbb{N} = \{1, 2, 3, \dots\}$. The symbol N alone always denotes a width.

Definition 2.1 (Parameters and physical geometry)**.** Fix a mass $m > 0$, a length $\ell > 0$ and a quartic coefficient $u > 0$. A *regulator* is a real a with $0 < a < 1/m$; put $\mu = 1/a$, $\kappa_a^2 = a^{-2} - m^2 > 0$ and $r_a = 1 - a^2 m^2 \in (0, 1)$; the value $a = 0$ denotes the unregulated case. A *stabilizer* is $\varepsilon \geq 0$.

(i) An *auxiliary square* is $\Omega = [-R, R]^2$, $R > 0$.

(ii) A *physical square* inside Ω is a centered **axis-parallel** square $\Lambda = (-l, l)^2$ or $\Lambda = [-l, l]^2$ with $0 < l \leq R$. Then $\bar{\Lambda} = [-l, l]^2 \subseteq \Omega$, and both Λ and Ω are invariant under the two central coordinate reflections $\theta_1(z_1, z_2) = (-z_1, z_2)$ and $\theta_2(z_1, z_2) = (z_1, -z_2)$.

(iii) The physical squares of the main theorem are $\Lambda_n = (-n\ell, n\ell)^2$, $n \geq 1$, with closures $\bar{\Lambda}_n = [-n\ell, n\ell]^2$.

Open and closed physical squares differ by the Lebesgue-null boundary $\partial\Lambda$; every integrated object below is unchanged when Λ is replaced by $\bar{\Lambda}$ (Lemma 9.1).

The axis-parallel requirement in (ii) is a hypothesis, not a convention: centering alone does not give the two reflection invariances used for finite-stage positivity (Remark 6.8). Rotated physical squares are not part of this definition; the general positivity theorem of Section 6 applies to every region and potential that are invariant under both central reflections, whatever their shape.

The Fourier convention is $\hat{f}(p) = \int e^{-ip\cdot z} f(z)\, dz$, extended to L^2 by Plancherel, with inversion $C(z) = (2\pi)^{-2} \int e^{ip\cdot z} \hat{C}(p)\, dp$ and $\int |f|^2 = (2\pi)^{-2} \int |\hat{f}|^2$; for $g \in L^1(\mathbb{R}^2)$ we write $\mathcal{I}g(x) = (2\pi)^{-2} \int e^{ip\cdot x} g(p)\, dp$. With $p = (w, q)$,

$$\hat{C_{\text{free}}}(p) = \hat{C}_0(p) = (m^2 + |p|^2)^{-1}, \qquad \hat{C}_a(w, q) = (m^2 + w^2 + q^2 + a^2 w^2 q^2)^{-1}, \tag{1}$$
$$C_\mu^{\text{end}}(z) = \tfrac{1}{4} e^{-\mu|z|_1}, \qquad \hat{C}_\mu^{\text{end}}(w, q) = \frac{\mu^2}{(\mu^2 + w^2)(\mu^2 + q^2)}.$$

$C_{\text{free}} = (-\Delta + m^2)^{-1}$ is the free covariance, C_a the regulated physical covariance and C_μ^{end} the endpoint covariance. The Wick variance c_a is the variance $C_a(0)$ of the regulated Gaussian physical reference field, the centered Gaussian field with covariance C_a (Theorem 7.9):

$$c_a = (2\pi)^{-2} \int_{\mathbb{R}^2} \hat{C}_a(p)\, dp = (2\pi)^{-1} \int_0^\infty \left[(m^2 + q^2)(1 + a^2 q^2)\right]^{-1/2} dq \in (0, \infty). \tag{2}$$

It is strictly decreasing in a, satisfies $c_a > 1/4$ for every regulator (compare the integrand in (2) with its value at $a = 1/m$, where the right-hand side is $1/4$; see Lemma C.2(K4)), and tends to $+\infty$ as $a \downarrow 0$ by monotone convergence. The endpoint variance is $C_\mu^{\text{end}}(0) = 1/4$; the two variances are never interchanged.

2.2 The mass-shift identity and the order of stages

The neurons produce the endpoint covariance C_μ^{end}, whereas the target is built on the massive free covariance C_{free}. The identity below shows how this gap is bridged for the Gaussian references:

subtracting κ_a^2 from the reciprocal of the endpoint multiplier gives the reciprocal of the multiplier of C_a, and $1/\hat{C}_a$ differs from $1/\hat{C_{\text{free}}}$ only by the term $a^2w^2q^2$ in (1). The negative quadratic term $-T_{\Omega,a}$ of the potential (Definition 2.8) implements this shift (stages (3) and (5) below). The numbered list records the order in which the approximations are removed and the result controlling each stage.

Since $1/\hat{C}_\mu^{\text{end}} = \mu^2 + w^2 + q^2 + a^2w^2q^2$ and $a\mu = 1$,

$$\frac{1}{\hat{C}_\mu^{\text{end}}} - \kappa_a^2 = \frac{1}{\hat{C}_a}, \qquad \hat{C}_a = \frac{\hat{C}_\mu^{\text{end}}}{1 - \kappa_a^2 \hat{C}_\mu^{\text{end}}} = \sum_{k\geq 0} \kappa_a^{2k} (\hat{C}_\mu^{\text{end}})^{k+1}. \tag{3}$$

This is an identity between Fourier multipliers of *Gaussian reference covariances*. It is used in the following precise order of stages; it is not a formula for the covariance of a finite-width law or of a quartically tilted law.

(1) *Finite width.* The neural law $P^b_{N,\varepsilon,\Omega,a;\Lambda}$ reweights the bounded, non-Gaussian field F_N by the full potential V, whose negative quadratic term acts on all of Ω.

(2) *Width limit.* As $N \to \infty$ these laws converge to the full Gibbs tilt $\nu^v_{\varepsilon,\Omega,a;\Lambda}$ of the Gaussian endpoint reference G_μ (covariance C_μ^{end}) by the same potential (Proposition 9.2); we assert neither that this tilt is Gaussian nor that it is not.

(3) *Pure quadratic tilt on finite* Ω. The tilt of G_μ by $e^{T_{\Omega,a}}$ alone is the centered Gaussian law $G'_{\Omega,a}$ on the whole plane with covariance $C'_{\Omega,a}$ (Theorem 7.4), and $C_\mu^{\text{end}} \leq C'_{\Omega,a} \leq C_a$ pointwise. The full tilt is the residual quartic tilt of $G'_{\Omega,a}$ (Proposition 9.3).

(4) *Stabilizer removal* $\varepsilon \downarrow 0$ at fixed a, Ω, Λ (Proposition 9.4).

(5) *Auxiliary-volume removal* $\Omega \uparrow \mathbb{R}^2$ at fixed a, Λ: $C'_{\Omega,a} \to C_a$ locally uniformly (Theorem 7.6); this is where the multiplier (3) becomes the covariance of the Gaussian reference, and the residual laws converge to the regulated physical law $\nu_{a,\Lambda}$ (Proposition 9.5).

(6) *Regulator removal* $a \downarrow 0$ at fixed Λ (Proposition 9.6); the anisotropy $a^2w^2q^2$ of the Gaussian reference disappears.

(7) *Physical volume* $\Lambda_n \uparrow \mathbb{R}^2$, through classical continuum input (Section 10).

The interacting laws at every stage are non-Gaussian in general, and their covariances are not given by (3).

2.3 Field spaces and localization

Two spaces are needed. At every finite stage the neural field is a bounded function, so the quartic potential is an ordinary integral of a function in E. The limiting law lives on distributions (Theorem 10.10(iii)), so convergence is stated in the space X of local H^{-2} distributions. Its seminorm p_j cuts the field off with χ_j, regards the result as a periodic distribution on the torus $\mathbb{T}_j$, and measures it by a weighted Fourier sum; every compact smooth smearing $v(f)$ is controlled by one such seminorm through (16).

Definition 2.2 (Field spaces)**.** $E = L^4_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{R})$ consists of real locally fourth-power integrable functions modulo null functions, with the metric $d_E(v,w) = \sum_{j\ge1} 2^{-j}\min(1, \|v-w\|_{L^4((-j,j)^2)})$. $X = H^{-2}_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{R})$ consists of the real distributions v on $\mathbb{R}^2$ with $\chi v \in H^{-2}(\mathbb{R}^2)$ for every $\chi \in C^\infty_c(\mathbb{R}^2)$. The map $\iota : E \to X$ sends a function class v to $f \mapsto \int vf$. The *localization system* consists of:

- real cutoffs $\chi_j \in C^\infty_c(\mathbb{R}^2)$, $j \ge 1$, with $0 \le \chi_j \le 1$, $\chi_j = 1$ on $[-j,j]^2$, $\operatorname{supp}\chi_j \subset (-j-\frac12, j+\frac12)^2$ and $\chi_{j+1} = 1$ on $\operatorname{supp}\chi_j$;
- the square tori $\mathbb{T}_j$ of side $2(j+1)$, $|\mathbb{T}_j| = 4(j+1)^2$, with fundamental cell $\mathcal{C}_j = (-(j+1), j+1)^2$; a distribution supported in $\operatorname{supp}\chi_j$ is regarded as a periodic distribution on $\mathbb{T}_j$;
- the orthonormal basis $e_{j,k}(z) = |\mathbb{T}_j|^{-1/2}\exp(i\pi k\cdot z/(j+1))$, $k \in \mathbb{Z}^2$, of $L^2(\mathbb{T}_j)$, and the weights $\lambda_{j,k} = 1 + \pi^2|k|^2/(j+1)^2$, so that $(1-\Delta)\overline{e_{j,k}} = \lambda_{j,k}\overline{e_{j,k}}$ and $\lambda_{j,-k} = \lambda_{j,k}$.

For $r \in \mathbb{R}$ and a periodic distribution h on $\mathbb{T}_j$ put $\|h\|^2_{H^r(\mathbb{T}_j)} = \sum_k \lambda^r_{j,k}|h(\overline{e_{j,k}})|^2 \in [0,\infty]$. The local seminorms, the metric and the bounded-Lipschitz distance are

$$\begin{aligned} p_j(v) = \|\chi_j v\|_{H^{-2}(\mathbb{T}_j)}, \qquad d(v,w) = \sum_{j\ge1} 2^{-j}\min(1, p_j(v-w)), \\ \beta(\rho,\eta) = \sup_{|H|\le1,\ \mathrm{Lip}_d(H)\le1} |\rho(H) - \eta(H)|, \end{aligned} \tag{4}$$

where $(\chi_j v)(g) = v(\chi_j g)$ for smooth periodic g, so that $p_j(v)^2 = \sum_k \lambda^{-2}_{j,k}|v(\chi_j\overline{e_{j,k}})|^2$, with v extended complex-linearly. X carries the topology of d and its Borel σ-algebra.

Proposition 5.1 proves that E and X are Polish, that d is a complete metric, that compact smooth evaluations $v \mapsto v(f)$ are continuous and generate the Borel σ-algebra of X, that ι is a continuous injection, and that β metrizes weak convergence. A *weak law statement on* X always means convergence of $\int H\,d\rho$ for every real bounded continuous $H : X \to \mathbb{R}$.

2.4 The two process-neuron architectures

Both architectures produce a random sign field with values $\pm1/2$, mean zero and the same endpoint covariance $C^{\mathrm{end}}_\mu(z-z') = \frac14 e^{-\mu|z-z'|_1}$ (Theorems 6.1(b) and 6.4(c)). The product neuron multiplies two independent telegraph signals, one along each coordinate, each flipping at the points of a Poisson process. The Mondrian neuron colors the cells of a random rectangular partition with independent signs; two points lie in a common cell with probability $e^{-\mu|z-z'|_1}$ (Theorem 6.2(e) with $s = \mu$). Because the covariances agree, both architectures have the same Gaussian reference in the width limit (Proposition 9.2).

The neurons are defined exactly; Section 6 states their properties, proved in Appendix A. Random objects live on a probability space $(\Gamma, \mathcal{A}, \mathbb{P})$; $\mathrm{Exp}(c)$ has density ce^{-ct} on $(0,\infty)$, $U(0,1)$ is uniform on $(0,1)$, and a *uniform sign* is uniform on $\{-1,1\}$.

Definition 2.3 (Product (telegraph) neuron)**.** Let $\mathcal{N}$ be a Poisson process on $\mathbb{R}$ of intensity $\mu/2$ and ϵ an independent uniform sign. The stationary symmetric telegraph process of flip rate $\mu/2$ is $\sigma(t) = \epsilon(-1)^{\mathcal{N}((0,t])}$ for $t \ge 0$ and $\sigma(t) = \epsilon(-1)^{\mathcal{N}((t,0])}$ for $t < 0$ (set $\sigma = \epsilon$ on the null event where $\mathcal{N}$ is not locally finite or charges 0). With τ an independent copy, the product neuron is $g(z) = \frac12\sigma(z_1)\tau(z_2)$.

Definition 2.4 (Boxes, colored partitions and the canonical Mondrian)**.** A *box* is $C = [a_1, b_1] \times [c_1, d_1]$ with $a_1 \le b_1$, $c_1 \le d_1$, and half-perimeter $p(C) = (b_1 - a_1) + (d_1 - c_1)$. For a box C with nonempty interior, $\mathrm{Part}^{\mathrm{col}}(C)$ is the set of finite sets of pairs (R, ξ), R a closed box with nonempty interior and $\xi \in \{-1, 1\}$, such that the boxes R have pairwise disjoint interiors and union C; it is encoded by finite counting measures on $\mathbb{R}^4 \times \{-1, 1\}$ (Lemma A.1). For boxes $B \subseteq C$ with nonempty interiors the restriction is $\mathrm{res}_B(P) = \{(R \cap B, \xi) : (R, \xi) \in P,\ \mathrm{int}(R \cap B) \neq \emptyset\}$; images under translations and under θ_1, θ_2 act on closed boxes. The *lower-left representative* of $R = [a_1, b_1] \times [c_1, d_1]$ is $R^{\mathrm{ll}} = [a_1, b_1) \times [c_1, d_1)$.

Let Tr be the set of finite words over $\{0, 1\}$ (root $\emptyset$) and let $(E_w, D_w, X_w, \xi_w)_{w \in \mathrm{Tr}}$ be independent, $E_w \sim \mathrm{Exp}(1)$, $D_w, X_w \sim U(0, 1)$, ξ_w a uniform sign. For a box C and budget $s \ge 0$ put $R_\emptyset = C$, $b_\emptyset = 0$; a born node w with box $R_w = [a_1, b_1] \times [c_1, d_1]$ dies at $d_w = b_w + E_w / p(R_w)$ ($d_w = \infty$ if $p(R_w) = 0$). If $d_w < s$, w splits: if $D_w < (b_1 - a_1)/p(R_w)$ at the vertical line $x = a_1 + X_w(b_1 - a_1)$, otherwise at the horizontal line $y = c_1 + X_w(d_1 - c_1)$; its children $w0, w1$ receive the two closed pieces (left/lower, right/upper) and are born at d_w. Born nodes with $d_w \ge s$ are leaves. The output $\Pi^{\mathrm{can}}(C, s) = \{(R_w, \xi_w) : w \text{ a leaf}\}$ has law $M^{\mathrm{col}}(C, s)$. Thus each cell with side lengths w', h' splits at rate $w' + h'$, orthogonally to the first axis with probability $w'/(w' + h')$, at a uniform position, and the daughters evolve independently until the budget is exhausted.

Definition 2.5 (Whole-plane Mondrian neuron; closed cells and half-open representatives)**.** Let $B_n = [-n, n]^2$. A *consistent colored Mondrian tessellation* of budget μ is a sequence $P = (P_n)_{n \ge 1}$ of random elements $P_n \in \mathrm{Part}^{\mathrm{col}}(B_n)$ on one probability space with $P_n \sim M^{\mathrm{col}}(B_n, \mu)$ for every n and, almost surely, $\mathrm{res}_{B_n}(P_{n+1}) = P_n$ for all n (the *consistency event*). For $z \in \mathbb{R}^2$ let $n_z = \min\{n \ge 1 : z \in [-n, n)^2\}$, and for $n \ge n_z$ let $(R_n(z), \xi_n(z))$ be the cell of P_n with $z \in R_n(z)^{\mathrm{ll}}$. The *closed geometric cell* is $R(z) = \bigcup_{n \ge n_z} R_n(z)$ and the *half-open representative* is $H(z) = \bigcup_{n \ge n_z} R_n(z)^{\mathrm{ll}}$. The *Mondrian neuron* is the field $g(z) = \frac{1}{2}\xi_{n_z}(z)$, defined for every z and every outcome (lower-left convention). An *admissible boundary convention* is any $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable g' such that on the consistency event $g'(z) = \xi/2$ for some cell (R, ξ) of P_{n_z+1} with $z \in R$. Maps act on tessellations through closed cells; half-open representatives of an image are obtained by applying the lower-left convention to its closed cells.

Definition 2.6 (Neural fields)**.** For $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, $N \ge 1$ and independent copies $g^b_{1,a}, \dots, g^b_{N,a}$ of the neuron of Definition 2.3 or 2.5 with $\mu = 1/a$, on the product $(\Gamma_b, \mathbb{P}_b)$ of N copies of its probability space, the neural field of width N is

$$F^b_{N,a} = N^{-1/2} \sum_{i=1}^{N} g^b_{i,a}, \qquad |F^b_{N,a}| \le \sqrt{N}/2. \tag{5}$$

Remark 2.7 (What the width counts). The width N counts process neurons. Each neuron uses countably many random variables; on every bounded region it has almost surely finitely many telegraph jump lines or Mondrian cell edges, but that number is random, unbounded and grows with the region. A finite-width stage is therefore not a random field encoded by finitely many global parameters for the whole plane. For background on the Mondrian process see [22]; for stationary tessellations stable under iteration see [18]. These are background attributions; every property used below is proved in Appendix A.

2.5 Potentials, Wick and coupling conventions

The potential has three parts with separate jobs. The Wick-ordered quartic $W_{a,\Lambda}$ on the physical square is the interaction that survives in the limit; its subtraction $-6c_a x^2 + 3c_a^2$ lets the interaction

converge as $a \downarrow 0$, although $c_a \to \infty$ (Theorem 8.2(a)). The negative quadratic $-T_{\Omega,a}$ implements the mass shift of Section 2.2. The stabilizing quartic $\varepsilon J_{\Omega,\Lambda}$ on $\Omega \setminus \Lambda$ keeps the weight e^{-V} bounded, see (18), which the width limit uses; it is removed at fixed a, Ω, Λ (Proposition 9.4).

Definition 2.8 (Integrated potentials)**.** For $v \in E$, a physical square Λ, an auxiliary square $\Omega \supseteq \bar{\Lambda}$, a regulator a and $\varepsilon \geq 0$, put

$$W_{a,\Lambda}(v) = u \int_\Lambda (v^4 - 6c_a v^2 + 3c_a^2)\, dz, \qquad J_{\Omega,\Lambda}(v) = \int_{\Omega\setminus\Lambda} v^4\, dz,$$
$$T_{\Omega,a}(v) = \frac{\kappa_a^2}{2} \int_\Omega v^2\, dz, \qquad V_{\varepsilon,\Omega,a;\Lambda} = W_{a,\Lambda} + \varepsilon J_{\Omega,\Lambda} - T_{\Omega,a}. \tag{6}$$

Equivalently $V_{\varepsilon,\Omega,a;\Lambda}(v) = \int_\Omega v_V(z, v(z))\, dz$ with the Borel integrand

$$v_V(z,x) = \mathbf{1}_\Lambda(z)\, u(x^4 - 6c_a x^2 + 3c_a^2) + \varepsilon \mathbf{1}_{\Omega\setminus\Lambda}(z)\, x^4 - \tfrac{1}{2}\kappa_a^2 x^2. \tag{7}$$

The negative quadratic term acts on all of Ω, including Λ. These are integrated potentials of ordinary functions; they are never evaluated as pointwise polynomials of an arbitrary distribution.

Remark 2.9 (Wick and coupling conventions). At a positive regulator, $x^4 - 6c_a x^2 + 3c_a^2$ is the fourth Wick polynomial at the variance $c_a = C_a(0)$ of the regulated Gaussian physical reference field (the centered Gaussian field with covariance C_a): own-mass, own-regulator Wick ordering of that Gaussian reference. The endpoint variance $1/4$ is never a Wick constant, and no finite-width or interacting law is asserted to have variance c_a. At $a = 0$ the interaction is the common-space fourth-chaos L^2 Wick limit $U_{0,\Lambda}$ constructed in Section 8; its precise construction statement is Theorem 8.2(b). The coefficient u multiplies the Wick quartic with no factorial divisor, as in [8] (polynomial λX^4) and [2] ($\lambda \int :P(\phi):_0$, $P(X) = X^4$); the finite-volume neural construction of [7] uses $\lambda/4!$, so its $\lambda/24$ plays the role of u. In natural units the field is dimensionless in two dimensions; m, μ, the telegraph flip rate $\mu/2$ and the Mondrian budget μ have units of inverse length; u and κ_a^2 of inverse length squared; u/m^2 is the dimensionless coupling, and the dilation of Section 10 preserves it.

2.6 The normalized laws

Definition 2.10 introduces the laws met along the stages of Section 2.2. The neural laws (a) are the objects of the main theorem, and the full Gaussian tilt (b) is their width limit. For $\varepsilon > 0$, the residual laws (c) rewrite that tilt as a quartic tilt of the Gaussian law $G'_{\Omega,a}$; they also define its stabilizer-free version at $\varepsilon = 0$. The physical laws (d) arise when the auxiliary square is removed; at $a = 0$ they are the continuum laws on a fixed square. Item (e) names the growing-volume laws and the target. Remark 9.7 lists the four fixed-volume arrows joining (a)–(d).

The following objects are well defined by results of Sections 6–8 (finiteness and positivity of normalizers, existence of Gaussian versions and of the L^2 Wick limit); the proofs of those results, in Appendices A–C, do not use this definition.

Definition 2.10 (Laws)**.** Fix a regulator a, a physical square Λ and an auxiliary square $\Omega \supseteq \bar{\Lambda}$.

(a) *Neural laws* ($\varepsilon > 0$). With $Z_N^b = \mathbb{E}_b \exp[-V_{\varepsilon,\Omega,a;\Lambda}(F_{N,a}^b)]$,

$$P_{N,\varepsilon,\Omega,a;\Lambda}^b(B) = \frac{\mathbb{E}_b\big[\mathbf{1}_B(\iota F_{N,a}^b)\, e^{-V_{\varepsilon,\Omega,a;\Lambda}(F_{N,a}^b)}\big]}{Z_N^b}, \qquad B \in \mathcal{B}(X). \tag{8}$$

These are *annealed* laws: the weight reweights the joint law of all neuron randomness. (The same formula is meaningful at $\varepsilon = 0$ for finite N, Theorem 6.5(a); that finite-width extension is not used below.)

(b) *Full Gaussian tilt* ($\varepsilon > 0$). Let G_μ be the law on $C(\mathbb{R}^2)$ of the continuous centered Gaussian field γ_μ with covariance C_μ^{end} (Theorem 7.1). With $Z_{\mathrm{full}} = \mathbb{E}_{G_\mu} e^{-V(\gamma_\mu)}$, $\nu^v_{\varepsilon,\Omega,a;\Lambda}(B) = \mathbb{E}_{G_\mu}[\mathbf{1}_B(\iota\gamma_\mu)e^{-V(\gamma_\mu)}]/Z_{\mathrm{full}}$.

(c) *Mass tilt and residual laws* ($\varepsilon \geq 0$). Let $\mathrm{Det}_{\Omega,a} = \mathbb{E}_{G_\mu} e^{T_{\Omega,a}(\gamma_\mu)}$ and $G'_{\Omega,a} = \mathrm{Det}^{-1}_{\Omega,a} e^{T_{\Omega,a}} G_\mu$, the centered Gaussian law on $C(\mathbb{R}^2)$ with covariance $C'_{\Omega,a}$ (Theorem 7.4). Put $z_{\varepsilon,\Omega,a;\Lambda} = \mathbb{E}_{G'_{\Omega,a}} e^{-W_{a,\Lambda}-\varepsilon J_{\Omega,\Lambda}}$, $d\mathrm{Res}_{\varepsilon,\Omega,a;\Lambda} = z^{-1}_{\varepsilon,\Omega,a;\Lambda} e^{-W_{a,\Lambda}-\varepsilon J_{\Omega,\Lambda}}\, dG'_{\Omega,a}$ and $\nu^{(2)}_{\Omega,a;\Lambda} = \iota_* \mathrm{Res}_{0,\Omega,a;\Lambda}$.

(d) *Physical laws* ($0 \leq a < 1/m$). On the common Gaussian space $(\Gamma_0, \mathcal{F}_0, \mathbb{P}_0)$ of Theorem 7.7, let $\Phi_a : \Gamma_0 \to X$ be the Borel version of Theorem 7.10 and, for $a > 0$, $\tilde{\phi}_a$ the continuous modification of Theorem 7.9. Put $U_{a,\Lambda} = W_{a,\Lambda}(\tilde{\phi}_a)$ for $a > 0$, $U_{0,\Lambda} = L^2(\mathbb{P}_0)\text{-}\lim_{a\downarrow 0} U_{a,\Lambda}$ (Theorem 8.2(a)), $Z_{a,\Lambda} = \mathbb{E}_0 e^{-U_{a,\Lambda}}$ and

$$\nu_{a,\Lambda}(B) = \mathbb{E}_0\big[\mathbf{1}_B(\Phi_a)\, e^{-U_{a,\Lambda}}\big]/Z_{a,\Lambda}. \tag{9}$$

(e) *Continuum finite-volume laws and the target.* $\nu_n = \nu_{0,\Lambda_n}$. The classical finite-volume laws $\nu_{m,u,\Lambda}$ and the classical target ν^S are probabilities on the real tempered distributions $\mathcal{S}'(\mathbb{R}^2)$ with the cylinder σ-algebra $\mathcal{C}$ (Definition 10.3, Theorem 10.8); $\nu = \nu^X_\infty$ is the X-valued version of ν^S (Theorem 10.10).

Symbol	Meaning	Bounds and traps
Z^b_N	neural normalizer	$e^{-\lvert\Omega\rvert M_N} \leq Z^b_N \leq e^{\lvert\Omega\rvert M_N}$, $M_N = \sup\{\lvert v_V(z,x)\rvert : z \in \Omega, \lvert x\rvert \leq \sqrt{N}/2\}$; not ≥ 1 in general
Z_{full}	full Gaussian normalizer	$Z_{\mathrm{full}} = \mathrm{Det}_{\Omega,a}\, z_{\varepsilon,\Omega,a;\Lambda}$ for $\varepsilon > 0$; no monotonicity in Ω is asserted (Remark 9.7)
$\mathrm{Det}_{\Omega,a}$	quadratic-tilt normalizer	$1 \leq \mathrm{Det}_{\Omega,a} \leq \exp(\kappa_a^2\lvert\Omega\rvert/(8a^2m^2))$; nondecreasing in Ω
$z_{\varepsilon,\Omega,a;\Lambda}$	residual normalizer	$e^{-3u\lvert\Lambda\rvert c_a^2 - 3\varepsilon\lvert\Omega\setminus\Lambda\rvert c_a^2} \leq z_\varepsilon \leq e^{6u\lvert\Lambda\rvert c_a^2}$
$Z_{a,\Lambda}$	physical normalizer	$1 \leq Z_{a,\Lambda} \leq \Xi_1(\Lambda)$, $0 \leq a < 1/m$; physical laws only; no uniformity in Λ

Table 1: Normalizers. The bound $Z \geq 1$ holds only for the physical laws on the common Gaussian space.

2.7 Tests, distributions and pairings

Definition 2.11 (Tests and complex-linear pairings)**.** For an open set U of a Euclidean space, $\mathcal{D}(U) = C^\infty_c(U;\mathbb{C})$ carries its standard inductive-limit (LF) topology over the Fréchet spaces $\mathcal{D}_K(U)$, $K \subset U$ compact, with seminorms $\|F\|_{C^M} = \max_{\lvert\alpha\rvert\leq M} \sup \lvert\partial^\alpha F\rvert$. The dual $\mathcal{D}'(U)$ consists of continuous *complex-linear* functionals; pairings are linear in the test. A real distribution v extends by $v(f+ig) = v(f) + iv(g)$. For $f_1, \ldots, f_k \in \mathcal{D}(\mathbb{R}^2)$ put $(f_1 \otimes \cdots \otimes f_k)(z_1, \ldots, z_k) = \prod_r f_r(z_r)$. For

$F \in \mathcal{D}(((-j,j)^2)^k)$, regarded as a smooth periodic function on $\mathbb{T}_j^k$ after extension by zero,

$$A_{j,k}(F) = \Big\| \prod_{r=1}^{k} (1 - \Delta_{z_r}) F \Big\|_{L^2(\mathbb{T}_j^k)}. \tag{10}$$

2.8 Reflections and half-plane observables

The positivity statements compare an observable with its mirror image. An observable is Σ_+^θ-measurable when it depends only on smearings of the field by tests supported in the open half-plane H_+^θ; then $F(\theta v)$ is the same observable applied to the reflected field, and so depends on the opposite half-plane. Inequality (11) requires the average of $\overline{F(\theta v)}F(v)$ to be nonnegative. The form $\Pi_\rho(f)$ is the analogous expression for polynomial observables built from compact joint tests, used in clause (E2) of Theorem 4.1.

Definition 2.12 (Central reflections, half-planes and positivity)**.** For $\theta \in \{\theta_1, \theta_2\}$ let H_+^θ be the open positive half-plane ($\{z_1 > 0\}$ for θ_1, $\{z_2 > 0\}$ for θ_2), define $(\theta v)(f) = v(f \circ \theta)$ for $v \in X$, and let Σ_+^θ be the σ-algebra on X generated by the evaluations $v \mapsto v(f)$, real $f \in C_c^\infty(H_+^\theta)$. A Borel probability ρ on X is *reflection positive across θ for bounded observables* if

$$\int_X \overline{F(\theta v)}\, F(v)\, \rho(dv) \geq 0 \tag{11}$$

for every bounded complex Σ_+^θ-measurable F. For $J \geq 0$, $f_0 \in \mathbb{C}$ and $f_r \in \mathcal{D}((H_+^\theta)^r)$, $1 \leq r \leq J$, put $(\theta_r \overline{f_r})(z_1, \dots, z_r) = \overline{f_r(\theta z_1, \dots, \theta z_r)}$ (value $\overline{f_0}$ at $r = 0$) and

$$\mathcal{O}_f(v) = \sum_{r=0}^{J} v^{\otimes r}(f_r), \qquad \Pi_\rho(f) = \sum_{r,s=0}^{J} \mathcal{S}_{r+s}^\rho((\theta_r \overline{f_r}) \otimes f_s), \tag{12}$$

where $v^{\otimes r}$ and $\mathcal{S}_k^\rho$ are the tensor evaluations and correlation distributions of Section 12, $v^{\otimes 0}(c) = c$ and $\mathcal{S}_0^\rho(c) = c$.

3 Classical inputs

The inputs answer distinct questions. C1 supplies the infinite-volume states, their Osterwalder–Schrader properties and local regularity, and convergence of free-boundary finite-volume laws once the pressure is differentiable. C2 gives uniqueness at weak coupling, which makes the two states coincide; Definition 3.2 turns its existential constants into the regime $0 < u/m^2 < g_*$. C3 provides reconstruction and a kernel theorem; C4 provides lattice approximation and correlation inequalities, and C5 a third-cumulant inequality, both used for the volume-uniform bounds of Theorem 10.10. C6 lists standard tools.

The following published results are used with attribution. Their proofs are not audited here. Each formulation records exactly the content used; its application to the present objects is verified at the point of use. Page references are to the printed pages; Appendix D records for each page whether the classical-input review read it as a page image or as supplied text.

Conventions shared by C1–C4. For a mass $M > 0$, P_M is the centered Gaussian probability on $(\mathcal{S}'(\mathbb{R}^2), \mathcal{C})$ with covariance $\langle f, (-\Delta + M^2)^{-1} g\rangle$ (the free measure μ_0 of [8, p. 496]); for open U, $\mathcal{C}_U$ is generated by $\psi \mapsto \psi(f)$ with real $f \in \mathcal{S}$, $\operatorname{supp} f \subset U$. The interaction is $\lambda \int_\Lambda :\phi^4:$ with whole-plane,

same-mass Wick ordering and no factorial; "free boundary" means the whole-plane Gaussian with the interaction restricted to the bounded region and zero external field [8, pp. 496–497], [2, (4), p. 685], [10, (IX.5a), pp. 311–312]. In these sources $:\phi^4:$ denotes Gaussian Wick normal ordering at the whole-plane reference covariance; for degree four this is the fourth-chaos variable realized as the L^2 limit of own-variance Wick quartics of mollified fields. Lemma 10.1 proves that this classical variable exists and has positive finite exponential normalizers using only the free Gaussian covariance and C6. The source finite-volume convention is applied only after these provisos are proved. The separate identification with the common-space regulator limit is Proposition 10.4; it is not an input to the restricted classical route.

Classical input C1 (Fröhlich–Simon)**.** Fix $M > 0$ and $P(X) = \lambda X^4$, $\lambda > 0$ [8].

(i) [8, Thm. 4.1, p. 509]: the Section-4 limits ν_+ and ν_- at zero external field are probabilities on $(\mathcal{S}', \mathcal{C})$ satisfying all Osterwalder–Schrader axioms except a priori clustering; used are proper Euclidean invariance and Osterwalder–Schrader positivity, in the measure form of [8, p. 495, (a)–(b)] or for moments of positive-time (possibly time-ordered) tests; both readings are treated in Section 10.

(ii) [8, §6 definition, pp. 513–514, Remarks 1–2 and Thm. 6.1; Thm. 7.2, p. 517; (1.2b), p. 498]: ν_+ and ν_- are ultraregular: for every bounded open B, the restriction of $\nu_\pm$ to $\mathcal{C}_B$ has a density f_B with respect to P_M lying in every $L^p(P_M)$, $p < \infty$, and $f_B = e^{-U_B} e^{-\alpha_\infty |B|} g_{\partial B}$ with $g_{\partial B} > 0$ almost everywhere, in every L^p, and measurable for the boundary σ-algebra, which by [8, p. 514, Remark 2] is either the intersection of $\mathcal{C}_R$ over open $R \supseteq \partial B$, or generated by $\phi(f)$ with $f \in H_{-1}$ of compact support in ∂B. Perimeter estimates are stated for rectangles.

(iii) [8, Cor. 4.3(ii), p. 510]: if $\nu_+ = \nu_-$, the pressure $\alpha_\infty(P - hX)$ is differentiable at $h = 0$ (only this implication is used).

(iv) [8, Thm. 5.2, p. 512]: if the pressure is differentiable at $h = 0$, the classical boundary-condition finite-volume measures, including the free ones, converge as Λ increases to $\mathbb{R}^2$ (ordinary exhaustion, not the iterated convention of [8, p. 503]) to $\nu_+ = \nu_-$. It is used only along centered squares, and only for the characteristic functions of finitely many compact smooth tests; the conclusion is the same under the moment or Laplace-functional readings of convergence (Lemma 10.6).

The two boundary algebras in C1(ii) are denoted BC1 (the neighborhood-intersection algebra) and BC2 (the boundary-supported H_{-1} algebra, with the completion specified below). The next remark specifies the two positivity readings.

Remark 3.1 (Literal alternative readings). The positivity alternatives above are retained separately. In the measure reading (b_M), the inequality is for every complex $F \in L^2(\mathcal{S}', \Sigma_+, \nu_\pm)$. In the weaker reading (b_P), it is for every polynomial $F = \sum_r c_r \prod_{i=1}^{n_r} \psi(f_{r,i})$, $c_r \in \mathbb{C}$, $n_r \geq 0$, real Schwartz $f_{r,i}$, whose individual product supports lie in $0 < (z_1)_2 < \cdots < (z_{n_r})_2$; empty products are constants. For C1(iv), the alternatives are convergence of the characteristic value for every real compact g, convergence of every moment of $\psi(g)$, or convergence of $\mathbb{E}e^{t\psi(g)}$ for every real t. The first supplies all frequencies by replacing g with tg. In BC2 the boundary algebra is completed by P_M-null sets; its variables for boundary-supported real H^{-1} tests are defined by their Gaussian L^2 limits of Schwartz approximants. Both BC1 and BC2, both positivity readings and all three convergence readings are treated below.

Classical input C2 (Albeverio–Høegh-Krohn–Zegarlinski)**.** [2] work on a set $\Omega_A \subset \mathcal{S}'(\mathbb{R}^2)$ with the traces of the weak-Borel sets; for closed D, $\mathcal{B}_D = \bigcap_{\Lambda' \supseteq D \text{ open}} \mathcal{B}_{\Lambda'}$ and $\Sigma_{\Lambda^c} = \mathcal{B}_{\Lambda^c} \cap \Omega_A$ [2, p. 684]; the specification consists of kernels $E^\eta_{\Lambda^c}$ with (a) each $E^\eta_{\Lambda^c}$ a probability with $E^\eta_{\Lambda^c}(F) = F(\eta)$ for Σ_{Λ^c}-measurable F, (b) $\eta \mapsto E^\eta_{\Lambda^c}(F)$ Σ_{Λ^c}-measurable, (c) compatibility; a Gibbs measure satisfies $\mu(E_{\Lambda^c}F) = \mu(F)$ for all $F \in L^1(\mu)$ [2, (7)–(8), p. 686]; $\mathcal{G}_{\mathrm{ur}}$ is the ultraregular Gibbs class [2, (10)–(12), p. 686].

(i) *Membership* [2, pp. 686–687], citing [8, Thm. 7.2]: both states ν_+ and ν_- of C1 at mass m_0 belong to $\mathcal{G}_{\mathrm{ur}}$ ([2] prints the maximal measures without sign suffix; [8, Thm. 7.2] names both signs).

(ii) *Regime and uniqueness* [2, Theorem, p. 687; after (47), p. 691]: there are $M_0, \lambda_0 > 0$ such that for every $m_0 > M_0$ and $0 < \lambda < \lambda_0$ the class $\mathcal{G}_{\mathrm{ur}}$ for λX^4 at mass m_0 is a singleton.

(iii) *Boundary-condition independence* [2, (13)–(14), p. 687; completed p. 691]: in that regime, for $\mu, \tilde{\mu} \in \mathcal{G}_{\mathrm{ur}}$ and every polynomial function F measurable for Σ_{Λ_0}, Λ_0 bounded,

$$\lim_{\Lambda \in F_0} (\mu \otimes \tilde{\mu}) \big| E^\eta_{\Lambda^c} F - E^{\tilde{\eta}}_{\Lambda^c} F \big| = 0,$$

where F_0 is the increasing sequence of centered squares. "Polynomial function" is read as $Q(\phi(f_1), \dots, \phi(f_k))$ with Q a real polynomial and real $f_i \in \mathcal{S}$ supported in a bounded open Λ_0; this reading is supported by [2, p. 689, after (31)], which localizes F in a finite union of unit cubes (Appendix D).

Definition 3.2 (Fixed admissible pair, threshold and regime)**.** Fix once and for all one pair (M_0, λ_0) of C2(ii), a reference mass $M_{\mathrm{ref}} > M_0$ and $0 < \lambda_* \le \lambda_0$, and put $g_* = \lambda_*/M^2_{\mathrm{ref}} > 0$. The *regime* consists of the pairs (m, u) with $m > 0$, $u > 0$ and $0 < u/m^2 < g_*$. This instantiates an existential statement: no numerical value of M_{ref}, λ_* or g_* is supplied and no larger coupling range is inferred. For (m, u) in the regime put $s = m/M_{\mathrm{ref}}$ and $\lambda = u/s^2 = uM^2_{\mathrm{ref}}/m^2$; then $0 < \lambda < \lambda_*$.

Classical input C3 (Reconstruction and tempered kernels)**.** (i) [8, Thm. 1.0, first assertion, and conditions (a)–(d), p. 495]: let μ be a probability on $(\mathcal{S}'(\mathbb{R}^2), \mathcal{C})$ with (a) invariance under proper Euclidean motions; (b) $\int \overline{\Theta F}\, F \, d\mu \ge 0$ for Σ_+-measurable F, where Θ reflects the last coordinate z_2 and Σ_+ is generated by $\phi(f)$, $f \in \mathcal{S}$, $\operatorname{supp} f \subset \{z_2 > 0\}$, and the inequality is required for every $F \in L^2(\Sigma_+, \mu)$; (c) a norm $|||\cdot|||$ continuous on $\mathcal{S}$ such that $f \mapsto \int e^{\phi(f)} d\mu$ is uniformly bounded and $|||\cdot|||$-continuous on $\{|||f||| \le 1\}$; (d) ergodicity of the translation group of $\mathbb{R}^2$. Then the moments of μ are the Schwinger functions of a unique relativistic quantum field theory satisfying all the Wightman axioms, listed on [8, p. 494] as temperedness, Poincaré invariance, positivity, spectrum condition, locality and clustering; by [8, p. 495, Remark 2], (d) implies uniqueness of the physical vacuum. [8] states this as a consequence of [19, 20], whose bodies were not available.

(ii) [8, p. 494]: the Wightman functions are boundary values of functions holomorphic in the permuted extended tube, which contains the Euclidean points with pairwise distinct arguments; the Schwinger functions are their values there and coincide with the moments. Uniqueness of a function holomorphic in a tube with vanishing boundary values is used as a standard attributed fact [24] (not available).

(iii) [17, Lecture 15, p. 2, Thm. 1]: $K \in \mathcal{S}'(\mathbb{R}^N \times \mathbb{R}^n)$ corresponds bijectively to linear maps $L : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^N)$ with $\psi \mapsto (L\psi)(\varphi)$ continuous for each φ, through $(L\psi)(\varphi) = K(\varphi \otimes \psi)$; the pairing is bilinear.

Not used: [8, Thm. 4.4], an identification of an Osterwalder–Schrader semigroup with a Wightman energy, the spectrum condition as an input, and tube uniqueness as an input to (d). Clustering, ergodicity and the vacuum statements of the target are obtained in the order of Theorem 10.16: Euclidean mixing first, reconstruction afterwards.

Below, C3(ii') is shorthand for the tube-uniqueness assertion contained in C3(ii).

Classical input C4 (Guerra–Rosen–Simon, free boundary condition)**.** Fix $m, u > 0$ and a centered square Λ of side L; take meshes $\delta = L/(2k+1)$, so that $\partial\Lambda$ lies midway between lattice sites [10, §IX.1, pp. 309–310]. Let $I_\delta = \{i \in \mathbb{Z}^2 : \delta i \in \Lambda\}$, $A_\delta = (m^2\delta^2 + 4)I -$ adjacency on $\mathbb{Z}^2$ [10, (IX.2)], $K_\delta = (A_\delta^{-1})|_{I_\delta \times I_\delta}$ and $B_\delta = K_\delta^{-1}$ (the free precision, [10, (IX.4c), p. 311], there written A_Λ^F), whose off-diagonal entries are nonpositive; the diagonal c_δ of K_δ is the whole-lattice variance. The normalized lattice law has density proportional to $\exp[-\frac{1}{2}q^T B_\delta q - u\delta^2 \sum_{i\in I_\delta}(q_i^4 - 6c_\delta q_i^2 + 3c_\delta^2)]$ [10, pp. 311–312]. For real $h \in C_c^\infty(\Lambda)$ put $Y_\delta(h) = \delta^2 \sum_i h(\delta i) q_i$ [10, (IX.19), p. 316].

(i) *Fixed-volume convergence* [10, (IX.19)–(IX.21), p. 316; Lemmas IX.3 (with the remark covering the sharp indicator of Λ) and IX.4, p. 317; review to p. 319]: on a common Gaussian realization of mass m, the lattice smearings converge to the continuum smearings and the lattice interaction densities to the continuum density $e^{-u\int_\Lambda :\phi^4:}$ in every finite Gaussian L^p, with positive normalizer limits; the smeared Schwinger functions converge.

(ii) *Correlation inequalities* [10, §IX.2(i),(iii), p. 324], boundary condition F, zero external field: for nonnegative real $h_1, \dots, h_r \in C_c^\infty(\Lambda)$, $\mathbb{E}_{\nu_\Lambda} \prod_i \phi(h_i) \geq 0$, and for nonnegative $g \in C_c^\infty(\Lambda)$ the truncated four-point function smeared with g is nonpositive.

Not used: [10, Thm. IX.5, p. 320] (periodic and Neumann) and Thm. IX.7, and any free–Neumann comparison. *Attribution limit.* [10, pp. 316–319] review the free case using its references [29], [62] and [28], and sketch a uniform exponential estimate (p. 317); these antecedents were not available, and the complete classical proof was not audited.

Classical input C5 (Simon–Griffiths)**.** [23, Thm. 1, p. 148; (4) and Thm. 2, (5), p. 151]: for finitely many real variables with density proportional to $\prod_i F_i(s_i) \exp(\sum_{i\neq j} a_{ij} s_i s_j + \sum_i h_i s_i)$, where $F_i(s) = e^{-a_i s^4 - b_i s^2}$ with $a_i > 0$ and $b_i \in \mathbb{R}$, the ordered-pair coefficients satisfy $a_{ij} > 0$, and $h_i \geq 0$, every mixed third cumulant $\langle s_i s_j s_k\rangle + 2\langle s_i\rangle\langle s_j\rangle\langle s_k\rangle - \langle s_i\rangle\langle s_j s_k\rangle - \langle s_j\rangle\langle s_i s_k\rangle - \langle s_k\rangle\langle s_i s_j\rangle$ (repeated indices allowed) is nonpositive. This is a finite-system statement.

Classical input C6 (Standard tools)**.** Standard analysis and probability, non-exhaustively and always with hypotheses checked where used, including: Fubini–Tonelli, dominated and monotone convergence, Fatou, Hölder, Minkowski, Jensen (also conditional), Markov, Chernoff, Paley–Zygmund, the portmanteau and continuous-mapping theorems, Lévy's continuity theorem, the method of moments, Prokhorov's theorem on Polish spaces, Dynkin's π–λ and the monotone-class theorems, Doob–Dynkin factorization and countable dependence of functions measurable for generated σ-algebras, the Lusin–Souslin Borel-image theorem, conditional expectations, martingale convergence for increasing filtrations, polynomial density in L^2 of a finite-dimensional law with finite exponential moments, Fourier series and Parseval on tori and Fejér kernels, compact Sobolev embeddings on a fixed torus, Banach–Steinhaus on Fréchet spaces, density of finite sums of product tests, Gaussian likelihoods and the Cameron–Martin shift for finite-dimensional Gaussian vectors, the Hermite expansion, real and complex L^2 duality on finite measure spaces, Bochner integration in L^2, approximate identities, the layer-cake formula, uniqueness of Fourier transforms of finite measures, and the Schur product theorem. The following standard formulations are stated explicitly:

(a) *Hypercontractivity* (Nelson): for Y in the k-th homogeneous Wiener chaos of a Gaussian Hilbert space and real $r \geq 2$, $\|Y\|_r \leq (r-1)^{k/2}\|Y\|_2$.

(b) *Wick products and chaos*: for centered jointly Gaussian ξ_i, η_i, $\mathbb{E}[:\xi_1 \cdots \xi_k: :\eta_1 \cdots \eta_l:] = \delta_{kl} \sum_{\pi \in S_k} \prod_i \mathbb{E}[\xi_i \eta_{\pi(i)}]$; the k-th chaos is the closed span of $:\xi_1 \cdots \xi_k:$, chaoses of different order are orthogonal, chaos of order ≥ 1 has mean zero, and finite products of jointly Gaussian variables with dense span generate a dense subspace of each chaos.

(c) *Hausdorff–Young and Plancherel* on $\mathbb{R}^2$ with the stated Fourier convention.

(d) *Continuity criterion* (Kolmogorov–Chentsov–Loève): a real process on $\mathbb{R}^d$ with $\mathbb{E}|Y(z) - Y(z')|^\alpha \leq c|z - z'|^{d+\beta}$ has a version with continuous paths for every outcome.

(e) *Kolmogorov extension* for consistent finite-dimensional laws, in particular centered Gaussian processes with a given positive semidefinite kernel, and for sequences of standard Borel spaces.

(f) *Isonormal processes*: every separable real Hilbert space carries an isonormal process.

(g) *Operators*: Neumann series; Hilbert–Schmidt operators and trace class ($\operatorname{tr} T^*T = \|T\|_{\mathrm{HS}}^2$); the spectral theorem for compact self-adjoint operators.

(h) *Point processes*: restrictions of a homogeneous Poisson process on $\mathbb{R}$ to disjoint Borel sets are independent; $t \mapsto -t$ preserves its law; a fixed point is almost surely not an atom; $\mathbb{E}(-1)^{\mathrm{Poi}(c)} = e^{-2c}$. Finite counting measures on a Polish space form a standard Borel space.

4 Main results

Two devices keep the main theorem compact. The enumerations (13) fix in advance the countable sets from which the regulator, the auxiliary half-side, the stabilizer and the width are chosen. The discrepancy dist_n of (14) adds to the bounded-Lipschitz distance β the differences in expected values of the first n powers of the first n local seminorms, so one number controls weak closeness and moment closeness at stage n. In Theorem 4.1, clause (S) asserts the sequence, (0) that its laws are well defined, (A)–(D) the convergence and identification statements, and (E) finite-stage positivity.

Fix once and for all the parameter enumerations

$$a(k) = \frac{1}{2mk}, \qquad R(k) = k, \qquad \varepsilon(k) = \frac{1}{k}, \qquad N(k) = k, \qquad k \in \mathbb{N}. \tag{13}$$

For $n \geq 1$ and probabilities ρ, η on X with finite moments of p_j^r for $j, r \leq n$, put

$$\mathrm{dist}_n(\rho, \eta) = \beta(\rho, \eta) + \sum_{j=1}^{n} \sum_{r=1}^{n} \big|\rho(p_j^r) - \eta(p_j^r)\big|. \tag{14}$$

Theorem 4.1 (Main theorem)**.** *Fix the admissible pair of Definition 3.2, and $m > 0$, $\ell > 0$, $u > 0$ with $0 < u/m^2 < g_*$. Assume the classical inputs C1–C5 in the formulations of Section 3 and the standard tools C6. No other hypothesis is made. Fix the localization system of Definition 2.2 and the enumerations* (13)*.*

(S) One sequence. *There is one deterministic sequence $(a_n, R_n, \varepsilon_n, N_n)_{n \geq 1}$, determined by m, ℓ, u, the localization system and the enumerations alone, and the same for both architectures, with*

$$0 < a_n < \min(1/n, 1/(2m)), \quad a_n < a_{n-1}/2; \qquad R_n \in \mathbb{N}, \quad R_n \geq \max(n, n\ell), \quad R_n > R_{n-1};$$

$$0<\varepsilon_n<1/n,\quad \varepsilon_n<\varepsilon_{n-1}/2;\qquad N_n\in\mathbb{N},\quad N_n>n,\quad N_n>N_{n-1}$$

(conditions with index $n-1$ only for $n>1$), such that with $\Omega_n=[-R_n,R_n]^2$ and $P_n^b=P^b_{N_n,\varepsilon_n,\Omega_n,a_n;\Lambda_n}$,

$$\operatorname{dist}_n(P_n^b,\nu_n)<2^{-n}\qquad \text{for both } b \text{ and every } n\ge 1. \tag{15}$$

Apart from the discrepancies dist_n, which are determined by the localization system, no observable, test function, polynomial or functional enters this choice; in particular none of the objects quantified in (A)–(E) does. The physical index n is never passed to a subsequence.

(0) Laws. *Every P_n^b is the Borel probability* (8) *on X with positive finite normalizer, and every power of every p_j is integrable under it. The limit ν is the X-valued version of the classical target ν^S.*

(A) Whole field and local moments. *For both b, $P_n^b\Rightarrow\nu$ weakly on X; for all $j,r\ge 1$, $\sup_{n\ge1,b}P_n^b(p_j^r)<\infty$ and $P_n^b(p_j^r)\to\nu(p_j^r)$.*

(B) Every later compact polynomial tuple. *For every $k\ge1$, all real $f_1,\dots,f_k\in C_c^\infty(\mathbb{R}^2)$ (repetitions allowed) and every real or complex polynomial Q on $\mathbb{R}^k$ of any degree, all expectations below are absolutely finite,*

$$\mathbb{E}_{P_n^b}Q(v(f_1),\dots,v(f_k))\to\mathbb{E}_\nu Q(v(f_1),\dots,v(f_k)),\qquad \sup_{n,b}\mathbb{E}_{P_n^b}|Q(v(f_1),\dots,v(f_k))|^q<\infty\ (q>0),$$

the same powers are ν-integrable, and the laws of $(v(f_1),\dots,v(f_k))$ converge weakly.

(C) Joint compact-test correlations. *Put $\mathcal{L}=\{\nu\}\cup\{P_n^b:n\ge1,\ b\in\{\mathrm{prod},\mathrm{Mon}\}\}$ and $M_{j,r}=\sup_{\rho\in\mathcal{L}}\int p_j^r\,d\rho$, $M_{j,0}=1$. For every $k\ge1$:*

- *(C1) for every $v\in X$ there is a unique $v^{\otimes k}\in\mathcal{D}'((\mathbb{R}^2)^k)$ with $v^{\otimes k}(f_1\otimes\cdots\otimes f_k)=\prod_r v(f_r)$ for $f_r\in\mathcal{D}(\mathbb{R}^2)$; it is symmetric, real in the sense $\overline{v^{\otimes k}(F)}=v^{\otimes k}(\overline{F})$, $v\mapsto v^{\otimes k}(F)$ is continuous on X, and for $\operatorname{supp}F\subset((-j,j)^2)^k$, $|v^{\otimes k}(F)|\le p_j(v)^kA_{j,k}(F)\le 3^k|\mathbb{T}_j|^{k/2}p_j(v)^k\|F\|_{C^{2k}}$;*
- *(C2) for $\rho\in\mathcal{L}$, $\mathcal{S}_k^\rho(F)=\int_X v^{\otimes k}(F)\,\rho(dv)$ is absolutely defined, lies in $\mathcal{D}'((\mathbb{R}^2)^k)$, is the unique distribution extending $(f_1,\dots,f_k)\mapsto\int\prod_r v(f_r)\,\rho(dv)$, and $\sup_{\rho\in\mathcal{L}}|\mathcal{S}_k^\rho(F)|\le M_{j,k}A_{j,k}(F)$ for $\operatorname{supp}F\subset((-j,j)^2)^k$, with $M_{j,k}<\infty$;*
- *(C3) with $\mathcal{S}_{n,k}^b=\mathcal{S}_k^{P_n^b}$ and $\mathcal{S}_{\infty,k}=\mathcal{S}_k^\nu$, $\mathcal{S}_{n,k}^b(F)\to\mathcal{S}_{\infty,k}(F)$ for every $F\in\mathcal{D}((\mathbb{R}^2)^k)$, that is, in $\sigma(\mathcal{D}'((\mathbb{R}^2)^k),\mathcal{D}((\mathbb{R}^2)^k))$.*

(D) Target identification. *$\mathcal{S}_{\infty,k}$ is the restriction to $\mathcal{D}((\mathbb{R}^2)^k)$ of the tempered Schwinger distribution Schw_k of ν^S (Theorem 10.16(v)).*

(E) Finite-stage central reflection positivity. *Fix $\theta\in\{\theta_1,\theta_2\}$.*

- *(E1) Every P_n^b satisfies* (11) *for every bounded complex Σ_+^θ-measurable F.*
- *(E2) For every $J\ge0$, $f_0\in\mathbb{C}$ and $f_r\in\mathcal{D}((H_+^\theta)^r)$, and every $\rho\in\mathcal{L}$: $\mathcal{O}_f$ and $\mathcal{O}_f\circ\theta$ lie in $L^2(\rho)$ with separate bounds, every term of $\Pi_\rho(f)$ is absolutely finite and $\Pi_\rho(f)=\int\overline{\mathcal{O}_f(\theta v)}\mathcal{O}_f(v)\,\rho(dv)$; for $\rho=P_n^b$ it is a nonnegative real number, and $\Pi_{P_n^b}(f)\to\Pi_\nu(f)\ge0$.*

Theorem 4.2 (Target properties and reconstruction; attributed classical results with verified hypotheses)**.** *Fix the admissible pair and (m, u) in the regime. Assume C1–C3 and C6. The target ν^S of Theorem 10.8 has the properties (i)–(vii) of Theorem 10.16: Euclidean and field-sign invariance; reflection positivity across every line for square-integrable positive-half-plane observables; strong mixing under translations, hence clustering in this Euclidean sense and translation ergodicity; the exponential regularity of C3(i)(c) for an explicit Schwartz-continuous norm; tempered symmetric Schwinger distributions; non-Gaussianity with local $L^2(B^n)$ kernels and a nonzero even truncated Schwinger distribution of order at least four on tests in disjoint Euclidean-time slabs; and, by [8, Thm. 1.0] applied only after ergodicity has been established, the scalar relativistic Wightman theory in $1+1$ dimensions that it determines, with the Wightman axioms listed in C3(i), a unique physical vacuum, and a field that fails Gaussian factorization. The classical existence and reconstruction theorems are attributed; the verification of their hypotheses for this target is project-derived and appears in Section 10.*

Every property of Theorem 4.2 is proved along the restricted classical route in Lemma 10.1, the source portion of Theorem 10.8, Lemmas 10.13–10.15 and Theorem 10.16. This route uses only C1–C3 and C6; the separate common-space identification for the neural construction uses the Gaussian and Wick foundations. Sign symmetry, including its use in the even-cumulant witness, has the same restricted source route.

Remark 4.3 (Quantifier order; one unchanged sequence)*.* The order of quantifiers is: fix the admissible pair, hence g_* and the regime; fix (m, ℓ, u) in the regime; fix the localization system and the enumerations; then the sequence of (S) is determined, once, for both architectures. Only afterwards are the following quantified, each independently and without reselection of parameters: every $H \in C_b(X)$; every j, r; every k, every finite real compact smooth tuple and every real or complex polynomial of any degree; every $q > 0$; every order k and every joint test $F \in \mathcal{D}((\mathbb{R}^2)^k)$; every reflection θ, every J and every list f. Clauses (A)–(E) use exactly the sequence of (S), with no subsequence. No countability restriction is placed on the later objects. The constants $M_{j,k}$ are suprema over all stages and over ν; they are finite by Proposition 11.9 and do not depend on n, b or F.

Remark 4.4 (What is not claimed)*.* No effective end-to-end rate or schedule for the full chain of limits and no parameter cost: the $\sqrt{a}$ bounds of Theorem 8.2(d),(f) and Proposition 9.6 are upper bounds at fixed physical volume, with volume-dependent constants, and they give no rate for the width, stabilizer, auxiliary-volume or physical-volume limits. No numerical value of g_*, and no algorithm deciding the selection predicates. No convergence on Schwartz or other noncompact tests at the neural stages, no strong-dual $\mathcal{S}'$ convergence or tightness, no compact-L^2 or bounded Borel local tests, no local total variation or uniform density bound, and no interchange of the ordered limits. No finite-stage Euclidean invariance or clustering statement and no finite-stage reflection positivity across lines other than the two central coordinate lines; nothing is asserted about those other lines either way. No bound uniform over all neural parameter choices: uniformity holds along the selected sequence only. No independent neural-estimate construction of the thermodynamic limit (the volume limit uses classical continuum input), no reconstruction of the neural approximants, no new interaction, dimension or architecture class, no scattering statement, no quantitative mass gap, no computational advantage, no training or sampling procedure, and no priority. "Interacting" means the non-Gaussian quantum field of Theorem 4.2.

5 Field spaces

Sections 5–8 contain the foundations; none of them uses an interacting law as an input (Remark 13.2). This section establishes that X supports the probabilistic arguments used later. It is Polish, so Prokhorov's theorem applies and β metrizes weak convergence; compact smooth smearings are continuous and controlled by one seminorm through (16); the potentials are continuous on E, the reflections on X, and the map $L : C(\mathbb{R}^2) \to X$ for uniform convergence on compacts. Readers mainly interested in the construction can treat Proposition 5.1 as a toolbox and return to it where it is cited.

Proposition 5.1 (Field-space facts)**.** *(i) (E, d_E) is Polish. If $F : \mathbb{R}^2 \times \Gamma \to \mathbb{R}$ is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable and every path is locally fourth-power integrable (for instance $|F| \le r$), then $\omega \mapsto [F(\cdot, \omega)]$ is a Borel E-valued random element.*

(ii) For $v \in X$ and $j \ge 1$, $p_j(v) < \infty$; d is a complete separable metric on X, so X is Polish.

(iii) Duality. *For $v \in X$, $j \ge 1$ and $f \in C_c^\infty((-j, j)^2; \mathbb{C})$, with $\|f\|_{H^2(\mathbb{T}_j)}$ computed for the zero extension,*

$$|v(f)| \le p_j(v)\, \|f\|_{H^2(\mathbb{T}_j)}. \tag{16}$$

Hence every compact smooth evaluation is d-continuous. The Borel σ-algebra of X equals the σ-algebra generated by the compact smooth evaluations, and equals the one generated by the countably many real and imaginary parts of $v \mapsto v(\chi_j \overline{e_{j,k}})$, $j \ge 1$, $k \in \mathbb{Z}^2$.

(iv) $\iota : E \to X$ is a continuous injection.

(v) β metrizes weak convergence of Borel probabilities on X.

(vi) For bounded measurable $D \subset \mathbb{R}^2$ and $k \in \{2, 4\}$, $v \mapsto \int_D v^k$ is continuous on E; hence $W_{a,\Lambda}$, $J_{\Omega,\Lambda}$, $T_{\Omega,a}$, $V_{\varepsilon,\Omega,a;\Lambda}$ and their exponentials are continuous on E, and also on $C(\mathbb{R}^2)$ with the topology of uniform convergence on compacts.

(vii) Reflections. *For $\theta \in \{\theta_1, \theta_2\}$ and $v \in X$, $\theta v \in X$, $\theta\theta v = v$, $(\theta v)(f) = v(f \circ \theta)$, and $p_j(\theta v) \le C_j p_{j+1}(v)$ with $C_j = \|\chi_j\|_\infty + 2\sum_{a=1,2} \|\partial_a \chi_j\|_\infty + \|\Delta\chi_j\|_\infty$; in particular θ is continuous on X. For $w \in E$, $\theta\iota w = \iota(w \circ \theta)$.*

(viii) The map L. *$(LG)(f) = \int G\,(1 - \Delta)f$ defines a linear map $L : C(\mathbb{R}^2) \to X$ (the distribution $(1 - \Delta)G$) with $p_j(LG) \le (2j + 1)C_j \sup_{[-j-1/2, j+1/2]^2} |G|$; L is continuous for uniform convergence on compacts.*

(ix) A countable separating family. *Let $\mathcal{D}_0$ be the set of real and imaginary parts of $\chi_n P$, $n \ge 1$, P a trigonometric polynomial on $\mathbb{T}_n$ with coefficients in $\mathbb{Q} + i\mathbb{Q}$. Two real distributions that agree on $\mathcal{D}_0$ are equal.*

(x) Countable factorization. *If F is bounded, complex and Σ_+^θ-measurable, there are real $f_1, f_2, \dots \in C_c^\infty(H_+^\theta)$ and a Borel $g : \mathbb{R}^{\mathbb{N}} \to \mathbb{C}$ with $|g| \le \sup |F|$ such that $F(v) = g((v(f_j))_j)$ and $F(\theta v) = g((v(f_j \circ \theta))_j)$ for every $v \in X$; in particular $v \mapsto \overline{F(\theta v)} F(v)$ is bounded and Borel, and $\Sigma_+^\theta \subset \mathcal{B}(X)$.*

Proof. (i) Restriction embeds E isometrically as the closed subspace of compatible families in $\prod_j L^4((-j, j)^2)$ with the product metric; each factor is a separable Banach space, so E is complete and separable. For a fixed Borel representative s_0 of an element of E, Tonelli makes $\omega \mapsto \int_{(-j,j)^2} |F(z, \omega) -$

$s_0(z)|^4 dz$ measurable, so $\omega \mapsto d_E([F(\cdot,\omega)], s_0)$ is measurable; in a separable metric space every open set is a countable union of open balls.

(iii), duality. Let $f \neq 0$ be supported in $(-j,j)^2$ and let $\tilde{f}$ be the $2(j+1)$-periodic extension of $f|_{\mathcal{C}_j}$; it is smooth on $\mathbb{T}_j$, and $f_k = \int_{\mathcal{C}_j} \overline{e_{j,k}} f$ satisfies $\sum_k \lambda_{j,k}^2 |f_k|^2 = \|(1-\Delta)\tilde{f}\|^2_{L^2(\mathbb{T}_j)} < \infty$. The Fourier series of $\tilde{f}$ converges in $C^\infty(\mathbb{T}_j)$, so $\chi_j \sum_{|k|\le K} f_k e_{j,k} \to \chi_j \tilde{f} = f$ in C^∞ with supports in $\operatorname{supp}\chi_j$ (because $\chi_j = 1$ on $[-j,j]^2 \supset \operatorname{supp} f$). Since v is a distribution, $v(f) = \lim_K \sum_{|k|\le K} f_k v(\chi_j e_{j,k})$, and $v(\chi_j e_{j,k}) = v(\chi_j \overline{e_{j,-k}})$. Cauchy–Schwarz with the weights $\lambda_{j,k}^{\pm 1}$ and $\lambda_{j,-k} = \lambda_{j,k}$ give (16), provided $p_j(v) < \infty$, which is proved next.

(ii), finiteness. Let $u = \chi_j v \in H^{-2}(\mathbb{R}^2)$ and choose a real $\zeta \in C_c^\infty((-j-\frac{1}{2}, j+\frac{1}{2})^2)$ with $\zeta = 1$ near $\operatorname{supp}\chi_j$. For a trigonometric polynomial $P = \sum_{|k|\le K} c_k e_{j,k}$, ζP is a compactly supported smooth function and $\sum_k c_k u(\zeta e_{j,k}) = u(\zeta P)$, with $u(\zeta e_{j,k}) = v(\chi_j e_{j,k})$. By the product rule $(1-\Delta)(\zeta P) = \zeta(1-\Delta)P - 2\nabla\zeta\cdot\nabla P - P\Delta\zeta$, and Parseval on $\mathbb{T}_j$ gives $\|\zeta P\|_{H^2(\mathbb{R}^2)} \le C(\zeta)(\sum_k \lambda_{j,k}^2 |c_k|^2)^{1/2}$ with $C(\zeta) = \|\zeta\|_\infty + 2\sup|\nabla\zeta| + \|\Delta\zeta\|_\infty$. Taking $c_k = \lambda_{j,k}^{-2}\overline{u(\zeta e_{j,k})}$ in $|u(\zeta P)| \le \|u\|_{H^{-2}}\|\zeta P\|_{H^2}$ gives $\sum_{|k|\le K} \lambda_{j,k}^{-2}|v(\chi_j e_{j,k})|^2 \le C(\zeta)^2\|u\|^2_{H^{-2}}$ for all K; reindexing $k \mapsto -k$ yields $p_j(v) \le C(\zeta)\|\chi_j v\|_{H^{-2}(\mathbb{R}^2)}$.

(ii), metric and completeness. Each p_j is a seminorm, so d is a pseudometric; if $d(v,w) = 0$ then $p_j(v-w) = 0$ for all j and (16) gives $v = w$. Let (v_n) be d-Cauchy. Then $(\chi_j v_n)_n$ is Cauchy in the Hilbert space of periodic distributions with finite $H^{-2}(\mathbb{T}_j)$ norm. For $f \in C_c^\infty((-j,j)^2)$, $v_n(f)$ is Cauchy by (16); define $v(f) = \lim_n v_n(f)$, which does not depend on j. On each $\mathcal{D}_K$, $|v(f)| \le \sup_n p_j(v_n)\,\|f\|_{H^2(\mathbb{T}_j)} \le 3|\mathbb{T}_j|^{1/2}\sup_n p_j(v_n)\|f\|_{C^2}$, so v is a distribution. Fatou for series gives $p_j(v_n - v)^2 \le \liminf_l p_j(v_n - v_l)^2 \to 0$. Finally $v \in X$: for $\chi \in C_c^\infty((-j,j)^2)$ and $g \in \mathcal{S}(\mathbb{R}^2)$, $(\chi v)(g) = v(\chi g)$ and (16) with the product rule give $|v(\chi g)| \le p_j(v) C(\chi)\|g\|_{H^2(\mathbb{R}^2)}$, so $\chi v \in H^{-2}(\mathbb{R}^2)$. Hence $d(v_n, v) \to 0$.

(ii), separability, and (iii), Borel generation. The map $J(v) = \big((\lambda_{j,k}^{-1} v(\chi_j \overline{e_{j,k}}))_k\big)_j$ takes values in $\prod_j \ell^2(\mathbb{Z}^2;\mathbb{C})$, and $p_j(v-w) = \|J_j(v) - J_j(w)\|_{\ell^2}$; thus $d(v,w) = \sum_j 2^{-j}\min(1, \|J_j(v) - J_j(w)\|)$ is the restriction of a product metric on a separable space, and X is separable. Let $\mathcal{E}$ be the σ-algebra generated by the real and imaginary parts of the localized Fourier evaluations. For fixed w, $v \mapsto d(v,w)$ is a countable expression in $\mathcal{E}$-measurable functions, so open balls, and by separability all open sets, lie in $\mathcal{E}$. The Fourier tests $\chi_j\overline{e_{j,k}}$ have compact support, so $\mathcal{E}$ is contained in the σ-algebra of compact smooth evaluations, which is contained in $\mathcal{B}(X)$ because each evaluation is continuous: if $d(v,w) < 2^{-j}$ then $p_j(v-w) \le 2^j d(v,w)$, and (16) applies for $\operatorname{supp} f \subset (-j,j)^2$.

(iv) For $v \in E$, $(\iota v)(\chi_j\overline{e_{j,k}}) = \langle e_{j,k}, \chi_j v\rangle_{L^2(\mathbb{T}_j)}$, so Parseval and $\lambda_{j,k} \ge 1$ give $p_j(\iota v) \le \|\chi_j v\|_{L^2} \le (2j+2)^{1/2}\|v\|_{L^4((-j-1,j+1)^2)}$ by Hölder; $\chi\iota v = \chi v \in L^2 \subset H^{-2}$. A locally integrable function with zero integral against every compact smooth test vanishes almost everywhere.

(v) Every Borel probability on the Polish space X is tight. If $\beta(\rho_n, \rho) \to 0$, the bounded Lipschitz functions $(1 - k\,\mathrm{dist}_d(\cdot, F))_+ / \max(1,k)$ give $\limsup \rho_n(F) \le \rho(F)$ for closed F, hence weak convergence. Conversely, let $\rho_n \Rightarrow \rho$ and fix $\delta > 0$; choose a compact K with $\rho(K) > 1 - \delta$; its open η-neighbourhood K^η has $\rho_n(K^\eta) > 1 - 2\delta$ for large n. Choose a finite η-net of K and quantize the values of functions H with $|H| \le 1$, $\mathrm{Lip}_d(H) \le 1$ at the net points to accuracy η; this yields finitely many classes, with one representative each. A function and its representative differ by at most 3η on K and 5η on K^η, so $\beta(\rho_n,\rho) \le 10\eta + 6\delta + \max|\rho_n(H_i) - \rho(H_i)|$ over the finitely many representatives; let $n \to \infty$ and then $\eta, \delta \to 0$.

(vi) $|x^k - y^k| \le k|x-y|(|x|+|y|)^{k-1}$ and Hölder on D; the inclusion $C(\mathbb{R}^2) \to E$ is continuous.

(vii) For a smooth periodic f on $\mathbb{T}_j$ put $\zeta = (\chi_j \circ \theta)(f \circ \theta)$, supported in $(-j-\frac{1}{2}, j+\frac{1}{2})^2$

where $\chi_{j+1} = 1$. Then $(\chi_j \theta v)(f) = v(\zeta) = (\chi_{j+1} v)(\zeta)$, and the product rule with $\|\partial^\alpha (f \circ \theta)\|_{L^2(\mathcal{C}_j)} = \|\partial^\alpha f\|_{L^2(\mathbb{T}_j)}$ and $\|f\|_2, \|\partial_a f\|_2 \le \|f\|_{H^2(\mathbb{T}_j)}$ give $|(\chi_j \theta v)(f)| \le C_j p_{j+1}(v) \|f\|_{H^2(\mathbb{T}_j)}$; taking the supremum over trigonometric polynomials f with $\|f\|_{H^2(\mathbb{T}_j)} \le 1$ gives the bound. For $\chi \in C_c^\infty$, $\chi \theta v = \theta((\chi \circ \theta) v)$ and the Fourier transform of θw is $\hat{w} \circ \theta$ with $|\theta p| = |p|$, so $\theta v \in X$. The identity for $w \in E$ is the change of variables $z \mapsto \theta z$.

(viii) For real $\chi \in C_c^\infty$, testing against f gives $\chi \cdot LG = (1 - \Delta)(\chi G) + 2 \sum_i \partial_i (G \partial_i \chi) - G \Delta \chi$, so $LG \in X$. With $\chi = \chi_j$, $(LG)(\chi_j \overline{e_{j,k}}) = \lambda_{j,k} \langle e_{j,k}, \chi_j G \rangle + 2i \frac{\pi}{j+1} \sum_i k_i \langle e_{j,k}, G \partial_i \chi_j \rangle - \langle e_{j,k}, G \Delta \chi_j \rangle$, and $\lambda_{j,k}^{-2} (\pi |k_i| / (j+1))^2 \le 1$, Minkowski in ℓ^2 and Parseval give the bound, since all functions involved are supported in $[-j - \frac{1}{2}, j + \frac{1}{2}]^2$ of area $(2j+1)^2$. Continuity follows from dominated convergence for the series defining d, with summable majorant 2^{-j}.

(ix) For real $f \in C_c^\infty$ choose n with $\operatorname{supp} f \subset [-n, n]^2$; the periodic extension $\tilde{f}$ of $f|_{\mathcal{C}_n}$ is smooth, its Fourier series converges in $C^\infty(\mathbb{T}_n)$, and truncation and rational rounding give trigonometric polynomials P_l with $\chi_n P_l \to \chi_n \tilde{f} = f$ in C^∞ with supports in $\operatorname{supp} \chi_n$; hence $\operatorname{Re}(\chi_n P_l) \to f$ in that sense and $v(\operatorname{Re}(\chi_n P_l)) \to v(f)$.

(x) The sets in Σ_+^θ that are measurable with respect to the evaluations at some countable subfamily of $C_c^\infty(H_+^\theta; \mathbb{R})$ form a σ-algebra containing the generators, hence all of Σ_+^θ. A bounded F is a uniform limit of simple functions; collecting countably many tests gives $F = \tilde{g} \circ (v \mapsto (v(f_j))_j)$ by Doob–Dynkin, and $g = \tilde{g} \mathbf{1}_{\{|\tilde{g}| \le \sup |F|\}}$ still represents F. By (vii), $F(\theta v) = g(((\theta v)(f_j))_j) = g((v(f_j \circ \theta))_j)$; both factors are Borel by (iii). □

Remark 5.2 (Use and scope of these facts). In (viii), LG is the distribution $(1 - \Delta)G$, not ιG. It is used only for the Gaussian versions $\Phi_a = L\tilde{\psi}_a$, where $\tilde{\psi}_a$ carries the extra spectral multiplier $(1 + |p|^2)^{-1}$, so that $\Phi_a(f) = \phi_a(f)$ (Theorem 7.10). Items (ii) (separability and completeness) and (iii) (Borel generation) are proved here directly; they re-express arguments of the earlier width record and of a Gaussian-foundations lemma whose Borel-law part lies outside that record's accepted scope, and they require current-version review in this text (Appendix D). Completeness is used for Prokhorov compactness and for (v).

6 Process neurons and finite-width central reflection positivity

Conditional factorization gives positivity for the interacting law at finite width. For product neurons, given the complete tuple of boundary data of all N neurons, the two half-plane sides are independent with equal conditional laws (Theorem 6.1(d)); for Mondrian neurons, the trace decomposition of Theorem 6.3, followed by approximation from finite boxes (Proposition A.15), plays this role. The reflection-invariant additive potential splits into half-plane contributions and is absorbed into the side observables (Lemma A.14). Theorem 6.5 states the result, and Corollary 6.6 applies it to the tilt of the main theorem.

This section states the properties of the neurons and the finite-width positivity theorem; all proofs are in Appendix A. For open $U \subseteq \mathbb{R}^2$, $L^1_{\mathrm{loc}}(U)$ is the Polish Fréchet space of classes of locally integrable functions on U, and $\mathrm{Ball}_r(U) = \{s \in L^1_{\mathrm{loc}}(U) : |s| \le r \text{ a.e.}\}$, a closed subset. For $\theta \in \{\theta_1, \theta_2\}$ write $H_+ = H_+^\theta$. The *one-neuron sides* of a neuron g across θ are $s_+^\theta = g|_{H_+}$ and $s_-^\theta = (g \circ \theta)|_{H_+}$; the *$N$-neuron sides* are $S_+^\theta = F_N|_{H_+}$ and $S_-^\theta = (F_N \circ \theta)|_{H_+}$, random elements of $\mathrm{Ball}_{\sqrt{N}/2}(H_+)$. For a product neuron the *boundary data* are $b^{\theta_1} = (\epsilon_\sigma, \tau)$ and $b^{\theta_2} = (\sigma, \epsilon_\tau)$, where ϵ_σ, ϵ_τ are the initial signs of σ, τ.

Theorem 6.1 (Product neuron). *(a) σ is càdlàg with $|\sigma| = 1$, $\mathbb{E}\sigma(t) = 0$, $\mathbb{E}\sigma(t)\sigma(t') = e^{-\mu|t-t'|}$,*

and its finite-dimensional laws are those of the stationary symmetric two-state Markov chain with flip rate $\mu/2$.

(b) g is $\mathcal{B}(\mathbb{R}^2)\otimes\mathcal{A}$-measurable, $|g|=1/2$ everywhere, $\mathbb{E}g(z)=0$, $\mathbb{E}g(z)g(z')=C_\mu^{\mathrm{end}}(z-z')$, and almost surely g is constant on each cell of the locally finite grid formed by the jump points of σ and of τ.

(c) Given b^{θ_1}, the one-neuron sides across θ_1 are conditionally independent with equal conditional laws; likewise across θ_2 given b^{θ_2}.

(d) For independent product neurons $g_1,\dots,g_N$, the N-neuron sides $S^\theta_\pm$ are conditionally independent with equal conditional laws given the complete *tuple $(b^\theta_1,\dots,b^\theta_N)$. Conditioning on the data of a single neuron does not suffice when $N\geq 2$.*

Theorem 6.2 (Box Mondrian)**.** *Let C be a box with nonempty interior and $s\geq 0$.*

(a) Almost surely finitely many nodes are born; the mean number of born nodes is at most $2e^{sp(C)}-1$; $\Pi^{\mathrm{can}}(C,s)\in\mathrm{Part}^{\mathrm{col}}(C)$ is a Borel function of (C,s) and the ingredients.

(b) $M^{\mathrm{col}}(C,s)$ satisfies the first-step recursion of Definition A.6, and every family of laws satisfying that recursion with an admissible daughter assignment coincides with M^{col}.

(c) $T_*M^{\mathrm{col}}(C,s)=M^{\mathrm{col}}(TC,s)$ for translations T and for $T\in\{\theta_1,\theta_2\}$.

(d) $(\mathrm{res}_B)_*M^{\mathrm{col}}(C,s)=M^{\mathrm{col}}(B,s)$ for every box $B\subseteq C$ with nonempty interior.

(e) For $z,z'\in C$, the probability that some cell contains both is $e^{-s|z-z'|_1}$; a fixed $z\in\operatorname{int}C$ is almost surely interior to a cell.

Theorem 6.3 (Core-exterior construction and trace decomposition)**.** *For boxes $K\subseteq C$ the core-exterior construction of Definition A.9 produces almost surely a finite colored partition of C with law $M^{\mathrm{col}}(C,s)$, whose restriction to a core K with nonempty interior is the canonical partition of K. For $C=[-L,L]\times J_0$ and $K=\{0\}\times J_0$, the colored partition of the right half-box and the θ_1-mirror image of that of the left half-box are, given the trace data (positions and auxiliary times of the horizontal cuts of straddling cells and colors of the final straddling cells), conditionally independent with the same conditional law; the same holds with coordinates exchanged. Times and positions are both needed in this conditioning (Appendix A).*

Theorem 6.4 (Whole-plane Mondrian neuron)**.** *(a) A consistent colored Mondrian tessellation of budget μ exists; the law of $(P_n)_n$ is unique and invariant under translations and under θ_1,θ_2 acting on closed cells; for a box B with nonempty interior, $\mathrm{res}_B(P_n)$ ($B\subseteq B_n$) does not depend on n almost surely and has law $M^{\mathrm{col}}(B,\mu)$. The lower-left representatives of the cells of any element of $\mathrm{Part}^{\mathrm{col}}(B_n)$ partition $[-n,n)^2$.*

(b) Almost surely each closed cell $R(z)$ is a bounded closed box with nonempty interior, equal to $R_n(z)$ for large n; the distinct closed cells cover $\mathbb{R}^2$, have disjoint interiors and are locally finite; $H(z)=R(z)^{\mathrm{ll}}$, and the distinct half-open representatives partition $\mathbb{R}^2$ and are locally finite.

(c) g is $\mathcal{B}(\mathbb{R}^2)\otimes\mathcal{A}$-measurable, $|g|=1/2$ everywhere, almost surely constant on each $H(z)$, $\mathbb{E}g(z)=0$ and $\mathbb{E}g(z)g(z')=C_\mu^{\mathrm{end}}(z-z')$.

(d) Every admissible convention agrees with g off the union of cell boundaries, a null set for every outcome of the consistency event; for open U the law of $g|_U$ in $L^1_{\mathrm{loc}}(U)$ is the same for every realization and every admissible convention; $g \circ \theta_j \overset{d}{=} g$ and $g(\cdot + x) \overset{d}{=} g$ in $L^1_{\mathrm{loc}}(\mathbb{R}^2)$.

For $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, $N \geq 1$, a bounded Borel Ω and an $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{B}(\mathbb{R})$-measurable $v : \mathbb{R}^2 \times \mathbb{R} \to \mathbb{R}$, bounded on each $\Omega \times [-r, r]$, put $U = \int_\Omega v(z, F_N(z))\, dz$, $Z = \mathbb{E}e^{-U}$, $P_N^v(B) = \mathbb{E}[\mathbf{1}_B(\iota F_N)e^{-U}]/Z$ and $M_N = \sup\{|v(z, x)| : z \in \Omega, |x| \leq \sqrt{N}/2\}$, with the convention $\sup \emptyset = 0$; if $|\Omega| = 0$, then $U = 0$ and $Z = 1$.

Theorem 6.5 (Finite-width annealed normalized reflection positivity)**.** *Let $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, $N \geq 1$, $\theta \in \{\theta_1, \theta_2\}$, Ω bounded Borel with $\theta\Omega = \Omega$, and v as above with $v(\theta z, x) = v(z, x)$ for all $z \in \Omega$, $x \in \mathbb{R}$. Then:*

(a) U is a real random variable, $|U| \leq |\Omega| M_N$ and $e^{-|\Omega| M_N} \leq Z \leq e^{|\Omega| M_N}$; P_N^v is a Borel probability on X determined by the law of one neuron, independently of the realization and of the admissible boundary convention;

(b) P_N^v is reflection positive across θ for bounded observables, in the sense of (11)*;*

(c) $\mathbb{E}[e^{-U}\overline{\Phi(S_-^\theta)}\Phi(S_+^\theta)] \geq 0$ for every bounded complex Borel Φ on $L^1_{\mathrm{loc}}(H_+)$; and for every bounded complex Y measurable for $\sigma(F_N(z) : z \in H_+)$, written $Y = \tilde{Y}((F_N(z_k))_{k \in \mathbb{N}})$, the reflected variable $Y^\theta = \tilde{Y}((F_N(\theta z_k))_k)$ is almost surely independent of the representation and $\mathbb{E}[e^{-U}\overline{Y^\theta} Y] \geq 0$.

Corollary 6.6 (Two central lines; the axis-parallel manuscript tilt)**.** *Let $\Lambda = [-l_1, l_1] \times [-l_2, l_2]$ and $\Omega = [-w_1, w_1] \times [-w_2, w_2]$ with $0 \leq l_i \leq w_i$: closed axis-parallel rectangles,* both centered*. Let $q_{\mathrm{in}}, q_{\mathrm{out}}$ be real polynomials, $\kappa \in \mathbb{R}$, and $v(z, x) = \mathbf{1}_\Lambda(z) q_{\mathrm{in}}(x) + \mathbf{1}_{\Omega \setminus \Lambda}(z) q_{\mathrm{out}}(x) - \kappa^2 x^2/2$. Then for both architectures and every $N \geq 1$ the conclusions (a)–(c) of Theorem 6.5 hold across θ_1 and across θ_2.* Manuscript tilt. *If Λ is a physical square in $\Omega = [-R, R]^2$ in the sense of Definition 2.1, the integrand v_V of* (7) *agrees off $\partial\Lambda \times \mathbb{R}$ with the displayed v for the rectangles $\Lambda = [-l, l]^2$, $\Omega = [-R, R]^2$, with $q_{\mathrm{in}}(x) = u(x^4 - 6c_a x^2 + 3c_a^2)$, $q_{\mathrm{out}}(x) = \varepsilon x^4$, $\kappa = \kappa_a$; for every outcome $U = V_{\varepsilon, \Omega, a; \Lambda}(F_N)$, and $P_N^{v_V} = P^b_{N, \varepsilon, \Omega, a; \Lambda}$. Hence every neural law* (8) *is reflection positive across both central coordinate lines; this covers $\Lambda_n = (-n\ell, n\ell)^2$ with $n\ell \leq R$.*

Corollary 6.7 (General potentials invariant under both reflections)**.** *If $\theta_1\Omega = \theta_2\Omega = \Omega$ and v is $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{B}(\mathbb{R})$-measurable, bounded on each $\Omega \times [-r, r]$ and invariant under both central reflections on Ω, then (a)–(c) of Theorem 6.5 hold across both central lines, for both architectures and every $N \geq 1$.*

Remark 6.8 (Geometry, measurability and scope). (i) *Centering alone is not enough.* For $O = 5^{-1/2}\left(\begin{smallmatrix} 2 & -1 \\ 1 & 2 \end{smallmatrix}\right)$, the centered rotated square $O((-1, 1)^2)$ lies in $\Omega = [-2, 2]^2$, and with $m = u = \varepsilon = 1$, $a = 1/2$ the integrand built on it satisfies $v_V(z, 0) = 3uc_a^2 > 0 = v_V(\theta_j z, 0)$ for both j on a disc of positive area. This shows only that the invariance hypothesis of Theorem 6.5 fails for that tilt; no failure of reflection positivity is claimed for it or for any rotated square. (ii) *The general clause keeps its own hypotheses.* A centered square rotated by an odd multiple of $\pi/4$, $\{|z_1| + |z_2| < c\}$, is invariant under both central reflections; a potential of the form (7) built on it, with closure inside an invariant Ω, satisfies the hypotheses of Corollary 6.7, which therefore applies to it, although it is not a physical square in the sense of Definition 2.1. (iii) *Measurability.* "Measurable potential" means $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{B}(\mathbb{R})$-measurable. Under the broader reading "Lebesgue measurable on $\mathbb{R}^3$" the integrated potential can be undefined: for a non-measurable $A \subset (0, 1)^2$ symmetrized under both reflections

and $v(z,x) = \mathbf{1}_A(z)\mathbf{1}\{x = x_0\}$ with $x_0 \in \{0, 1/(2\sqrt{N})\}$ of the parity of N, v is Lebesgue-null on $\mathbb{R}^3$, but on an event of positive probability $F_N = x_0$ on Ω and $z \mapsto v(z, F_N(z))$ is not measurable. (iv) *Scope.* Positivity is asserted across θ only when Ω and v are θ-invariant; for the two-line statements, invariance under both reflections of both the inner and the outer region is used. Nothing is asserted for shifted or rotated lines, for Euclidean invariance of tilted laws, or for limits in N, ε, Ω or a by this theorem.

7 Gaussian reference fields

Gaussian fields enter in two roles. First, they serve as references for the intermediate laws of the chain: the width limit replaces the untilted neural reference field by the continuous endpoint field γ_μ (Theorem 7.1); its quadratic tilt on a bounded Ω is again Gaussian, with a covariance $C'_{\Omega,a}$ between C^{end}_μ and C_a (Theorems 7.4 and 7.5); and $C'_{\Omega,a} \to C_a$ locally uniformly as Ω grows (Theorem 7.6). Second, Theorems 7.7–7.10 realize all regulated fields and the free field on one probability space, so that the regulator can be removed through L^2 estimates for random variables.

Proofs are in Appendix B. Throughout, a is a regulator unless $a = 0$ is stated, and Ω is a bounded measurable set.

Theorem 7.1 (Endpoint Gaussian)**.** *For every $\mu > 0$ there is a $\mathcal{B}(\mathbb{R}^2)\otimes\mathcal{F}$-measurable $\gamma_\mu : \mathbb{R}^2 \times \Gamma_\mu \to \mathbb{R}$ with continuous paths for every outcome, which is a centered Gaussian process with covariance $C^{\mathrm{end}}_\mu(z - z')$ and $\mathbb{E}|\gamma_\mu(z+h) - \gamma_\mu(z)|^2 = (1 - e^{-\mu|h|_1})/2 \le \mu|h|_1/2$. Its law G_μ on $C(\mathbb{R}^2)$ is the unique Borel probability under which the coordinate process is centered Gaussian with this covariance.*

Theorem 7.2 (Regulated covariance and mass shift)**.** *(a) $\hat{C}_a \le \hat{C}_0 \le m^{-2}$, $\hat{C}_a(w,q) \le m^{-2}(1 + a^2w^2)^{-1}(1 + a^2q^2)^{-1}$, $\hat{C}_a \in L^1$, and (2) holds with $c_a \in (0,\infty)$.*

(b) (3) holds pointwise, with $0 \le \kappa_a^2 \hat{C}^{\mathrm{end}}_\mu \le r_a$.

(c) $s_k = \kappa_a^{2k}(C^{\mathrm{end}}_\mu)^{*(k+1)}$ is continuous and positive with $\sup s_k \le r_a^k/4$ and $\|s_k\|_1 = \mu^{-2} r_a^k$; $\sum_k s_k$ converges uniformly and in L^1 to C_a. Hence C_a is continuous, $0 < C^{\mathrm{end}}_\mu \le C_a \le c_a$, $\|C_a\|_1 = m^{-2}$ and $\|C_a\|_2^2 \le 1/(4\pi m^2)$.

(d) As operators on $L^2(\mathbb{R}^2)$: $\|C^{\mathrm{end}}_\mu\| = \mu^{-2}$, $C_a = C^{\mathrm{end}}_\mu(I - \kappa_a^2 C^{\mathrm{end}}_\mu)^{-1}$, and $0 \le \langle f, C_a f\rangle \le m^{-2}\|f\|_2^2$.

(e) For every $h \in \mathbb{R}^2$: $2(c_a - C_a(h)) \le \varpi_a(h)^2 := (2\pi m^2 a^3)^{-1} \sum_{i=1,2} |h_i|[\log(1 + 4a^2/h_i^2) + 2]$ (summand 0 if $h_i = 0$).

(f) For $0 < \alpha < 1$: $2(c_a - C_a(h)) \le M'_{a,\alpha}|h|^\alpha$ for all h, with $M_{a,\alpha} = [2(2a)^{1-\alpha}/(1-\alpha) + 2]/(\pi m^2 a^3)$ and $M'_{a,\alpha} = \max(M_{a,\alpha}, 2c_a)$.

Theorem 7.3 (Restricted endpoint covariance)**.** *$K_\Omega f(x) = \int_\Omega C^{\mathrm{end}}_\mu(x-y) f(y)\,dy$ on $L^2(\Omega;\mathbb{R})$ is positive, self-adjoint and trace class with $\operatorname{tr} K_\Omega = |\Omega|/4$, $\|K_\Omega\| \le \mu^{-2}$ and $\kappa_a^2\|K_\Omega\| \le r_a < 1$; it has an orthonormal eigenbasis with eigenvalues $\lambda_j \in [0, \mu^{-2}]$.*

Theorem 7.4 (Pure quadratic tilt on finite Ω)**.** *(a) $\mathrm{Det}_{\Omega,a} = \mathbb{E}_{G_\mu} e^{T_{\Omega,a}} = \prod_j (1 - \kappa_a^2\lambda_j)^{-1/2} = \det_F(I - \kappa_a^2 K_\Omega)^{-1/2}$, with $1 \le \mathrm{Det}_{\Omega,a} \le \exp(\kappa_a^2|\Omega|/(8a^2m^2))$, nondecreasing in Ω.*

(b) $c_0^\Omega(z,z') = C^{\mathrm{end}}_\mu(z-z')$ and $c_k^\Omega(z,z') = \kappa_a^{2k}\int_{\Omega^k} C^{\mathrm{end}}_\mu(z-x_1)\cdots C^{\mathrm{end}}_\mu(x_k - z')\,dx$ are continuous, symmetric and $0 \le c_k^\Omega \le s_k(z-z')$; $C'_{\Omega,a} = \sum_k c_k^\Omega$ converges uniformly on $\mathbb{R}^2 \times \mathbb{R}^2$, and the operator with this kernel is $C^{\mathrm{end}}_\mu(I - \kappa_a^2 \mathbf{1}_\Omega C^{\mathrm{end}}_\mu)^{-1} = C^{\mathrm{end}}_\mu + \kappa_a^2 C^{\mathrm{end}}_\mu R^*(I - \kappa_a^2 K_\Omega)^{-1} R\, C^{\mathrm{end}}_\mu$.

(c) $G'_{\Omega,a} = \mathrm{Det}^{-1}_{\Omega,a} e^{T_{\Omega,a}} G_\mu$ *is the unique Borel probability on* $C(\mathbb{R}^2)$ *under which the coordinate process at* all *points of the plane is centered Gaussian with covariance* $C'_{\Omega,a}$*, and* $\mathbb{E}_{G_\mu}[Fe^{T_{\Omega,a}}] = \mathrm{Det}_{\Omega,a}\mathbb{E}_{G'_{\Omega,a}}[F]$ *for Borel* $F \geq 0$ *and for Borel* F *integrable under* $G'_{\Omega,a}$*.*

Theorem 7.5 (Covariance order)**.** $0 < C^{\mathrm{end}}_\mu(z-z') \leq C'_{\Omega,a}(z,z') \leq C_a(z-z') \leq c_a$ *pointwise;* $0 \leq C^{\mathrm{end}}_\mu \leq C'_{\Omega,a} \leq C_a$ *as operators;* $C'_{\Omega,a} \leq C'_{\Omega',a}$ *for* $\Omega \subseteq \Omega'$*.*

Theorem 7.6 (The covariance difference)**.** *Let* $\Delta_{\Omega,a} = C_a - C'_{\Omega,a}$ *and* $Q = \mathbf{1}_{\mathbb{R}^2\setminus\Omega}$*.*

(a) $\Delta = \kappa_a^2 C_a Q C' = \kappa_a^2 C' Q C_a = \kappa_a^2 C' Q C' + \kappa_a^4 C' Q C_a Q C'$ *as operators, with the corresponding kernel identities.*

(b) Δ *is continuous and positive semidefinite as an operator and as a kernel.*

(c) $0 \leq \Delta(z,z') \leq \kappa_a^2 \int_{\mathbb{R}^2\setminus\Omega} C_a(z-x)C_a(x-z')\,dx$*; if* Ω *contains every point at distance* $< r$ *from a compact* K*, then* $0 \leq \Delta(z,z') \leq \epsilon_a(r) = \kappa_a^2 \int_{|x|>r} C_a(x)^2 dx$ *for* $z, z' \in K$*, and* $\epsilon_a(r) \downarrow 0$*.*

(d) $\sup_{K\times K} \Delta_{\Omega,a} \to 0$ *along every family of bounded* Ω *that eventually contains each bounded set, for instance* $[-R,R]^2$*,* $R \to \infty$*.*

(e) $\Delta(z+h,z+h)+\Delta(z,z)-2\Delta(z+h,z) \leq 2(c_a - C_a(h))$*, and the same for* C'*; centered Gaussian processes with covariance* $\Delta_{\Omega,a}$ *or* $C'_{\Omega,a}$ *exist and have jointly measurable modifications with continuous paths.*

The spectral Gaussian space: $\mu_0(dp) = (2\pi)^{-2}\hat{C}_0(p)\,dp$, $H_{\mathbb{C}} = \{h \in L^2(\mu_0;\mathbb{C}) : h(-p) = \overline{h(p)}\}$ with the real inner product $\langle g,h\rangle = \int \overline{g}h\,d\mu_0$, $\tau_a = (\hat{C}_a/\hat{C}_0)^{1/2}$ ($\tau_0 = 1$), $e_z(p) = e^{-ip\cdot z}$ and $\rho(p) = (1+|p|^2)^{-1}$. An isonormal process on a real Hilbert space is a real-linear isometry into L^2 whose range consists of centered jointly Gaussian variables.

Theorem 7.7 (One common Gaussian space)**.** *There is a probability space* $(\Gamma_0, \mathcal{F}_0, \mathbb{P}_0)$ *carrying an isonormal process* W *on* $H_{\mathbb{C}}$*. With* $\phi_a(f) = W(\hat{f}\tau_a)$ *for* $0 \leq a < 1/m$ *and real* $f \in L^2(\mathbb{R}^2)$*,* $\phi = \phi_0$*, the whole family is centered and jointly Gaussian,* $f \mapsto \phi_a(f)$ *is real-linear, and* $\mathbb{E}[\phi_a(f)\phi_b(g)] = (2\pi)^{-2}\int \overline{\hat{f}}\hat{g}(\hat{C}_a\hat{C}_b)^{1/2}dp$*.*

Theorem 7.8 (Smearing bounds)**.** *For real* $f \in L^2$*:* $\|\phi_a(f)\|_2 \leq \|\phi(f)\|_2 \leq m^{-1}\|f\|_2$ *and* $\|\phi_a(f) - \phi(f)\|_2 \leq \frac{a}{2}\|f\|_2$*. For complex* f *and* $T_a(f) = \phi_a(\mathrm{Re}f) + i\phi_a(\mathrm{Im}f)$*:* $\mathbb{E}|T_a(f)|^2 \leq m^{-2}\|f\|_2^2$ *and* $\mathbb{E}|T_a(f) - T_0(f)|^2 \leq \frac{a^2}{4}\|f\|_2^2$*.*

Theorem 7.9 (Point fields and continuous modifications)**.** *For a regulator* a*,* $\phi_a(z) = W(e_z\tau_a)$ *is a centered Gaussian process with* $\mathbb{E}[\phi_a(z)\phi_b(z')] = C_{ab}(z-z')$*,* $C_{ab} = \mathcal{I}[(\hat{C}_a\hat{C}_b)^{1/2}]$*, and* $\mathbb{E}|\phi_a(z+h) - \phi_a(z)|^2 = 2(c_a - C_a(h)) \leq \varpi_a(h)^2$*. There is a* $\mathcal{B}(\mathbb{R}^2)\otimes\mathcal{F}_0$*-measurable* $\tilde{\phi}_a$ *with continuous paths for every outcome and* $\mathbb{P}_0(\tilde{\phi}_a(z) = \phi_a(z)) = 1$ *for every* z*; for real* $f \in L^1 \cap L^2$*,* $\int |f||\tilde{\phi}_a| < \infty$ *and* $\int f\tilde{\phi}_a = \phi_a(f)$ *almost surely. The exceptional null sets depend on* (a,z) *and on* (a,f)*; for countable families of regulators, points and tests there is one common null set; no common null set for uncountably many* a*, and no measurability or continuity in* a*, is asserted.*

Theorem 7.10 (Borel X-valued versions, including $a = 0$)**.** *For each* $0 \leq a < 1/m$*,* $\psi_a(z) = W(e_z\tau_a\rho)$ *has a modification* $\tilde{\psi}_a$ *with continuous paths, and* $\Phi_a := L(\tilde{\psi}_a(\cdot,\omega))$ *(the map* L *of Proposition 5.1(viii)) is a Borel map* $\Gamma_0 \to X$ *with:*

(a) $\Phi_a(f) = \phi_a(f)$ *almost surely for every real* $f \in C_c^\infty(\mathbb{R}^2)$*, and* $\Phi_a(f) = T_a(f)$ *for complex* f*;*

(b) $\Phi_a = \iota\tilde{\phi}_a$ *outside a null set depending on* a*, for* $a > 0$*;*

(c) if $\Phi' : \Gamma_0 \to X$ *satisfies* $\Phi'(f) = \phi_a(f)$ *almost surely for every* $f \in \mathcal{D}_0$ *(Proposition 5.1(ix)), then* $\Phi' = \Phi_a$ *almost surely;*

(d) for countable families of regulators and tests the identities in (a), (b) hold off one null set;

(e) $\mathbb{E}_0 p_j(\Phi_a)^2 \le m^{-2} B_j$ *and, for* $a > 0$, $\mathbb{E}_0 p_j(\Phi_a - \Phi_0)^2 \le \frac{a^2}{4} B_j$*, with* $B_j = (\|\chi_j\|_2^2/|\mathbb{T}_j|) \sum_k \lambda_{j,k}^{-2} < \infty$*; hence* $\|d(\Phi_a, \Phi_0)\|_{L^2(\mathbb{P}_0)} \le h(a) := \sum_j 2^{-j} \min(1, \frac{a}{2} B_j^{1/2}) \to 0$*;*

(f) for real $f_1, \dots, f_n \in C_c^\infty$ *the vector* $(\Phi_a(f_i))_i$ *is centered Gaussian with covariance* $(2\pi)^{-2} \int \overline{\hat{f}_i} \hat{f}_l \hat{C}_a \, dp$*.*

No interaction functional, Wick power, weight or normalizer enters these constructions.

Corollary 7.11 (Laws of the versions on X)**.** *For* $0 \le a < 1/m$ *the Borel law of* Φ_a *on* X *is the unique Borel probability whose finite-dimensional laws on compact smooth tests are those of (f); at* $a = 0$ *it is the centered Gaussian law with covariance* C_{free}*.*

Proof. By Proposition 5.1(iii) the finite-dimensional cylinder events of compact smooth evaluations form a $\cap$-stable generator of $\mathcal{B}(X)$; apply Dynkin's theorem. □

Remark 7.12 (Scope of the Gaussian foundations). Theorems 7.1–7.10 use only the Gaussian space; they assert nothing about interaction weights, Wick powers, normalizers, interacting laws or reflection positivity. Corollary 7.11 uses the Borel generation of Proposition 5.1(iii), which is proved in this paper. The interaction objects on the same space are supplied by Theorem 8.2 below. Theorem 7.2(e) is proved for every regulator $0 < a < 1/m$; nothing is claimed at $a = 1/m$.

8 Fixed-volume Wick theory

This section asks whether the interaction has a limit when the regulator is removed. The free field is distribution-valued, so its fourth power has no pointwise meaning, and the Wick constant c_a diverges; nevertheless the Wick-ordered integrals $U_{a,\Lambda}$ converge in L^2 to a fourth-chaos variable $U_{0,\Lambda}$ (Theorem 8.2(a),(b)). The key estimate is the bound $\mathbb{E}_0 e^{-pU_a} \le \Xi_p$, uniform in the regulator (Theorem 8.2(e)), which controls the weights and normalizers. These weight and regulator estimates have volume-dependent constants (Remark 8.4); the volume-uniform moment bounds needed later come from classical input (Section 10).

Proofs are in Appendix C. Fix $m, u > 0$, a physical square Λ with half-side $l > 0$ (open or closed; the boundary is null), and $A = |\Lambda| = 4l^2$. There is no weak-coupling restriction. For $c \ge 0$ let $:x^4:_c = x^4 - 6cx^2 + 3c^2$; for a centered Gaussian variable ξ put $:\xi^4: = :\xi^4:_{\operatorname{Var}\xi}$ (own-variance Wick power; zero if $\operatorname{Var}\xi = 0$). Let $H_1 = W(H_{\mathbb{C}})$ be the Gaussian Hilbert space of Theorem 7.7, $H^{:4:}$ its fourth Wiener chaos, $\mathcal{N}$ the $\mathbb{P}_0$-null sets and $\mathcal{F}_{\text{free}} = \sigma(\phi(f) : f \in C_c^\infty(\mathbb{R}^2; \mathbb{R})) \vee \mathcal{N}$.

Definition 8.1 (Interactions and admissible cutoffs)**.** For a regulator a, $U_a = U_{a,\Lambda} = u \int_\Lambda :\tilde{\phi}_a(z)^4:_{c_a} dz$ pathwise, and $U_0 = L^2(\mathbb{P}_0)\text{-}\lim_{a \downarrow 0} U_a$ through all real regulators (existence is part of Theorem 8.2). An *admissible cutoff family* is a family $(\chi_\epsilon)_{\epsilon > 0}$ of measurable functions with $\chi_\epsilon(-p) = \overline{\chi_\epsilon(p)}$, $\sup_\epsilon \|\chi_\epsilon\|_\infty < \infty$, $\chi_\epsilon \to 1$ almost everywhere and $\int |\chi_\epsilon|^2 \hat{C}_0 < \infty$; put $\phi_\epsilon^\chi(z) = W(e_z \chi_\epsilon)$ and $N_\epsilon^\chi = u \int_\Lambda :\phi_\epsilon^\chi(z)^4: dz$ (Bochner integral in $L^2(\mathbb{P}_0)$). Examples: smooth mollifiers $\chi_\epsilon(p) = \hat{\eta}(\epsilon p)$ with real $\eta \in C_c^\infty$, $\int \eta = 1$, for which $\phi_\epsilon^\chi(z) = \phi(\eta_\epsilon(z - \cdot))$;

and sharp cutoffs $\mathbf{1}_{\{|p|\le 1/\epsilon\}}$. The constants are $\kappa_0 = e(1+\sqrt{2})$, $\kappa_2 = (2\pi)^{-1}(17\pi/16)^{1/2}$, $M_6 = (2\pi)^{-5/3}(5\pi)^{5/6}m^{-1/3}$, $K_\Lambda = u(384A\kappa_2 M_6^3)^{1/2}$, $K_\sharp = \max\{12uA, (2/e)K_\Lambda(\kappa_0/m)^{1/2}\}$, $T(K) = \exp(-\exp((\pi/2)(K/(12uA))^{1/2}-1))$ and $\Xi_p = e^{pK_\sharp} + \int_{K_\sharp}^\infty pe^{pK}T(K)\,dK$.

For $\theta \in \{\theta_1, \theta_2\}$ let Θ be the reflection of the Gaussian space induced by $h \mapsto h \circ \theta$ on $H_{\mathbb{C}}$ (Appendix C): it is linear, multiplicative and law-preserving, with $\Theta\phi_a(f) = \phi_a(f \circ \theta)$ and $\Theta\phi_a(z) = \phi_a(\theta z)$. Put $\mathcal{A}_+^{\theta,a} = \sigma(\phi_a(f) : f \in C_c^\infty(H_+^\theta; \mathbb{R})) \vee \mathcal{N}$.

Theorem 8.2 (Fixed-volume Wick theory)**.** *On the common Gaussian space of Theorem 7.7:*

(a) $U_a \in L^2(\mathbb{P}_0)$ *lies in* $H^{:4:}$*, equals the Bochner integral of* $z \mapsto u{:}\phi_a(z)^4{:}$*, and satisfies* $U_a \ge -6uAc_a^2$ *pathwise; the limit* U_0 *exists and lies in* $H^{:4:}$*.*

(b) Construction statement. *For every admissible cutoff family,* $N_\epsilon^\chi \to U_0$ *in* $L^2(\mathbb{P}_0)$*;* U_0 *is the unique* $Y \in H^{:4:}$ *with* $\mathbb{E}_0[Y{:}\phi(f_1)\cdots\phi(f_4){:}] = 4!\,u\int_\Lambda \prod_{j\le 4}(C_{\text{free}} * f_j)(z)\,dz$ *for all real* $f_j \in C_c^\infty$*; and* U_0 *is* $\mathcal{F}_{\text{free}}$*-measurable. The construction uses only the Gaussian variables* $W(h)$*,* $h \in H_{\mathbb{C}}$*, their own-variance Wick polynomials, Bochner integrals in* $L^2(\mathbb{P}_0)$ *and* $L^2(\mathbb{P}_0)$*-limits; it uses no point variable at* $a = 0$ *and forms no pointwise product of the distribution-valued free field. Nothing is asserted about other representations of* U_0*.*

(c) $\mathbb{E}_0 U_a = 0$ *for* $0 \le a < 1/m$*.*

(d) $\|U_a - U_0\|_2 \le K_\Lambda\sqrt{a}$ *for every regulator* a*.*

(e) For every real $p \ge 1$*,* $\Xi_p < \infty$ *and* $\mathbb{E}_0 e^{-pU_a} \le \Xi_p$ *for* $0 \le a < 1/m$*, including* $a = 0$*.*

(f) For real $p \ge 1$ *and regulators* a*,* $\|e^{-U_a} - e^{-U_0}\|_p \le 2(2p-1)^2 K_\Lambda \Xi_{2p}^{1/(2p)}\sqrt{a}$*.*

(g) $1 \le Z_a := \mathbb{E}_0 e^{-U_a} \le \Xi_1$ *for* $0 \le a < 1/m$*.*

(h) For $\theta \in \{\theta_1, \theta_2\}$*, a regulator* a *and* $F \in L^2(\mathcal{A}_+^{\theta,a})$*:* $\mathbb{E}_0[\overline{\Theta F} F e^{-U_a}] \ge 0$*.*

(i) For $\theta \in \{\theta_1, \theta_2\}$ *and bounded* $\mathcal{A}_+^{\theta,0}$*-measurable* F*:* $\mathbb{E}_0[\overline{\Theta F} F e^{-U_0}] \ge 0$*.*

(j) For $0 \le a < 1/m$ *and any measurable* $\Phi' : \Gamma_0 \to X$ *with* $\Phi'(f) = \phi_a(f)$ *almost surely for real* $f \in C_c^\infty$*:* $B \mapsto \mathbb{E}_0[\mathbf{1}_B(\Phi')e^{-U_a}]/Z_a$ *is a Borel probability on* X*, reflection positive across* θ_1 *and* θ_2 *for bounded observables.*

(k) No weak-coupling restriction is used; no statement is uniform in Λ*.*

Moreover, for $K \ge K_\sharp$*:* $\sup_{0<a<1/m} \mathbb{P}_0(U_a \le -K) \le T(K)$ *and* $\mathbb{P}_0(U_0 < -K) \le T(K)$*.*

Corollary 8.3 (The physical laws)**.** *For* $0 \le a < 1/m$*,* $\nu_{a,\Lambda}$ *of Definition 2.10(d), built with* $\Phi_a = L\tilde{\psi}_a$ *of Theorem 7.10, is a Borel probability on* X *with* $1 \le Z_{a,\Lambda} \le \Xi_1(\Lambda)$*; it is reflection positive across both central coordinate lines for bounded observables; and any Borel* Φ' *satisfying the identity of Theorem 7.10(a) gives the same law. At* $a = 0$ *this is the finite-volume Wick-quartic law on* X*; no interface hypothesis remains.*

Proof. Theorem 7.10(a) supplies the hypothesis of Theorem 8.2(j) for every $0 \le a < 1/m$, including $a = 0$. Uniqueness: Theorem 7.10(c) gives $\Phi' = \Phi_a$ almost surely. □

Remark 8.4 (Scope of the Wick theory). Every constant depends on Λ through $A = |\Lambda|$: $K_\Lambda \propto uA^{1/2}m^{-1/2}$, $K_\sharp \geq 12uA$ and $\Xi_p \geq e^{12puA}$. The only volume-uniform statement is $Z_{a,\Lambda} \geq 1$. The $\sqrt{a}$ bounds are fixed-volume upper bounds, not practical convergence rates. The Gaussian input is used only through Theorems 7.7, 7.9 and 7.10; the weight, the normalizer and reflection positivity enter only after that input is fixed. Exceptional null sets are those of Theorems 7.9 and 7.10: one per fixed regulator value and test, one for countable families, none for uncountably many regulators. The classical inputs C6(a)–(c) are used in their stated forms; their original sources were not inspected.

9 The fixed-volume chain

Throughout this section a physical square Λ (Definition 2.1) is fixed; only the width, the stabilizer, the auxiliary square and the regulator move, one at a time, with all other parameters fixed. Every statement holds for all $m, u > 0$; no weak-coupling restriction is used, and no constant is uniform in the parameters held fixed. The inputs are the theorems of Sections 5–8. Appendix D records the argument lineage.

Lemma 9.1 (Open and closed physical squares)**.** *The neural, Gaussian, residual and physical laws built with $\Lambda = (-l, l)^2$ coincide with those built with $\bar{\Lambda} = [-l, l]^2$.*

Proof. For $v \in E$ the integrals of $v^4 - 6c_a v^2 + 3c_a^2$ and of v^4 over Λ and $\bar{\Lambda}$ (and over the corresponding parts of Ω) coincide because $\partial\Lambda$ is Lebesgue-null; $T_{\Omega,a}$ does not involve Λ. Hence every ordinary-field law coincides. On the common Gaussian space $U_{a,\Lambda} = U_{a,\bar{\Lambda}}$ for every regulator, so their $L^2(\mathbb{P}_0)$ limits coincide. □

The four arrows below remove the width, the stabilizer, the auxiliary square and the regulator, in that order. The width arrow combines a central limit theorem for finitely many smearings with compactness in E, obtained from the increment bounds (17); the bounded continuous weight e^{-V} then passes to the limit. The stabilizer arrow is a monotone limit of weights, the auxiliary-square arrow couples two Gaussian fields whose covariances differ by $\Delta_{\Omega,a}$, and the regulator arrow uses the estimates of Theorem 8.2. Remark 9.7 summarizes the resulting chain.

9.1 The width arrow

Proposition 9.2 (Width arrow)**.** *Fix $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, a regulator a, $\varepsilon > 0$, a physical square Λ and $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$. Then Z_N^b and Z_{full} are positive and finite, $Z_N^b \to Z_{\mathrm{full}}$, and $P^b_{N,\varepsilon,\Omega,a;\Lambda} \Rightarrow \nu^v_{\varepsilon,\Omega,a;\Lambda}$ weakly on X as $N \to \infty$. The limit is the same full Gibbs tilt of the Gaussian reference for both architectures.*

Proof. Write $F_N = F^b_{N,a}$, Q_N for its law on E and Q for the law of γ_μ on E; both are Borel by Proposition 5.1(i), since $|F_N| \leq \sqrt{N}/2$ and γ_μ is continuous and jointly measurable (Theorems 6.1, 6.4, 7.1).

Moments. By Theorems 6.1(b) and 6.4(c), each neuron has $|g| = 1/2$ everywhere, $\mathbb{E}g = 0$ and $\mathbb{E}g(z)g(z') = C^{\mathrm{end}}_\mu(z - z')$. For $r = |z - z'|_1$ put $b(r) = \frac{1}{2}(1 - e^{-\mu r}) \leq \mu r/2$. Then $d_i = g_i(z) - g_i(z') \in \{0, \pm 1\}$, $\mathbb{E}d_i = 0$ and $\mathbb{E}d_i^4 = \mathbb{E}d_i^2 = b(r)$; independence across neurons gives

$$\mathbb{E}|F_N(z) - F_N(z')|^4 = \frac{b(r)}{N} + 3\Big(1 - \frac{1}{N}\Big)b(r)^2 \leq \frac{\mu r}{2} + \frac{3\mu^2 r^2}{4}, \qquad \mathbb{E}F_N(z)^4 = \frac{3 - 2/N}{16} \leq \frac{3}{16}. \tag{17}$$

For γ_μ the increment has variance $b(r)$ and fourth moment $3b(r)^2$, and $\mathbb{E}\gamma_\mu(z)^4 = 3/16$, so the same bounds hold.

Compact containment in E. For a box B and the cell-average projection P_δ onto a partition of B into squares of ℓ^1-diameter at most δ, Jensen on each cell C gives $\int_C |v - |C|^{-1}\int_C v|^4 \le |C|^{-1}\int_C\int_C |v(x)-v(y)|^4$, so by (17) and Tonelli $\mathbb{E}\|v - P_\delta v\|^4_{L^4(B)} \le |B|(\mu\delta/2 + 3\mu^2\delta^2/4)$ and $\mathbb{E}\|v\|^4_{L^4(B)} \le 3|B|/16$ under every Q_N and under Q. Fix $\rho > 0$. For $B_j = (-j, j)^2$ choose R_j with Markov tail $\mathbb{P}(\|v\|_{L^4(B_j)} > R_j) \le \rho 2^{-j-1}$, and for $l \ge 1$ a dyadic projection $P_{j,l}$ with $\mathbb{P}(\|v - P_{j,l}v\|_{L^4(B_j)} > 2^{-l}) \le \rho 2^{-j-l-1}$, uniformly in N. The set $K_j = \{v : \|v\|_4 \le R_j,\ \|v - P_{j,l}v\|_4 \le 2^{-l}\ \forall l\} \subset L^4(B_j)$ is closed and totally bounded (each $P_{j,l}K_j$ is bounded in a finite-dimensional space), hence compact; $K = \{v \in E : v|_{B_j} \in K_j\ \forall j\}$ is compact in E, and $\inf_N Q_N(K) \ge 1 - \rho$, $Q(K) \ge 1 - \rho$.

Finite vectors. For real $f_1, \dots, f_r \in C_c^\infty$, the vectors $Y_i = (\int g_i f_1, \dots, \int g_i f_r)$ are independent, identically distributed, bounded and centered, with covariance $\Sigma_{kl} = \iint f_k(z) C_\mu^{\mathrm{end}}(z - z') f_l(z')\, dz\, dz'$ (Fubini). A Taylor expansion of $\mathbb{E}e^{it\cdot Y_i/\sqrt{N}}$ and Lévy's continuity theorem give $(F_N(f_1), \dots, F_N(f_r)) \Rightarrow (\gamma_\mu(f_1), \dots, \gamma_\mu(f_r))$.

Identification on E. Let $\mathcal{D}_E$ be a countable family of real compact smooth tests, dense in each $\mathcal{D}_K$; the evaluations $v \mapsto \int v d$, $d \in \mathcal{D}_E$, are continuous on E and separate its points. Let $H \in C_b(E)$, $\eta > 0$, and K as above. Around each $x \in K$ choose a relatively open $U_x \ni x$ in K on which $|H - H(x)| < \eta$. Some finite-coordinate cylinder $\{v \in K : |\int (v-x)d_i| < \epsilon_x,\ i \le r_x\}$ lies in U_x; otherwise a sequence in the closed set $K \setminus U_x$ would agree ever more closely with x on the first k coordinates, and a convergent subsequence would have the same coordinates as x, contradicting separation. Cover K by finitely many half-size cylinders, let π be the finite vector of all coordinates used, choose continuous $\alpha_i \ge 0$ on its range space supported in the full cylinders and positive on the half-size ones, put $S = \sum \alpha_i$, $A = \pi(K)$ and $G(y) = \sum_i H(x_i)\alpha_i(y)/(S(y) + \mathrm{dist}(y, A))$. Then G is continuous, $\|G\|_\infty \le \|H\|_\infty$ and $\sup_K |H - G\circ\pi| \le \eta$. Hence $|Q_N(H) - Q(H)| \le |Q_N(G\circ\pi) - Q(G\circ\pi)| + 2\eta + 4\|H\|_\infty\rho$, and the finite-vector convergence gives $Q_N \Rightarrow Q$ on E.

Full weight. Completing squares in x^2 gives $ux^4 - b_a x^2 + 3uc_a^2 \ge 3uc_a^2 - b_a^2/(4u)$ with $b_a = 6uc_a + \kappa_a^2/2$ on Λ, and $\varepsilon x^4 - \kappa_a^2 x^2/2 \ge -\kappa_a^4/(16\varepsilon)$ on $\Omega \setminus \Lambda$, so

$$0 < e^{-V_{\varepsilon,\Omega,a;\Lambda}} \le M_{\mathrm{full}} = \exp\{|\Lambda|[b_a^2/(4u) - 3uc_a^2] + |\Omega \setminus \Lambda|\kappa_a^4/(16\varepsilon)\}. \tag{18}$$

V is continuous on E (Proposition 5.1(vi)) and finite on every element. Applying $Q_N \Rightarrow Q$ to the bounded continuous functions e^{-V} and He^{-V} gives $Z_N^b \to Z_{\mathrm{full}} \in (0, \infty)$ and convergence of the normalized tilts on E; composing with the continuous ι gives the statement on X. □

9.2 The mass tilt and the residual laws

Proposition 9.3 (Whole-field mass-tilt identities)**.** *Fix a regulator a, a physical square Λ and $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$. For $\varepsilon \ge 0$, the residual weight satisfies $0 < e^{-W_{a,\Lambda} - \varepsilon J_{\Omega,\Lambda}} \le M_{\mathrm{res}} = e^{6u|\Lambda|c_a^2}$ on $C(\mathbb{R}^2)$, and $e^{-3u|\Lambda|c_a^2 - 3\varepsilon|\Omega\setminus\Lambda|c_a^2} \le z_{\varepsilon,\Omega,a;\Lambda} \le M_{\mathrm{res}}$. For $\varepsilon > 0$, $Z_{\mathrm{full}} = \mathrm{Det}_{\Omega,a} z_{\varepsilon,\Omega,a;\Lambda}$ and $\nu^v_{\varepsilon,\Omega,a;\Lambda} = \iota_* \mathrm{Res}_{\varepsilon,\Omega,a;\Lambda}$. These identities concern the whole field, including observations outside Ω.*

Proof. $x^4 - 6cx^2 + 3c^2 = (x^2 - 3c)^2 - 6c^2$ gives $W_{a,\Lambda} \ge -6u|\Lambda|c_a^2$, and $J \ge 0$. Under $G'_{\Omega,a}$ the coordinate $\gamma'(z)$ is centered Gaussian with variance $r(z) = C'_{\Omega,a}(z,z) \in [1/4, c_a]$ (Theorem 7.5), so $\mathbb{E}W_{a,\Lambda} = 3u\int_\Lambda (r - c_a)^2 \le 3u|\Lambda|c_a^2$ and $\mathbb{E}J = 3\int_{\Omega\setminus\Lambda} r^2 \le 3|\Omega \setminus \Lambda|c_a^2$; Jensen gives the lower bound. The functionals are continuous on $C(\mathbb{R}^2)$, hence Borel. Since $e^{-V} = e^{T_{\Omega,a}} e^{-W - \varepsilon J}$, Theorem 7.4(c)

with $F = e^{-W-\varepsilon J}\mathbf{1}_B \circ \iota \geq 0$ gives $\mathbb{E}_{G_\mu}[\mathbf{1}_B(\iota\gamma_\mu)e^{-V}] = \mathrm{Det}_{\Omega,a}\mathbb{E}_{G'}[\mathbf{1}_B(\iota\gamma')e^{-W-\varepsilon J}]$; $B = X$ gives the normalizer identity. □

9.3 The stabilizer and auxiliary-square arrows

Proposition 9.4 (Stabilizer arrow). *With a, Ω, Λ fixed, $\|\mathrm{Res}_{\varepsilon,\Omega,a;\Lambda} - \mathrm{Res}_{0,\Omega,a;\Lambda}\|_{\mathrm{var}} \to 0$ as $\varepsilon \downarrow 0$, where $\|\rho - \eta\|_{\mathrm{var}} = \sup_{|H|\leq 1 \text{ Borel}} |\rho(H) - \eta(H)|$; hence $\|\nu^{v}_{\varepsilon,\Omega,a;\Lambda} - \nu^{(2)}_{\Omega,a;\Lambda}\|_{\mathrm{var}} \to 0$, $z_\varepsilon \to z_0$ and $Z_{\mathrm{full}} \to \mathrm{Det}_{\Omega,a} z_0$.*

Proof. Under $G'_{\Omega,a}$ the weights $h_\varepsilon = e^{-W-\varepsilon J}$ increase pointwise to h_0 as $\varepsilon \downarrow 0$ and are bounded by M_{res}, so $\|h_\varepsilon - h_0\|_1 = z_0 - z_\varepsilon \to 0$ and $\|h_\varepsilon/z_\varepsilon - h_0/z_0\|_1 \leq 2\|h_\varepsilon - h_0\|_1/z_0$. The variation distance of the path laws equals this L^1 distance, and a measurable pushforward does not increase it. □

Proposition 9.5 (Auxiliary-square arrow). *With a regulator a and Λ fixed, along every increasing family of squares $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$ with $R \to \infty$, $\beta(\nu^{(2)}_{\Omega,a;\Lambda}, \nu_{a,\Lambda}) \leq e^{3u|\Lambda|c_a^2}(2e_\Omega + M_{\mathrm{res}} r_\Omega) \to 0$ and $|Z_{a,\Lambda} - z_{0,\Omega,a;\Lambda}| \leq e_\Omega \to 0$, where e_Ω and r_Ω are defined in the proof. For each Ω a separate coupling is used.*

Proof. Fix Ω and write $\Delta = \Delta_{\Omega,a}$. On a product space let γ' have law $G'_{\Omega,a}$ and let ψ_Ω be an independent centered Gaussian field with covariance Δ and continuous paths (Theorem 7.6(e)); independence of the two continuous paths follows from their countable dense point evaluations. Put $X_\Omega = \gamma' + \psi_\Omega$ and $Y_\Omega = \gamma'$. Then X_Ω is a continuous centered Gaussian field with covariance $C' + \Delta = C_a$, so its law on $C(\mathbb{R}^2)$ is that of $\tilde{\phi}_a$ (Theorem 7.9). On the common Gaussian space $\Phi_a = \iota\tilde{\phi}_a$ almost surely (Theorem 7.10(b)) and $U_{a,\Lambda} = W_{a,\Lambda}(\tilde{\phi}_a)$, so $\nu_{a,\Lambda}(B) = \mathbb{E}[\mathbf{1}_B(\iota X_\Omega)e^{-W(X_\Omega)}]/\mathbb{E}e^{-W(X_\Omega)}$ and $Z_{a,\Lambda} = \mathbb{E}e^{-W(X_\Omega)}$, while $\nu^{(2)}_{\Omega,a;\Lambda}(B) = \mathbb{E}[\mathbf{1}_B(\iota Y_\Omega)e^{-W(Y_\Omega)}]/z_0$.

Let $\delta_\Omega(K) = \sup_{z\in K} \Delta(z,z)$ for compact K. By Parseval, $p_j(\iota\psi) \leq \|\chi_j\psi\|_{L^2}$, so $\mathbb{E}p_j(\iota X_\Omega - \iota Y_\Omega)^2 \leq \|\chi_j\|_2^2 \delta_\Omega(\mathrm{supp}\,\chi_j)$ and $\mathbb{E}d(\iota X_\Omega, \iota Y_\Omega) \leq r_\Omega := \sum_j 2^{-j}\min(1, \|\chi_j\|_2 \delta_\Omega(\mathrm{supp}\,\chi_j)^{1/2})$, which tends to 0 by Theorem 7.6(d) and dominated convergence. For $p_c(x) = x^4 - 6cx^2 + 3c^2$, $|p_c(x) - p_c(y)| \leq |x-y|(|x|+|y|)(x^2+y^2+6c) \leq |x-y|[2(|x|^3+|y|^3) + 6c(|x|+|y|)]$. Each $X_\Omega(z)$, $Y_\Omega(z)$ has variance at most $c = c_a$, so the bracket has L^2 norm at most $L_c = (4\sqrt{15}+12)c^{3/2}$, and Cauchy–Schwarz gives $\mathbb{E}|W(X_\Omega) - W(Y_\Omega)| \leq u|\Lambda|L_c\,\delta_\Omega(\bar{\Lambda})^{1/2}$. Both interactions are at least $-6u|\Lambda|c^2$, where $x \mapsto e^{-x}$ is M_{res}-Lipschitz, so $\mathbb{E}|e^{-W(X_\Omega)} - e^{-W(Y_\Omega)}| \leq e_\Omega := M_{\mathrm{res}} u|\Lambda|L_c\delta_\Omega(\bar{\Lambda})^{1/2} \to 0$. For $|H| \leq 1$ and $\mathrm{Lip}_d(H) \leq 1$, adding and subtracting the cross terms gives $|\nu^{(2)}(H) - \nu_{a,\Lambda}(H)| \leq [2e_\Omega + \mathbb{E}(e^{-W(X_\Omega)}d(\iota X_\Omega, \iota Y_\Omega))]/z_0$, and $z_0 \geq e^{-3u|\Lambda|c^2}$ by Proposition 9.3. □

9.4 The regulator arrow

Proposition 9.6 (Regulator arrow). *With Λ fixed, $Z_{a,\Lambda} \to Z_{0,\Lambda} \geq 1$ and $\beta(\nu_{a,\Lambda}, \nu_{0,\Lambda}) \leq 2e_a + \Xi_2(\Lambda)^{1/2}h(a) \to 0$ as $a \downarrow 0$, where $e_a = \|e^{-U_{a,\Lambda}} - e^{-U_{0,\Lambda}}\|_{L^1(\mathbb{P}_0)} \leq 2K_\Lambda \Xi_2(\Lambda)^{1/2}\sqrt{a}$ and $h(a)$ is as in Theorem 7.10(e).*

Proof. Theorem 8.2(f) with $p = 1$ bounds e_a, so $|Z_a - Z_0| \leq e_a$, and $Z_a, Z_0 \geq 1$ by Theorem 8.2(g). For $|H| \leq 1$, $\mathrm{Lip}_d(H) \leq 1$, with $w_a = e^{-U_a}$, $|\mathbb{E}[H(\Phi_a)w_a - H(\Phi_0)w_0]| \leq e_a + \mathbb{E}[w_0 d(\Phi_a, \Phi_0)] \leq e_a + \Xi_2^{1/2}h(a)$ by Cauchy–Schwarz, Theorem 8.2(e) with $p = 2$ and Theorem 7.10(e); the normalizer correction is at most e_a. □

Remark 9.7 (The chain of laws). For fixed parameters the five laws $P^b_{N,\varepsilon,\Omega,a;\Lambda}$, $\nu^v_{\varepsilon,\Omega,a;\Lambda} = \iota_*\mathrm{Res}_\varepsilon$, $\nu^{(2)}_{\Omega,a;\Lambda} = \iota_*\mathrm{Res}_0$, $\nu_{a,\Lambda}$ and $\nu_{0,\Lambda}$ are joined by the four arrows of Propositions 9.2, 9.4, 9.5 and 9.6; there is no mass change, no factorial change, no omitted quadratic term and no extra normalization between them. The quadratic-tilt normalizer $\mathrm{Det}_{\Omega,a}$ is nondecreasing in Ω (Theorem 7.4(a)). No monotonicity in Ω is asserted for the full Gaussian normalizer $Z_{\mathrm{full}} = \mathrm{Det}_{\Omega,a} z_{\varepsilon,\Omega,a;\Lambda}$, and none holds in general at a fixed $\varepsilon > 0$. For $\Omega_1 = \bar{\Lambda}$ the outer integral $J_{\Omega_1,\Lambda}$ vanishes, so Z_{full} for Ω_1 does not depend on ε. For a strictly larger centered square Ω_2, the continuous endpoint Gaussian is almost surely nonzero at an interior point of $\Omega_2 \setminus \bar{\Lambda}$, so $J_{\Omega_2,\Lambda} > 0$ almost surely; since $e^{-W_{a,\Lambda}} e^{T_{\Omega_2,a}} \le e^{6u|\Lambda|c_a^2} e^{T_{\Omega_2,a}}$ is integrable, dominated convergence gives $Z_{\mathrm{full}} \to 0$ for Ω_2 as $\varepsilon \to \infty$. Only the residual normalizers $z_{0,\Omega,a;\Lambda}$, which satisfy $e^{-3u|\Lambda|c_a^2} \le z_{0,\Omega,a;\Lambda} \le e^{6u|\Lambda|c_a^2}$ for every Ω (Proposition 9.3), enter the auxiliary arrow. The identification of $\nu_{0,\Lambda}$ with the classical free-boundary law is Proposition 10.4.

10 The continuum target: reduced lemmas

The growth of the physical square is handled with classical input in three steps. Proposition 10.4 identifies the continuum law $\nu_{0,\Lambda}$ on compact smooth tests with the classical free-boundary law, so the classical theorems apply. Theorem 10.8 uses dilation (Lemma 10.5) to move (m, u) to the reference mass, where C2 makes the two classical states coincide and C1 gives convergence for finitely many smearings. Theorem 10.10 upgrades this to weak convergence on X with all compact moments, and Theorem 10.16 records the properties of the target.

This section proves, directly from C1–C6 and the foundations of Sections 5–9, the three continuum statements used by the construction: the continuum limit with its Wick identification (Theorem 10.8), the whole-field volume limit with all compact moments (Theorem 10.10) and the target properties (Theorem 10.16). Appendix D records the argument lineage.

10.1 Classical finite-volume laws and the Wick identification

For $\psi \in \mathcal{S}'(\mathbb{R}^2)$ and a real even nonnegative $\rho_1 \in C_c^\infty(\mathbb{R}^2)$ with $\int \rho_1 = 1$, put $\rho_\delta = \delta^{-2}\rho_1(\cdot/\delta)$ and $\psi_\delta(z) = \psi(\rho_\delta(z - \cdot))$.

Lemma 10.1 (Classical Wick existence and finite normalizers from standard tools)**.** *Let $r, b > 0$ and $B \subset \mathbb{R}^2$ be bounded measurable, with $A = |B|$. On the free Gaussian space $(\mathcal{S}', \mathcal{C}, P_r)$ the own-variance mollified quartics $b\int_B :\psi_\delta(z)^4: dz$ have a mollifier-independent limit $U^S_{r,b,B}$ in the fourth homogeneous Gaussian chaos in $L^2(P_r)$. It has mean zero and, for every real $t > 0$,*

$$\mathbb{E}_{P_r} e^{-tU^S_{r,b,B}} < \infty, \qquad 1 \le Z^S_{r,b,B} := \mathbb{E}_{P_r} e^{-U^S_{r,b,B}} < \infty.$$

The normalized density defines the classical finite-volume law in the convention of Section 3. If B is a centered axis-parallel square, this law is invariant under field sign and θ_1, θ_2. This lemma uses only the free covariance and C6; it does not use the Gaussian or Wick foundations of Sections 7–8. Every constant below is fixed-volume. If $A = 0$, the interaction is zero and all conclusions hold with $Z^S = 1$.

Proof. Assume $A > 0$ and write $c(p) = (r^2 + |p|^2)^{-1}$ and $C = \mathcal{I}c$. Polar integration and Hausdorff–Young give

$$\int_{\mathbb{R}^2} c(p)^{4/3} dp = 3\pi r^{-2/3}, \qquad \|C\|_4 \le M_4(r) := (2\pi)^{-3/2}(3\pi)^{3/4} r^{-1/2}. \tag{19}$$

For the real even mollifier fixed above, the cross-covariance of $\psi_\delta(z)$ and $\psi_\epsilon(z')$ is $C_{\delta\epsilon}(z-z') = (\rho_\delta * C * \rho_\epsilon)(z-z')$. Approximate-identity convergence gives $C_{\delta\epsilon} \to C$ in L^4; equivalently apply Hausdorff–Young and dominated convergence to $\hat\rho_1(\delta p)\hat\rho_1(\epsilon p)c(p)$. The multipliers are bounded uniformly, converge pointwise to one and are dominated by a constant times $c \in L^{4/3}$. The same argument works with two different real mollifiers of integral one. Since

$$\|f^4 - g^4\|_1 \le \|f-g\|_4(\|f\|_4^3 + \|f\|_4^2\|g\|_4 + \|f\|_4\|g\|_4^2 + \|g\|_4^3),$$

the fourth powers converge in L^1. The Wick covariance identity gives

$$\mathbb{E}J_\delta J_\epsilon = 4!b^2 \int_{B^2} C_{\delta\epsilon}(z-z')^4\,dz\,dz', \qquad J_\delta = b\int_B :\psi_\delta(z)^4:\,dz. \tag{20}$$

For integrable H, $\int_{B^2}|H(z-z')| \le A\|H\|_1$. Thus J_δ is Cauchy in L^2, with the same limit for every such mollifier. The integrands are continuous L^2 fourth-chaos variables; their Bochner integrals equal the pathwise integrals by Fubini, using finite Gaussian moments on B. Chaos closure proves fourth-chaos membership and mean zero. No point variable or product of distributions at zero cutoff is formed.

A cutoff with a lower bound. Fix a real radial $\chi \in C_c^\infty(\mathbb{R}^2)$ with $0 \le \chi \le 1$, $\chi = 1$ on the unit ball and $\chi = 0$ outside the ball of radius two. For $R \ge R_0 := \max(1, r)$, put $\chi_R(p) = \chi(p/R)$, $\psi_R(z) = \psi(\mathcal{I}\chi_R(z-\cdot))$ and $U_R = b\int_B :\psi_R(z)^4:_{c_R} dz$. The inverse transform is a real Schwartz function, so these are ordinary smoothed evaluations of ψ; their paths are smooth and jointly measurable. Their variance and lower bound are

$$c_R = (2\pi)^{-2}\int \chi_R^2 c \le \frac{1}{4\pi}\log(1+4R^2/r^2), \qquad U_R \ge -6bAc_R^2. \tag{21}$$

The last inequality is $(x^2-3c)^2 - 6c^2 \ge -6c^2$. Let $C_R^1 = \mathcal{I}(\chi_R c)$, $C_R^2 = \mathcal{I}(\chi_R^2 c)$ and $a_4 = (2\pi)^{-3/2}(3\pi)^{3/4}$. Polar integration of the tail gives, for $j = 1, 2$,

$$\|C - C_R^j\|_4 \le a_4 R^{-1/2}, \qquad \|C_R^j\|_4 \le M_4(r). \tag{22}$$

Indeed $|1-\chi_R^j| \le \mathbf{1}_{\{|p|>R\}}$ and $\int_{|p|>R} c^{4/3} \le 3\pi R^{-2/3}$. Taking the mollifier limit in the Gaussian cross-covariances yields

$$\begin{aligned}\|U_{r,b,B}^S - U_R\|_2^2 &= 4!b^2\int_{B^2}[C^4 + (C_R^2)^4 - 2(C_R^1)^4](z-z')\,dz\,dz' \\ &\le 288b^2AM_4(r)^3a_4R^{-1/2}.\end{aligned} \tag{23}$$

Here each fourth-power difference is bounded by four times $M_4(r)^3$ times its L^4 difference, and the coefficients of the two differences are one and two. Put $a_2 = b[288AM_4(r)^3a_4]^{1/2}$. Both interactions lie in the fourth chaos, so

$$\|U_{r,b,B}^S - U_R\|_p \le (p-1)^2 a_2 R^{-1/4} \quad (p \ge 2). \tag{24}$$

Negative tails and every needed normalization. The following explicit choices make the fixed-volume character of the constants unambiguous:

$$c_0 = \frac{1}{4\pi}\log(1+4R_0^2/r^2), \qquad d = \frac{\pi}{\sqrt{24bA}}, \qquad R(K) = R_0 e^{d\sqrt{K}},$$

$$K_0 = \max\{1, 32bAc_0^2, (16\log 2/d)^2, 2ea_2\}.$$

For $K \geq K_0$, the logarithmic variance bound implies $c_{R(K)} \leq c_0 + d\sqrt{K}/(2\pi)$ and hence $6bAc_{R(K)}^2 \leq 12bAc_0^2 + 3bAd^2K/\pi^2 \leq K/2$. On $\{U_{r,b,B}^S < -K\}$ the difference from $U_{R(K)}$ therefore has absolute value greater than $K/2$. Set $p = R(K)^{1/16} \geq 2$. Markov and (24) give

$$P_r(U_{r,b,B}^S < -K) \leq \left[\frac{2a_2}{K} R(K)^{-1/8}\right]^{R(K)^{1/16}} \leq \exp[-R(K)^{1/16}]. \tag{25}$$

The bracket is at most e^{-1} by the choice of K_0. For every fixed $t > 0$, the layer-cake formula bounds the exponential moment by

$$\mathbb{E}e^{-tU_{r,b,B}^S} \leq e^{tK_0} + \int_{K_0}^{\infty} te^{tK} \exp[-R_0^{1/16} e^{d\sqrt{K}/16}]\, dK < \infty. \tag{26}$$

The double exponential in $\sqrt{K}$ dominates every e^{tK}: for example $e^{d\sqrt{K}/16}/K \to \infty$. Jensen and $\mathbb{E}U_{r,b,B}^S = 0$ give $Z^S \geq 1$. Thus both the L^2 and finite-normalizer provisos in the source convention are proved before that convention is used. These estimates are separate from the constants K_Λ, Ξ_p of Theorem 8.2.

Finally P_r is invariant under field sign and Euclidean isometries, by its centered Gaussian covariance and cylinder-law uniqueness. With radial χ_R, the interaction U_R on a centered axis-parallel square is unchanged by sign and θ_1, θ_2. The same identities pass in L^2 to U^S, then to its exponential and its normalized law. This symmetry proof uses no foundation interaction or normalizer. □

Lemma 10.2 (Transfer between Gaussian realizations)**.** *Let $(\Gamma_0, \mathcal{F}_0, \mathbb{P}_0, W)$ be the common space of Theorem 7.7 at mass m and P_m the free measure on $(\mathcal{S}', \mathcal{C})$ of Section 3. There is a unique map Ξ from $L^0(\Gamma_0, \mathcal{F}_{\text{free}}^{\mathcal{S}}, \mathbb{P}_0)$, $\mathcal{F}_{\text{free}}^{\mathcal{S}} = \sigma(\phi(f) : f \in \mathcal{S}(\mathbb{R}^2;\mathbb{R})) \vee \mathcal{N}$, to $L^0(\mathcal{S}', \mathcal{C}, P_m)$ with $\Xi[g((\phi(f_j))_j)] = g((\psi(f_j))_j)$ for every sequence of real Schwartz f_j and every Borel $g : \mathbb{R}^{\mathbb{N}} \to \mathbb{C}$. It is linear and multiplicative, preserves joint laws of countable families, is an isometry of every L^r, commutes with L^r-limits and with Bochner integrals in L^2, and maps $:\xi^4:_{\text{Var}\,\xi}$ to $:(\Xi\xi)^4:_{\text{Var}\,\xi}$.*

Proof. For real Schwartz f, g, $\mathbb{E}_0[\phi(f)\phi(g)] = (2\pi)^{-2} \int \overline{\hat{f}}\hat{g}\hat{C}_0 = \mathbb{E}_{P_m}[\psi(f)\psi(g)]$ (Theorem 7.7), so for any two sequences the vectors $((\phi(f_j)), (\phi(f'_j)))$ and $((\psi(f_j)), (\psi(f'_j)))$ are centered Gaussian on $\mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$ with the same covariances, hence have the same law. If $g((\phi(f_j))) = g'((\phi(f'_j)))$ almost surely, the Borel set $\{g \neq g'\}$ therefore has measure zero for the ψ-vectors too, so Ξ is well defined. Every $Y \in L^0(\mathcal{F}_{\text{free}}^{\mathcal{S}})$ has such a representation: for a countable family $\mathcal{D}_S$ dense in $\mathcal{S}$, $\|\phi(f) - \phi(d)\|_2 \leq m^{-1}\|f - d\|_2$ (Theorem 7.8), so each $\phi(f)$ is an almost sure limit of $\phi(d_k)$, and Doob–Dynkin applies. Merging families gives linearity and multiplicativity; law preservation gives the isometries; a bounded linear map commutes with Bochner integrals; variances are preserved. □

Definition 10.3 (Classical finite-volume laws)**.** For a physical square Λ, the mollified own-variance Wick integrals $N_\delta^S = u\int_\Lambda :\psi_\delta(z)^4: dz$ (pathwise integrals of the continuous functions $z \mapsto \psi_\delta(z)$, equal to the corresponding Bochner integrals in $L^2(P_m)$) converge in $L^2(P_m)$ as $\delta \downarrow 0$ (Lemma 10.1); put $U_{m,u,\Lambda}^S = \lim_\delta N_\delta^S$, $Z_{m,u,\Lambda}^S = \mathbb{E}_{P_m} e^{-U^S}$ and $d\nu_{m,u,\Lambda} = (Z^S)^{-1} e^{-U^S} dP_m$. The normalizer is positive and finite by Lemma 10.1, independently of the project Wick foundations. By the convention paragraph of Section 3, this is the free-boundary finite-volume law of [8, (1.1), pp. 496–497] for $P(X) = uX^4$ at mass m, and the continuum law of [10, §IX] with boundary condition F.

Proposition 10.4 (Identification of the zero-regulator law)**.** *For every $m, u > 0$ and physical square Λ:*

(i) $U^S_{m,u,\Lambda}$ *exists, does not depend on* ρ_1*, and* $U^S = \Xi U_{0,\Lambda}$*; also* $Z^S_{m,u,\Lambda} = Z_{0,\Lambda} \in [1, \Xi_1(\Lambda)]$ *and* $\|d\nu_{m,u,\Lambda}/dP_m\|_{L^2(P_m)} \le \Xi_2(\Lambda)^{1/2}$*;*

(ii) for every finite tuple of real $f_1, \dots, f_r \in C_c^\infty(\mathbb{R}^2)$*, the law of* $(v(f_1), \dots, v(f_r))$ *under* $\nu_{0,\Lambda}$ *equals the law of* $(\psi(f_1), \dots, \psi(f_r))$ *under* $\nu_{m,u,\Lambda}$*;*

(iii) $\nu_{m,u,\Lambda}$ *is invariant under* $\psi \mapsto -\psi$ *and under* θ_1*,* θ_2*.*

Proof. (i) The family $\chi_\delta(p) = \overline{\hat\rho_1(\delta p)}$ is admissible (Definition 8.1), with $\phi^\chi_\delta(z) = \phi(\rho_\delta(z - \cdot))$, so by Theorem 8.2(b) the corresponding Bochner integrals N_δ on Γ_0 converge to $U_0 = U_{0,\Lambda}$ in $L^2(\mathbb{P}_0)$. Each N_δ and U_0 is $\mathcal{F}^{\mathcal{S}}_{\mathrm{free}}$-measurable (Theorem 8.2(b)), and $\Xi N_\delta = N^S_\delta$ by Lemma 10.2. Hence $N^S_\delta \to \Xi U_0$ in $L^2(P_m)$; the limit does not depend on ρ_1 because U_0 does not. Multiplicativity and law preservation give $\mathbb{E}_{P_m} e^{-pU^S} = \mathbb{E}_0 e^{-pU_0}$, and Theorem 8.2(e),(g) give the bounds. (ii) By Theorem 7.10(a), $\Phi_0(f_i) = \phi(f_i)$ almost surely; for bounded Borel G, $\mathbb{E}_0[G(\phi(f))e^{-U_0}] = \mathbb{E}_{P_m}[G(\psi(f))e^{-U^S}]$ by the lemma. (iii) P_m is invariant under $\psi \mapsto -\psi$ and under θ_j, and the mollified quartics are even. By (i) we may take ρ_1 radial; then $(\theta_j\psi)_\delta(z) = \psi_\delta(\theta_j z)$, and since Λ is θ_j-invariant (a centered axis-parallel square) the change of variables $z \mapsto \theta_j z$ shows that $N^S_\delta \circ \theta_j = N^S_\delta$; hence $U^S \circ \theta_j = U^S$ almost surely. □

Lemma 10.5 (Dilation)**.** *For* $s > 0$ *let* $(L_s f)(x) = s^{-2} f(x/s)$ *and* $(T_s\psi)(f) = \psi(L_s f)$ *on* $\mathcal{S}'$*. Then* $(T_s)_* P_M = P_{sM}$*; for* $s = m/M$*,* $\lambda = u/s^2$ *and every physical square* Λ*,* $U^S_{m,u,\Lambda} \circ T_s = U^S_{M,\lambda,s\Lambda}$ P_M*-almost surely and* $\nu_{m,u,\Lambda} = (T_s)_* \nu_{M,\lambda,s\Lambda}$*. With* $(A_b\psi)(f) = \psi(f(\cdot - b))$ *and* $(R\psi)(f) = \psi(f \circ R^{-1})$ *for orthogonal* R*:* $A_b T_s = T_s A_{sb}$ *and* $RT_s = T_s R$*;* T_s *commutes with* θ_1, θ_2 *and maps* $\mathcal{C}_{H^\theta_+}$ *into itself.*

Proof. This proof uses the classical L^2 limits and finite normalizers of Lemma 10.1, without the common-space identification. $\widehat{L_s f}(p) = \hat f(sp)$, so

$$\begin{aligned}\operatorname{Var}_{P_M} \psi(L_s f) &= (2\pi)^{-2} \int |\hat f(sp)|^2 (M^2 + |p|^2)^{-1} dp \\ &= (2\pi)^{-2} \int |\hat f(q)|^2 (s^2 M^2 + |q|^2)^{-1} dq.\end{aligned}$$

Next, $L_s[\rho_\delta(x - \cdot)] = \rho_{s\delta}(sx - \cdot)$, so $(T_s\psi)_\delta(x) = \psi_{s\delta}(sx)$; since $T_s\psi$ has law P_m under P_M, the Wick constants agree, $\operatorname{Var}_{P_m} \phi_\delta(x) = \operatorname{Var}_{P_M} \psi_{s\delta}(sx)$. The change of variables $y = sx$ in the Bochner integral gives $u \int_\Lambda {:}(T_s\psi)_\delta(x)^4{:}\, dx = \lambda \int_{s\Lambda} {:}\psi_{s\delta}(y)^4{:}\, dy$; as T_s is measure preserving from $(\mathcal{S}', P_M)$ to $(\mathcal{S}', P_m)$, the L^2 limits agree, and the normalized laws correspond. The intertwining relations follow from $L_s[f(\cdot - b)] = (L_s f)(\cdot - sb)$ and $L_s(f \circ R^{-1}) = (L_s f) \circ R^{-1}$; L_s preserves supports in $\{z_2 > 0\}$ and $\{z_1 > 0\}$. □

Lemma 10.6 (Readings of the convergence in C1(iv))**.** *Let* μ_n, μ *be probabilities on* $(\mathcal{S}', \mathcal{C})$ *and* $g \in C_c^\infty$ *real, and suppose* $\mathbb{E}_\mu e^{t\psi(g)} < \infty$ *for all real* t*. If* $\mathbb{E}_{\mu_n} e^{it\psi(g)} \to \mathbb{E}_\mu e^{it\psi(g)}$ *for every real* t*, or all moments of* $\psi(g)$ *converge, or* $\mathbb{E}_{\mu_n} e^{t\psi(g)} \to \mathbb{E}_\mu e^{t\psi(g)}$ *for all real* t*, then the laws of* $\psi(g)$ *converge weakly. For the states of C1 at mass* M*,* $\mathbb{E} e^{t\psi(g)} \le \|f_B\|_{L^2(P_M)} \exp(t^2 \langle g, (-\Delta + M^2)^{-1} g\rangle)$ *for every real* t *and every bounded open* $B \supseteq \operatorname{supp} g$*, by C1(ii) and Cauchy–Schwarz.*

Proof. Characteristic functions: Lévy's continuity theorem, applied with convergence at every real frequency t to the characteristic function of the law of $\psi(g)$ under μ. Moments: a law with finite exponential moments is determined by its moments, and moment convergence then implies weak convergence. Laplace transforms: convergence of moment generating functions on a neighbourhood of 0 implies weak convergence. □

Remark 10.7 (One frequency does not suffice). Convergence of $\mathbb{E}_{\mu_n} e^{i\psi(g)}$ at the single frequency $t = 1$ does not imply weak convergence of the laws of $\psi(g)$: for a nonzero real $g \in C_c^\infty$ and the regular tempered distribution $h = 2\pi g / \int g^2$, the probabilities $\mu_n = \delta_h$ and $\mu = \delta_0$ have equal characteristic values at frequency one, and μ has every exponential moment, while the laws of $\psi(g)$ are $\delta_{2\pi}$ and δ_0. An earlier version of Lemma 10.6 admitted that single-frequency alternative; it is withdrawn. Where the characteristic-function reading of C1(iv) is used (proof of Theorem 10.8), it is applied to $g = \sum_j t_j L_s f_j$ for every $t \in \mathbb{R}^r$, which supplies every real frequency.

Theorem 10.8 (Continuum limit and target; reduced lemma LR-R)**.** *Fix the admissible pair of Definition 3.2 and (m, u) in the regime; let $s = m/M_{\mathrm{ref}}$ and $\lambda = uM_{\mathrm{ref}}^2/m^2 \in (0, \lambda_*)$. The two states $\nu_\pm$ of C1(i) for λX^4 at mass M_{ref} coincide; call the common state v, and put $\nu^S = (T_s)_* v$. Then for every $\ell > 0$ and every finite tuple of real $f_1, \dots, f_r \in C_c^\infty(\mathbb{R}^2)$,*

$$\mathrm{Law}_{\nu_{m,u,\Lambda_n}}(\psi(f_1), \dots, \psi(f_r)) \Rightarrow \mathrm{Law}_{\nu^S}(\psi(f_1), \dots, \psi(f_r)), \qquad \Lambda_n = (-n\ell, n\ell)^2,$$

and therefore, by Proposition 10.4(ii), the laws of $(v(f_1), \dots, v(f_r))$ under $\nu_n = \nu_{0,\Lambda_n}$ converge to the same limit. The same ν^S serves all tuples and all ℓ.

Proof. At mass $M_{\mathrm{ref}} > M_0$ and $0 < \lambda < \lambda_* \le \lambda_0$, both ν_+ and ν_- belong to $\mathcal{G}_{\mathrm{ur}}$ (C2(i)), which is a singleton (C2(ii)); the states coincide as measures on $\mathcal{C}$ (only $\mathcal{C}$-measurable observables are used, so only the restrictions of the AHKZ members to $\mathcal{C}$ enter). By C1(iii) the pressure is differentiable at zero external field, and by C1(iv) the free measures $\nu_{M_{\mathrm{ref}},\lambda,\Lambda'}$ converge to v along every exhaustion by centered squares Λ', on characteristic functions of compact smooth tests (under the moment or Laplace readings, Lemma 10.6 gives weak convergence of the law of $\psi(g)$, hence convergence of $\mathbb{E}e^{i\psi(g)}$, for every real compact smooth g). For $t \in \mathbb{R}^r$ put $g = \sum_j t_j L_s f_j$, a real compact smooth test. By Lemma 10.5, $\mathbb{E}_{\nu_{m,u,\Lambda_n}} e^{i\sum_j t_j \psi(f_j)} = \mathbb{E}_{\nu_{M_{\mathrm{ref}},\lambda,s\Lambda_n}} e^{i\psi(g)} \to \mathbb{E}_v e^{i\psi(g)} = \mathbb{E}_{\nu^S} e^{i\sum_j t_j \psi(f_j)}$, since $s\Lambda_n = (-ns\ell, ns\ell)^2$ is an exhaustion by centered squares. Since t ranges over all of $\mathbb{R}^r$, this is convergence of the joint characteristic functions at every frequency; the limit is the characteristic function of $\mathrm{Law}_{\nu^S}(\psi(f_1), \dots, \psi(f_r))$, and Lévy's continuity theorem concludes. The classical finite-volume laws used here exist with finite normalizers by Lemma 10.1; that direct lemma, dilation and the readings argument use only C6 and the stated classical sources. Finally Proposition 10.4(ii) supplies the separate identification with ν_n on the common project space. Only this last bridge uses the Gaussian/Wick foundations. □

Remark 10.9 (Scope of Theorem 10.8). The classical target and finite-joint classical limit use only C1, C2, C6 and the direct classical Lemma 10.1. Its exponential normalizers follow from the explicit negative-tail proof, separately from L^2 convergence. The identification with the project common-space law additionally uses Proposition 10.4 and the Gaussian/Wick foundations. No neural law, stagewise neural bridge or midpoint approximation is used. The convergence asserted is finite-joint weak convergence for compact smooth tests; no moment, density, total-variation or rate conclusion is drawn from C1(iv).

10.2 The volume limit on X

Theorem 10.10 (Volume estimates and whole-field limit; reduced lemma V1-R)**.** *(i)* (E, fixed volume.) *For all $m, u > 0$, every physical square Λ, every nonnegative real $g \in C_c^\infty(\Lambda)$ and real t, $\mathbb{E}_{\nu_{m,u,\Lambda}} e^{t\psi(g)} \le \exp(t^2 \mathbb{E}_{\nu_{m,u,\Lambda}} \psi(g)^2/2)$.*

(ii) (E, uniform.) *Let (m,u) be in the regime, $g \geq 0$ real compact smooth and $N_g \geq 2$ an integer such that* $\operatorname{supp} g \subset \Lambda_n$ *for $n \geq N_g$ (the choice $N_g \geq 2$ makes the early-volume maximum below nonempty). With $R_g > 0$ such that* $\sup_{n \geq N_g} \nu_n(|v(g)| > R_g) < 1/12$ *and $V_g = 2R_g^2$: for $n \geq N_g$, real smooth f with $|f| \leq g$ and real t, $\mathbb{E}_{\nu_n} e^{tv(f)} \leq e^{V_g t^2/2}$, and $\mathbb{E}_{\nu_n}|v(f)|^p \leq 2(2V_g)^{p/2}\Gamma(1+p/2)$ for $p > 0$. Including the finitely many $n < N_g$, $\sup_{n \geq 1, |f| \leq g} \mathbb{E}_{\nu_n}|v(f)|^p \leq C_{p,g} < \infty$, with $C_{p,g} = \max\{2(2V_g)^{p/2}\Gamma(1+\frac{p}{2}),\ [2^p\Gamma(p+\frac{1}{2})/\sqrt{\pi}]^{1/2} m^{-p}\|g\|_2^p \max_{n<N_g} \Xi_2(\Lambda_n)^{1/2}\}$.*

(iii) (F.) *For $j \geq 1$ and $1 < s < 2$, $\sup_n \mathbb{E}_{\nu_n}\|\chi_j v\|^2_{H^{-s}(\mathbb{T}_j)} \leq B_{j,s} < \infty$, and every finite positive moment of this norm is bounded uniformly in n. The family (ν_n) is tight on X. There is a Borel probability $\nu = \nu_\infty^X$ on X and a $\mathcal{C}$-measurable map $\mathcal{R}$ from a set of full ν^S-measure into X with $\mathcal{R}(\psi)(f) = \psi(f)$ for all compact smooth f and $\mathcal{R}_*\nu^S = \nu$ (ν represents ν^S); and $\nu_n \Rightarrow \nu$ weakly on X. The target is almost surely locally H^{-s} for every $s > 1$.*

(iv) (M.) *For every finite tuple of real compact smooth tests and every real or complex polynomial Q, $\mathbb{E}_{\nu_n} Q(v(f_1), \dots) \to \mathbb{E}_\nu Q(v(f_1), \dots)$; all these expectations are finite, and every fixed positive power of $|Q|$ has expectation bounded uniformly in n.*

Proof. (i) Fix Λ of side L and meshes $\delta = L/(2k+1)$ as in C4. By C4 the lattice density is $\exp[-\sum_i (a_i q_i^4 + b_i q_i^2) + \sum_{i<j} J_{ij} q_i q_j]$ up to normalization, with $a_i = u\delta^2 > 0$, $b_i = (B_\delta)_{ii}/2 - 6u\delta^2 c_\delta \in \mathbb{R}$ and $J_{ij} = -(B_\delta)_{ij} \geq 0$. That the off-diagonal entries are nonpositive can be checked directly: with I the sites in Λ and O the others, $A_{OO}^{-1} = \sum_k (4 + m^2\delta^2)^{-1-k} \mathrm{adj}^k_{OO}$ has nonnegative entries, $A_{IO} \leq 0$, and $B_\delta = A_{II} - A_{IO}A_{OO}^{-1}A_{OI}$. In the ordered-pair form of C5 the pair coefficients are $a_{ij} = a_{ji} = J_{ij}/2$. For $\epsilon \in (0,1]$ replace them by $(J_{ij} + \epsilon)/2$; every fixed polynomial moment and the normalizer converge as $\epsilon \to 0$ by dominated convergence with the integrable majorant $|q|^k \prod_i e^{-a_i q_i^4 - b_i q_i^2} \exp(\sum_{i<j}(J_{ij}+1)|q_i||q_j| + \sum_i h_i|q_i|)$, so the third-cumulant inequality of C5 holds at $\epsilon = 0$. Let $Y_\delta = \delta^2 \sum_i g(\delta i) q_i$ and $F_\delta(t) = \log \mathbb{E}_\delta e^{tY_\delta}$, smooth by quartic confinement. For $t \geq 0$ the t-tilt adds the fields $h_i = t\delta^2 g(\delta i) \geq 0$, and $F_\delta'''(t)$ is the third cumulant of Y_δ, a nonnegative combination of mixed third cumulants, hence ≤ 0. Symmetry gives $F_\delta(0) = F_\delta'(0) = 0$ and $F_\delta''(0) = v_\delta = \mathbb{E}_\delta Y_\delta^2$; integrating three times, $F_\delta(t) \leq v_\delta t^2/2$ for $t \geq 0$, and for all t by evenness. By C4(i), on the common Gaussian realization the smearings and the interaction densities converge in every Gaussian L^p with positive normalizer limits; for bounded continuous H the difference of weighted expectations is controlled by the L^1 density error plus bounded convergence in probability, so the normalized lattice laws of Y_δ converge weakly to that of $\psi(g)$ under $\nu_{m,u,\Lambda}$ (Definition 10.3), and $v_\delta \to \mathbb{E}\psi(g)^2$. For the nonnegative continuous $y \mapsto e^{ty}$, the portmanteau inequality gives $\mathbb{E}e^{t\psi(g)} \leq \liminf_\delta \mathbb{E}_\delta e^{tY_\delta} \leq e^{t^2 \mathbb{E}\psi(g)^2/2}$.

(ii) By Theorem 10.8 the laws of $v(g)$ under ν_n converge, hence are tight, and R_g exists. Let $n \geq N_g$ and $v_n = \mathbb{E}_{\nu_n} v(g)^2$; the laws on compact smooth tests are those of ν_{m,u,Λ_n} (Proposition 10.4). By C4(ii) and the vanishing of odd moments, $\mathbb{E}v(g)^4 \leq 3v_n^2$. If $v_n > 0$, Paley–Zygmund gives $\nu_n(v(g)^2 > v_n/2) \geq \frac{1}{4}v_n^2/\mathbb{E}v(g)^4 \geq 1/12$; if $v_n > 2R_g^2$ this event lies in $\{|v(g)| > R_g\}$, a contradiction. So $v_n \leq V_g$, and (i) gives the bound for $f = g$. For $|f| \leq g$ put $a = (g+f)/2$, $b = (g-f)/2 \geq 0$; then $\mathbb{E}v(g)^{2r} - \mathbb{E}v(f)^{2r} = 2\sum_{k \text{ odd}} \binom{2r}{k} \mathbb{E}[v(a)^{2r-k} v(b)^k] \geq 0$ by the first Griffiths inequality C4(ii), and comparing the series of $\mathbb{E}\cosh(tv(f)) = \mathbb{E}e^{tv(f)}$ and $\mathbb{E}\cosh(tv(g))$ gives the bound for f. Chernoff gives $\nu_n(|v(f)| > z) \leq 2e^{-z^2/(2V_g)}$, and tail integration the moment bound. For $n < N_g$, Hölder with $\|d\nu_{m,u,\Lambda_n}/dP_m\|_2 \leq \Xi_2(\Lambda_n)^{1/2}$ (Proposition 10.4(i)) and the Gaussian moments with $\mathrm{Var}_{P_m}\psi(f) \leq m^{-2}\|g\|_2^2$ give the second entry of $C_{p,g}$.

(iii) Let $a_{j,k}(v) = v(\chi_j \overline{e_{j,k}})$ and $g_j = |\mathbb{T}_j|^{-1/2}\chi_j$; the real and imaginary parts of $\chi_j\overline{e_{j,k}}$ are dominated by g_j. Choose an integer $N_j \geq 2$ with $\operatorname{supp}\chi_j \subset \Lambda_n$ for $n \geq N_j$, and use $N_{g_j} = N_j$ in

(ii). For $n \geq N_j$, (ii) gives $\mathbb{E}_{\nu_n}|a_{j,k}|^2 \leq 2V_{g_j}$; for $n < N_j$, $\mathbb{E}_{\nu_n}|a_{j,k}|^2 \leq \sqrt{3}\,\Xi_2(\Lambda_n)^{1/2}m^{-2}\|\chi_j\|_2^2/|\mathbb{T}_j|$. Hence $\sup_{n,k}\mathbb{E}_{\nu_n}|a_{j,k}|^2 \leq A_j/|\mathbb{T}_j|$ with $A_j = \max\{2V_{g_j}|\mathbb{T}_j|, \sqrt{3}m^{-2}\|\chi_j\|_2^2 \max_{n<N_j}\Xi_2(\Lambda_n)^{1/2}\}$. The shell $|k|_\infty = r$ has $8r$ points with $\lambda_{j,k} \geq \pi^2 r^2/(j+1)^2$, so $S_{j,s} = \sum_k \lambda_{j,k}^{-s} < \infty$ exactly when $s > 1$, and Tonelli gives $\sup_n \mathbb{E}_{\nu_n}\|\chi_j v\|^2_{H^{-s}(\mathbb{T}_j)} \leq B_{j,s} := A_j S_{j,s}/|\mathbb{T}_j|$. For $p \geq 2$, (ii) gives $\|a_{j,k}\|_{L^p(\nu_n)} \leq 2C_{p,g_j}^{1/p}$ uniformly, and Minkowski in $L^{p/2}$ bounds $\mathbb{E}_{\nu_n}\|\chi_j v\|^p_{H^{-s}}$ by $2^p C_{p,g_j} S_{j,s}^{p/2}$; for $p < 2$ use Jensen.

Fix $\epsilon > 0$ and $K_\epsilon = \{v \in X : \|\chi_j v\|^2_{H^{-3/2}(\mathbb{T}_j)} \leq 2^j(B_{j,3/2}+1)/\epsilon \ \forall j\}$. Each extended norm is a supremum of continuous finite sums, hence lower semicontinuous, so K_ϵ is closed; Markov gives $\nu_n(X \setminus K_\epsilon) \leq \epsilon$. Let $(v_l) \subset K_\epsilon$. For each j the localized sequence is bounded in $H^{-3/2}(\mathbb{T}_j)$, and the embedding into $H^{-2}(\mathbb{T}_j)$ is compact (finite Fourier truncations, and a tail bounded by the $H^{-3/2}$ radius times $\sup \lambda^{-1/2}$ on the tail); a diagonal subsequence makes $(\chi_j v_l)_l$ convergent in $H^{-2}(\mathbb{T}_j)$ for every j, hence d-Cauchy, hence convergent in X by completeness (Proposition 5.1(ii)), with limit in K_ϵ. Thus K_ϵ is compact and (ν_n) is tight. By Prokhorov every subsequence has a further subsequence converging weakly to some μ; by continuity of evaluations and Theorem 10.8, μ has the finite-dimensional compact-test laws of ν^S.

Let $J_X : X \to \mathbb{R}^{\mathbb{N}}$ list the real and imaginary parts of all $a_{j,k}$; it is continuous and injective (Proposition 5.1(iii)), so by Lusin–Souslin $J_X(X)$ is Borel and J_X^{-1} is Borel on it. Let J_S be the same evaluations on $\mathcal{S}'$. Equal finite-dimensional laws give $(J_X)_*\mu = (J_S)_*\nu^S$, so $\nu^S(J_S \in J_X(X)) = 1$. Put $\mathcal{R} = J_X^{-1} \circ J_S$ on that set. For ψ in it, $\mathcal{R}(\psi)$ and ψ have the same localized Fourier coefficients, and the expansion in the proof of (16), valid for every distribution, gives $\mathcal{R}(\psi)(f) = \psi(f)$ for compact smooth f; and $\mathcal{R}_*\nu^S = \mu$. Since $\mathcal{B}(X) = \sigma(J_X)$, $(J_X)_*\mu$ determines μ; all subsequential limits coincide, and $\nu_n \Rightarrow \nu := \mathcal{R}_*\nu^S$. Lower semicontinuity and the portmanteau inequality transfer the H^{-s} bounds to ν.

(iv) Choose a real nonnegative compact smooth g with $|f_i| \leq g$ for all i. For a polynomial of degree d, $|Q(x)| \leq c_Q(1 + \sum_i |x_i|^d)$, so (ii) bounds $\sup_n \mathbb{E}_{\nu_n}|Q|^q$ for every $q > 0$. The finite-vector laws converge, the target has the corresponding moments by the portmanteau inequality, and continuous truncation at level T with tails at most a constant times T^{1-q} $(q > 1)$, uniformly in n and at the target, gives the convergence. □

Remark 10.11 (Scope of Theorem 10.10). Part (i) holds for all $m, u > 0$. Every volume-uniform statement uses the weak-coupling regime only through the tightness supplied by Theorem 10.8. The early volumes use their individual finite density norms; no bound uniform in n on the densities $d\nu_{m,u,\Lambda_n}/dP_m$ is used. The volume limit is classical continuum input to the construction, not a neural-estimate thermodynamic limit.

10.3 Target properties and reconstruction

Clause (D) of Theorem 4.1 identifies the limiting correlations with Schwinger distributions, and reconstruction C3(i) produces a relativistic quantum field from a measure satisfying four conditions: Euclidean invariance, time-reflection positivity, an exponential regularity bound and translation ergodicity. Theorem 10.16 verifies them in parts (i), (ii), (iv) and (iii), proves temperedness in (v) and non-Gaussianity in (vi), and applies reconstruction only afterwards, in (vii). Non-Gaussianity is the sense in which the reconstructed field is interacting (Remark 10.17).

Throughout, (m, u) is in the regime, s, λ, v and $\nu^S = (T_s)_*v$ are as in Theorem 10.8, and $M = M_{\mathrm{ref}}$. Time is z_2; Θ is the reflection $z_2 \mapsto -z_2$ acting on observables, $\Sigma_+ = \mathcal{C}_{\{z_2>0\}}$, and τ_r shifts an observable by r in the positive time direction.

Lemma 10.12 (Determining family)**.** *There is a countable $D \subset C_c^\infty(\mathbb{R}^2;\mathbb{R})$, independent of all parameters and dense in $\mathcal{S}(\mathbb{R}^2;\mathbb{R})$, with $\sigma\{T \mapsto T(d) : d \in D\} = \mathcal{C}$. Two probabilities on $\mathcal{C}$ with the same finite-dimensional D-laws coincide.*

Proof. Fix real smooth η, ζ with $\eta = 1$ on $[-1,1]^2$, $\operatorname{supp}\eta \subset (-2,2)^2$, $\zeta = 1$ on $[-2,2]^2$ and $\operatorname{supp}\zeta \subset (-3,3)^2$. Put $\eta_n(x) = \eta(x/n)$ and $\zeta_n(x) = \zeta(x/n)$. On the numerical torus with cell $(-3n,3n)^2$ let Q_n be the real trigonometric polynomials with rational coefficients in the sine/cosine basis of frequencies $\pi k/(3n)$, $k \in \mathbb{Z}^2$. Then $D = \bigcup_{n\ge1}\{\zeta_n P : P \in Q_n\}$ is countable, real compact smooth and independent of the model and localization parameters. For every real Schwartz f, the product rule and decay outside $[-n,n]^2$ give $\|(1-\eta_n)f\|_{(N)} \le C_N n^{-1}\|f\|_{(2N+1)}$ for $N \ge 1$. Thus $\eta_n f \to f$ in $\mathcal{S}$. At fixed n, the zero periodic extension of $\eta_n f$ is smooth; Fourier sums and rational approximation of their finite coefficients approximate it in C^∞. Multiplication by ζ_n fixes a compact support and the limit $\zeta_n\eta_n f = \eta_n f$. Choose P_n so that $\zeta_n P_n - \eta_n f$ is less than $1/n$ in the first n Schwartz seminorms. Then $d_n = \zeta_n P_n \to f$ in $\mathcal{S}$. Every $T \in \mathcal{S}'$ satisfies $T(f) = \lim_n T(d_n)$, so all Schwartz evaluations are measurable for $\sigma(D)$; the reverse inclusion is immediate. Finite D-cylinders form a generating π-system, so probability uniqueness follows by π–λ. □

Lemma 10.13 (Source-defined target and symmetries)**.** *The source-defined v and $\nu^S = (T_s)_* v$ of Theorem 10.8 are well defined from C1, C2, C6 alone, with the definitions of the admissible pair and dilation. They are invariant under field sign and the central coordinate reflections, and ν^S is invariant under the full Euclidean group. These conclusions hold under each admitted reading of C1. No foundation input is used here.*

Proof. At $M = M_{\mathrm{ref}} > M_0$ and $0 < \lambda = uM^2/m^2 < \lambda_* \le \lambda_0$, C2(i),(ii) make the two Gibbs extensions of the C1 states the same member μ of $\mathcal{G}_{\mathrm{ur}}$. Their cylinder restrictions therefore coincide as v. This identity uses neither a finite-volume neural law nor a volume-limit estimate. By C1(iii) the specified pressure is differentiable. Lemma 10.1 has independently proved the two provisos that license the classical free-boundary convention. Apply C1(iv) to those laws along centered axis-parallel square exhaustions. For its characteristic reading, apply the assertion to tg for every real t; Lévy's theorem gives the scalar weak limit. For the moment reading, C1(ii) and Gaussian Cauchy–Schwarz give

$$\mathbb{E}_v e^{t\psi(g)} \le \|f_B\|_2 e^{t^2\langle g,(-\Delta+M^2)^{-1}g\rangle} < \infty$$

for every real compact smooth g, all real t, and bounded open B containing its support. Moment determinacy and convergence then give that same scalar weak limit. For the Laplace reading, the moment-generating function theorem applies on a neighborhood of zero. Applying these arguments to every real linear combination of a fixed tuple, and then the finite-dimensional Lévy theorem, gives joint compact-test convergence in all three readings. This is exactly the classical portion of Theorem 10.8; the later common-space identification is not needed.

Lemma 10.1 gives finite-volume field-sign and central reflection invariance. Apply joint convergence to the union of a compact tuple and its transformed tuple; the symmetries pass to the corresponding v-laws. For real Schwartz tests, compact cutoffs converge in $\mathcal{S}$ and their evaluations converge pointwise on every tempered distribution. Bounded convergence of characteristic functions extends the identities to Schwartz cylinders; π–λ gives equality on $\mathcal{C}$. The positivity readings of C1 are unused in this step, and the three convergence readings were treated separately above.

Proper Euclidean invariance of v is directly C1(i)(a). The elementary identities $A_b T_s = T_s A_{sb}$ and $RT_s = T_s R$ of Lemma 10.5 transport it to ν^S. That lemma's Gaussian and Wick dilation identities use the direct construction of Lemma 10.1 and no foundation. Dilation commutes with

sign and θ_1, θ_2, so those symmetries transfer too. Proper motions and one central reflection generate every Euclidean isometry. In particular translation invariance has been proved before any translated local-density norm is used. □

Lemma 10.14 (Local densities of the target)**.** *For every bounded open B, the restriction of ν^S to $\mathcal{C}_B$ has density $F_B = f_{sB} \circ T_{1/s}$ with respect to P_m, and $\|F_B\|_{L^p(P_m)} = \|f_{sB}\|_{L^p(P_M)} < \infty$ for $1 \le p < \infty$. For an open centered axis-parallel square,*

$$F_B = e^{-U^S_{m,u,B} - \alpha_\infty |sB|} g_B, \qquad g_B = g_{\partial(sB)} \circ T_{1/s} > 0 \quad P_m\text{-a.s.}, \tag{27}$$

where g_B is measurable for either stated boundary convention, with that convention retained. Moreover $F_{B+k} = F_B \circ A_k$ P_m-a.s., so their finite L^p norms agree. For real $f_1, \dots, f_r$ supported in B and $t \in \mathbb{R}^r$, writing $q = \sum_i t_i f_i$,

$$\mathbb{E}_{\nu^S} e^{\psi(q)} \le \|F_B\|_2 e^{\langle q, C_{\mathrm{free}} q \rangle} < \infty. \tag{28}$$

Proof. T_r is a cylinder-measurable linear bijection with inverse $T_{1/r}$; L_r acts continuously on $\mathcal{S}$ and scales test supports from U to rU. For bounded $\mathcal{C}_B$-measurable H, Gaussian dilation gives

$$\int H F_B \, dP_m = \int (H \circ T_s) f_{sB} \, dP_M = \int H \circ T_s \, dv = \int H \, d\nu^S.$$

The same change of Gaussian law gives the exact equality of pth absolute integrals, with no probability-norm Jacobian. For a square, compose the C1(ii) factorization with $T_{1/s}$ and use $U^S_{M,\lambda,sB} \circ T_{1/s} = U^S_{m,u,B}$ from the direct mollified dilation in Lemma 10.5. The pressure factor is $\alpha_\infty |sB|$ at mass M, and positivity of g_B transports with Gaussian null sets.

BC1. For every open neighborhood V of ∂B, sV is an open neighborhood of $\partial(sB)$. The boundary factor pulled back by $T_{1/s}$ is therefore $\mathcal{C}_V$-measurable for every such V, hence measurable for their intersection. *BC2.* For real Schwartz h and $r > 0$,

$$\|L_r h\|^2_{H^{-1}} = (2\pi)^{-2} \int \frac{|\hat{h}(q)|^2}{r^2 + |q|^2} \, dq \le \max(1, r^{-2}) \|h\|^2_{H^{-1}}. \tag{29}$$

Thus dilation extends continuously on H^{-1} and scales distribution support by r. Gaussian evaluation at mass M is continuous $H^{-1} \to L^2(P_M)$, since its multiplier is at most $\max(1, M^{-2})(1 + |p|^2)^{-1}$; similarly at mass m. For h supported in $\partial(sB)$ and real Schwartz $h_j \to h$ in H^{-1}, the exact evaluation identity $\psi(h_j) \circ T_{1/s} = \psi(L_{1/s} h_j)$ passes in $L^2(P_m)$ to the extended Gaussian evaluations. The transformed test is supported in ∂B. Gaussian null sets also pull back to Gaussian null sets. This proves transport of the completed BC2 algebra, without identifying it with the uncompleted BC1 algebra.

Translated norms, after invariance. Lemma 10.13 already proves translation invariance of ν^S; the Gaussian law is stationary by its covariance. For bounded $\mathcal{C}_{B+k}$-measurable H,

$$\int H (F_B \circ A_k) \, dP_m = \int (H \circ A_{-k}) F_B \, dP_m = \int H \circ A_{-k} \, d\nu^S = \int H \, d\nu^S.$$

Local Radon–Nikodym uniqueness gives $F_{B+k} = F_B \circ A_k$ and hence the exact norm equality. No part of the preceding invariance proof used it. Finally monotone convergence extends the local density identity to the nonnegative exponential. Cauchy–Schwarz gives $\mathbb{E} F_B e^{\psi(q)} \le \|F_B\|_2 (\mathbb{E}_{P_m} e^{2\psi(q)})^{1/2} = \|F_B\|_2 e^{\langle q, C_{\mathrm{free}} q \rangle}$, with $\langle q, C_{\mathrm{free}} q \rangle \le m^{-2} \|q\|_2^2$. □

Lemma 10.15 (Local L^2 kernels)**.** *If a probability π on $(\mathcal{S}',\mathcal{C})$ has a density $F \in L^2(P_m)$ on $\mathcal{C}_B$, B bounded open, then for every integer $n \geq 1$ its local moment and truncated kernels on B^n are represented by real functions in $L^2(B^n)$, with complex-linear test pairings.*

Proof. Choose the local measurable version $F \geq 0$ with $\mathbb{E}_{P_m} F = 1$. Local moments exist by Cauchy–Schwarz and Gaussian polynomial moments; no moment outside B is required. Plancherel and polar integration give

$$\|C_{\mathrm{free}}\|^2_{L^2(\mathbb{R}^2)} = \frac{1}{4\pi m^2}, \qquad R_B^2 := \|C_{\mathrm{free}}(x-y)\|^2_{L^2(B^2)} \leq \frac{|B|}{4\pi m^2}. \tag{30}$$

The restricted covariance operator K on real $L^2(B)$ is positive self-adjoint with $\|K\| \leq m^{-2}$ by its Fourier form; its integral kernel is $C_{\mathrm{free}}(x-y)$ by Plancherel, first on smooth tests and then by continuity. Empty B is vacuous, so assume $|B| > 0$.

For a finite sum Q of real compact smooth product tests on B^k, define $W_k(f_1 \otimes \cdots \otimes f_k) = {:}\psi(f_1)\cdots\psi(f_k){:}$ linearly. The Wick covariance identity yields

$$\|W_k(Q)\|^2_{L^2(P_m)} = k!\langle \mathrm{Sym}\, Q, K^{\otimes k} \mathrm{Sym}\, Q\rangle \leq k! m^{-2k} \|Q\|^2_{L^2(B^k)}. \tag{31}$$

This also proves independence of the product-sum representation. Density extends W_k boundedly to $L^2(B^k)$. The functional $Q \mapsto \mathbb{E}_{P_m}[F W_k(Q)]$ has norm at most $\sqrt{k!}m^{-k}\|F\|_2$, so real L^2 duality produces a symmetric $h_k \in L^2(B^k)$ representing it, with that norm bound. Normalize the finite Lebesgue measure to a probability when applying C6 duality, then undo the normalization; the displayed bound is unchanged. Put $h_0 = 1$.

For a partial pairing P of $\{1,\dots,n\}$, write $U(P)$ for the unpaired indices. The ordinary moment kernel is

$$M_n(x) = \sum_P \left(\prod_{\{i,j\}\in P} C_{\mathrm{free}}(x_i - x_j) \right) h_{|U(P)|}(x_{U(P)}). \tag{32}$$

Each factor uses a disjoint set of coordinates, so its L^2 norm is $R_B^{|P|}\|h_{|U(P)|}\|_2$. In particular

$$\|M_n\|_2 \leq \|F\|_2 \sum_{p=0}^{\lfloor n/2 \rfloor} \frac{n!}{2^p p!(n-2p)!} R_B^p \sqrt{(n-2p)!}\, m^{-(n-2p)}.$$

To identify it, expand $e^{\sum_i t_i X_i} = e^{\frac{1}{2}\sum_{i,j} t_i t_j \mathbb{E} X_i X_j} e^{\sum_i t_i X_i - \frac{1}{2}\sum_{i,j} t_i t_j \mathbb{E} X_i X_j}$ for the finite Gaussian tuple $X_i = \psi(f_i)$ and compare the coefficient $t_1 \cdots t_n$. It gives each partial pairing once, with the remaining factors Wick ordered. Multiply by F, integrate and use the representation of h_k to obtain $\mathbb{E}_\pi \prod_i \psi(f_i) = \int M_n \prod_i f_i$ for product tests. Every integral is absolute by the displayed L^2 bounds. Product-test density gives the unique distribution on B^n.

For a set partition π_0 with blocks A, define

$$M_n^T(x) = \sum_{\pi_0} (-1)^{|\pi_0|-1}(|\pi_0|-1)! \prod_{A\in\pi_0} M_{|A|}(x_A). \tag{33}$$

Every block product is an L^2 tensor product on disjoint coordinates, of norm $\prod_A \|M_{|A|}\|_2$. The finite cumulant partition identity and Fubini identify its product-test pairing with the joint cumulant. Density again gives uniqueness. Real kernels extend complex-linearly, as required; no point field of the unregulated distribution is used. □

Theorem 10.16 (Target properties; reduced lemma CT-R). *(i) ν^S is invariant under the full Euclidean group of $\mathbb{R}^2$ and under $\psi \mapsto -\psi$.*

(ii) For every line L, with Θ_L the reflection in L and Σ_+^L generated by smearings supported in one open half-plane bounded by L: $\int \overline{\Theta_L F}\, F\, d\nu^S \geq 0$ for every $F \in L^2(\Sigma_+^L, \nu^S)$ and every Σ_+^L-measurable F with $\overline{\Theta_L F} F$ integrable.

(iii) For $G, G' \in L^2(\mathcal{S}', \mathcal{C}, \nu^S)$, $\mathbb{E}[G\,(G' \circ A_b)] \to \mathbb{E}G\,\mathbb{E}G'$ as $|b| \to \infty$ (strong mixing). Consequently every $E \in \mathcal{C}$ with $\nu^S(E \triangle A_b^{-1}E) = 0$ for all $b \in \mathbb{R}^2$ has $\nu^S(E) \in \{0, 1\}$ (translation ergodicity), and $\mathbb{E}[\overline{\Theta F}\, \tau_r G] \to \overline{\mathbb{E}F}\, \mathbb{E}G$ for $F, G \in L^2(\Sigma_+)$.

(iv) With a real nonnegative smooth partition of unity $\sum_{k\in\mathbb{Z}^2} \vartheta_k = 1$, $\vartheta_k = \vartheta_0(\cdot - k)$, $\mathcal{A}(f) = \sum_k \|\vartheta_k f\|_2$ is a norm continuous on $\mathcal{S}$, and with $K = \|F_B\|_{L^2(P_m)} \geq 1$ for a square $B \supseteq \operatorname{supp}\vartheta_0$ and $D = 3^{1/4}K^{1/2}/m$: $\mathbb{E}e^{t\psi(f)} \leq K\exp(t^2\mathcal{A}(f)^2/m^2)$, $\|\psi(f)\|_{L^2} \leq D\mathcal{A}(f)$, and $|\mathbb{E}e^{\psi(f)} - \mathbb{E}e^{\psi(g)}| \leq 2DK^{1/2}e^{2/m^2}\mathcal{A}(f-g)$ for $\mathcal{A}(f), \mathcal{A}(g) \leq 1$. Hence condition (c) of C3(i) holds with $|||\cdot||| = \mathcal{A}$.

(v) For every k there is a unique $\operatorname{Schw}_k \in \mathcal{S}'(\mathbb{R}^{2k})$ with $\operatorname{Schw}_k(f_1 \otimes \cdots \otimes f_k) = \int \prod_r \psi(f_r)\, d\nu^S$ for $f_r \in \mathcal{S}(\mathbb{R}^2; \mathbb{C})$ (complex-linear extension); it is symmetric and real on real tests.

(vi) ν^S is not Gaussian; its local moment and truncated kernels lie in $L^2(B^n)$ for bounded B; and some even truncated Schwinger distribution of order at least four is nonzero on real tests supported in pairwise disjoint time slabs.

(vii) By C3(i), applied with (i), (ii), (iv) and (iii): the moments of ν^S are the Schwinger functions of a unique relativistic quantum field theory in $1+1$ dimensions satisfying all the Wightman axioms (temperedness, Poincaré invariance, positivity, spectrum condition, locality, clustering), with a unique physical vacuum; its field fails Gaussian factorization.

Proof. All parts of this proof use only C1–C3 and C6, with the source-defined target identity. The proofs of Lemmas 10.1, 10.5, 10.13, 10.14 and 10.15 have supplied the needed classical objects directly. Their use here does not import the separate common-space Gaussian/Wick foundations.

(i), invariance. This is Lemma 10.13, proved without translated-density norms, local kernels, mixing or reconstruction.

(ii), full measurable reflection positivity. Under the measure reading C1(i)(b_M), the L^2 time-reflection inequality for v transfers directly by positive dilation: T_s commutes with Θ and its pullback preserves the positive-half-plane algebra. Under the weaker reading C1(i)(b_P), positive dilation first transfers exactly the inequality for polynomials whose individual monomials have strictly ordered positive-time supports. Reflection is an $L^2(\nu^S)$ isometry by (i).

Let B be bounded open and let $V_n(G)$ be the polynomial associated to a finite sum G of compact smooth product tests on $(B \cap H_+)^n$. The local $2n$-point moment kernel of Lemma 10.15 gives

$$\|V_n(G)\|^2_{L^2(\nu^S)} = M_{2n}(\overline{G} \otimes G) \leq \|M_{2n}\|_{L^2(B^{2n})}\|G\|^2_{L^2(B^n)}. \tag{34}$$

Thus this map extends boundedly from product sums to $L^2((B \cap H_+)^n)$. Products of real smooth tests with pairwise disjoint positive-time supports span a dense subspace there. Indeed the union of equal-time hyperplanes has Lebesgue measure zero; its complement has a countable cover by relatively compact product boxes with disjoint positive time projections. A function orthogonal to all smooth products on those boxes vanishes almost everywhere on each box, hence on the whole

complement. For each such product, reorder the commuting field factors into increasing time order. It is then exactly a monomial admitted in (b_P). The bounded map above therefore approximates every compact positive-half-plane polynomial in $L^2(\nu^S)$ by admitted polynomials.

Every finite vector of compact evaluations has finite exponential moments in all real directions by (28). The polynomial density theorem of C6 makes its polynomials dense in its L^2 space. For every positive-half-plane Schwartz test, compact spatial cutoffs converge in $\mathcal{S}$, keep support in the open half-plane, and converge in evaluation pointwise on $\mathcal{S}'$. Thus compact evaluations generate the stated Σ_+. Countable dependence supplies a countable family for each observable; conditional-expectation approximation for increasing finite families and truncation now make the compact polynomials dense in $L^2(\Sigma_+, \nu^S)$. If $F_j \to F$ there, reflection isometry and Cauchy–Schwarz give

$$\left|\mathbb{E}\overline{\Theta F_j} F_j - \mathbb{E}\overline{\Theta F} F\right| \le \|F_j - F\|_2(\|F_j\|_2 + \|F\|_2).$$

Positivity therefore holds for the full L^2 class under both readings. For a measurable F whose reflected product is integrable, use $F_j = F\mathbf{1}_{\{|F|\le j\}}$: the reflected products converge pointwise and are dominated by $|\Theta F|\,|F|$. Bounded positivity and dominated convergence prove the additional integrable-form clause. For any line, conjugate the time-line inequality by a Euclidean isometry using (i). This concerns the target; the finite neural stages retain only their two stated central coordinate lines.

(iii), Euclidean mixing and ergodicity before reconstruction. At mass M use the unique Gibbs extension μ of v supplied by C2. Let $D \in F_0$ be its increasing square family and $K_D F(\eta) = E^{\eta}_{D^c} F$. For an integrable local polynomial F, integrability follows from C1(ii) and Gaussian moments, and the Gibbs identity gives $\mu(K_D F) = \mu(F)$. If H is bounded and Σ_{D^c}-measurable, the kernel's point-fixing property applied to H and $|H|^2$ shows that H equals the constant $H(\eta)$ under each kernel probability (apply to real and imaginary parts). Hence $K_D(HF) = HK_D F$ wherever the integral of $|F|$ is finite. The Gibbs identity for $|F|$ gives that finiteness μ-a.e.; its identity for HF yields

$$\mu(HF) = \mu(HK_D F). \tag{35}$$

For a real local polynomial F, C2(iii) with both Gibbs measures equal to μ and Jensen imply

$$\|K_D F - \mu F\|_1 \le \iint |K_D F(\eta) - K_D F(\eta')|\,\mu(d\eta)\mu(d\eta') \longrightarrow 0. \tag{36}$$

Complex polynomials follow by real and imaginary parts. Thus

$$|\mu(FH) - \mu F\,\mu H| \le \|H\|_\infty \|K_D F - \mu F\|_1.$$

For a fixed bounded compact cylinder G, $H = G \circ A_b$ is Σ_{D^c}-measurable for all sufficiently large $|b|$: its finitely many translated compact supports lie outside $\bar{D}$, hence in every open neighborhood used in the closed-set outside algebra. This condition holds uniformly over the direction of b. First choose D sufficiently far along F_0, then let $|b|$ grow. Translation invariance of v gives $\mu(G \circ A_b) = \mu G$, so mixing holds for each such F, G. Local polynomials are dense in $L^2(v)$ by the local exponential moments, finite-dimensional polynomial density and increasing finite-coordinate conditional expectations. Compact evaluations generate $\mathcal{C}$ by Lemma 10.12. Bounded compact cylinders are also dense by the same conditional expectations and truncation. Translations preserve $L^2(v)$ norms, so Cauchy–Schwarz extends the limit to all complex $G, G' \in L^2(v)$, in every direction.

The identity $A_b T_s = T_s A_{sb}$ transfers this strong mixing to ν^S, since $s > 0$ implies $|sb| \to \infty$. For an invariant event E as in the statement, take both observables to be $\mathbf{1}_E$. The mixing product

equals $\nu^S(E)$ for every b and tends to $\nu^S(E)^2$; its probability is therefore zero or one. The half-plane clustering formula follows by taking the first observable $\overline{\Theta F}$ and translating the second in positive time; (i) gives $\mathbb{E}\Theta F = \mathbb{E}F$. No reconstruction theorem, tube uniqueness, spectrum condition, unique-vacuum assertion, FS Theorem 4.4 or OS-energy identification has entered this argument. Condition C3(i)(d) is established at this point.

(iv), the exact norm and exponential regularity. Fix a real nonnegative $\vartheta_0 \in C_c^\infty$ with $\sum_{k\in\mathbb{Z}^2} \vartheta_0(\cdot - k) = 1$, put $\vartheta_k = \vartheta_0(\cdot - k)$ and $\mathcal{A}(f) = \sum_k \|\vartheta_k f\|_2$. The partition identity gives a norm. For each integer $N > 2$, bounded support of ϑ_0 gives $\|\vartheta_k f\|_2 \le c_N(1+|k|)^{-N}\|f\|_{(N)}$; summation proves finiteness and Schwartz continuity. Bounded overlap of the translated supports, the product rule and arbitrarily higher Schwartz decay show that the partition sums converge to f in every Schwartz seminorm. Choose a bounded open square $B \supset \operatorname{supp}\vartheta_0$ and set $K = \|F_B\|_2 \ge \mathbb{E}F_B = 1$, $D = 3^{1/4}K^{1/2}/m$. The translated-density norm equality has already been proved after invariance. For real g supported in $B + k$, the local density and free Gaussian formulas give

$$\mathbb{E}e^{t\psi(g)} \le Ke^{t^2\|g\|_2^2/m^2}, \qquad \mathbb{E}\psi(g)^2 \le K(\mathbb{E}_{P_m}\psi(g)^4)^{1/2} \le \sqrt{3}Km^{-2}\|g\|_2^2. \tag{37}$$

Minkowski bounds each finite partition sum in L^2 by $D\sum_k \|\vartheta_k f\|_2$. The same estimate for its tails makes it Cauchy in L^2; its pointwise Schwartz-evaluation limit is $\psi(f)$. Thus $\|\psi(f)\|_2 \le D\mathcal{A}(f)$ for every real Schwartz f.

Let $a_k = \|\vartheta_k f\|_2$ and, when $\mathcal{A}(f) > 0$, put $\alpha_k = a_k/\mathcal{A}(f)$ for the nonzero terms. For any finite set J, generalized Hölder with weights α_k (including a factor one with the unused weight) and (37) imply

$$\begin{aligned}\mathbb{E}\exp\left(t\sum_{k\in J}\psi(\vartheta_k f)\right) &\le K^{\sum_J \alpha_k}\exp\left(t^2m^{-2}\sum_J a_k^2/\alpha_k\right)\\ &\le K\exp(t^2\mathcal{A}(f)^2/m^2).\end{aligned}$$

Fatou passes this to the pointwise full sum. The zero-norm case is immediate. For $\mathcal{A}(f), \mathcal{A}(g) \le 1$, use $|e^x - e^y| \le |x-y|(e^x + e^y)$ and Cauchy–Schwarz, with $\|e^{\psi(f)}\|_2 \le K^{1/2}e^{2/m^2}$, to obtain exactly

$$|\mathbb{E}e^{\psi(f)} - \mathbb{E}e^{\psi(g)}| \le 2DK^{1/2}e^{2/m^2}\mathcal{A}(f-g). \tag{38}$$

The unit ball has exponential bound Ke^{1/m^2} and this Lipschitz constant, so C3(i)(c) holds. For later use, $\mathbb{E}e^{|\psi(f)|/\mathcal{A}(f)} \le 2Ke^{1/m^2} =: K'$ when $f \ne 0$, whence $\|\psi(f)\|_k \le (k!K')^{1/k}\mathcal{A}(f)$ for integers $k \ge 1$.

(v), tempered Schwinger distributions. For a k-linear form M on $\mathcal{S}(\mathbb{R}^2;\mathbb{C})^k$ with $|\mathsf{M}(f_1,\dots,f_k)| \le C\prod_i \|f_i\|_{(N)}$, there is a unique tempered kernel. Here are the required reduction details. For $k = 1$ it is the definition. Inductively, at a fixed last test ϕ, apply the $(k-1)$-variable assertion to obtain $\mathsf{L}(\phi)$. Kernel uniqueness makes L linear. For a tensor sum Ψ_j, $\phi \mapsto \mathsf{L}(\phi)(\Psi_j)$ is continuous. Approximate an arbitrary Schwartz Ψ by these sums; the values converge pointwise because each $\mathsf{L}(\phi)$ is a distribution. Banach–Steinhaus on the Fréchet space $\mathcal{S}(\mathbb{R}^2)$ makes the limiting functional continuous. C3(iii), with $N = 2(k-1)$ and $n = 2$, now gives the joint tempered kernel. Product-test density gives uniqueness. Apply this to $\mathsf{M}_k(f_1,\dots,f_k) = \mathbb{E}\prod_i \psi(f_i)$. Hölder and (iv) give $|\mathsf{M}_k| \le k!K'\prod_i \mathcal{A}(f_i)$ for real tests, and $\mathcal{A}(f) \le C_{\mathcal{A}}\|f\|_{(3)}$. Splitting complex tests into real and imaginary parts gives the required complex multilinear bound. Kernel uniqueness transfers symmetry and reality of the moments to $\mathrm{Schw}_k \in \mathcal{S}'_{\mathbb{C}}(\mathbb{R}^{2k})$.

(vi), a local quartic conditional density excludes Gaussianity. Take an open centered axis-parallel square B and nonzero real $h \in C_c^\infty(B)$. Put

$$k = (-\Delta + m^2)h, \quad q = \int hk = \int (|\nabla h|^2 + m^2 h^2) > 0, \quad X = \psi(k)/q, \quad \eta = \psi - Xh.$$

Under P_m, $\operatorname{Var} X = 1/q$ and $\operatorname{Cov}(X, \psi(f)) = q^{-1} \int hf$, so $\operatorname{Cov}(X, \eta(f)) = 0$ for every real Schwartz f. Gaussianity and the monotone-class theorem make X independent of the full residual evaluation algebra. Its law is $\gamma_q(dx) = (q/(2\pi))^{1/2} e^{-qx^2/2} dx$.

The boundary factor g_B in (27) has a residual-measurable version. For BC1 choose a boundary neighborhood disjoint from $\operatorname{supp} h$; each generating evaluation there is unchanged by replacing ψ by η. For BC2, boundary-supported $f \in H^{-1}$ has $\langle f, h \rangle = 0$. If real Schwartz $f_j \to f$ in H^{-1}, then $\langle f_j, h \rangle \to 0$ by $h \in H^1$, so the defining $L^2(P_m)$ limit of $\psi(f_j)$ is also the limit of $\eta(f_j)$. Gaussian completion changes versions only on null sets. This covers each stated boundary algebra separately.

For the fixed mollifier, $\psi_\delta = \eta_\delta + X h_\delta$, $h_\delta = \rho_\delta * h$. Let $H_j(x; 1/q)$ be the monic Hermite polynomials for γ_q. At each positive mollifier scale,

$$J_\delta = u \int_B :\psi_\delta^4: = \sum_{j=0}^{4} A_{j,\delta}(\eta) H_j(X; 1/q).$$

Its residual coefficients are square integrable by finite Gaussian moments and Minkowski on bounded B. Independence and Hermite orthogonality give

$$\|J_\delta - J_\epsilon\|_2^2 = \sum_{j=0}^{4} j! q^{-j} \|A_{j,\delta} - A_{j,\epsilon}\|_2^2. \tag{39}$$

The direct L^2 Wick limit therefore gives residual coefficients A_j with

$$U^S_{m,u,B} = \sum_{j=0}^{4} A_j(\eta) H_j(X; 1/q), \qquad A_4 = u \int_B h^4 =: c > 0. \tag{40}$$

This is an actual degree-four polynomial in the scalar X, with deterministic leading coefficient c; it is obtained without multiplying unregulated distributions.

Suppose ν^S were Gaussian. The local measurable density F_B depends on a countable family $\psi(f_j)$, $f_j \in C_c^\infty(B; \mathbb{R})$; include k in this list. Write $Y = (\eta(f_j))_j$ and $F_B = b(X, Y)$. Both X and each Y_j are local linear evaluations. Under P_m, X is independent of Y, and the target law of (X, Y) has density $b(x, y)$ relative to $\gamma_q(dx) P_Y(dy)$, by the local Radon–Nikodym identity. The full residual variables used by $A_0, \dots, A_4, g_B, Y$ depend on a countable residual family too. They need not all be measurable for the smaller local family Y. Independence of X from that full residual family and Fubini give, for almost every full residual configuration v_0,

$$\log b(x, Y(v_0)) = -\sum_{j=0}^{4} A_j(v_0) H_j(x; 1/q) - \alpha_\infty |sB| + \log g_B(v_0) \tag{41}$$

for Lebesgue-a.e. x. This uses the square factorization, not a global target density. In particular $b > 0$ almost everywhere. The integral $\int b(x, y) \gamma_q(dx)$ is positive and finite for P_Y-a.e. y by positivity and $\mathbb{E} F_B = 1$; the target marginal of Y is equivalent to P_Y. The conditional target density of X at such y is proportional to $b(x, y) e^{-qx^2/2}$.

Under the assumed Gaussian target, every finite vector $(X, Y_1, \dots, Y_j)$ is Gaussian. Finite Gaussian regression gives conditional characteristic functions $e^{itm_j - t^2 v_j/2}$, where $m_j = \mathbb{E}_{\nu^S}[X \mid Y_1, \dots, Y_j]$ converges in L^2 by the increasing-filtration theorem and the deterministic $v_j \geq 0$ decrease. Applying the same conditional-expectation theorem to e^{itX} identifies the conditional characteristic function given Y as $e^{it\mathbb{E}[X|Y] - t^2 v/2}$, $v = \lim v_j \geq 0$, first for rational t and then by continuity. Thus it is a Gaussian conditional law, possibly degenerate. Its exceptional Y set is null for P_Y by equivalence. Intersect with the full-measure residual set in (41). A positive Lebesgue density excludes zero variance. For positive variance its log density is quadratic, whereas (41) plus $-qx^2/2$ has leading coefficient $-c < 0$ in degree four. Equality a.e. would make the polynomials identical, which is impossible. This proves non-Gaussianity while preserving the local/full-residual distinction and the fixed boundary trace.

The local moment and truncated L^2 kernels for every bounded open B and every order are Lemmas 10.14 and 10.15. Field-sign invariance annihilates every odd joint cumulant. If all even compact cumulants of order at least four vanished as well, the moment-cumulant formula would leave only pair partitions. Local exponential moments and moment determinacy would make every compact linear evaluation centered Gaussian, hence every finite compact vector jointly Gaussian. For any real Schwartz test take compact cutoffs converging in $\mathcal{S}$; their evaluations converge pointwise on $\mathcal{S}'$, hence weakly in law. Centered Gaussian laws are closed under such weak convergence: unbounded variances would force a discontinuous limiting characteristic function at zero; a bounded-variance subsequence has a Gaussian limit. Apply this to each real linear combination to obtain Gaussian Schwartz vectors, contradicting the preceding argument. This passage needs no global neural Schwartz estimate.

Consequently some even $n \geq 4$ has a nonzero compact joint cumulant. Choose bounded open B containing its supports. Its real kernel $M_n^T \in L^2(B^n)$ is nonzero. The union of hyperplanes $x_{i,2} = x_{j,2}$ is Lebesgue null. The complement has a countable cover by relatively compact product boxes $U_1 \times \cdots \times U_n$ whose time projections lie in pairwise disjoint bounded open intervals. At least one box has nonzero kernel restriction; otherwise the squared integral would vanish on that countable cover. Density of real smooth product tests in its L^2 space then gives individual $f_i \in C_c^\infty(U_i; \mathbb{R})$ with

$$S_n^T(f_1, \dots, f_n) = \int M_n^T(x) \prod_i f_i(x_i)\, dx \neq 0. \tag{42}$$

Thus the witness is a real tensor product in disjoint time slabs at the same even order $n \geq 4$, rather than only a general joint test or a distribution on coincident times.

(vii), reconstruction and its precise non-Gaussian consequence. For ν^S on $(\mathcal{S}', \mathcal{C})$, condition (a) of C3(i) is (i), condition (b) is the full L^2 time-line assertion of (ii) under either positivity reading, condition (c) is the exact norm and estimates of (iv), and condition (d) was proved by Euclidean mixing in (iii). Apply C3(i) only now. It gives the stated unique scalar relativistic theory in $1+1$ dimensions, the listed Wightman axioms and unique physical vacuum; (v) identifies its tempered symmetric Schwinger representatives.

Only after this application use C3(ii),(ii'). On a common ordered tube, form the truncated holomorphic n-point function by the finite cumulant partition sum of the holomorphic moment functions, preserving the order within each block. Every factor is holomorphic there and uses disjoint variables from the other block factors. Its distribution boundary is the truncated Wightman distribution: the block products are tensor products, so no undefined multiplication of coincident-variable distributions is involved. At pairwise distinct Euclidean arguments its values are the truncated Schwinger function by C3(ii) and the same partition identity. If its Wightman boundary

vanished, C3(ii') would force the holomorphic function to vanish and hence its analytic continuation at those Euclidean points. Permute the disjoint time slabs of (42) into one strict order; symmetry of the Schwinger kernels preserves that nonzero pairing. Vanishing analytic values would make the pairing zero, a contradiction. The truncated Wightman distribution at this same even order $n \geq 4$ is therefore nonzero. In particular the field fails Gaussian factorization at some order at least three, exactly the claimed consequence. No scattering or quantitative mass-gap conclusion follows from this argument. □

Remark 10.17 (Order of the argument). Ergodicity, condition (d), is established in (iii) by a Euclidean mixing argument from the Gibbs identity and the boundary-condition independence of C2, before [8, Thm. 1.0] is applied. The spectrum condition, clustering in the Wightman sense and uniqueness of the vacuum are consequences of the reconstruction, not inputs to (d). In particular neither a unique-vacuum statement nor an ergodic form of the reconstruction theorem is used to prove ergodicity. A historical route through [8, Thm. 4.4] and an identification of the Osterwalder–Schrader energy with a Wightman energy is recorded, with its first unsupported step, in Appendix G; it is not used. The main theorem uses only part (v) of Theorem 10.16, in clause (D). The word "interacting" means the non-Gaussian quantum field of (vii); no scattering or quantitative mass-gap statement is made.

11 Local moments and the common sequence

To turn the iterated limits into one sequence serving all later observables of Theorem 4.1, each arrow is first upgraded to carry all moments of the local seminorms (Proposition 11.6), using uniform moment bounds and Lemma 11.5. Proposition 11.7 then chooses at stage n parameters that make each of four discrepancies smaller than $2^{-n}/4$, and the triangle inequality gives (15). Since each compact smearing is controlled by one seminorm through (16), clause (B) follows (Proposition 11.10).

The moment-transfer and selection arguments use the field-space, Gaussian, Wick, fixed-volume and continuum-volume results proved above, including Theorem 10.10. Their provenance is recorded in Appendix D.

Throughout, $S_j = \sum_{k\in\mathbb{Z}^2} \lambda_{j,k}^{-2}$, which is finite because the shell $|k|_\infty = r \geq 1$ has $8r$ points with $\lambda_{j,k} \geq 1 + \pi^2 r^2/(j+1)^2$; thus $B_j = \|\chi_j\|_2^2 S_j/|\mathbb{T}_j|$ in Theorem 7.10(e). Put

$$K_j = \|\chi_j\|_2/2, \qquad \tau_j = m^{-2}B_j, \qquad \sigma_j = \frac{\|\chi_j\|_1^2 S_j}{4|\mathbb{T}_j|}, \qquad L_p = \sum_{q=1}^{p} q^{2p}, \qquad (2p-1)!! = \prod_{i=1}^{p}(2i-1).$$

11.1 Moment bounds for the reference fields

Lemma 11.1 (Hilbert sums)**.** *Let $(a_k)_{k\in\mathbb{Z}^2}$ be complex random variables, $w_k \geq 0$ and $p \geq 1$. Then $\mathbb{E}(\sum_k w_k|a_k|^2)^p \leq (\sum_k w_k\|a_k\|_{2p}^2)^p$. In particular, for a Borel random element Ξ of X, with $a_k = \Xi(\chi_j\overline{e_{j,k}})$, $\mathbb{E}p_j(\Xi)^{2p} \leq (\sum_k \lambda_{j,k}^{-2}\|\Xi(\chi_j\overline{e_{j,k}})\|_{2p}^2)^p$.*

Proof. Minkowski's inequality in L^p for finite partial sums, $\||a_k|^2\|_p = \|a_k\|_{2p}^2$, and monotone convergence. □

Corollary 11.2 (Neural reference moments)**.** *For $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, a regulator a, $N \geq 1$, $j \geq 1$ and $p \in \mathbb{N}$:*

$$\mathbb{E}_b p_j(\iota F_{N,a}^b)^{2p} \leq L_p K_j^{2p}. \tag{43}$$

The separate bound $\mathbb{E}_b p_j(\iota F^b_{N,a})^{2p} \le L_p \sigma_j^p$ also holds. Both are uniform over every finite N and every regulator $a > 0$; K_j and $\sigma_j^{1/2}$ are distinct constants.

Proof. For a fixed spatial point z, $X_i = g^b_{i,a}(z)$ are independent, centered and bounded by $1/2$. In the $2p$-slot expansion of $\mathbb{E}|N^{-1/2}\sum_i X_i|^{2p}$, every singleton label has zero contribution. A surviving tuple has $q \le p$ distinct labels; there are at most $N^q q^{2p}$ tuples with q labels, each term bounded by 2^{-2p} in absolute value. Since $N^{q-p} \le 1$,

$$\mathbb{E}_b |F^b_{N,a}(z)|^{2p} \le L_p / 4^p. \tag{44}$$

Joint measurability and $|F_N| \le \sqrt{N}/2$ make all spatial integrals finite for fixed N. Integral Minkowski in L^p, including $p = 1$, gives

$$\left\| \int \chi_j(z)^2 |F_N(z)|^2 dz \right\|_{L^p} \le \int \chi_j(z)^2 (\mathbb{E}|F_N(z)|^{2p})^{1/p} dz \le L_p^{1/p} \|\chi_j\|_2^2 / 4.$$

Parseval and $\lambda_{j,k} \ge 1$ give $p_j(\iota F_N)^2 \le \|\chi_j F_N\|_2^2$. Raising to power p proves (43) with the original K_j exactly. This uses only positive-regulator pointwise neurons; no point field at $a = 0$ appears.

For the separate σ_j bound, let $f \in L^1(\mathbb{R}^2; \mathbb{C})$, $Y_i = \int g_i f$ and $c = \|f\|_1/2$. The Y_i are independent, identically distributed, centered (Fubini and $\mathbb{E}g = 0$), and $|Y_i| \le c$ because $|g_i| = 1/2$; put $s^2 = \mathbb{E}|Y_1|^2 \le c^2$. Expanding $\mathbb{E}|\sum_{i \le N} Y_i|^{2p} = \mathbb{E}(\sum_i Y_i)^p (\sum_i \overline{Y_i})^p$ over index tuples, a term vanishes unless each index occurring in it occurs at least twice. A term with exactly q distinct indices, with multiplicities $n_i \ge 2$ summing to $2p$, is bounded by $\prod_i \mathbb{E}|Y_i|^{n_i} \le \prod_i c^{n_i - 2} s^2 = c^{2p-2q} s^{2q}$; there are at most $\binom{N}{q} q^{2p} \le N^q q^{2p}$ such tuples and $q \le p$. Hence $\mathbb{E}|N^{-1/2}\sum_i Y_i|^{2p} \le \sum_{q \le p} q^{2p} N^{q-p} c^{2p-2q} s^{2q} \le L_p c^{2p}$. For $f = \chi_j \overline{e_{j,k}}$ we have $(\iota F_N)(f) = N^{-1/2}\sum_i Y_i$ and $\|f\|_1 \le \|\chi_j\|_1 |\mathbb{T}_j|^{-1/2}$, so $\|(\iota F_N)(f)\|_{2p}^2 \le L_p^{1/p} \|\chi_j\|_1^2 / (4|\mathbb{T}_j|)$; Lemma 11.1 concludes. □

Lemma 11.3 (Gaussian localized trace)**.** *Let Ξ be a Borel random element of X such that for every complex $f \in C_c^\infty$ the real and imaginary parts of $\Xi(f)$ are centered and jointly Gaussian with $\mathbb{E}|\Xi(f)|^2 \le m^{-2}\|f\|_2^2$. Then $\mathbb{E}p_j(\Xi)^{2p} \le (2p-1)!!\, \tau_j^p$ for all $j, p \ge 1$. This applies to Φ_a under $\mathbb{P}_0$ for $0 \le a < 1/m$, to $\iota\gamma'$ under $G'_{\Omega,a}$ and to $\iota\gamma_\mu$ under G_μ.*

Proof. For $T = X' + iY'$ with X', Y' real centered Gaussian, $\|T\|_{2p}^2 = \|X'^2 + Y'^2\|_p \le \|X'\|_{2p}^2 + \|Y'\|_{2p}^2 = ((2p-1)!!)^{1/p} \mathbb{E}|T|^2$. With $\mathbb{E}|\Xi(\chi_j \overline{e_{j,k}})|^2 \le m^{-2}\|\chi_j\|_2^2 / |\mathbb{T}_j|$, Lemma 11.1 gives the bound. For Φ_a the hypothesis is Theorem 7.10(a) with Theorem 7.8. For a continuous centered Gaussian field γ with covariance kernel C and complex $f \in C_c^\infty$, $\int \gamma f$ is an L^2-limit of Riemann sums, so its real and imaginary parts are jointly Gaussian, and $\mathbb{E}|\int \gamma f|^2 = \langle f, Cf \rangle \le m^{-2}\|f\|_2^2$ when $0 \le C \le C_a$ as operators (Theorems 7.5 and 7.2(d)); this covers $C'_{\Omega,a}$ and C_μ^{end}. □

Lemma 11.4 (Density bounds)**.** *Fix a regulator a, a physical square Λ and $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$. For Borel $G : X \to [0, \infty]$:*

$$\mathbb{E}_{P^b_{N,\varepsilon,\Omega,a;\Lambda}} G \le \frac{M_{\mathrm{full}}}{Z^b_N} \mathbb{E}_b G(\iota F^b_{N,a}) \qquad (\varepsilon > 0,\ N \ge 1). \tag{45}$$

In particular, before any eventual-width denominator replacement, every finite N, $j, p \in \mathbb{N}$, both architectures and every fixed regulator satisfy

$$P^b_{N,\varepsilon,\Omega,a;\Lambda}(p_j^{2p}) \le \frac{M_{\mathrm{full}} L_p K_j^{2p}}{Z^b_N} \qquad (\varepsilon > 0). \tag{46}$$

Only after these exact finite-width bounds, the eventual-width and other density bounds are

$$\mathbb{E}_{P^b_{N,\varepsilon,\Omega,a;\Lambda}} G \le \frac{2M_{\mathrm{full}}}{Z_{\mathrm{full}}}\, \mathbb{E}_b\, G(\iota F^b_{N,a}) \quad (\varepsilon > 0,\ N \ge N_0), \tag{47}$$

$$\mathbb{E}_{\iota_* \mathrm{Res}_{\varepsilon,\Omega,a;\Lambda}} G \le M_{\mathrm{res}}\, e^{3u|\Lambda|c_a^2 + 3\varepsilon|\Omega\setminus\Lambda|c_a^2}\, \mathbb{E}_{G'_{\Omega,a}} G(\iota\gamma') \quad (\varepsilon \ge 0), \tag{48}$$

$$\mathbb{E}_{\nu_{a',\Lambda}} G \le \Xi_2(\Lambda)^{1/2} \big(\mathbb{E}_0 G(\Phi_{a'})^2\big)^{1/2} \quad (0 \le a' < 1/m), \tag{49}$$

where M_{full} *is given by* (18), $M_{\mathrm{res}} = e^{6u|\Lambda|c_a^2}$, *and* N_0 *depends on* $(b, \varepsilon, \Omega, a, \Lambda)$ *and is such that* $Z^b_N \ge Z_{\mathrm{full}}/2$ *for* $N \ge N_0$. *At* $\varepsilon = 0$ *the factor in* (48) *does not depend on* Ω.

Proof. (45) is the exact normalized expectation bound from $0 < e^{-V} \le M_{\mathrm{full}}$ and $Z^b_N > 0$, valid at every finite width. Combining it with (43) gives (46). Only afterwards, (47) uses $Z^b_N \to Z_{\mathrm{full}} > 0$ (Proposition 9.2). (48): the density of Res_ε with respect to $G'_{\Omega,a}$ is $e^{-W-\varepsilon J}/z_\varepsilon \le M_{\mathrm{res}}/z_\varepsilon$, and z_ε is bounded below in Proposition 9.3. (49): $Z_{a',\Lambda} \ge 1$ and Cauchy–Schwarz with Theorem 8.2(e) for $p = 2$. □

Lemma 11.5 (Unbounded moment transfer)**.** *Let* $\rho_k \Rightarrow \rho$ *be Borel probabilities on* X, $j, r \ge 1$, *and* $\sup_k \rho_k(p_j^{2r}) < \infty$. *Then* $\rho(p_j^{2r}) \le \liminf_k \rho_k(p_j^{2r})$, *and* $\rho_k(H) \to \rho(H)$ *for every continuous* $H : X \to \mathbb{C}$ *with* $|H| \le c(1 + p_j^r)$.

Proof. For $T > 0$, $\min(p_j^{2r}, T)$ is bounded and continuous; let $T \to \infty$ by monotone convergence. With $H_T = H \min(1, T/|H|)$, bounded and continuous, $|\rho_k(H) - \rho_k(H_T)| \le \rho_k(|H|^2)/T \le 2c^2(1 + \rho_k(p_j^{2r}))/T$, and the same for ρ; let $k \to \infty$, then $T \to \infty$. □

11.2 Moments along the arrows

Proposition 11.6 (Moments along the arrows)**.** *Fix* $j, r \ge 1$, *a physical square* Λ *and, where they occur,* b, *a regulator* a, $\varepsilon > 0$ *and* $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$. *Every law below has finite moments of every power of* p_j, *and:*

(i) width: $P^b_{N,\varepsilon,\Omega,a;\Lambda}(p_j^r) \to \nu^v_{\varepsilon,\Omega,a;\Lambda}(p_j^r)$ *as* $N \to \infty$;

(ii) stabilizer: $\nu^v_{\varepsilon,\Omega,a;\Lambda}(p_j^r) \to \nu^{(2)}_{\Omega,a;\Lambda}(p_j^r)$ *as* $\varepsilon \downarrow 0$;

(iii) auxiliary square: $\nu^{(2)}_{\Omega,a;\Lambda}(p_j^r) \to \nu_{a,\Lambda}(p_j^r)$ *as* $R \to \infty$, *and* $\sup_R \nu^{(2)}_{\Omega,a;\Lambda}(p_j^{2r}) \le M_{\mathrm{res}} e^{3u|\Lambda|c_a^2}(2r-1)!!\, \tau_j^r$;

(iv) regulator: $\nu_{a,\Lambda}(p_j^r) \to \nu_{0,\Lambda}(p_j^r)$ *as* $a \downarrow 0$, *and* $\sup_{0 \le a < 1/m} \nu_{a,\Lambda}(p_j^{2r}) \le \Xi_2(\Lambda)^{1/2}((4r-1)!!)^{1/2} \tau_j^r$;

(v) physical volume: $\nu_n(p_j^r) \to \nu(p_j^r)$, *with* $\sup_n \nu_n(p_j^{2r}) < \infty$.

Consequently, for every fixed n, dist_n *between consecutive laws of each arrow tends to* 0 *along the arrow.*

Proof. (i) By (47) and Corollary 11.2, $\sup_{N \ge N_0} P^b_N(p_j^{2r}) \le 2M_{\mathrm{full}} L_r K_j^{2r}/Z_{\mathrm{full}}$ by the exact finite-width inequality (46); for $N < N_0$ each law has $p_j(\iota F_N) \le \|\chi_j\|_2 \sqrt{N}/2$. Apply Lemma 11.5 with Proposition 9.2. (ii) By Proposition 9.3, $\nu^v_\varepsilon = \iota_* \mathrm{Res}_\varepsilon$; the densities $e^{-W-\varepsilon J}/z_\varepsilon$ converge pointwise to e^{-W}/z_0 and are bounded by $M_{\mathrm{res}} e^{3u|\Lambda|c_a^2 + 3|\Omega\setminus\Lambda|c_a^2}$ for $\varepsilon \le 1$ (Proposition 9.4), and

$p_j(\iota\gamma')^r \in L^1(G'_{\Omega,a})$ by Lemma 11.3; dominated convergence applies. (iii) (48) at $\varepsilon = 0$, Lemma 11.3, Proposition 9.5 and Lemma 11.5. (iv) (49) with $G = p_j^{2r}$, Lemma 11.3 for $\mathbb{E}_0 p_j(\Phi_a)^{4r}$, Proposition 9.6 and Lemma 11.5. (v) Since $\lambda_{j,k} \geq 1$, $p_j(v) \leq \|\chi_j v\|_{H^{-3/2}(\mathbb{T}_j)}$, whose moments are bounded uniformly in n by Theorem 10.10(iii); $\nu_n \Rightarrow \nu$ there, and Lemma 11.5 applies. The last sentence combines (i)–(v) with the convergences in β of Propositions 9.2, 9.4 (variation distance dominates β), 9.5, 9.6 and Theorem 10.10(iii) with Proposition 5.1(v). □

11.3 The common sequence

Proposition 11.7 (Selection of the common sequence)**.** *Put $\eta_n = 2^{-n}/4$ and $\Omega_k = [-k,k]^2$. Define recursively, for $n = 1, 2, \ldots$ (conditions involving index $n-1$ only for $n > 1$):*

(1) $a_n = a(k)$ for the least $k \in \mathbb{N}$ with $a(k) < \min(1/n, 1/(2m))$, $a(k) < a_{n-1}/2$ and $\mathrm{dist}_n(\nu_{a(k),\Lambda_n}, \nu_n) < \eta_n$;

(2) $R_n = R(k) = k$ for the least $k \in \mathbb{N}$ with $k \geq \max(n, n\ell)$, $k > R_{n-1}$ and

$$\mathrm{dist}_n(\nu^{(2)}_{\Omega_k,a_n;\Lambda_n}, \nu_{a_n,\Lambda_n}) < \eta_n;$$

(3) $\varepsilon_n = \varepsilon(k) = 1/k$ for the least $k \in \mathbb{N}$ with $1/k < 1/n$, $1/k < \varepsilon_{n-1}/2$ and

$$\mathrm{dist}_n(\nu^{v}_{1/k,\Omega_n,a_n;\Lambda_n}, \nu^{(2)}_{\Omega_n,a_n;\Lambda_n}) < \eta_n,$$

where $\Omega_n = \Omega_{R_n}$;

(4) $N_n = N(k) = k$ for the least $k \in \mathbb{N}$ with $k > n$, $k > N_{n-1}$ and

$$\mathrm{dist}_n(P^{b}_{k,\varepsilon_n,\Omega_n,a_n;\Lambda_n}, \nu^{v}_{\varepsilon_n,\Omega_n,a_n;\Lambda_n}) < \eta_n$$

for both b.

Each least index exists, and the resulting sequence satisfies all conditions and (15) *of Theorem 4.1(S). It is determined by m, ℓ, u, the localization system and the enumerations* (13)*.*

Proof. In each step the order constraints hold for all large k, and by Proposition 11.6 the discrepancy tends to 0 as $k \to \infty$ (in (1) because $a(k) \downarrow 0$, in (2) along the increasing squares Ω_k, in (3) because $1/k \downarrow 0$, and in (4) for each of the two architectures); so some k satisfies all conditions and the least one is well defined. In (2), $k \geq n\ell$ gives $\bar{\Lambda}_n \subseteq \Omega_k$, so every law in (2)–(4) is defined, and $a_n < 1/(2m)$ is a regulator. Every law entering dist_n has finite moments of p_j^r (Proposition 11.6). Since dist_n satisfies the triangle inequality, $\mathrm{dist}_n(P^b_n, \nu_n) < 4\eta_n = 2^{-n}$. The laws involved are determined by m, u, ℓ, the neuron definitions and the parameters, and dist_n by the localization system; no other object enters. □

Remark 11.8 (The selection is not an algorithm)*.* The least-index rule makes the sequence a well-defined deterministic function of the data listed in Proposition 11.7. It is not an algorithm: the predicates involve exact values of β and of moments of non-explicit laws, and no procedure deciding them, no bound on the selected indices and no end-to-end rate is given. Selecting the four parameters in the order $a_n, R_n, \varepsilon_n, N_n$ follows the order of the iterated limits of Section 9; no interchange of limits is used or claimed.

11.4 Whole-field convergence, moments and later tests

Proposition 11.9 (Clauses (0) and (A)). *For both architectures every P_n^b is a Borel probability on X with positive finite normalizer and $p_j(v) \le \|\chi_j\|_2\sqrt{N_n}/2$ on its support; $\beta(P_n^b,\nu)\to 0$; and for all $j,r\ge 1$, $P_n^b(p_j^r)\to\nu(p_j^r)$ and $M_{j,r}=\sup_{\rho\in\mathcal{L}}\int p_j^r\,d\rho<\infty$.*

Proof. Lemma 11.11 below and Theorem 6.5(a) give the Borel law and the normalizer bounds; Parseval gives $p_j(\iota F)\le\|\chi_j F\|_{L^2}$. Next, $\beta(P_n^b,\nu)\le \mathrm{dist}_n(P_n^b,\nu_n)+\beta(\nu_n,\nu)<2^{-n}+\beta(\nu_n,\nu)\to 0$ by Theorem 10.10(iii) and Proposition 5.1(v). For $n\ge\max(j,r)$, $|P_n^b(p_j^r)-\nu_n(p_j^r)|<2^{-n}$, and $\nu_n(p_j^r)\to\nu(p_j^r)<\infty$ by Proposition 11.6(v). A convergent sequence of finite numbers is bounded, and the finitely many stages with $n<\max(j,r)$ have finite moments. □

Proposition 11.10 (Clause (B)). *Let real $f_1,\dots,f_k\in C_c^\infty(\mathbb{R}^2)$ and a real or complex polynomial Q of degree d be given after the sequence. Then the expectations in Theorem 4.1(B) are absolutely finite, uniformly bounded, and converge; the laws of $(v(f_1),\dots,v(f_k))$ converge weakly.*

Proof. The evaluations are continuous on X (Proposition 5.1(iii)), so the continuous-mapping theorem gives weak convergence of the vectors, independently of Q. If Q is constant, the assertions involving Q follow directly from probability normalization. Otherwise its degree $d\ge 1$. Choose j with all supports in $(-j,j)^2$. By (16), $|v(f_i)|\le p_j(v)\|f_i\|_{H^2(\mathbb{T}_j)}$, so

$$|Q(v(f_1),\dots,v(f_k))|\le c_Q(1+p_j(v)^d)$$

and, for $q>0$, $|Q|^q\le c_{Q,q}(1+p_j^{\lceil dq\rceil})$; Proposition 11.9 bounds these uniformly over $\mathcal{L}$. Lemma 11.5 with $H=Q(v(f_1),\dots,v(f_k))$ and $r=d$ gives the convergence of expectations. □

11.5 Finite-stage reflection positivity

Lemma 11.11 (The stages satisfy the positivity hypotheses). *For every n, $\Lambda_n=(-n\ell,n\ell)^2$ is a physical square in $\Omega_n=[-R_n,R_n]^2$ (Definition 2.1); both $\bar\Lambda_n$ and Ω_n are centered closed axis-parallel squares invariant under θ_1 and θ_2; and $P_n^b=P_{N_n}^{v_V}$ with the integrand v_V of (7) for $(\varepsilon_n,\Omega_n,a_n;\Lambda_n)$, which agrees off the null set $\partial\Lambda_n\times\mathbb{R}$ with the integrand of Corollary 6.6 for the rectangles $\bar\Lambda_n=[-n\ell,n\ell]^2$ and Ω_n.*

Proof. $R_n\ge n\ell$ by Proposition 11.7; $\varepsilon_n>0$ and a_n is a regulator. The remaining assertions are the manuscript-tilt part of Corollary 6.6. □

Proposition 11.12 (Clause (E1)). *For $\theta\in\{\theta_1,\theta_2\}$, every n and both architectures, P_n^b satisfies (11) for every bounded complex Σ_+^θ-measurable F.*

Proof. Lemma 11.11 and Corollary 6.6, which gives Theorem 6.5(b) across both central lines; the integrand is θ-invariant because both the inner square and the outer square are. □

12 Joint compact-test correlations

Correlation functions involve products of field values at several points, which have no pointwise meaning for a distribution. This section uses instead the tensor power $v^{\otimes k}$, a distribution in k plane variables that equals $\prod_r v(f_r)$ on product tests; it is built from localized Fourier coefficients and

bounded by $p_j(v)^k$ times a norm of the test (Proposition 12.3). Averaging over a law gives the correlation distributions, the moment bounds of Section 11 give their convergence, and the same objects form the polynomial positivity expressions of clause (E2) (Proposition 12.9).

For $\kappa = (\kappa_1, \dots, \kappa_k) \in (\mathbb{Z}^2)^k$ put $e_{j,\kappa}(z_1, \dots, z_k) = \prod_r e_{j,\kappa_r}(z_r)$ and $\lambda_{j,\kappa} = \prod_r \lambda_{j,\kappa_r}$.

Lemma 12.1 (Fourier facts)**.** *Let $F \in \mathcal{D}(((-j,j)^2)^k)$, extended by zero to a smooth periodic function on $\mathbb{T}_j^k$, and $F_\kappa = \int_{\mathcal{C}_j^k} \overline{e_{j,\kappa}} F$. Then the Fourier series $\sum_\kappa F_\kappa e_{j,\kappa}$ converges to F in $C^\infty(\mathbb{T}_j^k)$,*

$$\sum_\kappa \lambda_{j,\kappa}^2 |F_\kappa|^2 = A_{j,k}(F)^2, \qquad A_{j,k}(F) \le 3^k |\mathbb{T}_j|^{k/2} \|F\|_{C^{2k}}. \tag{50}$$

Proof. $\prod_r (1 - \Delta_{z_r}) e_{j,\kappa} = \lambda_{j,\kappa} e_{j,\kappa}$, so Parseval on $\mathbb{T}_j^k$ gives the identity; repeated integration by parts makes F_κ decay faster than every power of $|\kappa|$, which gives convergence with all derivatives. Expanding the product of the operators $1 - \partial_1^2 - \partial_2^2$ gives 3^k terms, each a derivative of order at most $2k$, and $\|\cdot\|_{L^2(\mathbb{T}_j^k)} \le |\mathbb{T}_j|^{k/2} \|\cdot\|_\infty$. □

Lemma 12.2 (Support-preserving product density)**.** *Let $U \subseteq \mathbb{R}^2$ be open and $F \in \mathcal{D}(U^k)$. There are j, a real $\zeta \in \mathcal{D}(U \cap (-j,j)^2)$ and finite sums F_l of products $f_1 \otimes \cdots \otimes f_k$ with $f_r \in \mathcal{D}(U)$ and $\operatorname{supp} f_r \subseteq \operatorname{supp} \zeta$, such that $F_l \to F$ in C^M for every M, all supports lie in $(\operatorname{supp} \zeta)^k$, and $A_{j',k}(F_l - F) \to 0$ for every $j' \ge j$.*

Proof. Let $K_0 \subset U$ be the union of the coordinate projections of $\operatorname{supp} F$, choose j with $K_0 \subset (-j,j)^2$ and a real $\zeta \in \mathcal{D}(U \cap (-j,j)^2)$ with $\zeta = 1$ near K_0. Then $F = \zeta^{\otimes k} F$, and with the Fourier partial sums $S_l F$ on $\mathbb{T}_j^k$, $F_l = \zeta^{\otimes k} S_l F = \sum_{|\kappa|_\infty \le l} F_\kappa \bigotimes_r (\zeta e_{j,\kappa_r})$ converges to F in every C^M by Lemma 12.1 and the Leibniz rule; (50) gives the last assertion. □

Proposition 12.3 (Tensor evaluations; clause (C1))**.** *For $v \in X$ and $F \in \mathcal{D}(((-j,j)^2)^k)$ put $v^{\otimes k}(F) = \sum_\kappa F_\kappa \prod_r v(\chi_j e_{j,\kappa_r})$. The series converges absolutely, does not depend on j, and defines the unique $v^{\otimes k} \in \mathcal{D}'((\mathbb{R}^2)^k)$ with $v^{\otimes k}(f_1 \otimes \cdots \otimes f_k) = \prod_r v(f_r)$; it is symmetric, satisfies $\overline{v^{\otimes k}(F)} = v^{\otimes k}(\overline{F})$, $v \mapsto v^{\otimes k}(F)$ is continuous on X, and $|v^{\otimes k}(F)| \le p_j(v)^k A_{j,k}(F) \le 3^k |\mathbb{T}_j|^{k/2} p_j(v)^k \|F\|_{C^{2k}}$.*

Proof. Cauchy–Schwarz with the weights $\lambda_{j,\kappa}^{\pm 1}$, (50), $v(\chi_j e_{j,\kappa_r}) = v(\chi_j \overline{e_{j,-\kappa_r}})$ and $\lambda_{j,-\kappa_r} = \lambda_{j,\kappa_r}$ give absolute convergence and the bound. For $F = f_1 \otimes \cdots \otimes f_k$ the coefficients factor and the series is $\prod_r \sum_{\kappa_r} (f_r)_{\kappa_r} v(\chi_j e_{j,\kappa_r}) = \prod_r v(f_r)$ by the expansion in the proof of (16); this holds for every admissible j. For general F, Lemma 12.2 gives product sums $F_l \to F$ with $A_{j,k}(F_l - F) \to 0$ and $A_{j',k}(F_l - F) \to 0$, so the values for j and $j' \ge j$ agree. The bound shows continuity on each $\mathcal{D}_K$, so $v^{\otimes k} \in \mathcal{D}'$; uniqueness follows from the density of product sums. Symmetry and the reality relation hold on product tests (for the latter because v is real) and extend by density. For $v, w \in X$, telescoping gives $|v^{\otimes k}(F) - w^{\otimes k}(F)| \le k A_{j,k}(F)\, p_j(v - w) \max(p_j(v), p_j(w))^{k-1}$, and $p_j(v - w) \le 2^j d(v,w)$ when $d(v,w) < 2^{-j}$. □

Remark 12.4 (Meaning of the tensor evaluation). $v^{\otimes k}$ is the k-fold tensor power of the distribution v; for background on tensor products of distributions see [27], cited for orientation only. Every property used here is proved in Proposition 12.3. The evaluation is defined for every $v \in X$, with no exceptional set.

Lemma 12.5 (Complexification, reflection and products)**.** *Let $v \in X$, $\theta \in \{\theta_1, \theta_2\}$, $G \in \mathcal{D}((\mathbb{R}^2)^r)$ and $H \in \mathcal{D}((\mathbb{R}^2)^s)$. Then (a) $(\theta v)^{\otimes r}(G) = v^{\otimes r}(G \circ \theta^{\times r})$, where $(G \circ \theta^{\times r})(z_1, \dots, z_r) = G(\theta z_1, \dots, \theta z_r)$; (b) $v^{\otimes r}(G)\, v^{\otimes s}(H) = v^{\otimes (r+s)}(G \otimes H)$; (c) if $G \in \mathcal{D}((H_+^\theta)^r)$, then $v \mapsto v^{\otimes r}(G)$ is Σ_+^θ-measurable; (d) for $G \in \mathcal{D}((H_+^\theta)^r)$, $\overline{(\theta v)^{\otimes r}(G)} = v^{\otimes r}(\theta_r \overline{G})$ with $(\theta_r \overline{G})(z) = \overline{G(\theta z_1, \dots, \theta z_r)}$.*

Proof. On product tests, (a) is $(\theta v)(f) = v(f \circ \theta)$ (Proposition 5.1(vii), extended complex-linearly) and (b) is multiplicativity. Both sides of (a) are continuous in G for $A_{j,r}$ with supports in a fixed compact, since $p_j(\theta v) \le C_j p_{j+1}(v)$ and θ preserves $\mathbb{T}_j$ and the Laplacian; for (b), $A_{j,r+s}(G \otimes H) = A_{j,r}(G)A_{j,s}(H)$. Lemma 12.2 concludes. For (c), the approximants of Lemma 12.2 with $U = H^\theta_+$ give $v^{\otimes r}(G)$ as a pointwise limit of finite sums of products $\prod_i v(f_i)$ with $f_i \in \mathcal{D}(H^\theta_+)$, whose real and imaginary parts are real tests in $C^\infty_c(H^\theta_+)$. (d) combines (a) with the reality relation of Proposition 12.3. □

Proposition 12.6 (Correlation distributions; clause (C2))**.** *For $\rho \in \mathcal{L}$ and $F \in \mathcal{D}((\mathbb{R}^2)^k)$, $\mathcal{S}^\rho_k(F) = \int v^{\otimes k}(F)\,\rho(dv)$ is absolutely defined, $\mathcal{S}^\rho_k \in \mathcal{D}'((\mathbb{R}^2)^k)$ is the unique distribution with $\mathcal{S}^\rho_k(f_1 \otimes \cdots \otimes f_k) = \int \prod_r v(f_r)\,\rho(dv)$, and $\sup_{\rho\in\mathcal{L}} |\mathcal{S}^\rho_k(F)| \le M_{j,k}A_{j,k}(F)$ for $\operatorname{supp} F \subset ((-j,j)^2)^k$.*

Proof. $v \mapsto v^{\otimes k}(F)$ is continuous, hence Borel, with $|v^{\otimes k}(F)| \le p_j^k A_{j,k}(F)$, and $M_{j,k} < \infty$ (Proposition 11.9). Linearity is clear; the bound and (50) give continuity on each $\mathcal{D}_K$; uniqueness is the density of product sums (Lemma 12.2). □

Proposition 12.7 (Kernel convergence; clause (C3))**.** *For every k and $F \in \mathcal{D}((\mathbb{R}^2)^k)$, $\mathcal{S}^b_{n,k}(F) \to \mathcal{S}_{\infty,k}(F)$ for both architectures.*

Proof. $H(v) = v^{\otimes k}(F)$ is continuous with $|H| \le A_{j,k}(F)p_j^k$; $P^b_n \Rightarrow \nu$ and $\sup_n P^b_n(p_j^{2k}) < \infty$ (Proposition 11.9); apply Lemma 11.5. □

Proposition 12.8 (Target identification; clause (D))**.** *For every k, $\mathcal{S}_{\infty,k} = \mathrm{Schw}_k|_{\mathcal{D}((\mathbb{R}^2)^k)}$, with Schw_k the tempered Schwinger distribution of ν^S (Theorem 10.16(v)).*

Proof. For $f_r \in \mathcal{D}(\mathbb{R}^2)$, $\nu = \mathcal{R}_*\nu^S$ and $\mathcal{R}(\psi)(f) = \psi(f)$ (Theorem 10.10(iii), extended complex-linearly) give $\mathcal{S}_{\infty,k}(f_1 \otimes \cdots \otimes f_k) = \int \prod_r \psi(f_r)\,d\nu^S = \mathrm{Schw}_k(f_1 \otimes \cdots \otimes f_k)$. The restriction of Schw_k to $\mathcal{D}$ is a distribution because $\mathcal{D} \hookrightarrow \mathcal{S}$ is continuous; two distributions that agree on product tests agree (Lemma 12.2). □

Proposition 12.9 (Polynomial reflection positivity; clause (E2))**.** *Let $\theta \in \{\theta_1, \theta_2\}$, $J \ge 0$, $f_0 \in \mathbb{C}$, $f_r \in \mathcal{D}((H^\theta_+)^r)$ and $\rho \in \mathcal{L}$. Then $\mathcal{O}_f, \mathcal{O}_f \circ \theta \in L^2(\rho)$ with $\|\mathcal{O}_f\|_{L^2(\rho)} \le |f_0| + \sum_{r=1}^J A_{j,r}(f_r)M^{1/2}_{j,2r}$ and $\|\mathcal{O}_f \circ \theta\|_{L^2(\rho)} \le |f_0| + \sum_{r=1}^J C_j^r A_{j,r}(f_r)M^{1/2}_{j+1,2r}$ for j with all supports in the corresponding cubes; every term of $\Pi_\rho(f)$ is absolutely finite; $\Pi_\rho(f) = \int \overline{\mathcal{O}_f(\theta v)}\,\mathcal{O}_f(v)\,\rho(dv)$; $\Pi_{P^b_n}(f) \ge 0$; and $\Pi_{P^b_n}(f) \to \Pi_\nu(f) \ge 0$.*

Proof. The L^2 bounds follow from Proposition 12.3, $p_j(\theta v) \le C_j p_{j+1}(v)$ and Minkowski. By Lemma 12.5(b),(d), $\overline{(\theta v)^{\otimes r}(f_r)}\,v^{\otimes s}(f_s) = v^{\otimes(r+s)}((\theta_r \overline{f_r}) \otimes f_s)$ for every v; integrating gives the terms of (12), which are absolutely finite by Cauchy–Schwarz, and summing gives the integral formula. For $\rho = P^b_n$, $\mathcal{O}_f$ is Σ^θ_+-measurable (Lemma 12.5(c)), and the clipped observables $\mathrm{clip}_T\mathcal{O}_f = \mathcal{O}_f \min(1, T/|\mathcal{O}_f|)$ are bounded and Σ^θ_+-measurable, with $(\mathrm{clip}_T\mathcal{O}_f)(\theta v) = \mathrm{clip}_T(\mathcal{O}_f(\theta v))$. Proposition 11.12 gives $\int \overline{(\mathrm{clip}_T\mathcal{O}_f)(\theta v)}(\mathrm{clip}_T\mathcal{O}_f)(v)\,P^b_n(dv) \ge 0$; as $T \to \infty$ the integrands converge pointwise and are dominated by the integrable $|\mathcal{O}_f(\theta v)||\mathcal{O}_f(v)|$, so $\Pi_{P^b_n}(f) \ge 0$. Each term converges by Proposition 12.7, so $\Pi_{P^b_n}(f) \to \Pi_\nu(f)$, which is therefore real and nonnegative. □

Remark 12.10 (What the correlation statements mean). Convergence in (C3) is convergence of values on each fixed compactly supported smooth joint test, that is, in $\sigma(\mathcal{D}'((\mathbb{R}^2)^k), \mathcal{D}((\mathbb{R}^2)^k))$; no Schwartz or other noncompact test is used at the neural stages, and no strong-dual convergence is

claimed. The bounds $M_{j,k}$ hold along the selected sequence and at ν. The positivity of $\Pi_\nu(f)$ is obtained as a limit; it is consistent with, but not derived from, Theorem 10.16(ii).

13 Proof of the main theorem

Lemma 13.1 (One set of limiting objects)**.** *The law ν in clauses (0), (A)–(C) and (E) is $\nu = \mathcal{R}_*\nu^S$ of Theorem 10.10(iii), with ν^S the target of Theorem 10.8. Any Borel probability on X whose finite-dimensional laws on real compact smooth tests are those of ν^S equals ν. The law ν_n in* (15) *is ν_{0,Λ_n} of Definition 2.10(d),(e).*

Proof. The cylinder events of compact smooth evaluations form a $\cap$-stable generator of $\mathcal{B}(X)$ (Proposition 5.1(iii)); apply Dynkin's theorem. □

Proof of Theorem 4.1. The classical inputs enter only through Theorems 10.8, 10.10 and 10.16 and Lemma 10.14; the foundations of Sections 5–8 are proved in this paper.

Clause	Proved by
(S)	Proposition 11.7, from Propositions 9.2–9.6, 11.6 and Theorem 10.10
(0)	Proposition 11.9, Lemma 11.11, Theorem 6.5(a), Theorem 10.10(iii)
(A)	Proposition 11.9
(B)	Proposition 11.10
(C1)–(C3)	Propositions 12.3, 12.6, 12.7
(D)	Proposition 12.8, Theorem 10.16(v)
(E1)	Proposition 11.12 (Corollary 6.6)
(E2)	Proposition 12.9

Every clause uses the one sequence of Proposition 11.7 with no subsequence, and the objects quantified in (A)–(E) are chosen after it (Remark 4.3). Theorem 4.2 is Theorem 10.16. □

Remark 13.2 (No circularity). The order of proof is: field spaces (Section 5); neurons and their positivity, Gaussian references and the Wick theory (Sections 6–8, Appendices A–C), none of which uses an interacting law; the fixed-volume chain (Section 9); the continuum lemmas (Section 10), where Theorem 10.8 uses C1, C2 and the Wick identification, Theorem 10.10 uses C4, C5 and Theorem 10.8, and the restricted proof of Theorem 10.16 uses C1–C3, C6, the direct classical Wick lemma and the source portion of Theorem 10.8, without its common-space bridge; then Sections 11–12. Ergodicity of the target is proved in Theorem 10.16(iii) before [8, Thm. 1.0] is applied in (vii). No statement uses a later one.

14 Contribution and relation to prior work

14.1 Contribution

The quadratic term enters through a Gaussian identity. After the width limit, the quadratic tilt of the Gaussian endpoint reference over Ω is again a Gaussian law, with covariance $C'_{\Omega,a}$; as Ω grows, $C'_{\Omega,a}$ converges locally uniformly to C_a, whose Fourier multiplier is $(m^2 + |p|^2 + a^2 p_1^2 p_2^2)^{-1}$ by the mass-shift identity $1/\hat{C}_\mu^{\mathrm{end}} - \kappa_a^2 = 1/\hat{C}_a$. The regulator a therefore controls the anisotropy of this Gaussian reference covariance, and the anisotropy disappears as $a \to 0$. The quartic interaction is

separate from this identity: the finite-width laws and the interacting laws are non-Gaussian, and their covariances are not given by it.

For both architectures we prove the following. First, at fixed physical volume and for every $m, u > 0$, the laws converge successively in the width, the stabilizer, the auxiliary volume and the regulator to the free-boundary Wick-quartic law on the physical square, together with all positive integer powers of the local seminorms. Second, for $0 < u/m^2 < g_*$, one deterministic sequence of all parameters, fixed before any observable is chosen and never passed to a subsequence, converges weakly on $H^{-2}_{\mathrm{loc}}(\mathbb{R}^2)$ to the H^{-2}_{loc}-valued version of the classical infinite-volume weak-coupling Wick-quartic measure; the same sequence gives convergence, with uniform moment bounds, of every polynomial in finitely many compactly supported smooth smearings, and convergence of every compact joint-test correlation distribution. Third, every stage of this sequence is reflection positive across the two central coordinate lines $z_1 = 0$ and $z_2 = 0$, both for bounded half-plane observables and for compact joint-test polynomial forms. We prove no reflection-positivity statement for other lines, and we claim neither Euclidean invariance nor clustering of the finite stages.

These statements rest on attributed classical inputs and on the foundations proved here: the endpoint neurons and their finite-width central reflection positivity, the Gaussian reference constructions, and the fixed-volume Wick estimates. The infinite-volume target, its Osterwalder–Schrader properties and its reconstruction are classical results used with attribution.

14.2 Relation to prior work

Frank [7] gives a constructive neural-network approach to the finite-volume Wick-ordered ϕ^4_2 measure at every coupling. His networks are quenched random-feature shells with Gaussian coefficients and a smooth spatial cutoff; for almost every draw of the features, along a prescribed nested width schedule, the joint laws of finitely many Schwartz smearings converge in either order of the width and ultraviolet limits, with explicit width thresholds and width-uniform ultraviolet tail bounds. His coefficient $\lambda/4!$ corresponds to our u. Our construction differs in its architectures, in using annealed rather than quenched laws, in its sharp-square interaction with a stabilizer and a mass tilt, in its finite-stage reflection positivity, and in including a physical-volume limit, which we obtain by combining the fixed-volume neural limits with classical infinite-volume results. Frank's construction stops at finite volume and leaves open whether neural constructions reach infinite volume themselves; we do not claim that an analogous classical thermodynamic step could not be attached to it. At fixed volume, his width-uniform estimates imply, by a standard uniform-integrability argument (Remark 14.1), convergence of all polynomial moments of finitely many smeared fields for his laws, so we make no novelty claim based on moment convergence. Whether his estimates also yield fixed-volume convergence of joint compact-test kernels or of the field on H^{-2}_{loc} requires estimates that we have not established, chiefly a bound on localized smeared fields summed over all frequency shells uniformly in the localized Fourier mode, followed by tightness and an H^{-2}_{loc}-valued realization of his infinite-cutoff field. We base no novelty claim on the kernel or topology labels either.

The annealed reweighting is the parameter-density deformation $P = P_G\, e^{-\lambda \int \mathcal{O}}$ of Demirtas et al. [3]; Zhang [26] writes the same normalized reweighting of a free ensemble. In the parts of the former work that we consulted, the deformation is formal and at infinite width. We realize it at finite width for the two architectures above, with finite positive normalization and with the limits stated above.

The reflection-positivity argument uses conditional factorization. For product neurons, conditioning on the complete boundary-data tuple of all N neurons makes the two half-plane sides

independent with equal conditional laws. For Mondrian neurons, the square identity is first proved in symmetric finite boxes using the extended trace, including positions, auxiliary times and colors (Corollary A.12). Whole-plane half-plane positivity then follows by L^2 approximation of bounded observables (Proposition A.15). In both cases, a reflection-invariant additive potential is absorbed into the side observables, preserving positivity (Lemma A.14 and Theorem 6.5). This instantiates the conditional-factorization principle underlying the one-dimensional parameter-splitting mechanism of Ferko and Halverson [4]. They also show that symmetric Markov processes are reflection positive, a connection they attribute to Nelson and others, and that reflection positivity survives the linear combinations and pointwise neural maps they consider, for example of independent inputs and with random maps drawn independently of the input. Reflection positivity of Markov fields and lattice models is classical [9]. We instantiate the mechanism in two dimensions for these architectures, across the two central coordinate lines.

Harvey [13] shows that, for $d \geq 2$, a pointwise ensemble whose two-point function is Euclidean invariant, $S_2(gx, gy) = S_2(x, y)$ for all translations, rotations and reflections g, and continuous, and which is two-point reflection positive across one hyperplane, has a constant two-point function; he presents this as an answer, for that finite-variance class, to the question of reflection positivity in $d \geq 2$ raised in [6]. The invariance hypothesis concerns the two-point function, not the full law. Our finite stages have bounded values. We do not claim that the two-point functions of our normalized interacting finite-stage laws are Euclidean invariant, and we do not determine whether they are; we therefore assert neither that the theorem applies to these stages nor that it does not, and the statements we prove do not conflict with it. He notes that products of coordinate cosine networks have the product kernel $\prod_i e^{-m|x_i - y_i|}$ with two-point reflection positivity across coordinate hyperplanes, giving up exact rotation invariance. Our construction does not use rotation invariance either: the endpoint covariance $e^{-\mu|z|_1}/4$ of the neurons is not rotation invariant. This is a property of that reference covariance; it is not a statement about the two-point function of the interacting finite-stage laws. What we add at finite width is reflection positivity of the full normalized law for bounded observables across the two central coordinate lines, rather than two-point positivity alone.

Ferko and Mutchler [5] prove obstructions for specified finite-width classes: finite expansions in fixed deterministic features, including distributional neurons, cannot reproduce a nontrivial Källén–Lehmann covariance; ensembles with bounded mollified coincident variance, such as networks with locally bounded pointwise second moments, are incompatible with a nontrivial Källén–Lehmann representation, which temperedness, Euclidean invariance and reflection positivity would furnish; and finite single-layer ridge networks cannot reproduce all correlation functions of an Osterwalder–Schrader field with a nontrivial Källén–Lehmann measure, a result they state is specific to single-layer networks. They also stress that a parameter count alone cannot define finiteness, since a single Borel parameter can encode any field law. Our finite stages have bounded values and are not claimed to be Euclidean invariant, so they do not conflict with the variance obstruction, and the ridge theorem does not describe product-telegraph or Mondrian neurons. Ferko, Halverson and Mutchler [6] show that every law on $\mathcal{S}'(\mathbb{R}^d)$ admits a neural-network representation with countably many parameters, or formally with a single one; we therefore do not claim representability of the target as a contribution.

For fields built from neurons see Halverson, Maiti and Stoner [12] and Halverson [11]. Zhang [26] and Ageev and Ageeva [1] study adjacent architecture and cost questions that are not addressed here.

Remark 14.1 (Fixed-volume moments for the laws of [7]). The moment sentence above rests on the following short derivation, which uses only estimates stated in [7]; the conclusion is not stated there. Let X be the pairing of the cutoff field of [7] with a Schwartz function f. The normalizer there is

at least one [7, Lemma 2.8], so Cauchy–Schwarz gives $\mathbb{E}_\mu|X|^q \le (\mathbb{E}_\gamma e^{-2V})^{1/2}(\mathbb{E}_\gamma|X|^{2q})^{1/2}$, where γ is the Gaussian prior. The first factor is the normalizer at doubled coupling, which [7] bounds uniformly in the shell cutoff and the width index on the almost-sure feature event. Under the prior, X is centered Gaussian with variance bounded, for every feature realization, by the deterministic bound on the shell pairing coefficients [7, (6.38)]. Gaussian moments therefore bound $\sup \mathbb{E}|X|^q$ over the width index and the cutoff for every q, likewise for finite tuples, and Fatou passes the bounds to the limit laws. The weak convergence statements of [7, Lemma 7.5, Thm. 6.6(c), Thm. 7.6, Cor. 7.8], together with uniform integrability, then give convergence of every polynomial moment along each iterated limit. We do not claim the corresponding statements for joint compact-test kernels or for the topology of H^{-2}_{loc}; they would require the estimates named above.

15 Limitations

The construction is existential. It provides no effective end-to-end parameter schedule, computational cost, sampling guarantee or numerical weak-coupling threshold: the parameter sequence is shown to exist but is not computed. The fixed-volume Wick step includes explicit $\sqrt{a}$ bounds whose constants depend on the volume; these are fixed-volume upper bounds, not practical convergence rates.

This limitation describes the original common sequence. The private post-v1 Appendix H gives finite-observation bounds, conditional elementary selection and an explicit pressure-target example, with each result's separate hypotheses and source qualifications.

The infinite-volume limit, the target, its Euclidean invariance, clustering and reflection positivity across every line, and its $1 + 1$-dimensional Wightman reconstruction are classical results used with attribution. No new interacting theory and no statement in other dimensions is claimed. The classical inputs C1–C5 are used in the formulations of Section 3; their proofs were not audited, the bodies of [19, 20] and [24] and the antecedents of the review in [10, pp. 316–319] were not available, and the page-level access record is in Appendix D.

Finite-stage reflection positivity is proved across the two central coordinate lines, for the centered axis-parallel squares used here and, more generally, for every bounded auxiliary region and every measurable potential, bounded on bounded field ranges, that are invariant under both central reflections. Nothing is claimed for other lines. For a rotated physical square nothing is claimed beyond this invariant case; in particular, no failure of reflection positivity is asserted for any rotated square. The finite stages are not claimed to be Euclidean invariant, and no finite-stage clustering statement is made.

The laws are annealed; nothing is claimed about quenched or fixed-feature laws.

Neural convergence is proved for compactly supported test functions only. No global Schwartz or strong-dual convergence is claimed.

The reflection-positivity mechanism, the density-deformation mechanism and the endpoint covariance kernel all have prior instances in the cited works.

The Wick interaction at $a = 0$ is constructed on one common Gaussian space as an L^2 limit in the fourth chaos, for a fixed physical square; its exceptional null sets are fixed per regulator value and per test, or common to countable families. Nothing is asserted about other representations of that variable, about uncountable families of regulators, or about volume-uniform versions of the Wick estimates beyond $Z_{a,\Lambda} \ge 1$.

The review status of this exact version, including that of Theorem 4.1, is recorded in its

companion decision record, not in this text. Appendix D identifies the components that are retained, modified, re-expressed or newly written here, and the current-text review that the project's review policy requires for each. All review recorded for the antecedents was internal model review; none of it is independent expert review.

A Proofs for the process neurons

This appendix proves Theorems 6.1–6.5 and Corollaries 6.6, 6.7. Besides C6 it uses: (K1) Dynkin's π–λ theorem and the functional monotone-class theorem; (K2) Doob–Dynkin factorization, and the fact that a function measurable for the σ-algebra generated by a family of maps depends on countably many of them; (K3) if Y is Polish and continuous real $l_1, l_2, \dots$ separate its points, then $y \mapsto (l_k(y))_k$ is a continuous injection into $\mathbb{R}^{\mathbb{N}}$, hence (Lusin–Souslin) a Borel isomorphism onto its Borel image, so $\sigma(l_k : k \geq 1) = \mathcal{B}(Y)$; (K4) finite counting measures on a Polish space form a standard Borel space whose σ-algebra is generated by $\nu \mapsto \nu(A)$, and $\nu \mapsto \int f \, d\nu$ is Borel for bounded Borel f; (K5) Kolmogorov extension for sequences of standard Borel spaces; (K6) the Poisson facts of C6(h). For $\theta \in \{\theta_1, \theta_2\}$ write $H_+ = H_+^\theta$ and $Q_n = Q_n^\theta = H_+ \cap (-n, n)^2$, and $r_N = \sqrt{N}/2$.

A.1 Measurability

Lemma A.1 (Partition spaces)**.** *Let C be a box with nonempty interior. (i) $\mathrm{Part}^{\mathrm{col}}(C)$ is a Borel subset of the standard Borel space of finite counting measures on $\mathbb{R}^4 \times \{-1, 1\}$, hence standard Borel; restriction res_B, the union map $\cup(P_0, P_1) = P_0 \cup P_1$ (for daughters of a cut), and images under translations and θ_1, θ_2 are Borel. (ii) For open $U \subseteq \operatorname{int} C$, the field map $P \mapsto \mathrm{fld}(P)|_U$, $\mathrm{fld}(P) = \sum_{(R,\xi)\in P} (\xi/2)\mathbf{1}_{R^{\mathrm{ll}}}$, is Borel into $L^1(U)$. (iii) $(z, P) \mapsto \mathrm{fld}(P)(z)$ is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{B}(\mathrm{Part}^{\mathrm{col}}(C))$-measurable.*

Proof. (i) On the Borel set of counting measures with exactly k atoms of mass one, the atoms can be listed lexicographically by a Borel map into $(\mathbb{R}^4 \times \{\pm 1\})^k$. The conditions "each atom codes a box with nonempty interior contained in C, interiors pairwise disjoint, areas summing to $|C|$" are finitely many Borel conditions on the list, and they characterize the k-cell partitions: the union of the boxes is closed, and a missing point of C would have a missing relative neighbourhood of positive area. Restriction, union and the maps act coordinatewise on the list by Borel operations (maxima and minima of corners, discarding pieces of zero width or height). (ii) For bounded Borel f on U, $P \mapsto \int_U f \mathrm{fld}(P) = \int h_f \, dP$ with $h_f(a, b, c, d, \xi) = (\xi/2) \int_{U \cap [a,b] \times [c,d]} f$, Borel by (K4); the indicators of rational boxes with closure in U give continuous functionals on $L^1(U)$ that separate points (Lebesgue differentiation), so (K3) applies. (iii) The integrand $(z, a, b, c, d, \xi) \mapsto (\xi/2)\mathbf{1}\{a \leq z_1 < b,\ c \leq z_2 < d\}$ is Borel, and integrals of jointly Borel bounded functions against counting measures are jointly measurable (K1). □

Lemma A.2 (Random fields and parameter maps)**.** *Let $r \geq 0$, $U \subseteq \mathbb{R}^2$ open and $\mathrm{clip}_r(x) = \max(-r, \min(r, x))$. (i) If $F : \mathbb{R}^2 \times \Gamma \to \mathbb{R}$ is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable with $|F| \leq r$, then $\omega \mapsto [F(\cdot, \omega)|_U]$ is measurable into $\mathrm{Ball}_r(U)$, and for $U = \mathbb{R}^2$ into E. (ii) Two such fields with the same finite-dimensional distributions have the same law of $F|_U$. (iii) Let $(Y_{\mathrm{sp}}, \mathcal{Y})$ be a measurable space and $G : \mathbb{R}^2 \times Y_{\mathrm{sp}} \to \mathbb{R}$ be $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{Y}$-measurable. Then $\Gamma_G^r(y) = [\mathrm{clip}_r(G(\cdot, y))|_U]$ is defined for every y, takes values in $\mathrm{Ball}_r(U)$ and is measurable. If $Y : \Gamma \to Y_{\mathrm{sp}}$ is measurable and $F(z, \omega) = G(z, Y(\omega))$ for almost every $z \in U$ for every outcome of an almost sure event, then $[F|_U] = \Gamma_G^r(Y)$ almost surely, and the law of $[F|_U]$ is the image of the law of Y. If $|G| \leq r$, the clip changes nothing.*

Proof. (i) For $s_0 \in L^1_{\mathrm{loc}}(U)$ and compact $K \subset U$, $\omega \mapsto \int_K |F(z,\omega) - s_0(z)|dz$ is measurable by Tonelli; $L^1_{\mathrm{loc}}(U)$ is separable metric, so open sets are countable unions of balls; the same argument works in E. (ii) Indicators of rational boxes with closure in U generate the Borel σ-algebra (K3), finite cylinders form a π-system (K1), and the law of a bounded vector $(\int f_{k_1}F, \dots, \int f_{k_m}F)$ is determined by its mixed moments $\int\cdots\int \prod_j f_{k_j}(z_j)\mathbb{E}\prod_j F(z_j)\,dz$. (iii) Each section of G is Borel and clip_r is continuous, so the clipped section is Borel and bounded by r. For a Borel representative s_0 and compact K, $(z,y) \mapsto \mathbf{1}_K(z)|\mathrm{clip}_r(G(z,y)) - s_0(z)|$ is product measurable, so $y \mapsto \int_K |\mathrm{clip}_r G(z,y) - s_0(z)|dz$ is measurable (the measurability part of Tonelli needs σ-finiteness only in z: the sets D whose sections satisfy that $y \mapsto |D_y \cap K_m|$ is measurable form a Dynkin system containing rectangles); the metric of $L^1_{\mathrm{loc}}(U)$ and separability give measurability of Γ^r_G. If $F = G(\cdot, Y)$ almost everywhere on U and $|F| \le r$, then $\mathrm{clip}_r(G(\cdot,Y)) = F$ almost everywhere on U. Without the clip the map need not be defined: $G(z,1) = |z|^{-2}$ is not locally integrable at 0. □

Lemma A.3 (Integrated potentials of bounded fields)**.** *Let $U \subseteq \mathbb{R}^2$ be open, $A \subseteq U$ bounded and Lebesgue measurable, $r \ge 0$, and v an $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{B}(\mathbb{R})$-measurable function bounded on $A \times [-r,r]$. Then $\Psi_v(s) = \int_A v(z, s(z))\,dz$ is well defined and Borel on* $\mathrm{Ball}_r(U)$.

Proof. For a representative with values in $[-r,r]$, $z \mapsto (z, s(z))$ is $(\mathcal{L}, \mathcal{L}\otimes\mathcal{B})$-measurable, so $z \mapsto h(z,s(z))$ is Lebesgue measurable, and representatives differ on null sets. For $h = \mathbf{1}_{A'}(z)\varphi(x)$ with φ continuous, the map is continuous on $\mathrm{Ball}_r(U)$: if $s_k \to s$ in $L^1_{\mathrm{loc}}(U)$ within Ball_r and $K \subset U$ is compact with $|A \setminus K| < \delta$, then $\int_A |s_k - s| \le \int_K |s_k - s| + 2r\delta$, so $s_k \to s$ in measure on A and the integrals converge by uniform continuity of φ on $[-r,r]$ (the bound r handles sets A touching ∂U). The class of bounded h for which the map is Borel is a vector space closed under bounded monotone limits and contains a multiplicative generating class; (K1) applies to $v\mathbf{1}_{A\times[-r,r]}$. □

Lemma A.4 (Restrictions and point recovery)**.** *Fix θ and $r \ge 0$. (i) $r_n : \mathrm{Ball}_r(H_+) \to L^1(Q_n)$, $s \mapsto s|_{Q_n}$, is well defined and continuous, and $r_m = \rho_{n,m} \circ r_n$ for $m \le n$ with the continuous restriction $\rho_{n,m}$. (ii) The relative Borel σ-algebra of* $\mathrm{Ball}_r(H_+)$ *equals $\sigma(r_n : n \ge 1)$, and $\sigma(r_1, \dots, r_n) = \sigma(r_n)$ increases with n. (iii) For $z \in H_+$ with discs $D_j(z)$ of radius $1/j$, $L_z(s) = \limsup_j |D_j(z)|^{-1} \int_{D_j(z)} s$ is Borel on* $\mathrm{Ball}_r(H_+)$ *with values in $[-r,r]$, and $L_z(s) = c$ if s has a representative equal to c near z.*

Proof. (i) $|s| \le r$ and $|Q_n| < \infty$; if $s_j \to s$ in $L^1_{\mathrm{loc}}(H_+)$ within Ball_r, choose compact $K \subset Q_n$ with $|Q_n \setminus K| < \delta$. (Without the bound the map is undefined: $1/z_1$ is locally integrable on $\{z_1 > 0\}$ but not on $(0,n)\times(-n,n)$.) (ii) Each compact $K \subset H_+$ lies in some Q_n, and $s \mapsto \|s - s_0\|_{L^1(K)}$ is a continuous function of $r_n(s)$; relatively open sets are countable unions of such balls. (iii) For large j the closed disc lies in H_+, the averages are continuous, and a lim sup of countably many Borel maps is Borel. □

A.2 The canonical box Mondrian

For each finite binary tree τ (a finite prefix-closed set of words in which every non-leaf has both children), the event that the born nodes form τ is defined by finitely many Borel inequalities in (C, s) and the ingredients of the nodes of τ, and on it the output is a Borel function of finitely many variables; on the complement of the union of these countably many events (a null set, by the next lemma) set the output to $\{(C, \xi_\emptyset)\}$. For a box $C = [a,b]\times[c,d]$ with $p(C) > 0$, let κ_C be the law of the line which is vertical with probability $(b-a)/p(C)$ at a uniform position in (a,b) and horizontal

with probability $(d-c)/p(C)$ at a uniform position in (c,d); for a line λ meeting $\operatorname{int} C$, $C_0(\lambda)$ and $C_1(\lambda)$ are the closed left/lower and right/upper pieces.

Lemma A.5 (Nonexplosion)**.** *Almost surely finitely many nodes are born; the mean number of born nodes is at most $2e^{sp(C)}-1$; a fixed $z \in \operatorname{int} C$ is almost surely interior to a cell; and the event that z lies on a cell boundary is Borel in* $\mathrm{Part}^{\mathrm{col}}(C)$.

Proof. A daughter never has larger half-perimeter than its parent, so a born node w at depth k with strict ancestors $w_0, \dots, w_{k-1}$ has $s > b_w = \sum_j E_{w_j}/p(R_{w_j}) \geq \sum_j E_{w_j}/p(C)$. For a fixed word of length k the ancestors' clocks are independent $\mathrm{Exp}(1)$, so with $c = sp(C)$ the mean number of born nodes is at most $\sum_k 2^k \mathbb{P}(\Gamma_k \leq c) = \sum_k 2^k \mathbb{P}(\mathrm{Poi}(c) \geq k) = \sum_j \mathbb{P}(\mathrm{Poi}(c) = j)(2^{j+1}-1) = 2e^c - 1$, Γ_k a sum of k independent $\mathrm{Exp}(1)$. For a fixed node the cut position is $a + X_w(b-a)$ or $c + X_w(d-c)$ with X_w uniform and independent of the node's box, so it passes through z with probability 0; cell boundaries inside $\operatorname{int} C$ lie on ancestors' cut lines; there are countably many nodes. The boundary event is Borel by (K4). □

Definition A.6 (First-step recursion)**.** Let I be an index set and, for $i \in I$, C_i a box with nonempty interior, $s_i \geq 0$ and Q_i a probability on $\mathrm{Part}^{\mathrm{col}}(C_i)$. A *daughter assignment* gives, for every i, $t \in [0, s_i)$, line λ meeting $\operatorname{int} C_i$ and $\epsilon \in \{0,1\}$, a probability $Q_{i,t,\lambda,\epsilon}$ on $\mathrm{Part}^{\mathrm{col}}(C_\epsilon(\lambda))$ which is either Q_j for some $j \in I$ with $(C_j, s_j) = (C_\epsilon(\lambda), s_i - t)$ or $M^{\mathrm{col}}(C_\epsilon(\lambda), s_i - t)$, such that $(t,\lambda) \mapsto [Q_{i,t,\lambda,0} \otimes Q_{i,t,\lambda,1}](\cup^{-1} A)$ is Borel for Borel A. The family satisfies the *first-step recursion* if for every i and Borel A,

$$Q_i(A) = e^{-s_i p(C_i)} \mathbb{P}_\xi(\{(C_i,\xi)\} \in A) + \int_0^{s_i} p(C_i) e^{-p(C_i)t} \int \kappa_{C_i}(d\lambda)\, [Q_{i,t,\lambda,0} \otimes Q_{i,t,\lambda,1}](\cup^{-1} A)\, dt,$$

with ξ a uniform sign.

Lemma A.7 (First-step identity and characterization)**.** *(i) The laws $M^{\mathrm{col}}(C,s)$ satisfy the first-step recursion with daughter laws $M^{\mathrm{col}}(C_\epsilon(\lambda), s-t)$. (ii) Every family satisfying the first-step recursion with a daughter assignment has $Q_i = M^{\mathrm{col}}(C_i, s_i)$ for all i. (iii) For T a translation, θ_1 or θ_2, $T_* M^{\mathrm{col}}(T^{-1}C, s) = M^{\mathrm{col}}(C,s)$.*

Proof. (i) The root ingredients are independent of the two subfamilies indexed by $0w$ and $1w$, which are canonical families; $d_\emptyset = E_\emptyset/p(C) \sim \mathrm{Exp}(p(C))$ and $(D_\emptyset, X_\emptyset)$ produce $\lambda \sim \kappa_C$ independent of $d_\emptyset$. On $\{d_\emptyset \geq s\}$ the output is $\{(C,\xi_\emptyset)\}$; on $\{d_\emptyset < s\}$ the subtree rooted at ϵ, with births shifted by $d_\emptyset$, is the construction on $C_\epsilon(\lambda)$ with budget $s - d_\emptyset$; Fubini gives the formula, and joint Borel measurability of the output gives the measurability. (ii) Write $Q^{(k)}(A) = Q(A \cap \mathrm{Part}_k^{\mathrm{col}})$ for the k-cell part. Cells of the two daughters have disjoint interiors, so the k-cell part of the integrand involves only $Q^{(k_0)} \otimes Q^{(k_1)}$ with $k_0 + k_1 = k$, $k_\epsilon \geq 1$. Induct on k simultaneously for all i: for $k = 1$ only the first term contributes; for $k \geq 2$ every daughter law agrees with M^{col} on fewer than k cells by hypothesis, so $Q_i^{(k)}$ equals the k-cell part of the right side of (i). Both laws are carried by finite partitions (Lemma A.5). (iii) The family $T_* M^{\mathrm{col}}(T^{-1}C, s)$, indexed by all (C,s), satisfies the recursion: T commutes with $\cup$, preserves half-perimeters, maps $\kappa_{T^{-1}C}$ onto κ_C and daughters onto daughters, possibly exchanged, which does not change $[Q_0 \otimes Q_1](\cup^{-1} A)$. Apply (ii). □

Lemma A.8 (Competing clocks, conditional form)**.** *Let $A_1, \dots, A_k$ be independent, $A_j \sim \mathrm{Exp}(\alpha_j)$, and Z a random element independent of them; $\alpha = \sum \alpha_j$, $\tau = \min A_j$, $J = \operatorname{argmin} A_j$. Then $\mathbb{P}(J = j) = \alpha_j/\alpha$, and conditionally on $\{J = j\}$ the random elements τ, $(A_l - \tau)_{l \neq j}$ and Z are independent, with $\tau \sim \mathrm{Exp}(\alpha)$, $A_l - \tau \sim \mathrm{Exp}(\alpha_l)$ and Z distributed as unconditionally.*

Proof. $\mathbb{P}(J=j,\tau\in dt, A_l-\tau>r_l\ (l\neq j), Z\in G)=\alpha_j e^{-\alpha_j t}\prod_{l\neq j}e^{-\alpha_l(t+r_l)}\mathbb{P}(Z\in G)\,dt$. The residual vector is indexed by $l\neq J$ and is in general not independent of J: with $\alpha_1=1$, $\alpha_2=2$, the losing residual R has $\mathbb{P}(R>1\mid J=1)=e^{-2}$ and $\mathbb{P}(R>1\mid J=2)=e^{-1}$. □

A.3 The core-exterior construction

Definition A.9 (Core-exterior construction)**.** Let $K\subseteq C$ be boxes (K possibly degenerate), C with nonempty interior, $s\geq 0$. Ingredients, all independent: a canonical family $h=(E_w,D_w,X_w,\xi_w)_{w\in\mathrm{Tr}}$ (the *core*) and, for every $w\in\mathrm{Tr}$, side $\sigma\in\{L,R,D,U\}$ and $k\geq 1$, a triple $(E^\sigma_{w,k},X^\sigma_{w,k},G^\sigma_{w,k})$ with $E\sim\mathrm{Exp}(1)$, $X\sim U(0,1)$ and G a canonical family. Run the canonical construction on K with h (core boxes K_w, births b_w, deaths d_w, split lines); $e_w=\min(d_w,s)$. Each born core node carries an *enclosing box* $C_w(t)=[a_w(t),b_w(t)]\times[c_w(t),d_w(t)]\supseteq K_w=[a'_w,b'_w]\times[c'_w,d'_w]$, $t\in[b_w,e_w]$, with *gaps* $g^L=a'_w-a_w$, $g^R=b_w-b'_w$, $g^D=c'_w-c_w$, $g^U=d_w-d'_w$; initially $C_\emptyset(0)=C$. On $[b_w,e_w)$ side σ jumps at the times $t^\sigma_{w,k}=t^\sigma_{w,k-1}+E^\sigma_{w,k}/g^\sigma(t^\sigma_{w,k-1})$, $t^\sigma_{w,0}=b_w$ (no jumps while the gap is 0). At a jump time $t<e_w$ the gap g becomes $X^\sigma_{w,k}g$, and the removed slab (the *detached box*, of full current extent in the other coordinate) receives $\Pi^{\mathrm{can}}(\text{slab},s-t)$ built from $G^\sigma_{w,k}$: on the slab itself for $\sigma=R,U$, and on its image under θ_1 (for L) or θ_2 (for D), mapped back by the same reflection. If $d_w<s$, the core split line of K_w cuts $C_w(d_w)$ and each child receives the piece containing its core box (gap 0 on the side of the cut). The output is $\Pi^{\mathrm{ce}}(C,K,s)=\{(C_w(e_w),\xi_w): w\text{ a core leaf}\}\cup(\text{all detached colored partitions})$, with law $\mathcal{Q}(C,K,s)$.

Trace data. For $C=[-L,L]\times J_0$ ($L>0$, J_0 a compact interval of positive length) and $K=\{0\}\times J_0$, all core splits are horizontal, $K_w=\{0\}\times J_w$ and $g^D=g^U=0$. Write $Rr=(E^R_{w,k},X^R_{w,k},G^R_{w,k})_{w,k}$ and Ll likewise. The *straddling cells* are the enclosing boxes, and the trace data are $\mathrm{tr}(h)=((J_w,b_w,d_w)_{w\text{ born}},(\xi_w)_{w\text{ core leaf}})$: positions and auxiliary times of the horizontal cuts of straddling cells, and colors of the final straddling cells. For a side family Y (independent triples with the laws above) define widths $x_w(t)$: $x_\emptyset(0)=L$, children inherit $x_w(d_w)$, and on $[b_w,e_w)$ the width jumps from x to $X^Y_{w,k}x$ at times $t^Y_{w,k}=t^Y_{w,k-1}+E^Y_{w,k}/x$; for each jump with $t^Y_{w,k}<e_w$ put $S^Y_{w,k}=[X^Y_{w,k}x,x]\times J_w$, and

$$\Psi(h,Y)=\{([0,x_w(e_w)]\times J_w,\xi_w): w\text{ core leaf}\}\cup\bigcup_{t^Y_{w,k}<e_w}\Pi^{\mathrm{can}}(S^Y_{w,k},s-t^Y_{w,k};G^Y_{w,k}),$$

which depends on h only through $\mathrm{tr}(h)$. The same definitions with coordinates exchanged apply to $C=I_0\times[-L,L]$, $K=I_0\times\{0\}$.

Lemma A.10 (The core-exterior construction)**.** *(i) Almost surely the construction has finitely many events, its output is a finite colored partition of C and a Borel function of (C,K,s) and the ingredients; for every born w and $t\in[b_w,e_w]$, $K_w\subseteq C_w(t)$, $C_w(t)\cap K=K_w$, $C_w(t)$ has nonempty interior, and every detached box is disjoint from K. (ii) The family $\mathcal{Q}(C,K,s)$ satisfies the first-step recursion, with daughter laws $\mathcal{Q}(C_\epsilon(\lambda),K\cap C_\epsilon(\lambda),s-t)$ for a line meeting the relative interior of K in its cut coordinate, $\mathcal{Q}(C_\epsilon(\lambda),K,s-t)$ for the daughter containing K and $M^{\mathrm{col}}(C_\epsilon(\lambda),s-t)$ for the other daughter when the line lies in a gap, and M^{col} for the null set of lines through ∂K. (iii) $\mathcal{Q}(C,K,s)=M^{\mathrm{col}}(C,s)$.*

Proof. (i) The core is the canonical construction on K. A gap never exceeds $p(C)$, so inter-jump times are at least $E/p(C)$ and the number of jumps of one gap process in a window of length at most s is dominated by a $\mathrm{Poi}(sp(C))$ variable; each detached output is finite (Lemma A.5); there are countably many items. Measurability: decompose by the finite core tree, the finite

numbers and order of side events and the finite detached trees. Invariants by induction: a jump on side R removes the slab $[b'_w + Xg, b'_w + g] \times [c_w, d_w] \subset \{z_1 > b'_w\}$, which misses K because $C_w \cap K = K_w \subseteq \{z_1 \le b'_w\}$; a core split of K_w by $\{z_1 = x\}$ gives $C_{w0} \cap K = K_w \cap \{z_1 \le x\} = K_{w0}$; other sides and horizontal splits are analogous; core splits are strictly inside K_w's extent and gaps are multiplied by $X \in (0,1)$. (ii) The root clocks are the core clock $A_K = E_\emptyset / p(K)$ (if $p(K) > 0$) and the gap clocks $A_\sigma = E^\sigma_{\emptyset,1}/g^\sigma(0)$ (if positive), independent of each other and of all remaining ingredients Z; the rates sum to $p(K) + \sum_\sigma g^\sigma = p(C)$. By Lemma A.8, $\tau \sim \mathrm{Exp}(p(C))$, $\mathbb{P}(J = \text{core}) = p(K)/p(C)$, $\mathbb{P}(J = \sigma) = g^\sigma(0)/p(C)$. The first line is a function of J and one uniform position variable (a component of Z): a core line has law κ_K, a gap line on side R is uniform on (b', b), and so on; hence (τ, λ) has law $p(C)e^{-p(C)t}dt\,\kappa_C(d\lambda)$, and off a null set J is a function of λ. If $\tau \ge s$ the output is $\{(C, \xi_\emptyset)\}$. On a core event at $\tau < s$, the children's futures are the construction on $(C_\epsilon(\lambda), K_\epsilon, s - \tau)$ driven by distinct components of Z, conditionally independent of (τ, λ) with the ingredient law. On a gap event on side σ, the slab receives a canonical output of law $M^{\mathrm{col}}(S, s - \tau)$ (Lemma A.7(iii) for L, D), and the other daughter C' carries the continuation driven by the family Y' in which the core clock is replaced by $p(K)(A_K - \tau)$, side σ by its shifted sequence with the new gap, and each other side σ' by $g^{\sigma'}(0)(A_{\sigma'} - \tau)$; pathwise every later event of the original construction is the corresponding event of the construction on $(C', K, s - \tau)$ with ingredients Y', shifted by τ, and by Lemma A.8 the scaled residuals are fresh Exp(1) variables independent of τ and of Z. Integrating over (τ, λ) gives the recursion. (iii) Apply Lemma A.7(ii) to the family indexed by all triples. □

Corollary A.11 (Restriction and two points)**.** *(i) If $K = B$ has nonempty interior,*

$$\mathrm{res}_B(\Pi^{\mathrm{ce}}(C, B, s)) = \Pi^{\mathrm{can}}(B, s; h),$$

and $(\mathrm{res}_B)_* M^{\mathrm{col}}(C, s) = M^{\mathrm{col}}(B, s)$. (ii) For $z \ne z'$ in C and K the box they span, z and z' lie in one cell of $\Pi^{\mathrm{ce}}(C, K, s)$ if and only if the core root does not split before s; hence for $M^{\mathrm{col}}(C, s)$ the probability that some cell contains both is $e^{-s|z-z'|_1}$.

Proof. (i) $C_w(e_w) \cap B = K_w$ for core leaves and detached cells miss B (Lemma A.10(i)); apply Lemma A.10(iii). (ii) If the root does not split, $z, z' \in K \subseteq C_\emptyset(s)$; if it splits, the line lies strictly between the two points in one coordinate, and every final cell lies in a detached slab (disjoint from K) or in one closed side of the line. $p(K) = |z - z'|_1$, and the event is Borel. □

Corollary A.12 (Trace decomposition and box positivity)**.** *Let $C = [-L, L] \times J_0$ and $K = \{0\} \times J_0$. (i) $g^D = g^U = 0$ for all nodes and times; the right and left gaps evolve as the widths x_w of Definition A.9 driven by Rr and by Ll respectively. (ii) $\mathrm{res}_{[0,L]\times J_0}(\Pi^{\mathrm{ce}}) = \Psi(h, Rr)$ and $\theta_1 \mathrm{res}_{[-L,0]\times J_0}(\Pi^{\mathrm{ce}}) = \Psi(h, Ll)$. (iii) h, Rr, Ll are independent, $Rr \overset{d}{=} Ll$, and $\Psi(h, Y) = \hat\Psi(\mathrm{tr}(h), Y)$ for a Borel $\hat\Psi$. (iv) For N independent copies and bounded complex Borel Φ on $\mathrm{Part}^{\mathrm{col}}([0, L] \times J_0)^N$, $\mathbb{E}[\overline{\Phi((\Psi(h^i, Ll^i))_i)}\Phi((\Psi(h^i, Rr^i))_i)] = \mathbb{E}|\bar\Phi(h^1, \dots, h^N)|^2$ with $\bar\Phi = \int \Phi((\Psi(h^i, y^i))_i) \prod_i P_{Rr}(dy^i)$, a function of the trace data. (v) For independent $\Pi^1, \dots, \Pi^N$ with law $M^{\mathrm{col}}(C, s)$, $T^i_+ = \mathrm{res}_{[0,L]\times J_0}(\Pi^i)$ and $T^i_- = \theta_1 \mathrm{res}_{[-L,0]\times J_0}(\Pi^i)$, $\mathbb{E}[\overline{\Phi(T^1_-, \dots, T^N_-)}\Phi(T^1_+, \dots, T^N_+)] \ge 0$; the same holds across θ_2 for $C = I_0 \times [-L, L]$.*

Proof. (i) The vertical extents of C and K coincide initially, core lines are horizontal (K has zero width) and cut C_w and K_w at the same height, and side jumps L, R do not change vertical extents; at a horizontal straddling cut both daughters inherit the current widths and then use their own triples. (ii) Final straddling cells are $[-l_w(e_w), r_w(e_w)] \times J_w$ and restrict to $[0, r_w(e_w)] \times J_w$; R-slabs

$[Xr, r]\times J_w$ carry $\Pi^{\mathrm{can}}(\cdot; G^R)$ and L-slabs carry $\theta_1\Pi^{\mathrm{can}}([Xl, l]\times J_w; G^L)$; cells of one half-box meet the other in at most a segment and are discarded by the restriction. (iii) The ingredients are independent by definition, and Ψ uses h only through the born-node data and the leaf colors; measurability as in Lemma A.10(i). (iv) By Fubini, $\mathbb{E}[\Phi((\Psi(h^i, Rr^i))_i) \mid h] = \bar{\Phi}(h)$ and $\mathbb{E}[\overline{\Phi((\Psi(h^i, Ll^i))_i)} \mid h] = \overline{\bar{\Phi}(h)}$, and the two are conditionally independent given h. (v) By Lemma A.10(iii) the law of $(T^i_+, T^i_-)_i$ is that of N independent copies of $(\Psi(h^i, Rr^i), \Psi(h^i, Ll^i))$; apply (iv). Only the complete trace data are used; the finite-box hand checks of the endpoint record exhibit loss of conditional independence when either the auxiliary times or the positions are dropped from the conditioning. □

Proof of Theorems 6.2 and 6.3. Theorem 6.2: (a) Lemma A.5 and the measurability paragraph before it; (b) Lemma A.7(i),(ii); (c) Lemma A.7(iii); (d) Corollary A.11(i); (e) Corollary A.11(ii) and Lemma A.5. Theorem 6.3: Lemma A.10(iii), Corollary A.11(i) and Corollary A.12. □

A.4 The whole-plane Mondrian neuron

Proof of Theorem 6.4(a). The bonding maps $\pi_n = \mathrm{res}_{B_n}$ are Borel with $(\pi_n)_* M^{\mathrm{col}}(B_{n+1}, \mu) = M^{\mathrm{col}}(B_n, \mu)$ (Corollary A.11(i)); drawing $P_m \sim M^{\mathrm{col}}(B_m, \mu)$ and setting $P_n = \pi_n \circ \cdots \circ \pi_{m-1}(P_m)$ gives consistent finite-dimensional laws, and (K5) gives a probability on $\prod_n \mathrm{Part}^{\mathrm{col}}(B_n)$, under which the Borel set S of consistent sequences has full measure. For any consistent tessellation $(P_1, \dots, P_m)$ is almost surely the image of P_m under the bonding maps, so the law is unique (K1). Restriction is transitive ($(R \cap B_n) \cap B = R \cap B$, and a nonempty interior of $R \cap B$ forces one of $R \cap B_n$), so $\mathrm{res}_B(P_m) = \mathrm{res}_B(P_n)$ on S, with law $M^{\mathrm{col}}(B, \mu)$. For $T = \theta_j$, $(TP)_n = TP_n$ is consistent because $TB_n = B_n$, with marginals $M^{\mathrm{col}}(B_n, \mu)$ (Lemma A.7(iii)); for a translation by x and $m = n + \lceil |x|_\infty \rceil$, $(TP)_n = \mathrm{res}_{B_n}(TP_m)$ does not depend on the admissible m, is consistent, and has law $M^{\mathrm{col}}(B_n, \mu)$; uniqueness gives $TP \overset{d}{=} P$. The last sentence of (a) is part (A) of the next proof. □

Proof of Theorem 6.4(b),(c). (A) Let $P \in \mathrm{Part}^{\mathrm{col}}(D)$, $D = [a_D, b_D] \times [c_D, d_D]$ with nonempty interior, and $z \in [a_D, b_D) \times [c_D, d_D)$. The points $z + e(1, 1)$, $0 < e < e_0$, lie in $\mathrm{int}\, D$ and avoid the finitely many cell boundaries (an axis-parallel segment meets the open diagonal ray in at most one point), so they lie in the interior of one cell R, and then $z \in R^{\mathrm{ll}}$; conversely $z \in R^{\mathrm{ll}}$ forces $z + e(1, 1) \in \mathrm{int}\, R$ for small e. So the lower-left representatives partition $[a_D, b_D) \times [c_D, d_D)$. (B) On the consistency event, for $z \in [-n, n)^2$: $R_n(z) = R_{n+1}(z) \cap B_n$, $\xi_n(z) = \xi_{n+1}(z)$ and $R_n(z)^{\mathrm{ll}} \subseteq R_{n+1}(z)^{\mathrm{ll}}$, since z lies in the representative of $R_{n+1}(z) \cap B_n$, a cell of P_n with the same color; uniqueness in (A) gives the claim. (C) For every outcome $g(z) = \sum_n \mathbf{1}\{n_z = n\} \int hh(z, r)\, P_n(dr)$ with $hh(z, (a, b, c, d, \xi)) = (\xi/2)\mathbf{1}\{a \le z_1 < b, c \le z_2 < d\}$, exactly one atom contributing by (A); so g is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable (Lemma A.1(iii)) with $|g| = 1/2$. (D) If $R(z) \not\subseteq z + (-m, m)^2$, some $R_n(z)$ contains one of $z \pm me_1$, $z \pm me_2$; the events that some cell of P_n contains two fixed points increase in n and have probability $e^{-\mu|y-y'|_1}$ (Corollary A.11(ii)); so $\mathbb{P}(R(z) \not\subseteq z + (-m, m)^2) \le 4e^{-\mu m}$ and almost surely every $R(q)$, $q \in \mathbb{Q}^2$, is bounded. Every z has a rational q in the interior of $R_{n_z}(z)^{\mathrm{ll}}$, and then $R(q) = R(z)$, $H(q) = H(z)$. (E) If $R(z) \subseteq B_k$, $k \ge n_z$, then $R_n(z) = R_{n+1}(z)$ for $n \ge k$, so $R(z) = R_k(z)$ and $H(z) = R(z)^{\mathrm{ll}}$. If $w \in H(z) \cap H(z')$, then $R_n(z) = R_n(z') = R_n(w)$ for large n; so distinct representatives are disjoint and cover $\mathbb{R}^2$; $R(z) = \mathrm{cl}\, H(z)$ and $\mathrm{int}\, R(z) = \mathrm{int}\, H(z)$; a representative meeting B_k contains a representative of a cell of P_{k+1}, so finitely many meet B_k; $g = \xi(z)/2$ is constant on $H(z)$. (F) Under $M^{\mathrm{col}}(B_n, \mu)$ the colors are independent uniform given the uncolored partition, so $\mathbb{E}g(z) = 0$ and $\mathbb{E}g(z)g(z') = \frac{1}{4}\mathbb{P}(R_n(z) = R_n(z')) = \frac{1}{4}e^{-\mu|z-z'|_1}$, using that z, z' are almost surely interior to cells (Lemma A.5). □

Proof of Theorem 6.4(d). On the consistency event, if z is interior to a cell of P_{n_z+1}, that cell is the only one containing z, its color is $\xi(z)$, and every admissible convention gives $g'(z) = g(z)$; the exceptional points lie on countably many segments. Let $Y_{\mathrm{sp}} = \prod_n \mathrm{Part}^{\mathrm{col}}(B_n)$ and $G(z, (P'_n)_n) = \sum_n \mathbf{1}\{n_z = n\} \int hh(z, r)\, P'_n(dr)$. Then G is product measurable and $|G| = 1/2$ on all of $\mathbb{R}^2 \times Y_{\mathrm{sp}}$, consistent or not, because the representatives of any element of $\mathrm{Part}^{\mathrm{col}}(B_{n_z})$ partition $[-n_z, n_z)^2$ (part (A) above uses no consistency). So $\Gamma_G^{1/2}$ of Lemma A.2(iii) is a total measurable map with $[g(\cdot, \omega)|_U] = \Gamma_G^{1/2}((P_n(\omega))_n)$ for every outcome, and the law of $g|_U$ is the image of the unique law of $(P_n)_n$. An admissible g' is clipped to $F' = \mathrm{clip}_{1/2}(g')$, which equals g' on the consistency event and g almost everywhere there, so $g'|_U$ has the same law. For $T = \theta_j$, the field of TP equals $g \circ \theta_j$ off a null set on the consistency event and is $G(\cdot, (TP_n)_n)$, so $g \circ \theta_j \overset{d}{=} g$ because $TP \overset{d}{=} P$; translations are the same with $(TP)_n = \mathrm{res}_{B_n}(TP_m)$. □

A.5 The product neuron

Proof of Theorem 6.1. (a) The Poisson process $\mathcal{N}$ is almost surely locally finite, has no atom at 0 and has unbounded atoms in both directions; then σ is piecewise constant and càdlàg, and $|\sigma| = 1$. $\mathbb{E}\sigma(t) = 0$; for $t < t'$, $\sigma(t)\sigma(t') = (-1)^{\mathcal{N}((t,t'])}$ in each sign configuration, so $\mathbb{E}\sigma(t)\sigma(t') = e^{-2(\mu/2)|t-t'|}$ (K6); given $\sigma(t_1)$, parities on disjoint later intervals are independent, which gives the Markov chain with transition probabilities $(1 + e^{-\mu u})/2$.

(b) A right-continuous process is jointly measurable ($\sigma(t) = \lim_k \sigma(2^{-k}\lceil 2^k t\rceil)$), so g is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable, $|g| = 1/2$, $\mathbb{E}g = 0$ and $\mathbb{E}g(z)g(z') = \frac{1}{4} e^{-\mu|z_1 - z'_1|} e^{-\mu|z_2 - z'_2|}$; g is constant on the boxes $[t_i, t_{i+1}) \times [y_j, y_{j+1})$ of the jump points.

(c) Across θ_1 let $b = (\epsilon_\sigma, \tau)$, $M_+ = \mathcal{N}|_{(0,\infty)}$ and M_- the image of $\mathcal{N}|_{(-\infty,0)}$ under $t \mapsto -t$. These are random elements of the space N of $\{0, 1, \ldots, \infty\}$-valued measures on $(0, \infty)$ (with the σ-algebra of the maps $m \mapsto m(A)$); by (K6) they are independent with the same law P_{M} and independent of b. Let $D_\pm$ be the càdlàg functions $\mathbb{R} \to \{-1, 1\}$ with the evaluation σ-algebra; $(t, x) \mapsto x(t)$ is product measurable by right continuity. Put $Y_{\mathrm{prod}} = \{-1, 1\} \times D_\pm \times \mathsf{N}$, $\mathrm{Good}_+ = \{m : m((0, k]) < \infty\ \forall k\}$, $\mathrm{par}(m, t) = (-1)^{m((0,t])}$ on Good_+ and 1 otherwise, and $G_{\mathrm{prod}}(z, (e, x, m)) = \frac{1}{2} e\, \mathrm{par}(m, z_1) x(z_2)$ for $z_1 > 0$, 0 otherwise. Then G_{prod} is product measurable with $|G_{\mathrm{prod}}| \le 1/2$ everywhere, and Lemma A.2(iii) gives a total measurable $\Psi_{\mathrm{prod}} : Y_{\mathrm{prod}} \to \mathrm{Ball}_{1/2}(H_+)$. On the almost sure event of (a), $s_+ = G_{\mathrm{prod}}(\cdot, (\epsilon_\sigma, \tau, M_+))$ on H_+, and $s_-(z) = \frac{1}{2}\epsilon_\sigma (-1)^{\mathcal{N}((-z_1, 0])} \tau(z_2)$ differs from $G_{\mathrm{prod}}(z, (\epsilon_\sigma, \tau, M_-))$ only on countably many vertical lines; so $[s_\pm] = \Psi_{\mathrm{prod}}(b, M_\pm)$ almost surely, and Fubini gives $\mathbb{E}[f(s_+)f'(s_-) \mid b] = \bar{f}(b)\bar{f}'(b)$ with $\bar{f}(b) = \int f(\Psi_{\mathrm{prod}}(b, m)) P_{\mathsf{M}}(dm)$. Across θ_2 exchange σ and τ.

(d) For independent neurons, $(b_i)_i$, $(M^i_+)_i$, $(M^i_-)_i$ are independent families, the last two independent and identically distributed with the same law, and $S_\pm = N^{-1/2} \sum_i \Psi_{\mathrm{prod}}(b_i, M^i_\pm)$ almost surely; every value of $N^{-1/2} \sum_i \Psi_{\mathrm{prod}}(y_i)$ lies in $\mathrm{Ball}_{\sqrt{N}/2}(H_+)$. The same Fubini computation with the complete tuple $(b_1, \ldots, b_N)$ gives conditional independence and equal conditional laws. With only b_1 given and $N = 2$, neuron 2 is shared by both sides: $\mathrm{Cov}(F_2(t, 0), F_2(-t, 0) \mid b_1) = e^{-2\mu t}/8 > 0$ for $t > 0$, and point values are almost surely Borel functions of the sides (proof of Theorem 6.5(c) below), so the sides are not conditionally independent given b_1 alone. □

A.6 Finite-width annealed reflection positivity

Fix b, $N \ge 1$, θ, Ω and v as in Theorem 6.5; $|F_N| \le r_N$ everywhere.

Lemma A.13 (Projections)**.** *If $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \cdots$ are σ-algebras, $\mathcal{G}_\infty = \sigma(\bigcup_n \mathcal{G}_n)$ and $Y \in L^2(\mathcal{G}_\infty)$, then $\mathbb{E}[Y \mid \mathcal{G}_n] \to Y$ in L^2.*

Proof. The conditional expectations are orthogonal projections onto increasing closed subspaces V_n; the events $A \in \mathcal{G}_\infty$ with $\mathbf{1}_A$ in the closure of $\bigcup V_n$ form a Dynkin system containing the π-system $\bigcup \mathcal{G}_n$, so the closure is $L^2(\mathcal{G}_\infty)$; for $W \in V_m$ with $\|Y-W\| < e$ and $n \geq m$, $\|Y-\mathbb{E}[Y \mid \mathcal{G}_n]\| \leq \|Y-W\|$. □

Lemma A.14 (Weight factorization and reduction to the sides)**.** *Let $\Psi_+(s) = \int_{\Omega\cap H_+} v(z, s(z))\, dz$ on $\mathrm{Ball}_{r_N}(H_+)$ (Borel by Lemma A.3). Then $U = \Psi_+(S_+) + \Psi_+(S_-)$ for every outcome. For bounded complex Σ_+^θ-measurable F, with the representation of Proposition 5.1(x), $\Phi_F(s) = g((\int f_k s)_k)$ is bounded Borel on $\mathrm{Ball}_{r_N}(H_+)$, $F(\iota F_N) = \Phi_F(S_+)$, $F(\theta\iota F_N) = \Phi_F(S_-)$, and $\int_X \overline{F(\theta v)} F(v)\, P_N^v(dv) = Z^{-1}\mathbb{E}[\overline{\Phi(S_-)}\Phi(S_+)]$ with $\Phi = \Phi_F e^{-\Psi_+}$.*

Proof. The line ∂H_+ is null; the substitution $z' = \theta z$, $\theta(\Omega\cap H_+) = \Omega\cap H_-$ and $v(\theta z', x) = v(z', x)$ give $\Psi_+(S_-) = \int_{\Omega\cap H_-} v(z', F_N(z'))\, dz'$. For the second part, $\int f_k F_N = \int f_k S_+$ and $\int f_k (F_N \circ \theta) = \int f_k S_-$ because $\mathrm{supp}\, f_k \subset H_+$, and $\theta\iota F_N = \iota(F_N \circ \theta)$ (Proposition 5.1(vii)). □

Proposition A.15 (Unweighted side positivity)**.** *$(S_+, S_-) \overset{d}{=} (S_-, S_+)$, and $\mathbb{E}[\overline{\Phi(S_-)}\Phi(S_+)] \geq 0$ for every bounded complex Borel Φ on $\mathrm{Ball}_{r_N}(H_+)$.*

Proof. Product neurons. By Theorem 6.1(c),(d), $S_\pm = N^{-1/2}\sum_i \Psi_{\mathrm{prod}}(b_i, M_\pm^i)$ with independent families and equal laws of the last two, which gives the symmetry, and Fubini gives $\mathbb{E}[\overline{\Phi(S_-)}\Phi(S_+)] = \mathbb{E}|\bar\Phi(b_1, \ldots, b_N)|^2$ with $\bar\Phi(b) = \int \Phi(N^{-1/2}\sum_i \Psi_{\mathrm{prod}}(b_i, y_i)) \prod_i P_{\mathsf{M}}(dy_i)$.

Mondrian neurons. Step 1: $g_i \circ \theta \overset{d}{=} g_i$ (Theorem 6.4(d)), so $F_N \circ \theta \overset{d}{=} F_N$, and the continuous map $w \mapsto (w|_{H_+}, (w \circ \theta)|_{H_+})$ gives the symmetry. Step 2: take $\theta = \theta_1$, $B_n^+ = [0, n] \times [-n, n]$, $Q_n = (0, n) \times (-n, n)$. On the consistency events, off the null cell boundaries, $g_i = \mathrm{fld}(\mathrm{res}_{B_n^+}(P_n^i))$ and $g_i \circ \theta_1 = \mathrm{fld}(\theta_1 \mathrm{res}_{[-n,0]\times[-n,n]}(P_n^i))$ on Q_n; so $r_n(S_\pm) = \Lambda_n((T_{\pm,n}^i)_i)$ with the Borel map $\Lambda_n((P'^i)_i) = N^{-1/2}\sum_i \mathrm{fld}(P'^i)|_{Q_n}$, and Corollary A.12(v) with $C = B_n$ gives $\mathbb{E}[\overline{\Phi_n(r_n S_-)}\Phi_n(r_n S_+)] \geq 0$ for bounded Borel Φ_n on $L^1(Q_n)$. Step 3: by Lemma A.4(ii) the σ-algebras $\mathcal{G}_n = \sigma(r_n(S_+))$ increase and generate $\sigma(S_+)$; with $Y = \Phi(S_+)$ and $\mathbb{E}[Y \mid \mathcal{G}_n] = \Phi_n(r_n S_+)$ (K2; truncated to $|\Phi_n| \leq \sup|\Phi|$), Lemma A.13 gives $\Phi_n(r_n S_+) \to \Phi(S_+)$ in L^2, Step 1 gives the same for the minus side, and Cauchy–Schwarz passes the inequality to the limit. □

Proof of Theorem 6.5. (a) $(z, \omega) \mapsto v(z, F_N(z, \omega))$ is $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable, so U is a random variable with $|U| \leq |\Omega| M_N$ and the stated bounds on Z (no bound $Z \geq 1$ is claimed). F_N is a Borel random element of $\mathrm{Ball}_{r_N}(\mathbb{R}^2)$ and of E (Lemma A.2), and on Ball_{r_N} the inclusion into E is continuous ($|s_k - s|^4 \leq (2r_N)^3 |s_k - s|$), so ιF_N is Borel and P_N^v is a Borel probability. $U = \Psi_v(F_N)$ (Lemma A.3), so P_N^v depends only on the law of F_N, which is the image of the N-fold product of one neuron's law under the continuous sum map; that law is determined by the finite-dimensional laws for product neurons (Lemma A.2(ii)) and by Theorem 6.4(d) for Mondrian neurons. Admissible conventions change F_N only on a null set of z, on the almost sure intersection of the consistency events, so U and ιF_N are unchanged almost surely. (b) Lemma A.14 gives $e^{-U}\overline{\Phi(S_-)}\Phi(S_+) = \overline{\Phi'(S_-)}\Phi'(S_+)$ with $\Phi' = \Phi e^{-\Psi_+}$; apply Proposition A.15, and for X-observables Lemma A.14 and $Z > 0$. (c) The first assertion is the computation in (b). For each fixed z, almost surely F_N is constant near z (for Mondrian neurons z is interior to a cell of every P_n^i with $z \in \mathrm{int}\, B_n$; for product neurons z_1, z_2 are not atoms of the locally finite Poisson processes). Hence, almost surely for all k, $L_{z_k}(S_+) = F_N(z_k)$ and $L_{z_k}(S_-) = F_N(\theta z_k)$ (Lemma A.4(iii)), so $Y = \Phi(S_+)$ and

$Y^\theta = \Phi(S_-)$ with the bounded Borel $\Phi = \tilde{Y} \circ (L_{z_k})_k$. If Φ' comes from another representation, $\Phi(S_+) = \Phi'(S_+)$ almost surely, and $S_- \overset{d}{=} S_+$ gives $\Phi(S_-) = \Phi'(S_-)$ almost surely. □

Proof of Corollaries 6.6 and 6.7. Corollary 6.6: each coordinate interval of Λ and Ω is symmetric about 0, so both rectangles are invariant under θ_1 and θ_2; v is Borel, bounded on $\Omega \times [-r, r]$ by $\max_{|x| \le r}(|q_{\text{in}}| + |q_{\text{out}}| + \kappa^2 x^2/2)$ and invariant under both reflections on Ω; Theorem 6.5 applies to each θ separately. For a physical square, $\bar{\Lambda} = [-l, l]^2$ and $\Omega = [-R, R]^2$ are such rectangles; off $\partial\Lambda$, $\mathbf{1}_\Lambda = \mathbf{1}_{\bar{\Lambda}}$ and $\mathbf{1}_{\Omega\setminus\Lambda} = \mathbf{1}_{\Omega\setminus\bar{\Lambda}}$, so v_V agrees with the displayed v off the null set $\partial\Lambda \times \mathbb{R}$, and for every outcome $\int_\Omega v_V(z, F_N(z))\, dz = V_{\varepsilon,\Omega,\underline{a};\Lambda}(F_N)$, which is also the U of the displayed v. Axis-parallelism is used exactly in the identification $\bar{\Lambda} = [-l, l]^2$; for the rotated square of Remark 6.8(i) no conclusion, positive or negative, is drawn. The hypotheses "Ω centered" and "Λ centered" are both used: an outer rectangle that is invariant under θ_1 only does not satisfy the hypothesis for θ_2, and no claim is made for it. Corollary 6.7: the hypotheses of Theorem 6.5 hold for each θ. □

Remark A.16 (What the endpoint results do not contain). The conditional identities of Corollary A.12 are proved on the extended probability space of the core-exterior construction in a box; the auxiliary times are not functions of the final tessellation, and no whole-plane auxiliary-time history is constructed. Not needed and not proved: the Poisson law of trace cut points; reflection positivity across shifted or rotated lines; any Euclidean invariance of tilted laws; reflection positivity of the limits in N, ε, Ω or a by these arguments; and reflection positivity, or its failure, for tilts built on rotated squares that are not invariant under both reflections. The measurability hypothesis is used exactly in Lemma A.3 and in part (a) of Theorem 6.5; the broader reading of Remark 6.8(iii) fails.

B Proofs for the Gaussian reference fields

Besides C6(c)–(g) the following elementary Gaussian facts are used: an L^2-limit of centered jointly Gaussian vectors is centered jointly Gaussian; uncorrelated jointly Gaussian families are independent; for $\xi \sim N(0, \lambda)$, $\lambda > 0$, $t\lambda < 1$: $\mathbb{E}e^{t\xi^2/2} = (1 - t\lambda)^{-1/2}$, and the law with density proportional to $e^{tx^2/2}$ relative to $N(0, \lambda)$ is $N(0, \lambda/(1 - t\lambda))$; a centered Gaussian of variance v has $\mathbb{E}|Y|^{2n} = (2n - 1)!!v^n$; and *regression*: if $(Y, \xi_1, \dots, \xi_N)$ is centered jointly Gaussian, $Y \in \mathbb{R}^n$, with independent ξ_j of variances $\lambda_j > 0$, then $Y - \sum_j \mathbb{E}[Y\xi_j]\xi_j/\lambda_j$ is independent of (ξ_j) with covariance $\operatorname{Cov} Y - \sum_j \mathbb{E}[Y\xi_j]\mathbb{E}[Y\xi_j]^T/\lambda_j$. The path space $C(\mathbb{R}^2)$ carries the metric of uniform convergence on compacts; it is separable, its Borel σ-algebra is generated by the evaluations at rational points, and two Borel probabilities on it with the same finite-dimensional laws at rational points coincide.

Lemma B.1 (Spectral point fields)**.** *Let ν be an even σ-finite Borel measure on $\mathbb{R}^2$, $H_\nu = \{h \in L^2(\nu; \mathbb{C}) : h(-p) = \overline{h(p)}\}$ with $\langle g, h\rangle = \int \bar{g}h\, d\nu$, W an isonormal process on H_ν, $\sigma \ge 0$ even and Borel with $S_\sigma = \int \sigma^2 d\nu < \infty$, and $X(z) = W(e_z\sigma)$.*

(S0) H_ν is a separable real Hilbert space.

(S1) X is centered Gaussian with $\mathbb{E}[X(z)X(z')] = k(z - z')$, $k(x) = \int \cos(p \cdot x)\sigma^2 d\nu$, real, even, continuous, $|k| \le k(0) = S_\sigma$; and $\mathbb{E}[X(z)W(h)] = \int e^{ip\cdot z}\sigma h\, d\nu$.

(S2) $\mathbb{E}|X(z + h) - X(z)|^2 = \int 2(1 - \cos(p \cdot h))\sigma^2 d\nu = 2(k(0) - k(h))$.

(S3) If there are constants $M > 0$ and $\vartheta > 0$ with $2(k(0) - k(h)) \le M|h|^\vartheta$ for all $h \in \mathbb{R}^2$, there is a $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{F}$-measurable $\tilde{X}$ with continuous paths for every outcome, $\mathbb{P}(\tilde{X}(z) = X(z)) = 1$ for

every z, which is a Borel $C(\mathbb{R}^2)$-valued random element whose law is the unique Borel law with centered Gaussian coordinates of covariance $k(z-z')$.

(S4) Under (S3), for real $g \in L^1(\mathbb{R}^2)$: $\hat{g}\sigma \in H_\nu$, $\int |g||\tilde{X}| < \infty$ almost surely, and $\int g\tilde{X}\,dz = W(\hat{g}\sigma)$ almost surely.

Proof. (S0) $p \mapsto -p$ preserves ν, so $\langle g, h\rangle$ is real; H_ν is closed in $L^2(\nu)$ (an a.e.-convergent subsequence preserves the symmetry) and separable. (S1) $e_z\sigma \in H_\nu$; isometry gives $\mathbb{E}[X(z)X(z')] = \int e^{ip\cdot(z-z')}\sigma^2 d\nu$, whose sine part vanishes by evenness; continuity by dominated convergence; all $X(z)$ lie in the range of W. (S2) $|e^{-it}-1|^2 = 2(1-\cos t)$. (S3) Increments are Gaussian with $\mathbb{E}|X(z+h)-X(z)|^{2n} \le (2n-1)!!M^n|h|^{n\vartheta}$; for an integer $n > 2/\vartheta$ (which exists because $\vartheta > 0$) the continuity criterion C6(d) applies with $\alpha = 2n$, $\beta = n\vartheta - 2 > 0$. The applications below use $(M,\vartheta) = (\mu/\sqrt{2}, 1)$, $(M'_{a,1/2}, 1/2)$ and $(1/(4\pi), 2)$, all with $M, \vartheta > 0$. Joint measurability: the dyadic approximations $\tilde{X}(2^{-l}\lfloor 2^l z\rfloor, \omega)$ are product measurable and converge everywhere by continuity. Borel measurability and the law follow from the path-space facts. (S4) $|\hat{g}| \le \|g\|_1$ and $\hat{g}(-p) = \overline{\hat{g}(p)}$; $\mathbb{E}\int |g||\tilde{X}| \le \|g\|_1 S_\sigma^{1/2}$ and $\mathbb{E}(\int |g||\tilde{X}|)^2 \le \|g\|_1^2 S_\sigma$, which justify Fubini in: $\mathbb{E}I^2 = \iint g(z)g(z')k(z-z') = \int |\hat{g}|^2\sigma^2 d\nu$, $\mathbb{E}[IW(\hat{g}\sigma)] = \int g(z)\mathbb{E}[X(z)W(\hat{g}\sigma)]dz = \int |\hat{g}|^2\sigma^2 d\nu$ and $\mathbb{E}W(\hat{g}\sigma)^2 = \int |\hat{g}|^2\sigma^2 d\nu$, with $I = \int g\tilde{X}$; hence $\mathbb{E}|I - W(\hat{g}\sigma)|^2 = 0$. □

Proof of Theorem 7.1. Take $\nu = (2\pi)^{-2}\,dp$ and $\sigma = (\hat{C}_\mu^{\mathrm{end}})^{1/2}$. In one variable $\int e^{-iwt}e^{-\mu|t|}dt = 2\mu/(\mu^2+w^2)$, so $\hat{C}_\mu^{\mathrm{end}}$ is as in (1), $\int \hat{C}_\mu^{\mathrm{end}}\,dp = \pi^2$, $S_\sigma = 1/4$, and by Fourier inversion $k = C_\mu^{\mathrm{end}}$. Then $2(k(0)-k(h)) = (1-e^{-\mu|h|_1})/2 \le \mu|h|_1/2 \le \mu|h|/\sqrt{2}$, and Lemma B.1(S3) with $\vartheta = 1$ gives γ_μ and the law G_μ. □

Proof of Theorem 7.2. (a) $\hat{C}_a \le \hat{C}_0 \le m^{-2}$; since $a^2m^2 < 1$, termwise $m^2(1+a^2w^2)(1+a^2q^2) \le m^2+w^2+q^2+a^2w^2q^2$, which gives the product bound and $\int \hat{C}_a \le m^{-2}\pi^2/a^2$; $\int_{\mathbb{R}} dw/(A+Bw^2) = \pi/\sqrt{AB}$ with $A = m^2+q^2$, $B = 1+a^2q^2$ gives (2), whose integrand is at most $1/m$ and at most $1/(aq^2)$. (b) $1/\hat{C}_\mu^{\mathrm{end}} = (\mu^2+w^2)(\mu^2+q^2)/\mu^2 = \mu^2+w^2+q^2+a^2w^2q^2$, and $\hat{C}_\mu^{\mathrm{end}} \le \mu^{-2}$ gives $\kappa_a^2\hat{C}_\mu^{\mathrm{end}} \le r_a < 1$, so the geometric series converges pointwise. (c) $s_0 = C_\mu^{\mathrm{end}}$ is continuous, positive, $\sup s_0 = 1/4$, $\|s_0\|_1 = \mu^{-2}$; inductively $s_{k+1} = \kappa_a^2 C_\mu^{\mathrm{end}} * s_k$ has $\sup s_{k+1} \le r_a \sup s_k$, $\|s_{k+1}\|_1 = r_a\|s_k\|_1$ (Tonelli) and is continuous (continuity of translation in L^1); $\hat{s}_k = \kappa_a^{2k}(\hat{C}_\mu^{\mathrm{end}})^{k+1}$. The sum $S = \sum s_k$ converges uniformly and in L^1; $\hat{S} = \hat{C}_a$ by (b), and inversion gives $S = C_a$. Hence $C_a \ge C_\mu^{\mathrm{end}} > 0$, $|C_a| \le C_a(0) = c_a$, $\|C_a\|_1 = \hat{C}_a(0) = m^{-2}$, and $\|C_a\|_2^2 = (2\pi)^{-2}\int \hat{C}_a^2 \le (2\pi)^{-2}\int (m^2+|p|^2)^{-2}dp = 1/(4\pi m^2)$. (d) Convolution by C_μ^{end} is the multiplier $\hat{C}_\mu^{\mathrm{end}}$ with norm $\hat{C}_\mu^{\mathrm{end}}(0) = \mu^{-2}$; Young's inequality and the uniform-and-L^1 convergence give $C_a = \sum_k \kappa_a^{2k}(C_\mu^{\mathrm{end}})^{k+1}$ in operator norm, $= C_\mu^{\mathrm{end}}(I - \kappa_a^2 C_\mu^{\mathrm{end}})^{-1}$; the quadratic form is $(2\pi)^{-2}\int |\hat{f}|^2\hat{C}_a \in [0, m^{-2}\|f\|_2^2]$. (e) $2(c_a - C_a(h)) = (2\pi)^{-2}\int 2(1-\cos(p\cdot h))\hat{C}_a\,dp$. Use $1-\cos t \le \min(2,|t|)$, $\min(2,|\alpha+\beta|) \le \min(2,|\alpha|)+\min(2,|\beta|)$, the product bound of (a) and $\int(1+a^2q^2)^{-1}dq = \pi/a$: the right side is at most $(2\pi)^{-2}2m^{-2}(\pi/a)[I_a(h_1)+I_a(h_2)]$ with $I_a(s) = \int \min(2,|w||s|)(1+a^2w^2)^{-1}dw$. Splitting at $|w| = 2/|s|$, $I_a(s) \le (|s|/a^2)[\log(1+4a^2/s^2)+2]$, and the prefactor is $(2\pi m^2a^3)^{-1}$; this is $\varpi_a(h)^2$, for every h. (f) For $y \ge 0$ and $\beta \in (0,1]$, $\log(1+y) \le y^\beta/\beta$ (the difference has nonnegative derivative). With $\beta = (1-\alpha)/2$, $|h_i|\log(1+4a^2/h_i^2) \le 2(2a)^{1-\alpha}|h_i|^\alpha/(1-\alpha)$; for $|h| \le 1$, $2|h_i| \le 2|h_i|^\alpha$ and $\sum_i |h_i|^\alpha \le 2|h|^\alpha$, giving $M_{a,\alpha}|h|^\alpha$; for $|h| \ge 1$, $C_a(h) \ge 0$ gives $2(c_a - C_a(h)) \le 2c_a|h|^\alpha$. □

Proof of Theorem 7.3. Let B be the L^2 inverse transform of $(\hat{C}_\mu^{\mathrm{end}})^{1/2}$: real, even, $\|B\|_2^2 = 1/4$, and $B * B = C_\mu^{\mathrm{end}}$ pointwise. $(Tf)(u) = \int_\Omega B(u-x)f(x)\,dx$ is Hilbert–Schmidt from $L^2(\Omega)$ to $L^2(\mathbb{R}^2)$

with $\|T\|_{\mathrm{HS}}^2 = |\Omega|/4$, and Fubini gives $\langle Tf, Tg\rangle = \langle f, K_\Omega g\rangle$, so $K_\Omega = T^*T$ is positive and trace class with $\operatorname{tr} K_\Omega = |\Omega|/4$. $\langle f, K_\Omega f\rangle = \langle R^*f, C_\mu^{\mathrm{end}} R^* f\rangle \le \mu^{-2}\|f\|_2^2$, with R^* extension by zero. The kernel is real and symmetric, so the spectral theorem gives a real eigenbasis with eigenvalues in $[0, \mu^{-2}]$ summing to $|\Omega|/4$. □

Proof of Theorem 7.4. Work with $\gamma = \gamma_\mu$; $T = T_{\Omega,a}$ is continuous on $C(\mathbb{R}^2)$. Put $\xi_j = \int_\Omega \gamma e_j$ for the eigenbasis (e_j, λ_j) of $K = K_\Omega$; by Lemma B.1(S4) every finite vector of point values and ξ_j's is almost surely in the range of the isonormal process, hence centered jointly Gaussian, with $\mathbb{E}\xi_j\xi_l = \lambda_j\delta_{jl}$ and $\mathbb{E}[\gamma(z)\xi_j] = \langle g_z, e_j\rangle$, $g_z = C_\mu^{\mathrm{end}}(z - \cdot)|_\Omega$. The ξ_j are independent $N(0, \lambda_j)$ and, by Parseval for every outcome, $T(\gamma) = (\kappa_a^2/2)\sum_j \xi_j^2$. (a) With T_N the partial sums, $e^{T_N} \uparrow e^{T(\gamma)}$ and $\mathbb{E}e^{T_N} = \prod_{j\le N}(1 - \kappa_a^2\lambda_j)^{-1/2}$, so $\mathrm{Det}_{\Omega,a} = \prod_j (1 - \kappa_a^2\lambda_j)^{-1/2} \ge 1$; with $-\log(1-x) \le x/(1-r_a)$ on $[0, r_a]$, $1 - r_a = a^2m^2$ and $\sum \lambda_j = |\Omega|/4$, $\log \mathrm{Det} \le \kappa_a^2|\Omega|/(8a^2m^2)$; monotonicity in Ω is pointwise. (b) $0 \le c_k(z, z') \le \kappa_a^{2k}(C_\mu^{\mathrm{end}})^{*(k+1)}(z - z') = s_k(z - z')$; symmetry by relabelling; C_μ^{end} is Lipschitz with constant $\mu/4$ in $|\cdot|_1$, so c_k is Lipschitz; the operator $S_k = C_\mu^{\mathrm{end}}(\kappa_a^2 P C_\mu^{\mathrm{end}})^k$, $P = \mathbf{1}_\Omega$, has kernel c_k (Fubini, using $\int c_k(x, z')|f(z')|dz' \le \|f\|_2 r_a^k/(2\mu)$); $\|S_k\| \le \mu^{-2} r_a^k$, so $\sum S_k = C_\mu^{\mathrm{end}}(I - \kappa_a^2 P C_\mu^{\mathrm{end}})^{-1}$ in norm, and its kernel is $C' = \sum c_k$ (uniform limit). Since $PC_\mu^{\mathrm{end}} = R^* R C_\mu^{\mathrm{end}}$ and $(\kappa_a^2 R^* R C_\mu^{\mathrm{end}})^{k+1} = \kappa_a^2 R^*(\kappa_a^2 K)^k R C_\mu^{\mathrm{end}}$, the restriction form follows. (c) Fix points $z_1, \dots, z_n$ anywhere in the plane, $Y = (\gamma(z_l))_l$ and $b_j = \mathbb{E}[Y\xi_j]$. Regression on the ξ_j with $\lambda_j > 0$, $j \le N$, and the Gaussian tilting fact show that under $e^{T_N}\mathbb{P}/\mathbb{E}e^{T_N}$ the vector Y is centered Gaussian with covariance $\operatorname{Cov} Y + \kappa_a^2 \sum_{j\le N} b_j b_j^T/(1 - \kappa_a^2\lambda_j)$, and $\mathbb{E}[e^{it\cdot Y} e^{T_N}] = \mathbb{E}[e^{T_N}] \exp(-\frac{1}{2} t^T \Sigma_N t)$. Dominated convergence (majorant $e^{T(\gamma)}$) and absolute convergence of the series give, in the limit, the covariance $C_\mu^{\mathrm{end}}(z_l - z_{l'}) + \kappa_a^2\langle g_{z_l}, (I - \kappa_a^2 K)^{-1} g_{z_{l'}}\rangle$; expanding the resolvent and using the iterated kernels, this is $C'(z_l, z_{l'})$. Hence under $G'_{\Omega,a}$ the coordinates at all points are centered Gaussian with covariance C'; uniqueness is the path-space fact; the change-of-measure identity is the definition of G', extended to integrable F by splitting into positive and negative parts. □

Proof of Theorem 7.5. Pointwise: $c_0 = C_\mu^{\mathrm{end}} > 0$, $c_k \ge 0$ and $c_k \le s_k$ give $C_\mu^{\mathrm{end}} \le C' \le \sum s_k = C_a \le c_a$; each c_k is nondecreasing in Ω. As operators: $C' - C_\mu^{\mathrm{end}} = \kappa_a^2 C_\mu^{\mathrm{end}} R^*(I - \kappa_a^2 K_\Omega)^{-1} R C_\mu^{\mathrm{end}} \ge 0$ because the resolvent has spectrum in $[1, (a^2m^2)^{-1}]$; $C_a - C' = \Delta \ge 0$ by Theorem 7.6(b), whose proof uses only Theorem 7.4(b), the pointwise part and Theorem 7.2. □

Proof of Theorem 7.6. (a) For $x = (x_1, \dots, x_n)$, $1 - \prod_i \mathbf{1}_\Omega(x_i) = \sum_i [\prod_{l<i} \mathbf{1}_\Omega(x_l)]\mathbf{1}_{\Omega^c}(x_i)$ (first exit), so $s_n(z - z') - c_n(z, z') = \sum_{i=1}^n \kappa_a^2 \int_{\Omega^c} c_{i-1}(z, x) s_{n-i}(x - z')\, dx$ by Tonelli; summing, $\Delta(z, z') = \kappa_a^2 \int_{\Omega^c} C'(z, x) C_a(x - z')\, dx$, which is finite and symmetric, so also $= \kappa_a^2 \int_{\Omega^c} C_a(z - x) C'(x, z')\, dx$; substituting $C_a = C' + \Delta$ gives the third form. As operators, with $A = I - \kappa_a^2 C_\mu^{\mathrm{end}}$ and $B = I - \kappa_a^2 P C_\mu^{\mathrm{end}}$, $C_a = C_\mu^{\mathrm{end}} A^{-1}$, $C' = C_\mu^{\mathrm{end}} B^{-1}$ and $B - A = \kappa_a^2 Q C_\mu^{\mathrm{end}}$, so $C_a - C' = C_\mu^{\mathrm{end}} A^{-1}(B - A) B^{-1} = \kappa_a^2 C_a Q C'$; self-adjointness gives the other forms. (b) Continuity is that of C_a and C'. Operator positivity: $\langle f, \Delta f\rangle = \kappa_a^2\|QC'f\|^2 + \kappa_a^4\langle QC'f, C_a QC'f\rangle \ge 0$. Kernel positivity: with $F(x) = \mathbf{1}_{\Omega^c}(x)\sum_l c_l C'(z_l, x) \in L^1 \cap L^\infty$, $\sum c_l c_{l'} \Delta(z_l, z_{l'}) = \kappa_a^2\|F\|_2^2 + \kappa_a^4\langle F, C_a F\rangle \ge 0$. (c) From the last-exit form and $C' \le C_a$; if Ω contains the open r-neighbourhood of K, then $|z - x| \ge r$ for $z \in K$, $x \in \Omega^c$, and Cauchy–Schwarz gives $\epsilon_a(r)$, which decreases to 0 by dominated convergence since $C_a \in L^2$. (d) Choose r with $\epsilon_a(r) \le \eta$; every bounded Ω containing the r-neighbourhood of K has $\sup_{K\times K} \Delta \le \eta$; for $[-R, R]^2$ this holds once $R \ge r + \sup_K |z|_\infty$. (e) With $q_k = k(z + h, z + h) + k(z, z) - 2k(z + h, z)$, $q_{C_a} = q_{C'} + q_\Delta$ with both terms nonnegative, so each is at most $2(c_a - C_a(h)) \le M'_{a,1/2}|h|^{1/2}$; Kolmogorov extension gives Gaussian processes with covariances Δ and C', and the continuity criterion with $\mathbb{E}|Z(z) - Z(z')|^{12} \le 11!!(M'_{a,1/2})^6|z - z'|^3$ and the dyadic argument give jointly measurable continuous modifications. □

Proof of Theorems 7.7 and 7.8. μ_0 is even and σ-finite, so $H_{\mathbb{C}}$ is a separable real Hilbert space carrying an isonormal W (C6(f)). For real $f \in L^2$, $\hat{f}\tau_a$ is conjugate-symmetric with $\int |\hat{f}\tau_a|^2 d\mu_0 = (2\pi)^{-2}\int |\hat{f}|^2 \hat{C}_a \le m^{-2}\|f\|_2^2$; all $\phi_a(f)$ lie in the range of the single W, hence are jointly Gaussian; the covariance is the isometry, with $\tau_a\tau_b\hat{C}_0 = (\hat{C}_a\hat{C}_b)^{1/2}$. For Theorem 7.8: with $D = m^2 + |p|^2$, $0 \le \hat{C}_0 - \hat{C}_a = a^2w^2q^2/(D(D+a^2w^2q^2)) \le a^2w^2q^2/D^2 \le a^2/4$ (as $4w^2q^2 \le D^2$), and $(1-\tau)^2 \le 1-\tau^2$ on $[0,1]$, so $\|\phi_a(f) - \phi(f)\|_2^2 = (2\pi)^{-2}\int |\hat{f}|^2(1-\tau_a)^2\hat{C}_0 \le (2\pi)^{-2}\int |\hat{f}|^2(\hat{C}_0 - \hat{C}_a) \le \frac{a^2}{4}\|f\|_2^2$; complex tests split into real and imaginary parts. □

Proof of Theorem 7.9. Lemma B.1 with $\nu = \mu_0$ and $\sigma = \tau_a$, $\int \tau_a^2 d\mu_0 = c_a$ and $k = C_a$: (S1) gives the covariances, with $C_{ab} = \mathcal{I}[(\hat{C}_a\hat{C}_b)^{1/2}]$ where $\mathcal{I}g(x) = (2\pi)^{-2}\int e^{ip\cdot x} g(p)\,dp$ for $g \in L^1$ (here $(\hat{C}_a\hat{C}_b)^{1/2} \le \frac{1}{2}(\hat{C}_a + \hat{C}_b) \in L^1$); (S2) with Theorem 7.2(e) gives the increment; (S3) with Theorem 7.2(f), $\alpha = 1/2$, gives $\tilde{\phi}_a$; (S4) gives the smearing identity, the L^1 and Plancherel transforms agreeing on $L^1 \cap L^2$. Each modification and each smearing identity has its own null set; countable unions of null sets are null; $\tilde{\phi}_a$ is constructed separately for each a, so no common null set over uncountably many regulators arises, and no measurability in a is asserted. □

Proof of Theorem 7.10. $\sigma = \tau_a\rho$ has $\int \sigma^2 d\mu_0 \le (2\pi)^{-2}\int (m^2 + |p|^2)^{-1}(1 + |p|^2)^{-2} dp < \infty$, and $\mathbb{E}|\psi_a(z+h) - \psi_a(z)|^2 \le |h|^2(2\pi)^{-2}\int |p|^2 \hat{C}_0 (1 + |p|^2)^{-2} dp \le |h|^2/(4\pi)$, so Lemma B.1(S3) with $\vartheta = 2$ gives $\tilde{\psi}_a$, a Borel $C(\mathbb{R}^2)$-valued random element, and $\Phi_a = L\tilde{\psi}_a$ is Borel because L is continuous (Proposition 5.1(viii)). (a) For real $f \in C_c^\infty$ and $g = (1-\Delta)f$, $\hat{g} = (1+|p|^2)\hat{f}$ and, for every outcome, $\Phi_a(f) = \int \tilde{\psi}_a g$; by (S4) this is $W(\hat{g}\tau_a\rho) = W(\hat{f}\tau_a) = \phi_a(f)$ almost surely. (b) For $a > 0$, Φ_a and $\iota\tilde{\phi}_a$ agree on the countable family $\mathcal{D}_0$ of Proposition 5.1(ix) off one null set (by (a) and Theorem 7.9), hence everywhere off it. (c) Same argument with Φ'. (d) Countable unions of null sets. (e) By (a) for the countably many tests $\chi_j\overline{e_{j,k}}$, Tonelli and Theorem 7.8, $\mathbb{E}p_j(\Phi_a)^2 = \sum_k \lambda_{j,k}^{-2}\mathbb{E}|T_a(\chi_j\overline{e_{j,k}})|^2 \le m^{-2}B_j$, and the difference bound is the same with $\frac{a^2}{4}$; $B_j < \infty$ by the shell count. Since $\|\min(1,Y)\|_2 \le \min(1, \|Y\|_2)$ for $Y \ge 0$, Minkowski for the series gives $\|d(\Phi_a, \Phi_0)\|_{L^2} \le h(a)$, and $h(a) \to 0$ by dominated convergence. (f) is (a) with Theorem 7.7. □

Remark B.2 (Scope of the Gaussian proofs). These proofs use no interaction functional, Wick power, weight, normalizer or statement of Theorem 8.2; in particular the construction of Φ_0 does not use the existence of U_0. They establish exactly the Gaussian interface used by Theorem 8.2(j) and Corollary 8.3: the common isonormal space, the continuous modifications for regulators, and the Borel versions Φ_a for $0 \le a < 1/m$ with their compact-test identities. That the interacting laws exist on this space is a consequence of Theorem 8.2, not of this appendix. Nothing is asserted at the endpoint value $a = 1/m$.

C Proofs for the fixed-volume Wick theory

Throughout, Λ is a physical square with $A = |\Lambda|$, $0 < a < 1/m$ unless $a = 0$ is stated, $\|\cdot\|_r$ is the norm of $L^r(\mathbb{P}_0)$ or of $L^r(\mathbb{R}^2, dz)$, $D = m^2 + |p|^2$, $\sigma_a = \hat{C}_0 - \hat{C}_a$, $\hat{C}_{ab} = (\hat{C}_a\hat{C}_b)^{1/2}$ and $C_{ab} = \mathcal{I}\hat{C}_{ab}$. The standard inputs are C6(a)–(c), Gaussian characteristic functions $\mathbb{E}e^{iX} = e^{-\operatorname{Var} X/2}$ and $\mathbb{E}\mathrm{He}_4(N(0,1)) = 0$, Fourier series on tori, and Schur's product theorem.

C.1 Gaussian preliminaries

Lemma C.1 (Wick powers, reflection and the free σ-algebra)**.** *(i) For ξ in the first chaos H_1 with $c = \operatorname{Var}\xi \ge 0$, $:\xi^4: = :\xi^4:_c \in H^{:4:}$ (for $c = 0$, $\xi = 0$ almost surely and $:\xi^4: = 0$), $\mathbb{E}[:\xi^4::\eta^4:] =$*

$4!(\mathbb{E}\xi\eta)^4$ *for all* $\xi,\eta\in H_1$*, and* $z\mapsto{:}\phi_a(z)^4{:}$ *is continuous into* $H^{:4:}$*. For* $\xi\sim N(0,c)$ *and* $c'\geq 0$*,* $\mathbb{E}[\xi^4-6c'\xi^2+3c'^2]=3(c-c')^2$*, which vanishes exactly at the own variance. (ii) Let* T *be a surjective real-linear isometry of* H_1 *and* $\mathcal{F}_H=\sigma(H_1)\vee\mathcal{N}$*. There is a unique map* Θ_T *on* $L^0(\mathcal{F}_H)$ *with* $\Theta_T[g((\xi_j)_j)]=g((T\xi_j)_j)$ *for countable* $(\xi_j)\subset H_1$ *and Borel* g*; it is linear, multiplicative, commutes with Borel functions (in particular with conjugation and* $e^{-\cdot}$*), preserves laws, is an isometry of every* L^r *and commutes with* L^r*-limits and Bochner integrals in* L^2*. For* $\theta\in\{\theta_1,\theta_2\}$*,* $(T_\theta h)(p)=h(\theta p)$ *is orthogonal on* $H_{\mathbb{C}}$*, and with* $T=WT_\theta W^{-1}$*,* $\Theta=\Theta_T$ *satisfies* $\Theta\phi_a(f)=\phi_a(f\circ\theta)$*,* $\Theta\phi_a(z)=\phi_a(\theta z)$ *and* $\Theta{:}\phi_a(z)^4{:}={:}\phi_a(\theta z)^4{:}$*. (iii)* $\{\phi(f):f\in C_c^\infty(\mathbb{R}^2;\mathbb{R})\}$ *is dense in* H_1*, and* $\mathcal{F}_H=\mathcal{F}_{\mathrm{free}}$*.*

Proof. (i) C6(b) for positive variances; if a variance vanishes both sides vanish since $|\mathbb{E}\xi\eta|\leq(\operatorname{Var}\xi\operatorname{Var}\eta)^{1/2}$. Hence $\mathbb{E}|{:}\xi^4{:}-{:}\eta^4{:}|^2=4!(c_\xi^4-2(\mathbb{E}\xi\eta)^4+c_\eta^4)$, and $z\mapsto e_z\tau_a$ is continuous into $H_{\mathbb{C}}$. (ii) H_1 is separable; every $Y\in L^0(\mathcal{F}_H)$ is $g((\xi_j)_j)$ by Doob–Dynkin; if $g((\xi_j))=g'((\xi'_j))$ almost surely, the vectors $((\xi_j),(\xi'_j))$ and $((T\xi_j),(T\xi'_j))$ are centered Gaussian with the same covariances, hence have the same law, so the Borel set $\{g\neq g'\}$ is null for both. The listed properties follow by merging countable families. T_θ is orthogonal because Lebesgue measure and $\hat C_0$ are θ-invariant; $(f\circ\theta)\hat{}=\hat f\circ\theta$, $\tau_a\circ\theta=\tau_a$ and $T_\theta(e_z\tau_a)=e_{\theta z}\tau_a$. (iii) For $h\in H_{\mathbb{C}}$ and $R>0$, $\hat C_0\geq(m^2+R^2)^{-1}$ on $\{|p|\leq R\}$, so $h_R=h\mathbf{1}_{\{|p|\leq R\}}\in L^2(dp)$ is Hermitian, $f_R=\mathcal{I}h_R$ is real with $\phi(f_R)=W(h_R)$, and $\|h-h_R\|_{H_{\mathbb{C}}}\to 0$; $\|\phi(f)-\phi(f')\|_2\leq m^{-1}\|f-f'\|_2$ and density of C_c^∞ in L^2 conclude. Each element of H_1 is an almost sure limit of free-field smearings, and $\mathcal{N}\subset\mathcal{F}_{\mathrm{free}}$. □

Lemma C.2 (Kernel facts)**.** *(K1) For* $0\leq b\leq a<1/m$*:* $0<\hat C_a\leq\hat C_{ab}\leq\hat C_b\leq\hat C_0$*, and* $\hat C_b-\hat C_{ab}$*,* $\hat C_{ab}-\hat C_a$*,* $\hat C_b-\hat C_a$ *lie in* $[0,\sigma_a]$*. (K2)* $\sigma_a=a^2w^2q^2/(D(D+a^2w^2q^2))\in[0,\min(a^2/4,\hat C_0)]$*. (K3)* $\mathcal{I}$ *satisfies* $\|\mathcal{I}g\|_2=(2\pi)^{-1}\|g\|_{L^2(dp)}$ *and* $\|\mathcal{I}g\|_6\leq(2\pi)^{-5/3}\|g\|_{6/5}$*; for bounded Hermitian* k*,* $K[k]=\mathcal{I}(k\hat C_0)$ *is real,* $k\mapsto K[k]$ *is real-linear,* $\|K[k]\|_2=(2\pi)^{-1}\|k\hat C_0\|_{L^2(dp)}$ *and* $\|K[k]\|_6\leq\|k\|_\infty M_6$*; in particular* $\|C_{ab}\|_6\leq M_6$ *for* $a,b>0$*, and each of* C_b-C_{ab}*,* $C_{ab}-C_a$*,* C_b-C_a *has* L^2 *norm at most* $\kappa_2 a$ *for* $0<b\leq a$*. (K4)* c_a *is strictly decreasing,* $c_a\leq(2\pi)^{-1}\log(\kappa_0/(am))$*, and* $c_a>c_{1/m}=1/4$*. (K5)* $C_a(t,y)=(2\pi)^{-1}\int e^{iqy}e^{-E_a(q)|t|}(2E_a(q)(1+a^2q^2))^{-1}dq$ *with* $E_a(q)=[(m^2+q^2)/(1+a^2q^2)]^{1/2}$*, and* $\iint h(\theta z)C_a(z-z')h(z')\,dz\,dz'\geq 0$ *for real* $h\in C_c^\infty(H_+^\theta)$*. (K6)* C_{ab} *is bounded and continuous for* $a,b>0$*,* $\sup|C_a|=c_a$ *and* $C_a(\theta x)=C_a(-x)=C_a(x)$*.*

Proof. (K1) With $x=\hat C_a\leq y=\hat C_b$: $\sqrt{xy}\in[x,y]$, $y-\sqrt{xy}\leq y-x$, $\sqrt{xy}-x\leq y-x\leq\sigma_a$. (K2) Expand $1/D-1/(D+a^2w^2q^2)$ and use $w^2q^2\leq D^2/4$. (K3) For $s>1$, $\int(m^2+|p|^2)^{-s}dp=\pi m^{2-2s}/(s-1)$, so $\hat C_0\in L^2\cap L^{6/5}$ with $\|\hat C_0\|_{6/5}=(5\pi m^{-2/5})^{5/6}$; Hausdorff–Young for the unnormalized transform gives the factor $(2\pi)^{-2}(2\pi)^{1/3}$; the Plancherel and Hausdorff–Young extensions agree almost everywhere (truncations converge in both spaces); Hermitian symmetry gives reality. For the differences, $\|\sigma_a\|_{L^2(dp)}^2\leq\int_{|p|\leq 1/a}(a^2/4)^2+\int_{|p|>1/a}|p|^{-4}=17\pi a^2/16$. (K4) $\int_{\mathbb{R}}dw/(\alpha+\beta w^2)=\pi/\sqrt{\alpha\beta}$; the integrand of (2) is strictly decreasing in a for $q\neq 0$; $\int_0^{1/a}dq/\sqrt{m^2+q^2}=\operatorname{asinh}(1/(am))\leq\log((1+\sqrt2)/(am))$ and $\int_{1/a}^\infty dq/(aq^2)=1$; at $a=1/m$ the integral is $(2\pi)^{-1}\pi/2$ (used only for this comparison; $1/m$ is not a regulator). (K5) $\hat C_a=[(1+a^2q^2)(w^2+E_a^2)]^{-1}$ and $(2\pi)^{-1}\int e^{iwt}(w^2+E^2)^{-1}dw=e^{-E|t|}/(2E)$; for $\theta=\theta_1$ and $t,t'>0$ on $\operatorname{supp}h$, Fubini gives $(2\pi)^{-1}\int dq\,[2E_a(1+a^2q^2)]^{-1}|\int h(t,y)e^{-E_a(q)t}e^{iqy}dt\,dy|^2\geq 0$; $\hat C_a$ is symmetric in (w,q), which gives θ_2. (K6) $\hat C_{ab}\geq 0$ is integrable and invariant under $p\mapsto\theta p$ and $p\mapsto -p$. □

C.2 The interaction and its L^2 limit

Proof of Theorem 8.2(a), (c), (d) and the existence of U_0*. Pathwise equals Bochner.* $z\mapsto{:}\tilde\phi_a(z,\omega)^4{:}$ is continuous for every outcome, so U_a is a finite pathwise integral, measurable

by Fubini. $Y(z) = u{:}\phi_a(z)^4{:}$ is continuous into $H^{:4:}$ with $\mathbb{E}Y(z)^2 = 4!u^2c_a^4$, so the Bochner integral $B = \int_\Lambda Y$ exists in $H^{:4:}$; for bounded V, $\mathbb{E}[VU_a] = \int_\Lambda \mathbb{E}[VY(z)]dz = \mathbb{E}[VB]$, so $U_a = B$ almost surely. *Mean zero and lower bound.* $\mathbb{E}U_a = 0$ because the Wick constant is the own variance; $x^4 - 6cx^2 + 3c^2 = (x^2 - 3c)^2 - 6c^2$ gives $U_a \geq -6uAc_a^2$ for every outcome, and the same for $U_a^\pm = u\int_{\Lambda\cap H_\pm} {:}\tilde\phi_a^4{:}$. *$L^2$ estimate.* For $0 < b \leq a$, Fubini for Bochner integrals and Lemma C.1(i) give $\mathbb{E}|U_a - U_b|^2 = 4!u^2 \iint_{\Lambda^2}[C_a^4 - 2C_{ab}^4 + C_b^4](z - z') \leq 4!u^2A(\|C_a^4 - C_{ab}^4\|_1 + \|C_b^4 - C_{ab}^4\|_1)$; $|X^4 - Y^4| \leq 4|X - Y|(|X|^3 + |Y|^3)$, Cauchy–Schwarz and (K3) give $\|C_a^4 - C_{ab}^4\|_1 \leq 8\kappa_2 M_6^3 a$, so $\mathbb{E}|U_a - U_b|^2 \leq 384u^2A\kappa_2M_6^3a = K_\Lambda^2 a$. *Limit.* Along any sequence of regulators tending to 0 the U_{a_n} are Cauchy; interlacing shows that all limits coincide; letting $b \downarrow 0$ gives $\|U_a - U_0\|_2 \leq K_\Lambda\sqrt{a}$ for every regulator; $H^{:4:}$ is closed, and $\mathbb{E}U_0 = \lim \mathbb{E}U_a = 0$. A Wick constant $c' \neq c_a$ would give the nonzero mean $3uA(c_a - c')^2$. □

Proof of Theorem 8.2(b). Continuity of Q. For bounded Hermitian k put $Q(k) = \iint_{\Lambda^2} K[k](z - z')^4$. If $k_n \to k$ almost everywhere with $\sup \|k_n\|_\infty \leq \kappa$, then $|Q(k_n) - Q(k)| \leq (4A/\pi)\kappa^3M_6^3\|(k_n - k)\hat C_0\|_{L^2(dp)} \to 0$ by dominated convergence with majorant $4\kappa^2\hat C_0^2$. *All admissible cutoffs.* Since $\int \hat C_0 = \infty$, Fatou gives $c_\epsilon^\chi = \operatorname{Var}\phi_\epsilon^\chi(z) \to \infty$, so $c_\epsilon^\chi > 0$ for small ϵ (if $c_\epsilon^\chi = 0$, then $\chi_\epsilon = 0$ almost everywhere and $N_\epsilon^\chi = 0$). For admissible χ, ψ, $\mathbb{E}[\phi_\epsilon^\chi(z)\phi_\delta^\psi(z')] = K[\overline{\chi_\epsilon}\psi_\delta](z - z')$; the regulator family τ is admissible with $N_a^\tau = U_a$; so $\mathbb{E}|N_\epsilon^\chi - U_a|^2 = 4!u^2[Q(|\chi_\epsilon|^2) - 2Q(\overline{\chi_\epsilon}\tau_a) + Q(\tau_a^2)]$. Let $a \downarrow 0$ at fixed ϵ (existence of U_0 and continuity of Q), then $\epsilon \downarrow 0$ along any sequence: the limit is $4!u^2[Q(1) - 2Q(1) + Q(1)] = 0$. For a mollifier, $\chi_\epsilon(p) = \hat\eta(\epsilon p)$ gives $\phi_\epsilon^\chi(z) = \phi(\eta_\epsilon(z - \cdot))$, with $\sup|\chi_\epsilon| \leq \|\eta\|_1$. *Pairings and uniqueness.* For $a > 0$, $\mathbb{E}[U_a{:}\phi(f_1)\cdots\phi(f_4){:}] = 4!u\int_\Lambda \prod_j (C_{a0} * f_j)(z)\,dz$; since $|1 - \tau_a|\hat C_0 \leq \sigma_a \leq a^2/4$ and $\hat f_j \in \mathcal{S}$, $C_{a0} * f_j \to C_{\text{free}} * f_j$ uniformly, and the formula passes to U_0. If $Y \in H^{:4:}$ has the same pairings, $Y - U_0$ is orthogonal to the Wick monomials of the dense set of Lemma C.1(iii), which span a dense subspace of $H^{:4:}$ (C6(b)); so $Y = U_0$. *Measurability and scope.* $U_0 \in H^{:4:} \subset L^2(\mathcal{F}_H) = L^2(\mathcal{F}_{\text{free}})$. The construction uses only the variables $W(h)$, their own-variance Wick polynomials, Bochner integrals and L^2-limits; it never uses a point variable at $a = 0$ (indeed $e_z \notin L^2(\mu_0)$ because $\int \hat C_0 = \infty$) and forms no pointwise product of the distribution-valued free field; nothing is asserted about other representations of U_0. □

C.3 Nelson tail, weights and normalizers

Proof of the tail bound and of Theorem 8.2(e). By C6(a) with $k = 4$, $\|X\|_r \leq (r - 1)^2\|X\|_2$ for $X \in H^{:4:}$ and real $r \geq 2$. Fix $K \geq K_\sharp$ and put $s = (K/(12uA))^{1/2} \geq 1$ and $a' = (\kappa_0/m)e^{-2\pi s}$. (D1) $\kappa_0 = e(1+\sqrt{2}) < e^2 < e^{2\pi}$, so $a'm < 1$ and a' is a regulator. (D2) $c_{a'} \leq s$ by (K4), so $U_{a'} \geq -6uAs^2 = -K/2$ for every outcome. (D3) For $a' \leq a < 1/m$, $c_a \leq c_{a'}$, so $U_a \geq -K/2$ and $\mathbb{P}_0(U_a \leq -K) = 0$. (D4) For $0 < a < a'$, $\{U_a \leq -K\} \subseteq \{|U_{a'} - U_a| \geq K/2\}$. (D5) $X = U_{a'} - U_a \in H^{:4:}$ has $\|X\|_2 \leq K_\Lambda a'^{1/2}$, so for real $r \geq 2$, $\mathbb{P}_0(|X| \geq K/2) \leq [(2K_\Lambda/K)(\kappa_0/m)^{1/2}(r - 1)^2e^{-\pi s}]^r$. (D6) With $r = 1 + e^{\pi s/2 - 1} \geq 2$, $(r - 1)^2 = e^{\pi s - 2}$ and the bracket is $(2K_\Lambda/K)(\kappa_0/m)^{1/2}e^{-2} \leq e^{-1}$ because $K \geq (2/e)K_\Lambda(\kappa_0/m)^{1/2}$. (D7) $\mathbb{P}_0(U_a \leq -K) \leq e^{-r} \leq \exp(-\exp((\pi/2)(K/(12uA))^{1/2} - 1)) = T(K)$. The coarse regulator depends on K only, so the bound is uniform in a. By the layer-cake formula, $\mathbb{E}_0e^{-pU_a} = \int p\,e^{pK}\mathbb{P}_0(U_a < -K)\,dK \leq \Xi_p$. $\Xi_p < \infty$: with $c = (\pi/2)(12uA)^{-1/2}$ and $x = c\sqrt{K} - 1$, for $K \geq 4/c^2$ one has $e^x \geq x^4/24 \geq c^4K^2/384$, so the integrand is at most pe^{-pK} for $K \geq \max(K_\sharp, 4/c^2, 768p/c^4)$. At $a = 0$ choose $a_j \downarrow 0$ with $U_{a_j} \to U_0$ almost surely; Fatou gives $\mathbb{E}_0e^{-pU_0} \leq \Xi_p$ and $\mathbb{P}_0(U_0 < -K) \leq \liminf \mathbb{P}_0(U_{a_j} < -K) \leq T(K)$ (open event only). □

Proof of Theorem 8.2(f), (g). $|e^{-x} - e^{-y}| \leq |x - y|(e^{-x} + e^{-y})$ and Hölder with exponents $(2p, 2p)$ give $\|e^{-U_a} - e^{-U_0}\|_p \leq \|U_a - U_0\|_{2p}\|e^{-U_a} + e^{-U_0}\|_{2p} \leq (2p - 1)^2K_\Lambda\sqrt{a}\cdot 2\Xi_{2p}^{1/(2p)}$. Jensen and $\mathbb{E}U_a = 0$

give $Z_a \geq 1$, and (e) with $p = 1$ gives $Z_a \leq \Xi_1$; the lower bound concerns the physical laws on the common Gaussian space only. □

C.4 Central reflection positivity at fixed volume

Proof of Theorem 8.2(h), (i). Half-plane localization. For $z \in H_+$, the mollified tests $\rho_n(\cdot - z)$ lie in $C_c^\infty(H_+)$ for large n and $\phi_a(\rho_n(\cdot - z)) \to \phi_a(z)$ in L^2, so $:\phi_a(z)^4: \in L^2(\mathcal{A}_+^{\theta,a})$ and the Bochner integral U_a^+ lies in $L^2(\mathcal{A}_+^{\theta,a})$; $U_a = U_a^+ + U_a^-$ (the line is null), and $\Theta U_a^+ = U_a^-$ because Θ commutes with Bochner integrals and Λ is θ-invariant. *Exponentials are dense.* The span of $e^{i\phi_a(f)}$, $f \in C_c^\infty(H_+;\mathbb{R})$, is dense in $L^r(\mathcal{A}_+^{\theta,a})$, $1 \leq r < \infty$: an annihilating $g \in L^{r'}$ gives finite complex measures with vanishing Fourier transforms on every finite cylinder, hence $\mathbb{E}[g\mathbf{1}_S] = 0$ on a generating π-system and, by Dynkin, on $\mathcal{A}_+^{\theta,a}$. *Gaussian positivity.* For $G = \sum_k c_k e^{i\phi_a(f_k)}$, $\mathbb{E}[\overline{\Theta G} G] = \sum_{j,k} \overline{d_j} d_k \exp(M_{jk})$ with $d_k = c_k e^{-\langle f_k, C_a f_k\rangle/2}$ and $M_{jk} = \langle f_j \circ \theta, C_a f_k\rangle$; M is real symmetric and positive semidefinite by (K5), (K6), so $\exp \circ M = \sum_n M^{\circ n}/n!$ is positive semidefinite by Schur's theorem; density extends this to $G \in L^2(\mathcal{A}_+^{\theta,a})$. *(h)* With $e^{-U_a^\pm} \leq e^{6uAc_a^2}$, $G = Fe^{-U_a^+} \in L^2(\mathcal{A}_+^{\theta,a})$ and $\Theta G = (\Theta F)e^{-U_a^-}$, so $\mathbb{E}[\overline{\Theta F} F e^{-U_a}] = \mathbb{E}[\overline{\Theta G} G] \geq 0$. *(i)* For $G = \sum_j c_j e^{i\phi(f_j)}$ and $G_a = \sum_j c_j e^{i\phi_a(f_j)}$, (h) gives $\mathbb{E}[\overline{\Theta G_a} G_a e^{-U_a}] \geq 0$, and with $c = \sum |c_j|$, $|e^{ix} - e^{iy}| \leq |x - y|$, Theorem 7.8, (e) with $p = 2$ and (f) with $p = 1$, the difference from $\mathbb{E}[\overline{\Theta G} G e^{-U_0}]$ is at most $c\,\Xi_2^{1/2}(a \sum_j |c_j| \|f_j\|_2 + 2cK_\Lambda\sqrt{a}) \to 0$. For bounded $\mathcal{A}_+^{\theta,0}$-measurable F, approximate in L^4 by such G_n and use Hölder with exponents $(4, 4, 2)$ and $\|e^{-U_0}\|_2 \leq \Xi_2^{1/2}$. □

Proof of Theorem 8.2(j) and of the identification with Definition 2.10(d). For Borel B, $\mathbf{1}_B(\Phi')$ is measurable and $Z_a \in [1, \Xi_1]$, so the formula defines a Borel probability with $\int H\, d\nu = Z_a^{-1}\mathbb{E}_0[H(\Phi')e^{-U_a}]$ for bounded Borel H. For bounded Σ_+^θ-measurable F, Proposition 5.1(x) gives real $f_j \in C_c^\infty(H_+^\theta)$ and a bounded Borel g with $F(v) = g((v(f_j))_j)$ and $F(\theta v) = g((v(f_j \circ \theta))_j)$. Off one null set, $\Phi'(f) = \phi_a(f)$ for the countably many f_j and $f_j \circ \theta$; there $F(\Phi') = G := g((\phi_a(f_j))_j)$, which is $\mathcal{A}_+^{\theta,a}$-measurable, and $F(\theta\Phi') = g((\phi_a(f_j \circ \theta))_j) = \Theta G$ almost surely by Lemma C.1(ii). So $\int \overline{F(\theta v)} F(v)\, \nu(dv) = Z_a^{-1}\mathbb{E}_0[\overline{\Theta G} G e^{-U_a}] \geq 0$ by (h) for $a > 0$ and by (i) for $a = 0$. On the common space, Theorem 7.10(a) supplies the hypothesis on $\Phi' = \Phi_a$ for every $0 \leq a < 1/m$, including $a = 0$; two continuous modifications of ϕ_a agree everywhere off one null set (rational points and continuity), so the pathwise U_a does not depend on the modification; and any other Borel Φ' with the same compact-test identities equals Φ_a almost surely (Theorem 7.10(c)), hence gives the same law. This is Corollary 8.3. □

Remark C.3 (Proof of Theorem 8.2(k)). No step above uses a restriction on u/m^2, and every constant is computed for the fixed square Λ.

Remark C.4 (Volume dependence and unused routes). $K_\Lambda = u(384\kappa_2 M_6^3)^{1/2} A^{1/2}$, $K_\sharp \geq 12uA$ and $\Xi_p \geq e^{12puA}$: every quantitative bound is for a fixed physical square, and the only volume-uniform statement is $Z \geq 1$. A frequency-split tail argument, which gives another family of regulator-uniform moment bounds, was checked in the antecedent record and is not used. The proofs use the Gaussian input only through Theorems 7.7, 7.9 and 7.10 and Proposition 5.1; the weight, the normalizer and positivity enter only afterwards.

D Provenance, review status and source access

Authorship of this text. The manuscript and corrections A, AC and AC2 were written by Claude (Opus 5.5, Anthropic), with retained earlier Codex argument lineage as listed below, in a project directed by Cody Mitchell. The AC3 integration author is GPT-6.1 Sol (max effort, OpenAI), adopting the Sol-authored SB5 and SB6 proof arguments and SMH numerical repair, expanding their complete derivations and writing the dependency/source separation. Older Astra-line H1/K1 lineage and Opus authorship remain attributed. The authorship map records every adopted or rewritten argument; a reviewer is an author only when its own proof argument is adopted, never merely for its review verdict. Historical A/AC/AC2 and adverse review bytes are preserved; all earlier verdicts remain bound to the versions they read. The exact-version review status of this text is recorded in the companion decision. Every review mentioned below was an internal review by language models; none is an independent expert review, and no record is a formal (machine-checked) proof.

Historical PV4 provenance. The following publication-revision paragraph describes the retained 23 October baseline. The new private 26 October supplement records its own argument adoptions, source premises and required exact-version reviews in Appendix H and the accompanying proof map. The original mathematical blocks are unchanged.

Publication revision. The present revision adds explanatory prose by Claude (Opus 5.5), with title, opening, integration and precision clarifications by GPT-6 Astra under Cody Mitchell's direction. All numbered mathematical statements and proof bodies are retained verbatim from the accepted construction manuscript. The companion decision binds the internal revision reviews to the exact source and PDF.

Argument lineage in the construction chain. The fixed-volume arguments of Section 9 retain those of two earlier Codex-authored records (F1 and F2 in the table below): the local-L^4 compact containment and finite-vector identification of the width record in Proposition 9.2, and the residual normalization, covariance-defect coupling, estimates and constants of the fixed-volume record in Propositions 9.3–9.6.

Theorems 10.8, 10.10 and 10.16 have their own proofs; no earlier conditional record is cited for them. Their arguments retain forms from earlier Codex-authored records (LR, V1 and CT in the table below): the dilation and characteristic-function route of Theorem 10.8; the lattice inequality, Paley–Zygmund bound, Fourier moments, tightness and version of Theorem 10.10; and the invariance, regularity, non-Gaussianity, kernel and determining arguments of Theorem 10.16 other than part (iii). The Gaussian-realization transfer (Lemma 10.2) and corrected readings lemma (Lemma 10.6) were newly written in the Opus text. The AC3 integration adopts the Sol-authored SB5/SB6 derivations: the direct classical Wick/normalizer lemma, explicit determining/density/kernel proofs, source-only symmetries, full positivity, Euclidean mixing, exact regularity, and the local conditional-density non-Gaussianity and reconstruction route. The mixing argument also retains the earlier Claude classical-input lineage. These are manuscript argument adoptions, not attribution to the reviewers.

Sections 11 and 12 re-express, for the present text, the moment-transfer, selection and joint-correlation arguments of two earlier project records (H1 and K1 in the table below). In those records the fixed-volume arrows, the Gaussian and Wick estimates and the volume limit were premises;

here each input is a theorem proved in the main text or Appendices A–C, and every place where an input is used is cited. The arguments are otherwise of the same form.

Antecedent records. The following records were supplied with their acceptance decisions; an acceptance decision covers only its literal scope, and a lemma appearing in a file is not covered merely by being there.

Record	Author family	Use in this text
Endpoint foundations (E)	Claude (Opus 5.5)	Accepted as endpoint foundations under stated classical field-space premises; re-expressed in Section 6 and Appendix A.
Gaussian foundations (G)	Claude (Opus 5.5)	Accepted for its statements G1–G11, the interaction consequence of G11 only conditionally; its Borel-law lemma for X is outside that scope and is replaced by Proposition 5.1(ii),(iii), proved here. Re-expressed in Section 7 and Appendix B.
Wick theory (W)	Claude (Opus 5.5)	Accepted; re-expressed in Section 8 and Appendix C, including the precise construction statement of Theorem 8.2(b).
Classical inputs (C)	Claude (Opus 5.5)	Accepted formulations of C1–C6, used verbatim in content in Section 3. The application question on the reading of "polynomial function" in C2(iii) was closed by a later reading of the AHKZ supplement in the reconstruction-coverage review (RC), not by the C author.
Dependency route (D)	Claude (Opus 5.5)	Accepted route "candidate B": remove four project premises by new direct proofs and prove the other five; implemented here (Appendix E).
Contribution wording (N)	Claude (Opus 5.5)	Accepted wording, used in Sections 1, 14 and 15, including the derivation of Remark 14.1, which is N-authored; the Codex-family review RN reviewed that derivation and is not one of its authors.
Older records F1 (width), F2 (fixed volume), LR (continuum limit), V1 (volume), CT (target)	Codex	Conditional on the nine project premises; their premise-conditional statements are not cited for any conclusion here. Their argument forms are retained in the current text, which is therefore jointly authored (Codex argument lineage; historical Opus integration, current Sol adoption): F1's local-L^4 compact containment and finite-vector identification in Proposition 9.2; F2's field-space construction in Proposition 5.1 and its residual normalization, covariance-defect coupling, estimates and constants in Propositions 9.3–9.6; LR's dilation and characteristic-function route in Lemma 10.5 and Theorem 10.8; V1's lattice inequality, Paley–Zygmund bound, Fourier moments, tightness and version in Theorem 10.10; CT's invariance, regularity, non-Gaussianity, kernel and determining arguments in Theorem 10.16 other than part (iii) and Lemmas 10.12–10.15.

Older records H1 (moments), K1 (kernels)	Codex (requested model of the Astra line)	Reviewed by Claude (Opus 5.5) reviewers in hostile, source, scope, blind and matching roles. Their arguments are re-expressed, with rewritten proofs of the inputs, in Sections 11–12; their authorship is unchanged by this integration. AC2 replaced the original bounded Hilbert-sum constant by a distinct scalar σ_j bound. SMH recorded that literal mismatch. AC3 adopts SMH's Sol-authored pointwise-scalar/spatial-Minkowski repair to restore the exact original K_j inequality and its finite-width Z_N^b consequence, alongside the separate σ_j estimate. The common sequence and compact K1 arguments are unchanged.

Components and the review they require. The retained Opus/Astra-line components and joint claims keep their actual authorship. The adopted SB5/SB6/SMH arguments and new AC3 integration text add Sol authorship. Their current-version qualifying review must therefore cover each actual author model, with fresh contexts and the model-specific gate required by the activated project policy; an author model cannot qualify its own contribution. The exact allocations, distinct-model fallback status, findings, actual-proof matches and acceptance decision belong in the companion records. Historical family labels and earlier review requirements are preserved there; this paragraph does not promote a historical verdict or claim independent expert review. Theorems 4.1 and 4.2 require coverage of every component chain.

Source access. The authors of this text (tasks A, AC, AC2 and AC3) read no original classical source. AC3 read only the supplied frozen manuscript, statement, companion maps, proof modules and internal review records. The page-level access below is that recorded by the classical-input and reconstruction-coverage reviews; "image" means a page image was viewed, "text" means only supplied text of the page was read. Access recorded by later reviews of version A is recorded in their own records and is not credited to the authors of this text.

Source	Page images viewed	Supplied text only
[8]	pp. 495–498, 503, 512–514, 517, 518	pp. 494, 509–511
[2]	pp. 684, 685 (with a detail of eq. (1)), 686–690	p. 691 and a supplement
[10]	pp. 310–312, 316, 324	pp. 317–320
[23]	p. 151	p. 148
[17]	—	excerpts

Further page images of a source of Fröhlich were viewed for an alternative route that is not used here; that source is not cited. Not available: the bodies of [19, 20], [24], the antecedents of the review in [10, pp. 316–319], and the originals behind the Markov and lattice positivity attributions of Section 14. The neural-network literature of Section 14 was read by the contribution analysts, partly in excerpts (see the notes in the bibliography); the authors of this text did not read it.

E Disposition of the earlier premises

An earlier conditional assembly of this construction assumed C1–C6 together with nine project premises P1–P9. Their status here is as follows; C1–C6 are the classical inputs of Section 3, in their final accepted formulations.

Premise	Status	Where
P1 endpoint neurons	proved	Theorems 6.1–6.4; Appendix A
P2 central reflection positivity, all finite parameters (with the general invariant-potential form)	proved, without the earlier degree and mass hypotheses	Theorem 6.5, Corollaries 6.6, 6.7
P3 width moments at fixed parameters	removed; not used	local seminorm moments along the width arrow are proved in Proposition 11.6(i)
P4 finite-region Gaussian mass tilt and increment bound	proved	Theorems 7.2–7.6, 7.9; the endpoint value $a = 1/m$ is excluded
P5 common Gaussian space	proved, with Borel versions at $a = 0$	Theorems 7.7–7.10
P6 fixed-volume Wick integrability	proved, with the precise construction statement	Theorem 8.2
P7 regulator products	removed; not used	regulator arrow proved directly (Propositions 9.6, 11.6(iv))
P8 stagewise finite-vector bridge	removed by new proofs	Section 9 (arrows on X) and Theorem 10.8 (classical continuum limit); the earlier transfer claim is not used
P9 exact-cell midpoint convergence	removed; not used	no midpoint law occurs
C1–C5	classical, attributed	Section 3
C6	standard tools	Section 3

F Notation

Symbol	Meaning
m, ℓ, u; a, $\mu = 1/a$	mass, length, quartic coefficient (no factorial); regulator and endpoint rate
$\kappa_a^2 = a^{-2} - m^2$, $r_a = 1 - a^2 m^2$	mass-tilt coefficient; contraction constant
C_μ^{end}, C_a, C_{free}, $C'_{\Omega,a}$, $\Delta_{\Omega,a}$	endpoint, regulated, free, tilted covariances; $C_a - C'_{\Omega,a}$
$c_a = C_a(0)$	Wick constant: variance of the regulated Gaussian physical reference field (covariance C_a); not the endpoint variance $1/4$, and not asserted to be a variance of any finite-width or interacting law
Λ, $\Lambda_n = (-n\ell, n\ell)^2$, $\Omega = [-R, R]^2$	physical squares (axis-parallel, centered); auxiliary square
$F^b_{N,a}$, $b \in \{\mathrm{prod}, \mathrm{Mon}\}$	width-N neural field
$V = W + \varepsilon J - T$, v_V	full potential and its integrand
$P^b_{N,\varepsilon,\Omega,a;\Lambda}$, ν^v, Res, $\nu^{(2)}$, $\nu_{a,\Lambda}$, ν_n	neural, full-tilt, residual, zero-stabilizer, physical and continuum laws
$\nu_{m,u,\Lambda}$, ν^S, ν	classical finite-volume law, target on $\mathcal{S}'$, its X-valued version
E, X, p_j, d, β, dist_n	field spaces, local seminorms, metric, bounded-Lipschitz distance, selection discrepancy
$(\Gamma_0, \mathcal{F}_0, \mathbb{P}_0, W)$, ϕ_a, $\tilde{\phi}_a$, Φ_a	common Gaussian space, smeared and point fields, continuous modification, Borel version

$U_{a,\Lambda}$, $Z_{a,\Lambda}$, K_Λ, Ξ_p	Wick interaction, normalizer, fixed-volume Wick constants
$K_j = \|\chi_j\|_2/2$, σ_j	distinct neural reference-moment constants; original K_j bound restored in AC3
θ_1, θ_2, H_+^θ, Σ_+^θ	central reflections, half-planes, half-plane σ-algebras
$v^{\otimes k}$, $\mathcal{S}_k^\rho$, $A_{j,k}$, Schw_k	tensor evaluation, correlation distributions, coefficient norm, Schwinger distribution

G Historical records and paths not used

The historical ergodicity route. An earlier target record obtained ergodicity of the target through [8, Thm. 4.4] and an identification of the Osterwalder–Schrader semigroup with a Wightman energy, using the spectrum condition and uniqueness of the vacuum in the course of proving condition (d) of C3(i). Its first unsupported step was that identification. The route is not used: condition (d) is proved in Theorem 10.16(iii) by Euclidean mixing from [2, (7)–(8), (13)], before [8, Thm. 1.0] is applied.

Replaced sentences of the earlier assembly. (i) A compressed sentence in the setting described the mass-shift identity as if it applied to the neural laws; it is replaced by the stage order of Section 2.2, where the identity concerns Gaussian reference covariances only, the pure quadratic tilt on a finite auxiliary square has covariance $C'_{\Omega,a}$, and C_a appears only in the auxiliary-volume limit. (ii) A universal sentence stating that the interaction is never a pointwise fourth power is replaced by the precise construction statement of Theorem 8.2(b) and Remark 2.9. (iii) The description of the zero-regulator interaction as "the conventional Wick integral" is replaced by the same construction statement and by Proposition 10.4.

Geometry. An earlier formulation required only that the physical square be centered; a rotated centered square then breaks the invariance hypothesis of Theorem 6.5 (Remark 6.8(i)). The definition is restricted to axis-parallel squares, which contain the main sequence unchanged; no failure of positivity is asserted for any rotated square, and the general invariant-potential clause is unchanged. A proposed counterexample with an outer rectangle invariant under only one reflection lies outside the hypotheses, which require both rectangles to be centered.

Historical SMH comparison and the AC3 additions. SMH's comparison of AC2 with original H1 returned MISMATCH: the original K_j bounds (2.2), (2.4) had been replaced by different constants, and the generic centered-square wording had been narrowed to axis-parallel squares. That exact historical verdict remains unchanged. AC3 writes the separate proposed numerical repair in (43) and (46); its current-version review and historical statement match remain companion obligations. The geometry disposition above remains: the original generic orientation wording is not recovered, the actual Λ_n and two central-line conclusions are unchanged, and no new rotated-square claim is made. OB4's earlier PROVED WEAKER verdict and OB4-003/OB4-010 remain historical. The source-only Wick/normalizer, sign, kernel and conditional density proofs now written in this text address their mathematical issues; the author does not close those findings or accept the integrated result.

Corrections of version A. This version corrects the following statements of the earlier version A, whose bytes are preserved unchanged. (i) The readings lemma admitted convergence of a characteristic function at a single frequency; that alternative is false (Remark 10.7) and is replaced by convergence at every real frequency, which the proof of Theorem 10.8 already supplies. (ii) A remark asserted that the full Gaussian normalizer grows with the auxiliary square; that assertion is withdrawn (Remark 9.7); monotonicity of $\mathrm{Det}_{\Omega,a}$ and the residual bounds are unchanged. (iii) The empty-set convention $\sup \emptyset = 0$ is restored for M_N, and the early-volume cutoffs of Theorem 10.10 are chosen at least 2. (iv) The positive constants M, ϑ are restored in the continuity hypothesis of Lemma B.1. (v) The Wick variance c_a is identified throughout as the variance of the regulated Gaussian physical reference field. (vi) Absolute "no rate" wording is narrowed to the absence of an effective end-to-end rate or schedule, with the fixed-volume $\sqrt{a}$ bounds kept. (vii) Authorship lineage is recorded per component (Appendix D). (viii) An undefined macro and several overfull displays are corrected without change of any formula.

Withdrawn formulations in the antecedents. An unclipped parameter-map clause (false as a total map; replaced by Lemma A.2(iii)); an unconditional independence assertion for competing clocks (replaced by Lemma A.8); the reading "Lebesgue measurable on $\mathbb{R}^3$" of the potential hypothesis (fails; Remark 6.8(iii)); a sentence extending the increment bound to $a = 1/m$ (not part of the proved statements and not used); the stagewise transfer and midpoint premises P8, P9 (not used); and a frequency-split Nelson route (checked, not used).

H Quantitative observation control and the pressure target

The construction above reaches one interacting field through an observable-independent sequence. The subsequent analysis asks a more focused question: for specified smooth measurements and a requested tolerance, which finite stages suffice? The answer proceeds through locality, a finite interaction-box estimate, and five compatible errors. An explicit example then supplies one finite tuple for both process-neuron architectures. A separate pressure argument identifies that example's target and permits an application of classical reconstruction at an explicit coupling cap.

This appendix records applications of the complete companion results, rather than introducing a further synthesis theorem. Its mathematical premises are the literal accepted statements and restrictions in the packets named below. The public claim-to-proof map identifies the arguments contained in this edition. The companion index distinguishes this public manuscript source from the original private proof packets, decisions and software. Those original packets are not exported here; their full named statements remain premises where explicitly invoked below. Historical candidate headers describe their authoring stage. Internal acceptance and public source availability are separate qualifications.

H.1 Two parameter routes and one field convention

The original main theorem retains $0 < u/m^2 < g_*$ and its fixed admissible reference pair. Its universal sequence, selected before observations, remains the sequence proved above. The locality and budget packets CL1, FV, EB1 and EC1 were initially proved on $0 < u/m^2 < \min(g_*, 1/8000)$. No numerical lower bound for that originally fixed g_* is inferred here.

The distinct packet ADMIT (ADMIT-PRESSURE-v1) has the literal premises P1–P6. P1 supplies the finite ferromagnetic DSS/SK-q interfaces, including Gaussian sites and added mass at every finite positive scale. P2 is the finite-scale P* bound at its exact coupling, cutoff and torus restrictions. P3 gives fixed-square common-Gaussian Wick and source transfer with finite local exponential moments. P4 is fixed-mesh periodization. P5 is the original Fröhlich–Simon pressure/state/exhaustion theorem, with its converse pressure-differentiability implication. P6 lists the additional physical-law, field-space, local-density, DSS/SKQ/P*/H1/MIX/REG/FLOW/TP/T and fixed-stage NR1/L0 interfaces. These are named mathematical premises with the qualifications in their complete statements, not conclusions of arithmetic verification.

In the inclusive range $0 < u/m^2 \le 1/8000$, ADMIT AP2–AP5 identify the common original Section-4 maximal state of [8], denoted here by μ, and its same-distribution X version μ^X. For the quantitative neural budget, AP6 requires the strict range $0 < u/m^2 < 1/8000$ and every literal P6 interface. AP6 rebuilds applicability of the actual five-error argument to that same state; it does not invoke an old global theorem outside its g_* domain. Its original private proof is ADMIT, Sections 1–12 as indexed in the private acceptance map. This is the route used in the numerical example and in the strict-cap target interpretation below.

All routes use the whole-plane covariance $(-\Delta + m^2)^{-1}$, Fourier measure $dp/(2\pi)^2$, the same-mass coefficient-one fourth Wick integral J_B, and the potential uJ_B with no factorial. A bounded square restricts the interaction while the field occupies the plane. In particular the finite law is

$$d\mu_B = Z_B^{-1} e^{-uJ_B}\, dP_m. \tag{51}$$

AP4 identifies the original reference/dilation expression as the same target. Equality of these two specific maximal states is distinct from uniqueness in a larger class of Gibbs states.

H.2 Local response and the finite interaction box

Fix real $f = (f_1, \dots, f_r) \in C_c^\infty(\mathbb{R}^2)^r$, with support union $K \subset [-R, R]^2$, and let $Y_f(v) = (v(f_i))_i$. The observation distance throughout this appendix is

$$d_{\mathrm{BL}}(\rho, \eta) = \sup_{\substack{\|G\|_\infty \le 1 \\ \mathrm{Lip}_2(G) \le 1}} |\rho(G) - \eta(G)|. \tag{52}$$

The two constraints are separate, and the Lipschitz norm is Euclidean on $\mathbb{R}^r$. Repeated or linearly dependent tests need no density assumption.

For $\Lambda_j = (-j\ell, j\ell)^2$ set $S_j = \Lambda_{j+1} \setminus \Lambda_j$, $\Delta A_j = 4\ell^2(2j+1)$ and $d_j = \operatorname{dist}(K, S_j)$. On the common free Gaussian space put

$$U_{j,t} = u[(1-t)J_{\Lambda_j} + tJ_{\Lambda_{j+1}}], \quad W_j = u(J_{\Lambda_{j+1}} - J_{\Lambda_j}), \quad d\mu_{j,t} = Z_{j,t}^{-1} e^{-U_{j,t}} dP_m. \tag{53}$$

The shell insertion is a difference of fourth-chaos variables, so it uses no pointwise quartic of a distribution at zero regulator.

Application CL1. Under CL1's original classical hypotheses and $0 < u/m^2 < \min(g_*, 1/8000)$, choose a positive integer n_0 with $K \subset \Lambda_{n_0}$. One constant $C_f < \infty$ works for every $j \geq n_0$, every $t \in [0,1]$, and every real $G \in C_b^1(\mathbb{R}^r)$ with the two separate unit bounds:

$$|\mathrm{Cov}_{\mu_{j,t}}(G(Y_f), W_j)| \leq C_f u \Delta A_j e^{-\kappa d_j}, \qquad \kappa = m/256. \tag{54}$$

The complete CL1 proof, Sections 1–6, checks the common finite-lattice parameter intersection, confinement classes and all actual A1–A6/FLOW premises. It first proves a common torus estimate, then takes the fixed-mesh torus limit using TP and finally the plane mesh limit using T. The common bound precedes both pointwise limits. No separate unproved continuum response or spatial mixing assumption is added.

For reference, the literal coefficient used for the numerical work is the CL1 coefficient below. Here $\epsilon_* = \min(1/4, 1/m)$ is an auxiliary *lattice cutoff bound*, distinct from the physical stabilizer ε; s is a covariance scale, and C_H, C_{H2} are the H1 and MIX coefficients after their actual premises have been discharged:

$$\begin{aligned}
\bar{F} &= (2R + 2\epsilon_*)\Big(\sum_i \|f_i\|_\infty^2\Big)^{1/2}, & \bar{\delta} &= \frac{2}{m^2\ell^2}\log\frac{1}{1-e^{-4m\ell}}, \\
g(s) &= 1 + \bar{\delta} + \frac{\pi}{16}\log(1 + (m^2 s)^{-1}), & b_{CT} &= 1 + \frac{81\kappa}{5m}, \\
C_* &= \frac{486\kappa}{5m} + \frac{6561\kappa^2}{25m^2} + b_{CT}^2\left(\frac{4\sqrt{2\pi}\kappa}{m} + \frac{3300u}{m^2}\right), \\
M(s) &= 32\ell^{-2}\left(\frac{C_H^2 g(s)^3}{12} + \frac{u^2 C_{H2}^2 g(s)^7}{9}\right), \\
C_f &= e^{8\sqrt{2}\kappa/m} b_{CT}^2 \bar{F}\left[e^{C_*}\int_0^1 \sqrt{M(s)}\,ds + \frac{e^{2C_*}\sqrt{M(1)}}{m^2(1+m^2)}\right].
\end{aligned} \tag{55, 56}$$

The constant $81/5 = 16.2$ retains the smoothed Combes–Thomas weight restriction. All source, Gaussian/Wick, DSS/BD/BBD/Griffiths–Simon, SK-q and P* qualifications are those of the companion statements.

Application FV (CL1-FV1, version 2). Let ρ_j be the law of Y_f under (51) for $B = \Lambda_j$, and let ρ be the observation law of the original target. On the same old coupling range, for $n \geq n_0$ and $n\ell \geq R$,

$$\begin{aligned}
q &= e^{-\kappa\ell}, \qquad T_n = q^n\left[\frac{2n+1}{1-q} + \frac{2q}{(1-q)^2}\right], \\
E_n &= 4C_f u \ell^2 e^{\kappa R} T_n, \qquad d_{\mathrm{BL}}(\rho_n, \rho) \leq E_n = O_{m,u,\ell,K,f}(ne^{-m\ell n/256}).
\end{aligned} \tag{57}$$

The enclosure $K \subset [-R, R]^2$ is required when using $d_j \geq j\ell - R$.

For clarity, the FV application is the full proof of FV1–FV6, Sections 1–7. At each fixed j, exponential moments give an integrable envelope for $e^{-U_{j,t}}$ and $W_j e^{-U_{j,t}}$, including both endpoints. Normalized differentiation is therefore $\partial_t \mathbb{E}_{\mu_{j,t}} G(Y_f) = -\mathrm{Cov}_{\mu_{j,t}}(G(Y_f), W_j)$. The endpoints agree as actual densities on the same Gaussian space. Finite telescoping uses (54) and the exact sum $\sum_{j=n}^{M-1}(2j+1)q^j = T_n - T_M$.

Nonnegative unit-mass Euclidean mollification preserves both norms in (52); it extends the smooth estimate to every admitted Lipschitz G. The identified target limit is taken for each G before taking the supremum. These are the precise steps in the companion argument, including its physical/classical target identification for every $\ell > 0$. Empty/all-zero tuples have zero error directly; their arbitrary-R zero-error convention does not remove the enclosure from a geometric distance assertion. The literal earlier FV geometry statement remains a rejected version in the companion record.

H.3 Five errors between actual normalized laws

Application EB1 uses the old coupling range and the exact inherited Gaussian/Wick, finite-stage NR1/L0 and FV premises. Fix a physical box $\Lambda = \Lambda_n$, $0 < a < 1/m$, a positive stabilizer ε and $\Omega = [-R_{\mathrm{aux}}, R_{\mathrm{aux}}]^2 \supset \bar{\Lambda}$. For one architecture the adjacent whole-field laws are

$$P^b_{N,\varepsilon,\Omega,a;\Lambda} \longrightarrow Q_\varepsilon \longrightarrow Q_0 \longrightarrow Q_a \longrightarrow Q_{\mathrm{phys}} \longrightarrow \nu. \tag{58}$$

Here $Q_\varepsilon = \nu^v_{\varepsilon,\Omega,a;\Lambda}$ is the full endpoint-Gaussian tilt and equals the pushed-forward residual law by the mass-tilt identity. $Q_0 = \nu^{(2)}_{\Omega,a;\Lambda}$ is the residual zero-stabilizer endpoint, $Q_a = \nu_{a,\Lambda}$ is the physical regulated law and $Q_{\mathrm{phys}} = \nu_{0,\Lambda}$ is its fourth-chaos limit. Every arrow compares the same observation map under normalized probabilities.

Write $A = |\Lambda|$, $B_{\mathrm{ext}} = |\Omega \setminus \Lambda|$, $c = c_a = C_a(0)$, $\kappa_a^2 = a^{-2} - m^2$ and $F_p = (\sum_i \|f_i\|_p^2)^{1/2}$. The physical covariance C_a has multiplier $(m^2 + w^2 + q^2 + a^2 w^2 q^2)^{-1}$; its variance c differs from the endpoint variance $1/4$. Put $\Delta_{\Omega,a} = C_a - C'_{\Omega,a}$, $\delta_\Omega(H) = \sup_{x \in H} \Delta_{\Omega,a}(x,x)$ and $L_c = (4\sqrt{15} + 12)c^{3/2}$. Then EB1.b–d give

$$\begin{aligned} S &= \min\{2, 6\varepsilon B_{\mathrm{ext}} c^2 e^{9uAc^2}\}, \\ A_{\mathrm{aux}} &= \min\{2, e^{9uAc^2}[2uAL_c\sqrt{\delta_\Omega(\bar{\Lambda})} + F_1\sqrt{\delta_\Omega(K)}]\}, \\ R_{\mathrm{phys}} &= \min\{2, \Xi_2(\Lambda)^{1/2}[4K_\Lambda\sqrt{a} + (a/2)F_2]\}. \end{aligned} \tag{59}$$

The constants K_Λ, Ξ_p are exactly those of the original Wick theory and EC1 below. Variation in the stabilizer bound is the convention $\sup_{|H| \le 1}$ with maximum two. The auxiliary proof couples independent Gaussian covariance defects and controls both required compact sets. The regulator proof uses the actual common-space Wick weights and positive normalizers at fixed physical volume.

For completeness the exact sufficient width coefficient is given here. Choose a bounded box D containing Ω and every test support, put $O = |\Omega|$, $d_D = |D|$, $F = (\sum_i \|f_i\|^2_{4/3,D})^{1/2}$, $\alpha = \min(u, \varepsilon)$, $U = \max(u, \varepsilon)$ and $B_N = \kappa_a^2/2 + 6uc$. In EB1's equal-square case $\Lambda_0 = \Lambda_1 = \Lambda$, set

$$\begin{aligned} L_N &= B_N^2 O/(2\alpha), & v_1(x) &= 4Ux^3 + 2B_N\sqrt{O}\,x, \\ v_2(x) &= 12Ux^2 + 2B_N\sqrt{O}, & v_3(x) &= 24Ux, \quad v_4(x) = 24U, \\ e_0 &= 1, \quad e_1 = v_1, \quad e_2 = v_1^2 + v_2, & e_3 &= v_1^3 + 3v_1v_2 + v_3, \\ e_4 &= v_1^4 + 6v_1^2v_2 + 3v_2^2 + 4v_1v_3 + v_4. \end{aligned} \tag{60}$$

For a nonnegative-coefficient polynomial $P(x) = \sum p_k x^k$, define

$$\begin{aligned} b_0 &= 1, \quad b_k = (k/(2\alpha e))^{k/4} \ (k > 0), & \mathcal{T}(P) &= e^{L_N}\sum p_k b_k, \\ M_Z &= \mathcal{T}(e_4), & M_A &= \mathcal{T}\left(\sum_{j=0}^{4}\binom{4}{j}F^j e_{4-j}\right), \\ z &= e^{-3UO/16 - 3uc^2A}, & C_{\mathrm{law}} &= \frac{d_D(M_A + M_Z)}{96z}, \\ D_{\mathrm{law}} &= \frac{4}{3}(3\sqrt{6}C_{\mathrm{law}})^{1/4}(2\sqrt{r})^{3/4}. \end{aligned} \tag{61}$$

For $r \geq 1$ the width term is $W = \min(2, D_{\mathrm{law}} N^{-1/4})$. Its complete proof is EB1/NLAW, Sections 1–7: fourth-order Banach replacement uses full sign symmetry, matching covariance and the exact coefficient $d_D/(96N)$; positivity gives a normalization cost $1/z$; L0 smooths in ambient observation space. It proves both the C_b^4 law bound $\min(2, C_{\mathrm{law}}/N)$ and this all-width BL bound. NR1 itself is a fixed-stage response theorem; the law consequence is proved in EB1, with NR1's sharper-author/larger-blind-core qualification retained.

With E_n from (57), the complete EB1.f application is

$$d_{\mathrm{BL}}\big(\mathcal{L}_f(P^b_{N,\varepsilon,\Omega,a;\Lambda_n}), \mathcal{L}_f(\nu)\big) \leq \min\{2, W + S + A_{\mathrm{aux}} + R_{\mathrm{phys}} + E_n\}. \tag{62}$$

The EB1 integration paragraph and PHYS P1–P6 contain the full adjacent-law proofs, the local-H^{-2} metric bridge and the original-sequence application. In particular the unchanged universal sequence satisfies $\min(2, L_{f,J} 2^{-n} + E_n)$ in this observation distance, where $L_{f,J} = 2^J \max(2, (\sum_i \|f_i\|^2_{H^2(\mathbb{T}_J)})^{1/2})$ and every support lies in $(-J, J)^2$. This gives no relation between N_n and n. For the explicit-cap use, AP6 supplies the same actual-law budget with ν replaced by the identical pressure-identified target μ^X, under its own strict cap and every P1–P6 premise.

H.4 Elementary enclosures and conditional selection

The five clauses of EC1-v2 give elementary enclosures and conditional selection, with their complete proofs in GROWTH Sections 1–5. For example,

$$c_a \leq \bar{c}(a) = \frac{\operatorname{asinh}(1/(ma)) + 1}{2\pi}, \tag{63}$$

$$\int_{|x|>s} C_a(x)^2 \, dx \leq \min\left\{ \frac{1}{4\pi m^2}, \frac{e^{-\sqrt{2}ms}}{2\pi m^2} \right\}. \tag{64}$$

These estimates keep $a > 0$ fixed in the auxiliary-volume step. Multiplication of (64) by κ_a^2 bounds both covariance defects when Ω contains the Euclidean s-neighborhood of the two required compacts; the centered-square sufficient condition is $0 < s < R_{\mathrm{aux}} - \max(n\ell, R)$.

The literal Nelson constants admit the following enclosure for $A > 0$ and every fixed $p \geq 1$:

$$\begin{aligned}
K_\Lambda^2 &= \frac{75\sqrt{85}}{2\pi^3 m} u^2 A, & K_\sharp &= \max\{12uA, (2/e)K_\Lambda \sqrt{e(1+\sqrt{2})/m}\}, \\
\beta &= \pi^2/(48uA), & H_p &= \max\{K_\sharp, (4/\beta)[\log_+(32p/(e\beta))]^2\}, \\
\Xi_p &= e^{pK_\sharp} + \int_{K_\sharp}^\infty p e^{pH} e^{-\exp(\sqrt{\beta H}-1)} \, dH \\
&\leq e^{pH_p} + e^{-pH_p} \leq 2e^{pH_p}.
\end{aligned} \tag{65}$$

Here $\log_+ x = \max(0, \log x)$. The symbol β in this local formula is a positive scalar, distinct from the whole-field probability metric. No volume-uniform exponential moment is asserted.

EC1.e is conditional computability for *one fixed finite tuple*. Its additional numerical inputs are computable m, u, ℓ, an actual certificate for the original coupling gate, a valid n_0, a certified upper bound for C_f, a support enclosure and upper bounds for F_1, F_2, F. With positive allocations to the five errors it selects, in order,

$$n, \qquad a, \qquad R_{\mathrm{aux}}, \qquad \varepsilon, \qquad N. \tag{66}$$

This reverses the order in which limits are taken. Its strict interval searches terminate without exact-real equality decisions. EC1 by itself supplies neither the old g_* certificate nor a numerical locality coefficient; ADMIT and SCALE/OPT furnish the separate fixed-example inputs. The literal width majorant contains $\exp(B_N^2 O/(2\alpha) + 3UO/16 + 3uc_a^2 A)$ and the full positive polynomial. EC1.d characterizes growth of this sufficient rule. It supplies no necessary width or complexity lower bound.

H.5 One explicit nonzero observation and two sufficient tuples

Fix $m = 1$, $u = 10^{-10}$ and $\ell = 1000$. The test is the single nonzero compact smooth product bump

$$f(x_1, x_2) = b(x_1)b(x_2), \qquad b(t) = \begin{cases} \exp[-1/(1-t^2)], & |t| < 1, \\ 0, & |t| \geq 1. \end{cases} \tag{67}$$

It has support $[-1, 1]^2$, $R = 1$, $n_0 = 1$ and the certified norm bounds $F_1 \leq 4$, $F_2 \leq 2$, $F \leq 3$. The numerical ratio is strictly below $1/8000$. The target is precisely μ^X from AP3–AP6. All finite interfaces, $K_{\mathrm{cov}} = 1$, source qualifications and both process-neuron architectures are retained.

NS1 uses SCALE's $C_{H2} \leq 2^{1800}$ and $C_f \leq 2^{1810}$. OPT proves a changed bound for the same finite-interface H2 row, $C_{H2} \leq 2^{384}$, and the *literal* coefficient in (56) satisfies $C_f \leq 2^{360}$. The proof is OPT Sections 1–9: finite original-law moment and shift estimates, mixed real finite differences through derivative order eight, connected Bell partitions with both overlap charges and every bridge binomial, the actual source integral, and a separate H1 coefficient evaluation. The complete finite domain includes all allowed lattice parameters, $0 < s \leq 1$, both profile endpoints, coincident and zero-profile sites. The coefficients are conclusions on the literal conditional interfaces, with AP6 supplying their target applicability.

One displayed Wick-factor estimate was omitted from the frozen OPT author proof. The current OPTREF/OPTMATCH derivations complete that step from the supplied moments. Explicitly, with $g = 1 + \bar{\Gamma}_s$ and $|\eta| \leq g$, those moments give $\mathbb{E}\phi^{2q} \leq (2q-1)!!g^q$. Since $7!! = 105 < 2^8$, $15!! = 2027025 < 8^8$ and $23!! = 316234143225 < 32^8$, the triangle inequality in L^8 yields

$$H_0 = 1, \quad \|H_1\|_8 < 2\sqrt{g}, \quad \|H_2\|_8 < 9g, \quad \|H_3\|_8 < 38g^{3/2}, \qquad \|H_j\|_8 \leq 8^j g^{j/2} \ (0 \leq j \leq 3). \tag{68}$$

Here $H_1 = \phi$, $H_2 = \phi^2 - \eta$ and $H_3 = \phi^3 - 3\eta\phi$. The largest required degree is three. Thus the relation uses an explicit argument and no added moment premise; the original omission remains visible in the frozen author proof.

Table 5: Two sufficient tuples for the same observation, target and requested $1/5$ tolerance. Each nonspatial error is bounded by $1/256$ in both rows. The optimized tuple uses a looser spatial allocation.

Quantity	NS1 baseline	OPT sufficient tuple
C_{H2} upper bound	2^{1800}	2^{384}
C_f upper bound	2^{1810}	2^{360}
n	512	74
a	$2^{-16777216}$	2^{-16384}
R_{aux}	18446744073710063617	137439027473
ε	$2^{-4611686018427387904}$	$2^{-68719476736}$
N	$2^{(2^{4611686018494496904})}$	$2^{(2^{68719542354})}$
Spatial error upper	2^{-736}	$1/4096$
Certified total	$1/64 + 2^{-736} < 1/5$	$65/4096 < 1/5$

For OPT use $\Omega = D = [-R_{\mathrm{aux}}, R_{\mathrm{aux}}]^2$ and $\Lambda_0 = \Lambda_1 = \Lambda_{74}$. Sections 10–17 of its complete proof establish, in the order of (66),

$$E_{74} \leq \frac{1}{4096}, \qquad R_{\mathrm{phys}}, A_{\mathrm{aux}}, S, W \leq \frac{1}{256}, \qquad d_{\mathrm{BL}}(\mathcal{L}_f(P^b), \mathcal{L}_f(\mu^X)) \leq \frac{65}{4096} < \frac{1}{5}. \tag{69}$$

The NS1 complete proof gives the other row, including its five actual errors. Both applications retain the normalizers, physical variance, improper Nelson integral, two covariance defects and full width polynomial. The new tuple is smaller at the same requested tolerance; its certified error sum is slightly larger. This comparison asserts neither improved accuracy, optimality nor runtime speedup.

Every displayed power denotes one finite mathematical number. The certificate language has canonical integers, exact dyadics 2^z and recursive integer-valued powers of two. It represents $2^{(2^{68719542354})}$ without expanding the integer. Width counts whole-plane process neurons with their internal randomness; it remains astronomical and supplies no practical sampler.

Software assurance and its resource scope. The accepted OPT executable verifies selected finite arithmetic and frozen source bindings. The binding resource erratum qualifies the original "whole-command alarm" wording: API deadline checks are cooperative; the POSIX alarm starts after argument parsing and covers the protected verification phase (including a selection certificate write), then is cancelled before a final verification result-file write, serialization and standard output. Output-device latency can therefore exceed the configured phase limit. No hard whole-process time or memory guarantee is supplied. The selected author monitor's 385-bit watched values differ from the independently implemented expanded-polynomial calculation's 131349-bit high-water value. Neither computation materializes a giant power or neuron field. Code, pin tables and stable source files remain external trust roots; the package is not a formal proof kernel or an atomic adversarial filesystem guarantee. The NS1 rational representation attempt that exceeded its original 500000-bit ceiling remains a recorded inconclusive attempt. Historical candidate flags inside frozen code/certificates are governed by the separate current decisions and the binding erratum.

H.6 Reconstruction at the explicit pressure cap

The original reconstruction theorem above retains its original hypotheses. Here the exact application is REC-EXPLICIT-CAP-v2 (packet REC2), clauses (a)–(g), with its complete proof Sections 1–10. For the core clauses (a)–(f), assume $0 < u/m^2 \le 1/8000$, AP-source/AP-version and the literal classical inputs C1/C3/C6 in REC2's statement. These include the two specific original FS Section-4 states; their local positive Gaussian L^p densities under either boundary-algebra reading BC1 or BC2; proper Euclidean invariance; either the full measure OS reading M or the strictly time-ordered polynomial reading P; qualitative polynomial clustering from [8, Thm. 4.4, p. 511]; [8, Thm. 1.0, p. 495]'s reconstruction theorem; and the named kernel and tube-uniqueness inputs C3(ii),(iii). BC1 is the intersection of neighborhood algebras of the boundary. BC2 uses boundary-supported compact H^{-1} evaluations obtained by Gaussian L^2 approximation and Gaussian-null completion. These source premises and their access limits remain literal; no new numerical density norm is assumed.

Regularity, mixing and positivity before reconstruction. For every real Schwartz f and real t, REC2.b gives

$$1 \le \mathbb{E}_\mu e^{t\phi(f)} \le e^{3t^2\|f\|_2^2/(2m^2)}, \qquad N_m(f) = \frac{\sqrt{3}}{m}\|f\|_2, \tag{70}$$
$$\mathbb{E}_\mu \phi(f) = 0, \qquad \|\phi(f)\|_{L^2(\mu)} \le N_m(f).$$

This continuous Schwartz norm satisfies, on $N_m(f), N_m(g) \le 1$,

$$\mathbb{E}_\mu e^{\phi(f)} \le e^{1/2}, \qquad |\mathbb{E}_\mu e^{\phi(f)} - \mathbb{E}_\mu e^{\phi(g)}| \le 2eN_m(f-g). \tag{71}$$

For nonzero real f and integers $k \ge 1$,

$$\mathbb{E}_\mu|\phi(f)|^k \le 2e^{1/2}k!N_m(f)^k. \tag{72}$$

These are target estimates. They make no global Schwartz error assertion for finite neural fields.

REC2 Sections 1–4 match the exact pressure state to the classical clustering theorem, extend compact source bounds to Schwartz tests by cutoffs and Fatou, and use finite exponential moments to prove compact cylinder-polynomial density in full $L^2(\mu)$. Classical polynomial clustering and translation isometry therefore imply

$$\mathbb{E}_\mu[F(G \circ A_b)] \longrightarrow \mathbb{E}_\mu F\, \mathbb{E}_\mu G \quad (|b| \to \infty), \qquad F, G \in L^2(\mu). \tag{73}$$

Invariant events then have probability zero or one. This establishes ergodicity before reconstruction; it uses no reconstructed vacuum as a premise. The same mixing statement holds on X under μ^X.

For any line L and either one of its open half-planes let Σ_{L+} be its evaluation algebra. REC2.d is exactly

$$\mathbb{E}_\mu[\overline{\Theta_L F} F] \geq 0, \qquad F \in L^2(\Sigma_{L+}, \mu). \tag{74}$$

It also holds for Σ_{L+}-measurable F whose reflected product is absolutely integrable, and on the corresponding X algebras. The *same-half-plane measurability* is essential for both assertions. Under reading M it begins with the measure inequality. Under reading P, REC2 Sections 5–6 first construct local L^2 moment kernels, approximate positive-time polynomial tensors away from null equal-time hyperplanes, and then use full half-plane L^2 density. Reflection isometry passes positivity to L^2 limits. Truncation and dominated convergence give the stated same-half-plane integrable extension. Euclidean invariance transfers the result to any line. These complete arguments replace no hypothesis by a model verdict.

Kernels and the attributed quantum consequence. REC2.e supplies symmetric tempered Schwinger kernels at every order, with local moment and truncated kernels in $L^2(B^k)$ for each bounded B. Its full Sections 5 and 7 use the Gaussian Wick map and local L^2 density, the ordinary pairing and cumulant partition identities, and the stated Schwartz-kernel theorem. AP5's non-Gaussianity implies some even truncated pairing of order $k \geq 4$ is nonzero. Local L^2 regularity rules out a witness supported solely on coincident-time hyperplanes: real product tests can be chosen in pairwise disjoint Euclidean-time slabs.

At this point REC2 Section 8 has established the four hypotheses of [8, Thm. 1.0]: proper Euclidean invariance, full half-plane L^2 OS positivity, the continuous norm with (71), and translation ergodicity. That attributed consequence of [19, 20] reconstructs the unique scalar relativistic Wightman theory for these Schwinger data in $1+1$ dimensions, with its unique physical vacuum. The retained analytic tube-identification and boundary-uniqueness input then transfers the separated nonzero even truncated Schwinger pairing to a nonzero truncated Wightman distribution. Interaction here means failure of Gaussian factorization. No quantitative mass gap or scattering result is inferred.

REC2.g additionally requires the *strict* cap $0 < u/m^2 < 1/8000$ and every unchanged AP6/P1–P6 interface. It identifies the numerical budget's target as this reconstructed field's Euclidean law. Hence it applies to both rows of Table 5, with their existing errors, metric and widths. It supplies no extension of the old universal neural sequence.

Retained correction and source boundaries. The unrestricted measurable-F integrable OS clause in REC-v1 was false. Choose nonzero real h supported in the positive-time half-plane and put $q = h - h \circ \theta$. The retained two-half-plane observable $F = \phi(q)$ satisfies $\Theta F = -F$ and has form $-\mathbb{E}F^2 < 0$. Its bounded sign observable has reflected form -1 almost surely. Positive local Gaussian densities justify nondegeneracy; these witnesses show why integrability alone is insufficient. The version-2 half-plane restriction is a genuine domain correction. The old GAP/PROVED WEAKER assessments and CLOSED historical gate remain in the companion provenance; only REC2's current exact statement is used here. Source access is statement/application verification, with the original classical bodies and antecedents attributed as recorded. The kernel/tube premise and BC1/BC2 readings remain explicit. Neither arbitrary Gibbs-class uniqueness nor the AHKZ boundary estimate is newly proved.

H.7 Positioning and proof navigation

The comparison is architecture specific. Frank's version 2 already gives a rigorous finite-volume Wick-quartic neural construction with quenched features, a finite width per shell, quantitative ultraviolet and finite-observation bounds, commuting width/UV limits and target identification [7]. Removing its UV cutoff retains infinitely many shells. Its coefficient-marginal total-variation statement concerns a fixed finite-dimensional coefficient marginal; it does not say that all ultraviolet/infinite-dimensional field laws converge in total variation. The original Remark 14.1 records the smeared-polynomial-moment overlap. Generic smeared-moment convergence is therefore not used as a novelty separator.

Generic neural representations and interacting network/path-integral reparametrizations have predecessors [6, 14]. Finite-width non-Gaussianity and density deformation also precede this work [3, 11, 12]. Roberts and Yaida's effective-network treatment, based on research with Boris Hanin, relates a small quartic deformation and connected four-point function in a perturbative inverse-width expansion, and gives the leading quartic coupling for its specified second-layer marginal [21, Secs. 1.3 and 4.2]. This is bounded effective-network context and supplies no construction premise. The preprint metadata names Roberts, Yaida and Hanin; the book cover names Roberts and Yaida and credits Hanin's collaboration. The bibliography retains that credit distinction, with fresh manuscript source adjudication still required for this proposed addition.

Conditional-factorization and symmetric-Markov positivity mechanisms have prior instances [4, 9]. Harvey's coordinate-product example already has the product-exponential reference kernel and coordinate two-point positivity [13]. The construction here verifies the full normalized annealed law's positivity for every bounded half-plane observable at the two central coordinate lines, under its invariant geometry. The finite-stage interacting two-point law's Euclidean invariance is not established by anisotropy of the free reference. The exact obstruction hypotheses in the cited neural no-go results retain their separate scope.

Quantitative stochastic-quantization approximations and fixed-torus equilibrium convergence also have established results [15, 16, 25]. Their norms, initial data, dynamical/equilibrium targets and fixed volumes differ from the five-error infinite-volume observation problem here. The bounded contribution is the specified annealed product/Mondrian process ensembles, their full central-coordinate law positivity, and this traceable sufficient observation certificate under named premises. The comparison establishes neither priority nor efficiency.

Complete companion map and actual attribution. Table 6 summarizes the applications and historical complete-proof locations. The adjacent public PROOF-MAP and machine-readable JSON index this edition's source blocks. The original 23-packet index, complete private statement/proof/decision bindings and archive paths are not public attachments. SOURCE is a corrected application/source audit, not an additional theorem. Multi-result packets retain exclusions: in particular SM's E4/r_0 supplement and CR Corollary, blanket P3 $p = p_1$ wording, blind-only REG extensions and stronger blind-only coefficients are not adopted. A later theorem's separate acceptance does not retroactively accept an earlier excluded claim.

The baseline authorship and source-access record in Appendix D is preserved as historical PV4 provenance. This supplement's integration is by GPT-6.1 Sol at max effort under Cody Mitchell's direction. It adopts the exact complete packet arguments, retaining their individual Sol/Astra/Opus lineage, including the OPTREF/OPTMATCH Wick-factor derivation. Verdicts alone confer no argument authorship. The packet decisions record internal model review under their actual policies, including fresh-session Sol-only policy 1.7 where used; they are not independent expert review or formal machine-checked proofs. That 26 October integration was an authored private candidate at the time. The public REVIEW document records the later accepted manuscript baseline and the scope of this edition's review. Version 1 remains separately archived.

Table 6: Applications and historical complete-proof locations. Original path/hash/line bindings are retained in the private research archive; the public PROOF-MAP indexes this edition's source blocks.

Packet	Application in this appendix	Complete argument to apply
CL1	(54)–(56)	CL1-two-limit assembly, Sections 1–6; every prerequisite substitution and both ordered limits.
FV	(57)	CL1-FV1-v2, FV1–FV6, Sections 1–7; normalization, endpoints, telescoping, BL bridge and exact target.
EB1	(59)–(62)	Integration clause map; complete NLAW Sections 1–7 and PHYS P1–P6.
EC1	(63)–(66)	EC1-v2/GROWTH a–e; complete Sections 1–5.
ADMIT	Explicit-cap target, AP2–AP6	Complete AP1–AP6 pressure/source/ version/budget proof with literal P1–P6.
SCALE; NS1	Baseline row of Table 5	SCALE's coefficient proof; NS1 Sections 1–10, especially 3–9 for every actual error.
OPT	Optimized row and (69)	OPT Sections 1–17; OPTMATCH's explicit missing-Wick-display derivation; resource erratum.
REC2	(70)–(74), kernels and reconstruction	REC2(a)–(g), Sections 1–10, using only its version-2 half-plane domain.
Support	Literal finite interfaces	NR1, SM, SKQ, WFSP, H1, H1PREQ, GQ, H2RED, T, REG, MIX, TP, FLOW and qualified SOURCE, at their exact indexed scopes.

I 27 October construction, vector accuracy and linkage

This appendix brings the construction and the later quantitative results to the same field law. The central distinction is between fixing a sequence before measurements and selecting a sufficient finite tuple for specified measurements. The first gives one construction of the field; the second gives explicit, extremely large approximation bounds. Their comparison is a theorem about probability laws, even when the parameters of the first sequence have not been computed.

The seven inputs MOM, SEQ, NORM, SELECT, VEC, ACC and LINK have separate current internal acceptance decisions at their literal conditional scopes. The present integration is a candidate for fresh review. Original author headers in their frozen proofs record the time of writing; they are preserved alongside the later decisions. This appendix adopts their proof arguments, with GPT-6.1 Sol integration authorship; the earlier Opus, Astra-line and Sol argument lineage in Appendix D remains. No internal review is described as an outside expert review or as a machine-checked proof.

I.1 Sources, laws and the two coupling domains

All field spaces, Fourier conventions, neuron laws, own-variance Wick quartics, potentials, annealed probabilities and central reflections are those of Definitions 2.10 and 2.12 and the preceding construction. In particular u multiplies the Wick quartic without a factorial, the free covariance is $(-\Delta + m^2)^{-1}$, and a finite interaction square restricts the interaction of a whole-plane field. The endpoint neuron covariance $e^{-|z|_1/a}/4$ has point variance $1/4$; it is distinct from the physical covariance C_a and its variance c_a. Let $\nu^S = v_{m,u}$ be the common original Fröhlich–Simon Section-4 state, and put $\nu = \mu^X = R_*\nu^S$ on X. The realization R preserves every compact smooth evaluation simultaneously on one ν^S-full measurable set. In this appendix $\mu = 1/a$ in the neuron definitions continues to mean inverse regulator, rather than a probability law. We use $\nu_n = \nu_{0,\Lambda_n}$, $\Lambda_n = (-n\ell, n\ell)^2$, for the actual original physical laws throughout.

The source assumptions behind the explicit cap are mathematical premises, not consequences of a numerical certificate. For clarity their operational content is recorded here. P1–P6 below are the accepted ADMIT premises; the named interfaces retain their full source qualifications.

P1–P2: finite positive scales. On the shifted torus $\epsilon(\mathbb{Z} + 1/2)^2/L\mathbb{Z}^2$, use the inner product $\epsilon^2 \sum$, $A = -\Delta^\epsilon + m^2$, the *plane* lattice Wick variance c_ϵ, and profiles $q_x \in [0, 1]$. P1 supplies DSS/SK-q for the density proportional to $\exp[-u\epsilon^2 \sum q_x{:}\psi_x^4{:}_{c_\epsilon}]$ relative to $P_{A^{-1}}$, including Gaussian sites. At every finite real field H it supplies

$$0 \le \mathrm{Cov}_{\nu^H}(\psi_x, \psi_y) \le E_\nu(\psi_x\psi_y). \tag{75}$$

The same statements hold after adding physical mass $1/s$, for every finite $s > 0$. P2 assumes $0 < \epsilon \le \min(1/4, 18/(5m))$, $L \ge \max(1, \sqrt{2}/m)$, $L/\epsilon \in 2\mathbb{N}$, $L/\epsilon \ge 4$, and $0 < u \le m^2/8000$. Core P* with $S_{\max} = \infty$, $\Phi = 1$, $\kappa = 0$ gives

$$\max_x \epsilon^2 \sum_y S_{s,q}(x, y) \le \frac{s}{1 + m^2 s} + \frac{2s}{2 + m^2 s}, \qquad d\mu_{s,q} \propto e^{-u\epsilon^2 \sum q_x {:}\psi_x^4{:}_{c_\epsilon}} dP_{(A+1/s)^{-1}}. \tag{76}$$

This is a finite-positive-scale statement, with no original-law endpoint included as an assumption.

P3–P4: fixed-square transfer. For each fixed $B = (-b, b)^2$, the commensurate shifted plane mesh $\epsilon = b/N \downarrow 0$ is on a common Gaussian space with

$$\epsilon^2 \sum_{x \in B} {:}\psi_x^4{:}_{c_\epsilon} \longrightarrow J_B, \qquad \epsilon^2 \sum_x f(x)\psi_x \longrightarrow \phi(f) \quad \text{in Gaussian } L^2. \tag{77}$$

Here f is any real compact smooth test. For each fixed finite $p \ge 1$, the expectations of $\exp(-puJ_B^\epsilon)$ are uniformly finite for sufficiently small ϵ. The continuum normalizer is positive finite, with finite Gaussian L^p density at every finite p. At fixed mesh P4 gives entrywise torus-to-plane Gaussian covariance convergence on finite site sets through admissible even periods; the plane Wick variance is unchanged. These are fixed-square statements for all $m, u > 0$.

P5: pressure and the original source states. For $P_h(X) = uX^4 - hX$, use the normalized free pressure of [8, Proposition 2.1 and the remark after (2.3), pp. 502–503]. Its actual Corollary 4.3(ii), p. 510, equates differentiability at zero in h with equality of the two original maximal states. The converse is part of this premise; reversing the narrower old C1(iii) would not supply it. Theorem 5.2, p. 512, then gives ordinary free-boundary exhaustion convergence to that common state, with compact-test generating functionals read at every real frequency. These are attributed source facts whose full original proofs are not reproved here.

P6: the additional physical and finite interfaces. Retain the all-$m, u > 0$ physical/classical Wick-law identity, the complete X field-space and Borel-determining interfaces, and the original Gaussian/common-space Wick laws proved in the paper. Retain the all-semibounded-polynomial local-density and positive boundary factorization of C1(ii), at mass m:

$$dv_{m,u}|_{\mathcal{C}_B} = e^{-uJ_B - \alpha_m(uX^4)|B|} g_{\partial B}\, dP_m, \qquad g_{\partial B} > 0, \tag{78}$$

with the two recorded boundary-algebra readings and finite local Gaussian L^p densities. Also retain the exact DSS/SKQ/P*/H1-PREQ/H1/MIX/REG/ smoothed-CT/FLOW/TP/T interfaces, including the later source-normalization correction; FLOW has its own cap $u \le m^2/8000$ and TP/T are fixed-square transfers. Finally retain the raw positive-stage NR1/L0 interfaces,

$$\delta_Z \le \frac{d_D M_Z}{96N}, \qquad \delta_A \le \frac{d_D M_A}{96N}, \qquad Z_Q \ge z, \tag{79}$$

the full bilinear reference covariance matching, independent Gaussian covariance-defect coupling, and physical common-space Wick bounds and normalizers. L0 has its full ambient Euclidean operator-norm meaning stated below. NR1's exact sharper author constant and larger isolated-proof core keep their separate accepted scopes. Old globally g_*-qualified CL1/FV/EB1 conclusions are not used as expanded-domain assumptions. P6 includes neither full AHKZ-class uniqueness nor its averaged boundary estimate C2(iii).

The accepted consequences used in the following proofs are precisely these. AP1 supplies the finite original-law row bound $3/m^2$ and source bound. AP2 transfers that bound to the actual classical free law for every real compact smooth source h, supported anywhere in the plane, every $t \in \mathbb{R}$, and every physical square:

$$0 \le \log E_{\widehat{\nu_B}} e^{t\phi(h)} \le \frac{3t^2}{2m^2}\|h\|_2^2. \tag{80}$$

It also holds for the Gaussian L^2 source extension. AP3 gives $0 \le \alpha_m(uX^4 - hX) - \alpha_m(uX^4) \le 3h^2/(2m^2)$, pressure differentiability and equality of the original FS states, with their actual compact-joint free-volume limit. AP4 identifies the same state as $(T_s)_* v_{M,u/s^2}$ for every $M > 0$, $s = m/M$, using the original dilation. AP5 gives the physical/classical joint-law identity, the same-distribution realization $\nu = R_* v_{m,u}$, full $\nu_n \Rightarrow \nu$ on X, fixed compact polynomial moment/UI convergence, and non-Gaussianity from the positive local quartic density. These AP1–AP5 premises have domain

$$0 < u/m^2 \le 1/8000. \tag{81}$$

They do not assume any p_j moment conclusion. AP6 separately ports the five actual-law error formulas of Appendix H to this same target under P1–P6 and the *strict* domain

$$0 < u/m^2 < 1/8000. \tag{82}$$

The original fixed g_* remains numerically unknown; no statement here deduces $10^{-10} < g_*$.

For target kernels and reconstruction we retain exactly REC2.a–f, as applied in Appendix H: source-defined target, Schwartz regularity, qualitative L^2 mixing before reconstruction, half-plane positivity, tempered/local kernels and attributed reconstruction. Its added FS Theorem 4.4 clustering premise is distinct from the old restricted C1–C3,C6 route, whose mixing used C2(iii). Both OS M/P readings, both boundary readings, local-density, half-plane, kernel/tube/boundary and reconstruction premises remain literal. REC.g and quantitative uses of AP6 remain strict. Hashes and internal reviews certify neither unavailable primary classical bodies nor stronger source assertions.

I.2 All local moments and the original common sequence at the cap

Theorem I.1 (MOM and the separately derived SEQ-HS finiteness)**.** *Assume the original field-space, fixed-volume and physical/classical interfaces and AP2–AP5 under P1–P6, at* (81)*. Use the original p_j, modes $e_{j,k}$, weights $\lambda_{j,k}$ and volume $V_j = 4(j+1)^2$. For $s > 1$ put*

$$h_{j,s}(v)^2 = \sum_k \lambda_{j,k}^{-s} |v(\chi_j \overline{e}_{j,k})|^2, \qquad c_j = \|\chi_j\|_2^2 / V_j,$$
$$S_{j,s} = \sum_k \lambda_{j,k}^{-s} \le 1 + 8((j+1)/\pi)^{2s} \left(1 + \frac{1}{2s-2}\right), \qquad D_p = \frac{6}{m^2}(2\Gamma(p+1))^{1/p} \quad (p \ge 1). \tag{83}$$

For every fixed integer $j \ge 1$, real $s > 1$ and real $r > 0$, with $p = \max(1, r/2)$,

$$\sup_{\rho \in \{\nu_n : n \ge 1\} \cup \{\nu\}} \rho(h_{j,s}^r) \le (D_p c_j S_{j,s})^{r/2} < \infty. \tag{84}$$

In the formal MOM scope $s = 2$, $h_{j,2} = p_j$. Put $B_j = c_j S_{j,2}$ and $C_{j,r} = (D_{\max(1,r/2)} B_j)^{r/2}$. Then, for every real $p \ge 1$ and $r > 0$,

$$\sup_n \nu_n(p_j^{2p}) \le 2\Gamma(p+1)(6B_j/m^2)^p, \qquad \nu_n(p_j^r) \longrightarrow \nu(p_j^r),$$
$$\sup_\rho \rho(p_j^r 1_{\{p_j^r > L\}}) \le C_{j,2r}/L, \qquad \nu_n(H) \longrightarrow \nu(H) \tag{85}$$

for every continuous complex H with $|H(v)| \le c(1 + p_j(v)^r)$. All expectations are absolutely finite. Every early physical volume is included. The all-$s > 1$ assertion is SEQ-HS finiteness; moment convergence at $s \ne 2$ is not asserted, and $h_{j,s}$ need not be finite or continuous on all of X when $1 < s < 2$.

Proof. The exact scalar physical/classical law identity transfers (80), including its unbounded nonnegative exponential. With $\sigma_h^2 = 3\|h\|_2^2/m^2$, Chernoff at $t = x/\sigma_h^2$ and at the opposite sign gives

$$\nu_n(|v(h)| > x) \le 2e^{-x^2/(2\sigma_h^2)}, \qquad \nu_n(|v(h)|^q) \le 2(2\sigma_h^2)^{q/2}\Gamma(1 + q/2). \tag{86}$$

The latter follows by Tonelli and $y = x^2/(2\sigma_h^2)$ in the tail integral for any $q > 0$. A zero source is zero directly, without division. In particular $\|v(h)\|_{2p}^2 \le D_p \|h\|_2^2$.

The real and imaginary sources of $a_{j,k} = v(\chi_j \overline{e}_{j,k})$ are $V_j^{-1/2}\chi_j \cos(\pi k \cdot x/(j+1))$ and $-V_j^{-1/2}\chi_j \sin(\pi k \cdot x/(j+1))$. Their squared L^2 norms sum to c_j at every frequency, including the zero mode. Minkowski on their two nonnegative squares gives $\|a_{j,k}\|_{2p}^2 \le D_p c_j$, without an independence assumption. There are $8l$ modes on the $|k|_\infty = l$ shell; integral comparison of $\sum l^{1-2s}$ proves (83).

For the finite nonnegative sum $H_M = \sum_{|k|_\infty \le M} \lambda_{j,k}^{-s} |a_{j,k}|^2$, finite Minkowski gives $\nu_n(H_M^p) \le (D_p c_j S_{j,s})^p$. Increase M by monotone convergence to obtain the physical bound. To obtain the target bound, keep M finite first: $\min(L, H_M^p)$ is bounded continuous on X, so AP5 gives the same upper bound at ν. Increase L, then M. This uses no continuity of the infinite $h_{j,s}$. Concavity at $r/(2p) \le 1$ gives (84). In particular the same laws are concentrated on all real H_{loc}^{-s}, $s > 1$, by intersecting over countably many rational $s > 1$ and using $h_{j,s} \le h_{j,t}$ for $t < s$.

At $s = 2$ the original Hilbert norm is continuous, since $|p_j(v) - p_j(w)| \le p_j(v - w)$ and $d(v, w) < 2^{-j}\delta$ implies $p_j(v - w) < \delta$ for $0 < \delta < 1$. All positive powers are continuous. The order-$2r$ bound gives the tail inequality in (85). Apply AP5 to $\min(p_j^r, L)$, then let L increase, to prove moment convergence. For the complex-growth conclusion use radial truncation $\pi_L(z) = z \min(1, L/|z|)$, $\pi_L(0) = 0$. The uniform second moment of H is at most $2c^2(1 + C_{j,2r})$, so the truncation error is bounded by that quantity divided by L. Bounded-continuous convergence and then $L \to \infty$ prove the last assertion. No physical subsequence or uniform-volume Gaussian density bound has been used. □

Theorem I.2 (CAPSEQ: one unchanged sequence)**.** *Under the premises of Theorem I.1, the original accepted all-positive-m, u fixed-volume interfaces and the full REC2.a–f premises for target identification, all clauses*

(S),(0),(A)–(E) of Theorem 4.1 hold at (81)*, with the target* $\nu = R_* v_{m,u}$*. The sequence is exactly the original four least-index recursion of Proposition 11.7, with* $a(k) = 1/(2mk)$*,* $R(k) = k$*,* $\varepsilon(k) = 1/k$*,* $N(k) = k$*,* $\eta_n = 2^{-n}/4$ *and the unchanged*

$$\mathrm{dist}_n(\rho,\eta) = \beta_X(\rho,\eta) + \sum_{1\le j,r\le n} |\rho(p_j^r) - \eta(p_j^r)|. \tag{87}$$

It obeys, for every physical n*, both architectures simultaneously,*

$$\begin{aligned}
&0 < a_n < \min(1/n, 1/(2m)), \quad a_n < a_{n-1}/2,\\
&R_n \in \mathbb{N}, \quad R_n \ge \max(n, n\ell), \quad R_n > R_{n-1},\\
&0 < \varepsilon_n < 1/n, \quad \varepsilon_n < \varepsilon_{n-1}/2, \quad N_n \in \mathbb{N}, \quad N_n > n, \quad N_n > N_{n-1},\\
&\mathrm{dist}_n(P_n^b, \nu_n) < 2^{-n}, \qquad \Omega_n = [-R_n, R_n]^2,
\end{aligned} \tag{88}$$

with predecessor conditions omitted at $n = 1$*.*

Every selected law is a Borel probability with positive finite normalizer and every positive local moment. Both full sequences converge weakly on X*, their* p_j^r *expectations converge for every real* $r > 0$*, and their joint compact-test correlations converge in* $\sigma(\mathcal{D}', \mathcal{D})$ *at every order, including nondecomposable joint tests. Every real/complex polynomial of every later real compact smooth tuple has converging expectations and uniformly bounded, uniformly integrable positive powers. The deterministic tensors, correlation distributions, target restrictions of tempered Schwinger kernels, and all constants* $M_{j,r}$ *are exactly those in (C),(D) and Section 12. Finite laws are reflection positive across each central coordinate line, for bounded half-plane observables and the full polynomial forms (E2), with both* L^2 *estimates of that clause. These choices occur after the sequence, without reselection, countability restriction on later tests, or a subsequence.*

On the overlap with the original fixed $0 < u/m^2 < g_*$*, every selected index equals the original index. All clauses here preserve that original sequence. The quantitative premise AP6 still requires* (82)*.*

Proof. The fixed-volume arrows in Section 9 hold at every $m, u > 0$. Their moment arrows follow from the reference bounds and exact normalizer domination in Section 11: at the neural stage the density bound is M_{full}/Z_N^b for every finite N, becoming $2M_{\mathrm{full}}/Z_{\mathrm{full}}$ only eventually at fixed parameters. Residual and physical bounds keep their fixed-volume dependence. Truncate a continuous function of p_j growth, apply the weak arrow, and use a higher uniform moment to remove the truncation, as in Lemma 11.5. Theorem I.1 supplies the previously missing physical-volume seminorm-power arrow at the new cap.

At fixed n, each arrow therefore tends to zero in the full dist_n. In the original order, first choose the least regulator index satisfying its discrepancy and size/predecessor rules; then the least auxiliary integer; then the least positive reciprocal stabilizer; finally the least width satisfying the discrepancy for *both* architectures. At each step the admissible integers contain an entire tail. The intersection of the two width tails still contains a tail, as do the finitely many extra size constraints. Thus every least index exists. The four strict errors are each $< \eta_n$, and their triangle gives (88) with the full moment sum.

Each early neural law has $p_j(\iota F_N^b) \le \|\chi_j\|_2 \sqrt{N}/2$ by Parseval and the field bound. Its potential is bounded on the bounded field range in finite Ω, so its exact Z_N^b is positive finite. For any fixed integer j, r, the corresponding term of $\mathrm{dist}_n(P_n^b, \nu_n)$ is $< 2^{-n}$ once $n \ge \max(j, r)$. Physical moment convergence and the finitely many early laws give finite $M_{j,r}$ over both sequences and ν. Also $\beta_X(P_n^b, \nu) \le 2^{-n} + \beta_X(\nu_n, \nu) \to 0$. The same continuous-growth truncation extends moments to every real positive order. Compact duality $|v(f)| \le \|f\|_{H^2(\mathbb{T}_j)} p_j(v)$ then proves all later polynomial/power assertions, including complex, constant, vanishing and dependent cases.

For completeness the correlation mechanism is deterministic and has no coupling restriction. For $F \in \mathcal{D}(((-j, j)^2)^k)$ its Fourier tensor series is absolutely bounded by $A_{j,k}(F) p_j(v)^k$ through weighted Cauchy–Schwarz. Moreover

$$|v^{\otimes k}(F) - w^{\otimes k}(F)| \le k A_{j,k}(F) p_j(v - w) \max(p_j(v), p_j(w))^{k-1}. \tag{89}$$

The $k = 1$ last factor is one. The product-density construction and gluing in Section 12 give all arbitrary joint tests, conjugations and reflected products, as well as $A_{j,k}(F) \le 3^k V_j^{k/2} \|F\|_{C^{2k}}$. Uniform higher moments

allow integration and continuous-growth transfer of (89). REC2's same-target tempered kernels agree on compact products; product density gives (D) on all compact joint tests. Compact evaluations generate Borel(X), so no second target law is introduced.

The centered axis-parallel squares in (88) obey the original E.P2/E.T5 hypotheses. Replacing the open physical square by its closure changes no bounded-field potential integral. Hence bounded half-plane RP holds at each stage across θ_1 and θ_2. For $\mathcal{O}_f = \sum_{r=0}^{J} v^{\otimes r}(f_r)$, the two separate bounds are

$$\|\mathcal{O}_f\|_{L^2(\rho)} \le |f_0| + \sum_{r=1}^{J} A_{j,r}(f_r) M_{j,2r}^{1/2},$$
$$\|\mathcal{O}_f \circ \theta\|_{L^2(\rho)} \le |f_0| + \sum_{r=1}^{J} C_j^r A_{j,r}(f_r) M_{j+1,2r}^{1/2}. \tag{90}$$

The reflection estimate is the deterministic one of the original field space; no invariance of ρ is assumed. Half-plane product density gives the required measurability. Apply bounded RP to the radial truncation of $\mathcal{O}_f$, then let the truncation increase using these two L^2 bounds and Cauchy–Schwarz. Every reflection-form term is absolutely integrable; the tensor identities identify it with the corresponding compact correlation. Their finite sum converges to the real nonnegative target form. At $J = 0$ the form is $|f_0|^2$ directly.

Finally AP3/AP4 and compact-cylinder uniqueness identify the original target on the old-domain overlap. The four predicates have the identical laws, normalizers, enumerations, discrepancies, allocations and predecessor rules there. Induction in n, and within each n in a, R, ε, N order, makes the admissible sets and their least members identical. No bound for a least index, decision procedure or new value of g_* is obtained. □

I.3 A finite family with full ambient vector geometry

For the numerical results fix $m = 1$, $u = 10^{-10}$, $\ell = 1000$. The strict gate follows from $8000 < 10^{10}$ through AP6; the old fixed g_* is not part of that numerical check. Define

$$b(t) = \begin{cases} e^{-1/(1-t^2)}, & |t| < 1, \\ 0, & |t| \ge 1, \end{cases} \qquad f_h(x,y) = b(x-h_1)b(y-h_2). \tag{91}$$

The supported tuple is ordered, unweighted and has $1 \le r \le 4$. Every center coordinate is the canonical dyadic $p/2^k \in [-8,8]$, $0 \le k \le 16$: zero and integers have $k = 0$, and p is odd when $k > 0$. Decimal numerators have no plus sign, leading zero or negative zero. Repetition preserves order and multiplicity. There are no input amplitudes, signs, dilations, rotations or custom tests. Write

$$K = \bigcup_i (h_i + [-1,1]^2), \quad R = 1 + \max_{i,j} |h_{ij}| \le 9, \quad c_r = \lceil \sqrt{r} \rceil, \quad F_q = \left(\sum_i \|f_{h_i}\|_q^2 \right)^{1/2}. \tag{92}$$

The observation law $\mathcal{L}_f(\rho)$ is the pushforward by $O_f(v) = (v(f_{h_i}))_i$ into the original ambient $\mathbb{R}^r$. Write $\beta_X = \beta$ for the original field-law metric. The observation distance is (52), with separate value and Euclidean Lipschitz bounds. Let $\mathcal{U}_4(r)$ consist of real C^4 functions on $\mathbb{R}^r$ with all full multilinear operator norms of orders 0 through 4 at most one.

Proposition I.3 (SELECT geometry, source and covariance statements)**.** *Under the literal SELECT/AP2–AP6 and Gaussian/coupling premises, $K \subset [-9,9]^2 \subset \Lambda_1$ with open containment, and*

$$F_1 \le 4\sqrt{r} \le 4c_r, \quad F_2 \le 2\sqrt{r} \le 2c_r, \quad F = F_{4/3} \le 3\sqrt{r} \le 3c_r \le 6. \tag{93}$$

Use the identical chosen H1 coefficient $C_H < 2^{16}$ and $C_{H2} = 2^{384}$ in (55)–(56). At these physical parameters its literal coefficient satisfies

$$C_f(f;R) = \frac{\sqrt{r}(4R+1)}{5} C_f(f_0;1), \qquad C_f(f_0;1) \le 2^{360}, \quad C_f(f;R) \le \bar{C}_f := \frac{2^{360} c_r (4R+1)}{5} < 2^{364}. \tag{94}$$

The separate spatial factor $e^{R/256}$ remains in E_n.

Let $H_{ij} = \langle f_{h_i}, f_{h_j} \rangle_{L^2}$. Writing $h_*(d) = \int b(t)b(t-d)dt$, one has $H_{ij} = h_*(h_{i1} - h_{j1})h_*(h_{i2} - h_{j2})$ and $0 \le H \le 4rI$. An entry is positive exactly when both coordinate separations are < 2, and is zero if either is ≥ 2. Under every physical free law and the same target, for all $t \in \mathbb{R}^r$,

$$1 \le Ee^{t\cdot O_f} \le e^{3t^T Ht/2}, \qquad EO_f = 0, \qquad 0 \le \mathrm{Cov}(O_f) \le 3H \le 12rI. \tag{95}$$

These are quadratic-form inequalities. They are not entrywise covariance formulas for the interacting law, or source bounds for finite neural laws.

For the Gaussian observation covariances, $0 \le \Sigma_a \le \Sigma_0 \le H$ and $0 \le \Sigma_{\mathrm{end}} \le a^2 H$. For a positive covariance defect Δ with $\delta_\Omega(K) = \sup_{x\in K} \Delta(x,x)$, its observation covariance H_Δ obeys

$$|(H_\Delta)_{ij}| \le \|f_i\|_1 \|f_j\|_1 \delta_\Omega(K), \qquad \mathrm{tr}\, H_\Delta \le F_1^2 \delta_\Omega(K). \tag{96}$$

The actual Gaussian coupling therefore has mean observation displacement at most $F_1\sqrt{\delta_\Omega(K)}$. The physical regulator coupling has mean displacement at most $(a/2)F_2$. Repeated or dependent observations are included with their original ambient dimension throughout.

Proof. Every derivative of the interior bump is a polynomial in $(1-t^2)^{-1}$ times its exponential, which tends to zero at the two endpoints. Thus the zero extension is smooth, has support $[-1,1]$ and supremum e^{-1}. Translation preserves the Lebesgue norms; area four and $|f_h| \le 1$ give the individual bounds $4, 2, 4^{3/4} < 3$. Sum their squares over all coordinates. Hölder gives the full observation operator bound $|O_f(v)|_2 \le F\|v\|_4$.

In the literal CL1 expression all factors except $(2R+1/2)\sqrt{r}e^{-2}$ are identical to the base bump's $(5/2)e^{-2}$. Their ratio proves (94), even if their common factor vanishes. The upper uses $c_r \le 2$, $R \le 9$ and $74 < 80$. This compares the displayed coefficients with identical H1/H2 choices; it does not compare independently minimized constants.

Fubini gives the Gram formula. The overlap integrand is positive on an open rectangle exactly in the stated positive case. For any t, $t^T Ht = \|\sum t_i f_i\|_2^2$, while $\mathrm{tr}\, H \le 4r$. Apply AP2 to this real compact source. The upper passes to the target by bounded truncations of the positive exponential and weak convergence. Sign symmetry and the resulting exponential moments give zero mean and Jensen's lower bound. Differentiating the scalar MGF at zero in direction t, justified by its two-sided exponential bound, gives the covariance inequality. A zero Gram entry describes deterministic support geometry; it supplies no interacting independence.

The physical Fourier multipliers satisfy $0 \le \widehat{C}_a \le \widehat{C}_0 \le 1$, proving their Loewner bounds by Plancherel on $\sum t_i f_i$. The positive endpoint kernel integrates to a^2; Young's inequality proves its bound. Positivity of Δ gives $|\Delta(x,y)| \le \sqrt{\Delta(x,x)\Delta(y,y)}$ and hence (96). Sum the diagonal and use Cauchy–Schwarz in the actual coupling. The physical common-space bound is $E|(\Phi_a - \Phi_0)(f_i)|^2 \le (a^2/4)\|f_i\|_2^2$; summing gives the last assertion. Identical centers give identical coordinates, possibly singular laws; none of these proofs uses an inverse, density or rank. □

I.4 Relative normalization and the sharper fixed-accuracy width

We now give the full argument behind NORM and VEC's change to the first arrow. Its improvement comes from normalizing each replacement locally through a common zero-amplitude partition. It uses the actual joint law of background and replacement direction.

Take either of the following outer tuples, keeping their separate observation domains:

case	n	p $(a = 2^{-p})$	R_{aux}	t $(\varepsilon = 2^{-t})$
scalar NORM	74	16384	137439027473	68719476736
vector VEC	75	16384	137439028473	68719476736

(97)

The scalar case has the single unshifted f_0; the vector case has the whole typed family above, at accuracy $1/5$. In each case $D = \Omega = [-R_{\mathrm{aux}}, R_{\mathrm{aux}}]^2$, $\Lambda = (-1000n, 1000n)^2$, $A = |\Lambda|$, $O = d_D = |\Omega|$, and $E_D = L^4(D;\mathbb{R})$. Restrict the original neuron and endpoint Gaussian to D. They are strongly measurable, Bochner centered,

globally sign symmetric, have covariance C_{end}, and match in expectation for *every bounded bilinear form* on E_D. This last full matching is the literal finite-reference premise; point-pair matching alone is not substituted for it. Their point variances are $1/4$, and fourth moments are at most $3/16$.

Let V be the exact potential in Definition 2.8 on D, with physical $c = c_a$, and write $w = e^{-V}$, $x = \|v\|_4$. For independent neuron/Gaussian copies set

$$S_j = N^{-1/2}\left(\sum_{i=1}^{j}\gamma_i + \sum_{i=j+1}^{N-1} g_i^b\right),$$
$$v = S_j + \theta h/\sqrt{N}, \quad Z_{j,h}(\theta) = Ew(v), \qquad d\Pi_{j,h,\theta} = w(v)\, d\,\mathrm{Law}(S_j,h)/Z_{j,h}(\theta), \quad 0 \le j < N, \quad 0 \le \theta \le 1. \tag{98}$$

The direction h is independent of S_j *before* tilt and has either neuron or Gaussian law. Under Π it is generally dependent on v. Using the v_k, e_k of (60), with $B = \kappa_a^2/2 + 6uc$, define

$$\begin{aligned}
&\alpha = \min(u,\varepsilon), \quad U = \max(u,\varepsilon), && L = B^2O/(2\alpha), \quad J_z = 3UO/16 + 3uc^2A,\\
&H = L + J_z, \quad X = [2(H+4)/\alpha]^{1/4}, && Y = 2d_D^{1/4}(H+10)^{1/2},\\
&P_F(x) = 2e_4(x) + \sum_{q=1}^{4}\binom{4}{q}F^q e_{4-q}(x), && K_1 = v_1(X)Y,\\
&N_0 = \max(1, \lceil 16K_1^2\rceil), && K_4(F) = P_F(X)Y^4/6.
\end{aligned} \tag{99}$$

The scalar Y here bounds direction moments; O_f denotes the observation. All polynomials have nonnegative coefficients and $\deg P_F \le 12$. An upper for c or F can be used in these majorants without changing the actual potential.

Theorem I.4 (NORM/VEC relative replacement at the specified stages)**.** *Assume their literal reference, Wick, finite-interface and strict AP6 premises. Every amplitude bridge in* (98) *has positive finite Z, point variance at most* $1/4$*, fourth moment at most* $3/16$*, and*

$$\begin{aligned}
&w(v) \le e^{L-\alpha x^4/2}, \quad Z \ge e^{-J_z}, && \frac{d\Pi}{d\,\mathrm{Law}(S_j,h)} \le e^H,\\
&\Pi(x^4 > z) \le \min(1, e^{H-\alpha z/2}), && \|x\|_{L^q(\Pi)} \le X, \quad \|\|h\|_4\|_{L^q(\Pi)} \le Y \quad (1 \le q \le 16).
\end{aligned} \tag{100}$$

Before tilt $E\exp(\|\gamma\|_4^2/(4\sqrt{d_D})) < 4$. The logarithmic partition has a C^1 extension past the endpoints and

$$(\log Z)' = -N^{-1/2}E_\Pi DV(v)[h], \quad |(\log Z)'| \le K_1/\sqrt{N}, \quad \frac{Z_{j,h}(\theta)}{Z_{j,g}(1)} < 2 \quad (N \ge N_0). \tag{101}$$

More precisely the neuron ratio is at most $e^{K_1/\sqrt{N}}$, and the Gaussian ratio at most $e^{2K_1/\sqrt{N}}$; the zero-amplitude partitions agree.

For $G \in \mathcal{U}_4(r)$, center at its normalized Gaussian endpoint η_j and put $\Phi_j(v) = [G(O_f(v)) - \eta_j]w(v)$. Then $|\eta_j| \le 1$, $\Phi_j \in C_b^4(E_D)$ and

$$\|D^4\Phi_j(v)\|_{\mathrm{op}} \le w(v)P_F(x), \qquad E_\Pi[P_F(x)\|h\|_4^4] \le P_F(X)Y^4. \tag{102}$$

Each adjacent normalized replacement costs at most $K_4(F)/N^2$. Consequently the actual neural and full Gaussian tilted laws satisfy, for $N \ge N_0$,

$$\begin{aligned}
&|P_N^b(G\circ O_f) - Q_\varepsilon(G\circ O_f)| \le \min(2, K_4(F)/N),\\
&d_{\mathrm{BL}}(\mathcal{L}_f(P_N^b), \mathcal{L}_f(Q_\varepsilon)) \le \min\left(2, \frac{\sqrt{6}K_4(F)}{N\delta^3} + 2\sqrt{r}\,\delta\right) \quad (0 < \delta \le 1),\\
&d_{\mathrm{BL}}(\mathcal{L}_f(P_N^b), \mathcal{L}_f(Q_\varepsilon)) \le \min(2, D_r N^{-1/4}), \qquad D_r^4 = \frac{2048\sqrt{6}}{27}r^{3/2}K_4(F).
\end{aligned} \tag{103}$$

This includes all singular and repeated observation tuples in the stated family; their dimension factors remain.

Proof. At almost every point a bridge is a sum of independent centered variables, with coefficients $1/\sqrt{N}$ and $\theta/\sqrt{N}$. Their squared coefficient sum is $(N-1+\theta^2)/N \le 1$. Expanding the fourth moment leaves the single-index and two-pair terms, bounded by $(3/16)(\sum a_i^2)^2 \le 3/16$. This includes $N=1$, $\theta=0,1$ and both extreme j. Tonelli on bounded D gives the needed potential integrability. Hölder and completing the square give $V(v) \ge \alpha x^4 - B\sqrt{O}x^2 \ge \alpha x^4/2 - L$. Dropping the negative quadratic only from an upper bound gives $EV \le J_z$; Jensen gives $Z \ge e^{-J_z}$, hence the density and tail bounds. No neural or full-tilt normalizer is assumed at least one.

If T has tail $\le \min(1, e^{a-z})$ for $a \ge 0$, its moments are bounded by those of $a + E_1$ with a scalar unit exponential E_1. Tail integration proves the comparison and $EE_1^k = k!$. Apply it to $T = \alpha x^4/2$. Minkowski in L^4 gives $\|T\|_4 \le H + (4!)^{1/4} < H + 4$, hence $\|x\|_{16} \le X$. For a neuron $\|h\|_4 = d_D^{1/4}/2$ deterministically, also after tilt. For Gaussian h, spatial/probability Minkowski and its variance $1/4$ give $E\|h\|_4^{2k} \le d_D^{k/2}(k/2)^k$ for integers $k \ge 2$; at $k=1$, $E\|h\|_4^2 \le \sqrt{3d_D}/4 \le \sqrt{d_D}/2$. The nonnegative exponential series and $k! \ge (k/e)^k$ give

$$Ee^{\|h\|_4^2/(4\sqrt{d_D})} \le \sum_{k\ge 0}(e/8)^k < 4. \tag{104}$$

The actual joint density cap therefore yields the tilted tail $\Pi(T_h > z) \le \min(1, e^{H+2-z})$ for $T_h = \|h\|_4^2/(4\sqrt{d_D})$. The same scalar comparison in L^8 gives $\|T_h\|_8 < H + 10$, proving the Y bound. Lyapunov gives every real order through sixteen. Dependence after tilt is fully allowed.

Quartic Hölder gives the full Fréchet derivative bounds $\|D^k V\|_{\mathrm{op}} \le v_k(x)$, $k \le 4$. The exponential chain rule gives $\|D^k w\|_{\mathrm{op}} \le w e_k(x)$, with the exact fourth Bell coefficients $1, 6, 3, 4, 1$ in (60). Each weighted derivative is globally bounded: its majorant is $e^{L-\alpha x^4/2}$ times a finite polynomial, and $\sup x^k e^{-\alpha x^4/2} = (k/(2\alpha e))^{k/4}$ for $k > 0$. Dominated differentiation under expectation is thus bounded by a constant times $\|h\|_4/\sqrt{N}$, integrable before tilt. It gives a continuous derivative on a real neighborhood of $[0,1]$. Under the same joint Π, Hölder at orders four and two gives $E_\Pi |DV(v)[h]| \le (4UX^3 + 2B\sqrt{O}X)Y = K_1$. Integrate along the neuron branch, and along both branches through their equal zero-amplitude partitions. Their costs are respectively one and two copies of $K_1/\sqrt{N}$. For $N \ge N_0$ the latter exponential is at most $\sqrt{e} < 2$, proving (101).

The observation operator bound is F. The fourth product rule, and $|G - \eta_j| \le 2$, give (102); the two e_4 copies are essential. If $P_F(x) = \sum_{k=0}^{12} a_k x^k$, Hölder at total order $k + 4 \le 16$ gives $E_\Pi x^k \|h\|_4^4 \le X^k Y^4$, including $k = 0$ directly. This uses joint moments, never a product of dependent marginal expectations.

Taylor expansion at S_j through order three has fourth remainder

$$N^{-2}\int_0^1 \frac{(1-\theta)^3}{6} D^4\Phi_j(S_j + \theta h/\sqrt{N})[h,h,h,h]\,d\theta. \tag{105}$$

Globally bounded derivatives and finite untilted fourth moments justify all conditional expectations and interchanges. Given S_j, the two directions have matching constant terms, zero first and third terms by Bochner centering/global sign symmetry, and matching second terms by the full bounded-bilinear premise. Divide their difference by $Z_{j,g}(1)$; Gaussian-endpoint centering makes this exactly the difference of normalized expectations. Each of the two remainders uses its own partition ratio ≤ 2. Since the integral weight is $1/24$, their sum is at most $4P_F(X)Y^4/(24N^2) = K_4(F)/N^2$. Successive normalized endpoints agree, and the final normalized Gaussian sum has the original Gaussian law. The N terms telescope to K_4/N.

Literal L0 uses a standard Gaussian in the *ambient* $\mathbb{R}^r$. It gives $\|G - G_\delta\|_\infty \le \sqrt{r}\delta$ and full derivative bounds through four at most $\sqrt{6}\delta^{-3}$ for $0 < \delta \le 1$. Apply the smooth estimate to $G_\delta/(\sqrt{6}\delta^{-3})$ and keep both smoothing errors. With $A_0 = \sqrt{6}K_4$, $B_0 = 2\sqrt{r}$, the interior minimizer is $(3A_0/(B_0 N))^{1/4}$ when it is at most one. Its value has fourth power $(2048\sqrt{6}/27)r^{3/2}K_4/N$. Otherwise that expression exceeds $(4/3)B_0 > 2$ and the probability cap suffices. Both cases retain $N \ge N_0$. No covariance is inverted. □

Proposition I.5 (Exact scalar and vector sufficient widths)**.** *At the separate stages* (97)*, the accepted NORM and VEC enclosures are*

	N_0 *strictly below*	K_4 *strictly below*	N *sufficient*	*target upper*
NORM, scalar	$2^{274878071042}$	$2^{549756142097}$	$2^{549756142138}$	65/4096
VEC, $1 \le r \le 4$	$2^{274878071042}$	$2^{549756142100}$	$2^{549756142144}$	129/8192

(106)

Both target uppers are $< 1/5$ for both architectures. The vector result uses $n = 75$ and all its ambient factors; it does not transplant the scalar $n = 74$ constant. At the identical vector outer stages the old SELECT width $2^{2^{68719542354}}$ is sufficient with the same upper $129/8192$. The comparison is between specified sufficient integers; it states neither a necessary width nor an execution-time improvement.

Proof. Put $p = 16384$, $t = 2^{36}$, $l = t + 4p + 78$, $x_* = t/2 + p + 21$, $y_* = t/2 + 2p + 61$, $k_* = 2t + 5p + 127$. The variance bound in (63) gives $c \le \bar{c} < (p+3)/6 \le p$ from $\pi > 3$, $\operatorname{asinh}(2^p) < \log(3\,2^p) < p + 2$. In either case $R_{\mathrm{aux}} < 2^{38}$, $O < 2^{78}$, $6up < 1$, $B \le 2^{2p}$ and $\alpha = \varepsilon$, $U = u$. Consequently

$$L < 2^l, \quad J_z < 2^{79} \le 2^l, \quad H + 10, H + 4 < 2^{l+2}, \quad X < 2^{x_*}, \quad Y < 2^{y_*}, \quad K_1 < 2^{k_*-2}. \tag{107}$$

The strong last inequality gives $16K_1^2 < 2^{2k_*}$, so its ceiling is at most that integer; $N_0 \le 2^{2k_*} < 2^{2k_*+4} = 2^{2748780710 42}$. This explicitly retains the ceiling step.

Set $W_0 = 2^{3x_*+5}$. All four $v_k(X) < W_0$: $v_1(X) < 2^{3x_*+1}$, the terms of v_2 have upper powers $4 + 2x_*$ and $2p + 40$, and v_3, v_4 have upper powers $5 + x_*$ and 5. The Bell bounds are $1, W_0, 2W_0^2, 5W_0^3, 15W_0^4$. Thus $P_F(X) \le 16W_0^4[1 + (1+F)^4]$. For scalar $F \le 3$ its coefficient is $4112 < 2^{13}$; for vector $F \le 6$ it is $38432 < 2^{16}$. Multiplying by $Y^4/6$ and relaxing the divisor upward gives respectively

$$K_4 < 2^{12x_*+4y_*+33} = 2^{549756142097}, \qquad K_4(F) < 2^{12x_*+4y_*+36} = 2^{549756142100}. \tag{108}$$

Scalar smoothing at $\delta = 2^{-10}$ with its width in (106) costs $< 3\,2^{-11} + 2^{-9} = 7/2048 < 1/256$. Vector smoothing at $\delta = 2^{-11}$ costs $< 3\,2^{-11} + 4\,2^{-11} = 7/2048 < 1/256$. The width threshold is satisfied in both cases.

Here is the outer-arrow check, retaining its actual volume and vector factors. At $n = 75$, $A = 22500000000$, $uA = 9/4$; the scalar $n = 74$ values are smaller. In (65), $K_\Lambda^2 < 14u^2A < 1$, $K_\sharp < 32$ and $\beta > 1/12$. Since $e^6 > 67591/175 > 384$, $H_2 < 2048$ and $\sqrt{\bar{\Xi}_2} < 2^{4097}$. The vector regulator term is therefore $< 2^{4097}(4\,2^{-8192} + 2\,2^{-16384}) \le 1/256$; the scalar norm is smaller. With $j_0 = 2^{34}$, $J_0 = 9uA\bar{c}^2 < (81/4)2^{28}$ and $e^{J_0} < 2^{j_0}$, using $\log 2 > 1/2$. For auxiliary radius $s = 2^{37}$ the strict geometric margin is $R_{\mathrm{aux}} - 1000n - s = 1$ for both compacts. Equation (64) gives $\sqrt{\rho_a(s)} < 2^{p-s/2}$. As $L_{\bar{c}} < 32p^2$ and vector $F_1 \le 8$, the full bracket is $< 256p^2$. Thus

$$A_{\mathrm{aux}} < 2^{-51539591132} \le 1/256, \qquad S < 2^{-51539607443} \le 1/256. \tag{109}$$

The latter retains $B_{\mathrm{ext}} < O < 2^{78}$ and the full common Wick exponential. For the spatial tail, the degree-eight exponential partial sum at $125/32$ is $432528188408407789/8866461766385664 > 32$; hence $q < 1/32$. For vector $R \le 9$, $e^{R/256} \le 256/247 < 2$, and the $n = 75$ bracket at $q = 1/32$ is $149856/961 < 256$. Therefore $E_{75} < 2^{364-10+8-375} = 2^{-13}$. For scalar $C_f \le 2^{360}$, $R = 1$, the $n = 74$ bracket is < 256, giving $E_{74} < 2^{-12}$. Adding the four inner $1/256$ allocations gives the two exact totals in (106).

For comparison, the old vector polynomial has coefficient sum $P(1) \le 38432V^4$ at $V = 2^{2p+41}$, degree at most twelve, and $b_k \le 2^{3t+12}$. Its full coefficient obeys $C_{\mathrm{law}} < 2^{2^{l+2}+q_0+78}$ with $q_0 = 4(2p+41) + 3t + 28 = 206158561472$. The full exponential normalizer is retained in this inequality. The ambient prefactor in D_{law}^4 is $< 2^{11}$, so $N \ge 2^{43}C_{\mathrm{law}}$ suffices for width error $\le 1/256$. Since $q_0 + 78 + 43 < 2^{38} < 2^{l+2}$, the SELECT width with outer exponent $l + 4 = 68719542354$ satisfies that bound. The new vector exponent lies strictly between 2^{39} and 2^{40} and is smaller than $2^{68719542354}$. This proves the exact sufficient-width comparison. □

I.5 Supported accuracy search with the old vector backend

ACC selects stages for the typed family in (91), using the *old full-normalizer vector width coefficient* (60)–(61). Its supported tolerance strings are exactly

$$\{\texttt{1/5}\} \ \cup \ \{\texttt{2\^{}-q} : 3 \le q \le 12\}. \tag{110}$$

The sharper VEC backend above is a separate fixed-stage, fixed-1/5 result. This section does not apply it to ACC's changing stages.

For a positive rational x, write $\operatorname{clog}_2(x)$ for the least nonnegative integer h with $x \le 2^h$. The following gives the complete deterministic selection rule; all displayed operations are integer or rational operations. Let

$$\begin{aligned} d &= \min\{i \ge 0 : 5\,2^{-i} \le \tau\}, \qquad \delta = 2^{-d}, \\ \bar{E}(n) &= \frac{4}{10000}\bar{C}_f \frac{256}{256-R}\,2^{-5n}\left[\frac{(2n+1)32}{31} + \frac{64}{961}\right], \\ n &= \min\{1 \le i \le 80 : \bar{E}(i) \le \delta\}, \quad A = 4000000n^2, \quad U_A = uA, \quad b_N^0 = \frac{3}{16U_A}, \\ L_{10}(k) &= \sum_{j=0}^{10} \frac{k^j}{j!}, \quad k = \min\{1 \le i \le 7 : L_{10}(i) > 32/b_N^0\}, \\ H_N &= 1 + \left\lceil \max(12U_A, 4, 4k^2/b_N^0) \right\rceil . \end{aligned} \tag{111}$$

Here b_N^0 is a Nelson lower bound, distinct from the width polynomial weight $b_0 = 1$. Next select

$$\begin{aligned} p &= \min\{i \ge 4 :\ i \text{ even}, \quad 2H_N + 3 - i/2 \le -d-1, \quad 2H_N + 2 - i \le -d-1\}, \\ a &= 2^{-p}, \quad j_e = \lceil 18U_A p^2 \rceil, \quad h_A = \operatorname{clog}_2(64U_A p^2 + 4c_r), \\ h_s &= \min\{i \ge 1 : j_e + h_A + p - 2^{i-1} \le -d\}, \quad s = 2^{h_s}, \\ R_{\text{aux}} &= 1000n + s + 1, \quad O = 4R_{\text{aux}}^2, \quad o = \operatorname{clog}_2(O), \\ h_S &= \operatorname{clog}_2(6Op^2), \quad t = \max(34, j_e + h_S + d), \quad \varepsilon = 2^{-t}. \end{aligned} \tag{112}$$

Finally put

$$\begin{aligned} h_R &= \operatorname{clog}_2(2R_{\text{aux}}), \quad v_e = 2p + h_R + 2, \\ q_0 &= 4v_e + 3t + 28, \quad l = t + 4p + o, \quad H_{\text{small}} = q_0 + o + 11 + 4d. \end{aligned} \tag{113}$$

Require $\text{bitlength}(H_{\text{small}}) \le l+2$. Start $w = l+2$ and increase until this guard holds and $w > l+2$; successful output has $w = l+3$. Set $N = 2^{2^w}$. The powers defining a, ε, N are inert mathematical expressions; the program does not expand their denominators or widths. These minima refer to the displayed sufficient predicates, rather than canonical CAPSEQ least indices.

Theorem I.6 (ACC termination and accuracy at its literal domain)**.** *Under the accepted SELECT/OPT/EB1/EC1/AP6 and P1–P6 premises, every typed tuple and every tolerance in* (110) *has the selection above. It satisfies $n \le 80$, $k \le 7$, $H_N < 4096$, even $p < 20000$, $h_s \le 40$, $t < 2^{40}$ and $w < 2^{40}$. All selected watched integer/rational witnesses fit the stated* 1024*-bit ceiling and fewer than* 20000 *logical guard steps suffice for the specified arithmetic/source-inventory loops. These are finite operation and named-witness assertions, rather than hardware-time or process-memory bounds. The actual adjacent-law errors satisfy*

$$E_n \le \bar{E}(n) \le \delta, \quad R_{\text{phys}}, A_{\text{aux}}, S, W \le \delta, \qquad d_{\text{BL}}(\mathcal{L}_f(P^b), \mathcal{L}_f(\nu)) \le \bar{E}(n) + 4\delta \le 5\delta \le \tau. \tag{114}$$

The same selected stages work for both architectures and all the supported repeated, overlapping, touching, separated and singular tuples.

Proof. At $\tau = 1/5$ the allocation gives $d = 5$; at $\tau = 2^{-q}$ it gives $d = q+3$, so $5 \le d \le 15$. The exponential partial sum used in the preceding proof gives $q_{\text{spatial}} < 1/32$. Also $e^{R/256} \le 256/(256-R) < 2$. Positivity in the shell formula therefore proves $E_n \le \bar{E}(n)$. At $n = 80$ its bracket is $159776/961 < 256$, giving $\bar{E}(80) < 2^{364-10+8-400} = 2^{-38}$. Thus the box search stops, with $1000n \ge R$.

Here $U_A = n^2/2500 \le 64/25 < 3$ and $K_\Lambda^2 < 14u^2A < 1$. Consequently $K_\sharp \le \max(12U_A, 4)$ and the actual Nelson $\beta = \pi^2/(48U_A) > b_N^0 > 1/16$. Since $L_{10}(7) > 512 > 32/b_N^0$, the k search succeeds. For its chosen k, $e^k > L_{10}(k) > 32/b_N^0$ bounds the logarithm in H_2 by k. Thus $H_2 < H_N \le 3137 < 4096$ and $\sqrt{\Xi_2} < 2^{2H_N+1}$. The full vector regulator expression is then $< 2^{2H_N+3-p/2} + 2^{2H_N+2-p}$. The two predicates bound each term by 2^{-d-1}. The even value $4H_N + 2d + 8$ suffices, so $p \le 12586 < 20000$. As above $c_a \le \bar{c} < p$ and $6up < 1$.

The complete Wick exponential obeys $e^{9U_A\bar{c}^2} < 2^{j_e}$, using $\log 2 > 1/2$; $L_{\bar{c}} < 32p^2$. Both required compact sets have strict auxiliary margin $R_{\mathrm{aux}} - 1000n - s = 1$, so the defect tail gives $\sqrt{\rho_a(s)} < 2^{p-s/2}$. The full auxiliary expression is therefore $< 2^{j_e+h_A+p-s/2} \le 2^{-d}$. For termination, $j_e \le 21600000000 < 2^{35}$, $64U_Ap^2 + 4c_r < 2^{37}$, and $j_e + h_A + p + d < 2^{35}$. Hence $h_s \le 36$ already suffices, and $R_{\mathrm{aux}} < 2^{37}$, $O < 2^{76}$, $o \le 76$. As $6Op^2 < 2^{109}$, $h_S \le 109$ and $t < 2^{36}$. The choice $t \ge 34$ gives $\varepsilon < u$; retaining $B_{\mathrm{ext}} < O$, the stabilizer is $< 2^{j_e+h_S-t} \le 2^{-d}$.

For the old width backend use $B_N < 2^{2p}$ and $\sqrt{O} = 2R_{\mathrm{aux}} \le 2^{h_R}$. Its v_1, v_2 coefficient sums are at most 2^{v_e}; those of v_3, v_4 are smaller. The Bell sums are bounded by $1, V, 2V^2, 5V^3, 15V^4$ for $V = 2^{v_e}$. Keeping the extra pure e_4 and all five observation terms gives $P(1) \le 38432V^4 < 2^{16}V^4$. For each degree $1 \le j \le 12$, the scalar weight $(j/(2\alpha e))^{j/4} < 2^{3t+12}$; $b_0 = 1$ obeys the same upper. Thus the full $Q_{\mathrm{poly}} < 2^{q_0}$.

The normalizer exponent has all three positive terms: $L_N < 2^l$, $3uO/16 \le 2^{l-1}$ and $3U_A\bar{c}^2 \le 2^{l-1}$. The two latter comparisons are the explicit guards $o \le l - 1$ and $\mathrm{clog}_2(3U_Ap^2) \le l - 1$. They hold since $l \ge 51$, while $3U_Ap^2 < 2^{32}$. Therefore

$$C_{\mathrm{law}} < 2^{2^{l+2}+q_0+o}, \qquad D^4_{\mathrm{law}} < 2^{11}C_{\mathrm{law}}. \tag{115}$$

Here $(2048\sqrt{6}/27)r^{3/2} < 2^{11}$ retains ambient $r \le 4$. It is enough to have $N \ge 2^{11+4d}C_{\mathrm{law}}$. The proven bounds give $0 < H_{\mathrm{small}} < 2^{38}$ and $\mathrm{bitlength}(H_{\mathrm{small}}) \le 38 \le l + 2$. Thus the width search succeeds at $w = l + 3$, since

$$\log_2(2^{11+4d}C_{\mathrm{law}}) < 2^{l+2} + H_{\mathrm{small}} < 2^{l+3} = \log_2 N. \tag{116}$$

Also $l < 2^{37}$ and $w < 2^{40}$. This proves the width allocation with the full old normalizer and polynomial. The triangle on the same five actual observation laws proves (114).

For the stated finite-resource theorem, there are at most $16, 80, 7, 6292, 36, 2$ allocation, box, Nelson, regulator, auxiliary and width predicates, respectively. Each clog_2 loop has at most 110 iterations; fixed checks add fewer than 100 guards and the fixed source/core inventory adds 203. The resulting bound is below 8000, hence below 20000. Center denominators are at most 2^{16}; $\bar{C}_f$ has numerator below 384 bits and denominator at most $5\,2^{16}$. The factors in $\bar{E}(n)$ have denominator bit bounds $19, 25, 10, 14, 401$, with the last from 2^{5n}, so even their unreduced product is below 469 bits and its numerator below 450 bits. Adding 4δ keeps the final components below 512 bits. The fixed Nelson partial sums use fewer than 128 bits, the other named selected quantities fewer than 256, and the largest guarded logarithm shift at most 110 bits. The coefficient power itself uses 365 bits. These prove the named 1024-bit sufficiency, with no assertion about every uninstrumented intermediate or the work of hashing and I/O. □

For the four-corner request $(-8,-8), (-8,8), (8,-8), (8,8)$ and $\tau = 2^{-12}$, the accepted selected arithmetic records

$$\begin{aligned} &n = 75, \quad p = 6954, \quad s = 4294967296, \quad t = 1958503808, \quad w = 1958531694, \\ &\bar{E}(n) + 4\delta = \frac{919201379}{6076595200000} < \frac{1}{4096}. \end{aligned} \tag{117}$$

This is the historical ACC tuple, with $N = 2^{2^{1958531694}}$. Its old backend remains unchanged. The later same-stage sharper width is in (144); VEC retains its separate fixed-1/5 stages.

I.6 Actual same-law linkage and what selection computes

Write $C^b_n = P^b_n$ for the canonical law of Theorem I.2. For an independently selected admissible inner tuple $\vartheta = (a, R_{\mathrm{aux}}, \varepsilon, N)$ at the same physical box, let $F^c_n(\vartheta)$ be its neural law, and set $I_n(\vartheta) = W + S + A_{\mathrm{aux}} + R_{\mathrm{phys}}$. The architecture choices b, c are independent. The physical label n specifies Λ_n; it is distinct from a width and from the four enumeration indices in the least-index recursion.

Theorem I.7 (LINK metric, same-box comparison and physical-label modulus)**.** *For a real compact smooth tuple f choose J so all supports are compactly contained in $(-J, J)^2$, and define*

$$A_{f,J} = \left(\sum_i \|f_i\|^2_{H^2(\mathbb{T}_J)} \right)^{1/2}, \qquad L_{f,J} = 2^J \max(2, A_{f,J}). \tag{118}$$

For all Borel probabilities on X,

$$d_{\mathrm{BL}}(\mathcal{L}_f(\rho),\mathcal{L}_f(\eta)) \le \min(2, L_{f,J}\beta_X(\rho,\eta)) \le \min(2, L_{f,J}\,\mathrm{dist}_n(\rho,\eta)) \tag{119}$$

when the latter is defined. The first inequality needs no moment assumption or $J \le n$. Empty/all-zero tuples have distance zero. There is no reverse bound tending to zero with this one finite-observation discrepancy for either field metric, even among moment-finite smooth-field point masses.

Under the literal CAPSEQ and strict AP6 premises, for $n \ge n_0$ and $n\ell \ge R$, all target symbols below denote the identical original FS state version ν:

$$\begin{aligned} d_{\mathrm{BL}}(\mathcal{L}_f(C_n^b),\mathcal{L}_f(\nu)) &\le \min(2, L_{f,J}2^{-n} + E_n),\\ d_{\mathrm{BL}}(\mathcal{L}_f(F_n^c(\vartheta)),\mathcal{L}_f(\nu)) &\le \min(2, I_n(\vartheta) + E_n),\\ d_{\mathrm{BL}}(\mathcal{L}_f(C_n^b),\mathcal{L}_f(F_n^c(\vartheta))) &\le \min(2, L_{f,J}2^{-n} + I_n(\vartheta)). \end{aligned} \tag{120}$$

Under the full REC2 interpretation premises this is also its Euclidean law. At equality in the qualitative cap, CAPSEQ and REC2.a–f remain available but this AP6/REC.g quantitative use is refused.

With the additional certified computable positive m, u, ℓ, an admissible supplied n_0, a computable support enclosure R, finite computable uppers $\bar{C}_f \ge C_f$ and $\bar{A} \ge A_{f,J}$, and positive rational τ, one can compute a sufficient physical label n_τ so that every $n \ge n_\tau$ has

$$\bar{L}2^{-n} + 4\bar{C}_f u\ell^2 e^{\kappa R} q^n \left[\frac{2n+1}{1-q} + \frac{2q}{(1-q)^2}\right] < \tau, \quad \bar{L} = 2^J \max(2,\bar{A}), \quad q = e^{-m\ell/256}. \tag{121}$$

This output is a sufficient box label, without a claim of leastness; it does not compute $a_n, R_n, \varepsilon_n, N_n$ or their least indices.

Proof. Duality in the original torus convention gives $|O_f(v)-O_f(w)|_2 \le A_{f,J}p_J(v-w)$. For a unit BL function the difference after composition is at most $\min(2, A_{f,J}p_J(v-w))$, which is at most $\max(2, A_{f,J})\min(1, p_J(v-w)) \le L_{f,J}d(v,w)$. Thus $(G\circ O_f)/L_{f,J}$ is in the unit BL class on X. The definition of β_X and probability diameter two prove (119); nonnegative moment discrepancies give its last inequality for every n.

For the exact reverse obstruction choose $r+1$ nonzero smooth bumps with disjoint compact supports inside $(-1,1)^2$. The linear map from their coefficients to their r integrals against the tests has a nonzero kernel. Its combination h is nonzero, smooth compact, $O_f(h) = 0$ and $p_1(h) > 0$. For $\rho = \delta_0$, $\eta_M = \delta_{Mh}$ all local moments are finite and the observed laws coincide, while

$$\beta_X(\delta_0,\delta_{Mh}) = d(0,Mh) > 0, \qquad \mathrm{dist}_n(\delta_0,\delta_{Mh}) \ge Mp_1(h) \quad (n \ge 1). \tag{122}$$

The equality uses the admissible test $v \mapsto d(v,0) \in [0,1]$. These point masses demonstrate a general metric non-implication; they are not asserted to be physical-model laws or to prove undecidability.

AP3–AP5 identify the physical/classical finite joint laws and their target with those in CAPSEQ. Compact evaluations determine Borel probabilities on X, so the AP6 target is the same law. CAPSEQ gives $\beta_X(C_n^b,\nu_n) < 2^{-n}$; apply the pushforward bound and the AP6 spatial arrow. For the selected tuple, sum its four inner arrows to ν_n, then its last arrow to ν. For the last inequality of (120), insert $\mathcal{L}_f(\nu_n)$ directly between the two neural laws. Its spatial-to-target error is absent because this path ends at the same finite physical law. No upper bounds have been subtracted. The old g_* is neither estimated nor used to enlarge an old globally qualified theorem.

For the computable modulus choose certified rational $q \le \bar{q} < 1$, $L_0 \ge \bar{L}$, $B_0 \ge 4\bar{C}_f u\ell^2 e^{\kappa R}$. Strict positive bounds for ℓ and an upper for R give a safe integer $n_{\mathrm{start}} \ge n_0$ with $n_{\mathrm{start}}\ell \ge R$. The geometric identity $T_n(z) = \sum_{j=n}^{\infty}(2j+1)z^j$ shows it is increasing in z, decreasing in n and tends to zero for $0 < z < 1$. Search exact rational $L_0\,2^{-n} + B_0T_n(\bar{q}) < \tau$ from this starting label. It terminates, and monotonicity proves the assertion for every subsequent label. No equality of computable reals has to be decided. The canonical law at that label remains the existential least-index law; the label calculation does not evaluate its inner parameters or field discrepancy. □

Corollary I.8 (Literal bump at physical label 74)**.** *For the unshifted scalar bump at the fixed physical parameters, $J = 2$ is admissible and $A_{f,2} \le 374$, $L_{f,2} \le 1496$. Write P^c_{OPT} and P^c_{RB4} for the two separate accepted scalar tuples at Λ_{74}, with their original and relative widths. For all independent architecture choices b, c,*

$$\begin{aligned} d_{\mathrm{BL}}(\mathcal{L}_f(C^b_{74}), \mathcal{L}_f(\nu)) &\le 1/4096 + 1496\, 2^{-74} < 1/2048, \\ d_{\mathrm{BL}}(\mathcal{L}_f(P^c_{\mathrm{OPT/RB4}}), \mathcal{L}_f(\nu)) &\le 65/4096 < 1/5, \\ d_{\mathrm{BL}}(\mathcal{L}_f(C^b_{74}), \mathcal{L}_f(P^c_{\mathrm{OPT/RB4}})) &\le 1/64 + 1496\, 2^{-74} < 1/32, \\ d_{\mathrm{BL}}(\mathcal{L}_f(P^b_{\mathrm{OPT}}), \mathcal{L}_f(P^c_{\mathrm{RB4}})) &\le 1/128. \end{aligned} \tag{123}$$

No concrete inner tuple here is claimed to equal the canonical record.

Proof. For $r_t = (1 - t^2)^{-1}$ direct differentiation gives $b'' = (4t^2 r_t^4 - 2r_t^2 - 8t^2 r_t^3)e^{-r_t}$. The maxima of $r^k e^{-r}$ give $\|b''\|_\infty \le 8/e^2 + 216/e^3 + 1024/e^4 < 93$. On the area-four support, $\|f\|_2 \le 2$ and each pure second derivative has L^2 norm at most 186. The exact Fourier convention yields $\|f\|_{H^2(\mathbb{T}_2)} = \|(1 - \Delta)f\|_2 \le 374$, and hence $L_{f,2} \le 4\,374 = 1496$. The supports are compactly contained in $(-2, 2)^2$, as required. The scalar tuples share all four outer parameters. Each first-arrow error is at most $1/256$, with RB4 satisfying its $N \ge N_0$ threshold by (106); the other three inner errors are at most $1/256$ each, and $E_{74} \le 1/4096$. Apply (120). Since $1496 < 2^{11}$, its small term is $< 2^{-63}$, proving both strict bounds. For the last comparison insert the identical actual Q_ε between the two laws and add just their two first-arrow errors. This remains valid for different architectures. □

I.7 Comparison, software domains and proof navigation

Table 7 keeps each certificate at its literal domain. A smaller upper bound is a statement about these sufficient rules. All widths remain astronomical process-neuron counts with countably many internal spatial variables.

Table 7: Exact sufficient-bound comparisons at $m = 1$, $u = 10^{-10}$, $\ell = 1000$. The NS1 and OPT rows retain their baseline scope. ACC's example is one supported request.

Result	Observations / accuracy	Outer stages	Sufficient width	Target upper
NS1	scalar base / 1/5	$n = 512$, $p = 2^{24}$	$2^{2^{4611686018494496904}}$	$1/64 + 2^{-736}$
OPT	scalar base / 1/5	$n = 74$, $p = 16384$	$2^{2^{68719542354}}$	65/4096
NORM	scalar base / 1/5	same OPT outer tuple	$2^{549756142138}$	65/4096
SELECT	typed 1–4 / 1/5	$n = 75$, $p = 16384$	$2^{2^{68719542354}}$	129/8192
VEC	same typed family / 1/5	same SELECT outer tuple	$2^{549756142144}$	129/8192
ACC	four corners / 2^{-12}	$n = 75$, $p = 6954$	$2^{2^{1958531694}}$	$\frac{919201379}{6076595200000}$

Supported and refused software domains. The NORM executable is a fixed scalar-case verifier. SELECT and the separately versioned VEC executable support only the canonical ordered dyadic bump family at exactly 1/5; ACC supports that family and exactly the eleven tolerance strings in (110). Their strict schemas refuse altered model data, custom/weighted/signed tests, noncanonical dyadics, unsupported coordinates/counts or tolerances, floating/nonfinite values, duplicate keys and out-of-contract expression trees. Geometry, multiplicity, coefficient enclosures and stages are derived rather than accepted from caller assertions. Required source bytes and current decisions are hash-bound; a relocated root must supply those identical bytes. Missing source, changed source, failed witnesses and resource exhaustion supply no successful

numerical certificate. This is an account of the accepted software contract, not a claim that the integration author reran their original full source trees.

VEC binds 174 scientific source files; ACC binds 195. Those original source inventories and their recorded selected arithmetic/replays remain in the frozen companion packets. The present author verified all 189 assigned input files, including every one of the seven statement/proof/ decision triples. Source inventories inside a packet are evidence of its original run, rather than a claim that all their historical paths are present in this smaller integration packet. Complete unavailable primary classical bodies remain attributed premises.

The independently implemented ACC arithmetic verifier does not call its stage search; it shares the declared strict parser/source/guard utilities. The accepted repaired implementation checks the entire success envelope and refuses false candidate/availability labels. A well-typed historical timing record is metadata, without authenticated historical measurement. Its cooperative core timing excludes request parsing, includes verification-certificate parsing, and excludes serialization and the whole process. Named watched values and logical guards provide no all-intermediate, memory, hostile-filesystem-atomicity or hard latency guarantee. Default ACC ceilings are 1024 watched bits, 30000 core milliseconds, 20000 guards, $n \le 80$, $p \le 20000$, $h_s \le 40$, and $t, w \le 2^{40}$. Stricter ceilings may stop valid requests. Its six outcomes distinguish supported success, unsupported input, numerical source gap, source-binding refusal, resource exhaustion and refused certificate; every non-success has no certificate. VEC has corresponding invalid-input, source-mismatch, missing-source, inequality-failure and resource-limit refusals. Reviewed code, manifests and external freezes remain trust roots, rather than a formal proof kernel.

The accepted ACC decision records 110 actual search/verify pairs, maximum 4081 search guards, 307 verifier guards and 393 watched bits. These are observed finite-run facts, distinct from the theorem's supported-domain termination and limits. The sharper vector and variable accuracy backends have not been silently combined.

Retained adverse evidence and boundaries. The original MOM actual-match overall MISMATCH is retained: its mathematics matched, while two external-openness metadata flags were separately repaired and fresh MOMMETAFIX resolved that defect. The old SELECTNUM disagreement record, 314/315 checks, and its binding ancillary manifest refusal-label correction remain. ACC's original hostile GAP, float serialization and outer-envelope gaps remain with their fixes; the later three-file timing repair changed the contract/code wording while its proof, statement, schema, tests and 195 source records stayed unchanged. Earlier rejected REC-v1, witnesses and CLOSED gate, partial/incomplete versions, source-access exceptions, the rational resource-inconclusive route, and all old author/reviewer scope distinctions remain historical evidence. They are neither erased nor substituted for the current decisions.

The Gaussian four-corner obstruction is still useful: the product neuron at $(0,0),(a,0),(0,a),(a,a)$ has fourth product $1/16$, while its Gaussian covariance-pairing sum is $(2e^{-2}+e^{-4})/16 < 1/16$. The excess persists as $1/N$ for the iid normalized neuron sum. Thus Gaussian joint-moment domination of finite neural laws cannot replace the joint coercivity/density argument above.

The established neural topology is weak X convergence and compact smooth joint-correlation convergence in $\sigma(\mathcal{D}', \mathcal{D})$. Target Schwartz regularity does not supply global neural Schwartz or strong-dual control. The effective-selection obstruction consists of the missing all-field/moment discrepancy predicates and leastness decisions: one finite observation law cannot determine them by (122). This is a limitation of the proved implications, not a model-specific impossibility theorem. Full AHKZ-class uniqueness, the old boundary estimate, practical/necessary/minimal complexity, sampling/training, other models/dimensions, historical priority and public research release remain separate. Finite-stage RP is exactly across the two central coordinate lines in the centered axis-parallel geometry. The target's full symmetry, arbitrary-line half-plane positivity and quantum reconstruction retain their separate classical premises and attribution.

How to audit this revision. The public proof map gives this edition's source block locations and hashes. The original private audit retains the exact frozen statement/proof line ranges, packet hashes and baseline reuse. The full new proofs are in this appendix and the earlier paper sections it explicitly invokes; the private source packets give additional original details and provenance. Mathematical reasoning is not

replaced by an operational ledger. The baseline bibliography is unchanged. All 277 original mathematical environments, including nested displays, are byte-identical, and a reversible source delta recovers the exact baseline after removing the two added text regions and restoring its date. The preservation audit, compilation/reference checks and all-page render review are separate local evidence. The current integration's fresh source/quantifier/editorial review, actual-proof match and coordinator decision remain required.

J Sharper relative accuracy at the selected stages

The accepted SVA-ACC-RELATIVE-v1 mathematics carries the relative estimate through all supported accuracy-selected stages. The separately accepted SVC-relative-accuracy-v1 implementation realizes the symbolic rule. This appendix states their precise content and mathematical reasoning; it preserves the earlier results as dated checkpoints. Its mathematical implications are conditional on the literal source and finite-law interfaces below. The integrated presentation awaits its own fresh review.

J.1 Domain, notation and the actual unchanged laws

Fix $m = 1$, $u = 10^{-10}$, $\ell = 1000$, $K_{\mathrm{cov}} = 1$ and the original coefficient-one, same-mass, own-variance Wick quartic, with no factorial. The field state space is the original complete $X = H^{-2}_{\mathrm{loc}}$ and $\nu^X = \nu$ is the same Fröhlich–Simon target of Section I.1. All P1–P6/AP, Gaussian/Wick, DSS/BD/BBD/Griffiths–Simon/SK-q/P*, H1/MIX/REG/smoothed-CT/FLOW/TP/T, source/classical/coefficient and NR1 qualifications there retain their original scopes. Numerical target identification uses strict AP6, $0 < u/m^2 < 1/8000$. The original fixed g_* and $10^{-10} < g_*$ remain numerically unknown. The sharper NR1 author result and larger isolated blind-proof core remain distinct; excluded NR2 E4/r0 is not used.

The bump and ordered observations are exactly (91): $b(x) = \exp[-1/(1-x^2)]$ for $|x| < 1$, zero otherwise, and $f_h(z) = b(z_1 - h_1)b(z_2 - h_2)$. There are $r \in \{1, 2, 3, 4\}$ centers. Each coordinate is a canonical $P/2^k \in [-8, 8]$, $0 \le k \le 16$, with $k = 0$ for integers and zero, and odd numerator when $k > 0$. Decimal numerators have no plus, leading zero or negative zero. The tolerance strings are exactly (110). Additional observation amplitudes or signed coefficients, dilations and rotations are absent. Order and multiplicity are retained. Put

$$\begin{aligned} K &= \bigcup_i [h_{i1} - 1, h_{i1} + 1] \times [h_{i2} - 1, h_{i2} + 1], & R &= 1 + \max_{i,j} |h_{ij}|, \quad c_r = \lceil \sqrt{r} \rceil, \\ F_q &= \left(\sum_i \|f_{h_i}\|_q^2 \right)^{1/2}, & F &= F_{4/3}, \quad \bar{F} = 3c_r, \\ \bar{C}_f &= 2^{360} c_r (4R + 1)/5, & \eta &= 2^{-d}. \end{aligned} \tag{124}$$

The full ambient observation is $O_f(v) = (v(f_{h_i}))_{i=1}^r$. The metric is the supremum over real G with $\|G\|_\infty \le 1$ and Euclidean Lipschitz constant at most one separately on $\mathbb{R}^r$. The class $\mathcal{U}_4(r)$ uses full Euclidean multilinear operator norms through order four, all at most one, including absolute value at order zero. Hölder gives $\|O_f\| \le F$.

Use exactly the outer minima (111)–(112), with their δ renamed η here. Thus H_N is H_{Nel}, j_e is j_{exp}, h_A is h_{aux} and h_S is h_{stab}. All of $n, p, a, s, R_{\mathrm{aux}}, O, o, t, \varepsilon$ keep their original meanings. Define $\Lambda = (-1000n, 1000n)^2$, $D = \Omega = [-R_{\mathrm{aux}}, R_{\mathrm{aux}}]^2$, $A = 4000000n^2$ and $E = L^4(D; \mathbb{R})$. Here $O = |D|$, not the observation operator. The accepted SVA proof, applied to this unchanged ACC selection, gives the following whole-domain bounds; their derivation is recalled in the proof of Theorem J.2 below:

$$\begin{aligned} 5 \le d \le 15, \quad & 1 \le n \le 80, \quad & U_A = uA \le 64/25 < 3, \\ k \le 7, \quad & H_N \le 3137, \quad & 4 \le p \le 12586 < 20000 \quad (p \text{ even}), \\ h_s \le 36, \quad & R_{\mathrm{aux}} < 2^{37}, \quad & O < 2^{76}, \quad o \le 76, \\ h_{\mathrm{stab}} \le 109, \quad & 34 \le t < 2^{36}, \quad & \varepsilon < u, \quad 6up < 1. \end{aligned} \tag{125}$$

Also $1 \le R \le 9$, $K \subset (-1000, 1000)^2$ openly, $F_1 \le 4c_r \le 8$, $F_2 \le 2c_r \le 4$, $F \le \bar{F} \le 6$. With identical chosen H1 and H2 coefficients, the literal CL1 formula (56) gives $C_f(\mathrm{tuple}; R) = \sqrt{r}(4R+1)C_f(\mathrm{base}; 1)/5 \le \bar{C}_f < 2^{364}$; its separate factor $\exp(R/256)$ remains separate. The source flow variable in that formula is distinct from the auxiliary radius s.

Full selected-stage reference premise. At every selected (a, D) assume the original EB1/NR1 finite-reference interface in its complete form. The physical multiplier and variance are

$$\frac{1}{1+w^2+q^2+a^2w^2q^2}, \qquad c = c_a = \frac{1}{2\pi}\int_0^\infty \frac{dq}{\sqrt{(1+q^2)(1+a^2q^2)}}, \qquad \kappa_a^2 = a^{-2}-1, \tag{126}$$

with Fourier measure $dw\,dq/(2\pi)^2$ and $c \le \bar{c} = [\operatorname{asinh}(1/a)+1]/(2\pi) < p$. The endpoint γ is a centered, strongly measurable Gaussian E-valued random element with zero Bochner mean, global sign symmetry and covariance $e^{-|z-z'|_1/a}/4$. For each original architecture, $g^b|_D$ is strongly measurable, Bochner centered, globally sign symmetric, with $|g^b(z)| = 1/2$ a.e. almost surely and that covariance. The product neuron uses two independent stationary symmetric telegraph processes of rate $1/(2a)$ and their product divided by two. The Mondrian neuron uses the projectively consistent whole-plane axis-parallel partition at budget $1/a$, independent uniform cell signs, half-valued cells and the original lower-left half-open representative. A rectangle of sides w, h splits at rate $w+h$, vertically with probability $w/(w+h)$, at a uniform position, and daughters evolve independently for the remaining budget. Original existence, consistency and measurability are premises; boundary changes agree as E elements. For *every* bounded bilinear $B_0 : E \times E \to \mathbb{R}$ assume

$$EB_0(g^b, g^b) = EB_0(\gamma, \gamma). \tag{127}$$

Independent copies exist on product spaces. This full assumption is separate from point covariance. It is assumed at selected stages rather than inferred by applying the fixed-$n = 75$ VEC theorem elsewhere.

The actual potential and full normalized laws are

$$V(v) = u\int_\Lambda (v^4 - 6cv^2 + 3c^2) + \varepsilon\int_{\Omega\setminus\Lambda} v^4 - \frac{\kappa_a^2}{2}\int_\Omega v^2, \quad w(v) = e^{-V(v)},$$
$$F_N^b = N^{-1/2}\sum_{i=1}^N g_i^b, \quad P_N^b = \frac{w\ \mathrm{Law}(F_N^b)}{Ew(F_N^b)}, \qquad Q_\varepsilon = \frac{w\ \mathrm{Law}(\gamma)}{Ew(\gamma)}. \tag{128}$$

These are restrictions of the original whole-plane laws, not a new reference model. The actual V always retains c_a. All downstream laws and their common Gaussian/Wick couplings in (58)–(59) remain literal premises: $Q_\varepsilon \to Q_0 \to Q_a \to Q_{\mathrm{phys}} \to \nu^X$ has error bounds $S, A_{\mathrm{aux}}, R_{\mathrm{phys}}, E_n$. The physical Wick normalizer $Z_a = Ee^{-U_{a,\Lambda}} \ge 1$ uses $EU_{a,\Lambda} = 0$ on its physical Gaussian space. It is separate from each full normalizer in (128). AP5/AP6 identifies the same physical-to-target limit. Neither source hashes nor this restatement reprove these inputs.

Literal ambient L0 uses a standard Gaussian ξ on the original $\mathbb{R}^r$: $G_\zeta(y) = EG(y-\zeta\xi)$ has approximation error $\sqrt{r}\zeta$, absolute value and first derivative bounded by one, and $\|D^qG_\zeta\|_{\mathrm{op}} \le \sqrt{(q-1)!}\zeta^{1-q}$ for $q \ge 2$. For $0 < \zeta \le 1$, all four norms are bounded by $\sqrt{6}\zeta^{-3}$. Singular covariances retain this ambient dimension.

J.2 Relative comparison under that interface

For any $c_* \ge c$ used only in positive majorants define

$$\alpha = \varepsilon, \quad U = u, \quad B = \kappa_a^2/2 + 6uc_*, \qquad L = \frac{B^2O}{2\alpha}, \quad J_z = \frac{3UO}{16} + 3uc_*^2A, \quad H = L + J_z,$$
$$v_1(x) = 4Ux^3 + 2B\sqrt{O}x, \qquad v_2(x) = 12Ux^2 + 2B\sqrt{O},$$
$$v_3(x) = 24Ux, \qquad v_4(x) = 24U,$$
$$e_0 = 1, \quad e_1 = v_1, \quad e_2 = v_1^2 + v_2, \qquad e_3 = v_1^3 + 3v_1v_2 + v_3,$$
$$e_4 = v_1^4 + 6v_1^2v_2 + 3v_2^2 + 4v_1v_3 + v_4, \qquad P_F(x) = 2e_4(x) + \sum_{q=1}^4 \binom{4}{q} F^q e_{4-q}(x),$$
$$X_{\mathrm{rel}} = [2(H+4)/\alpha]^{1/4}, \qquad Y_{\mathrm{rel}} = 2O^{1/4}(H+10)^{1/2},$$
$$K_1 = v_1(X_{\mathrm{rel}})Y_{\mathrm{rel}}, \qquad N_0 = \max(1, \lceil 16K_1^2\rceil), \quad K_4(F) = P_F(X_{\mathrm{rel}})Y_{\mathrm{rel}}^4/6. \tag{129}$$

The scalar moment bounds $X_{\rm rel}, Y_{\rm rel}$ are distinct from $X = H_{\rm loc}^{-2}$ and any physical coupling fields. All polynomial coefficients are nonnegative and $\deg P_F \le 12$.

Theorem J.1 (SVA.a–c at every selected stage)**.** *Under the complete premises above, fix either architecture, integer $N \ge 1$ and $0 \le j < N$. Form S_j, v, $Z_{j,h}(\theta)$ and the actual joint law $\Pi_{j,h,\theta}$ exactly as in* (98)*, with h independent before tilt and of neuron or Gaussian law. Then the untilted point variance is at most $1/4$, its fourth moment at most $3/16$, every Z is positive finite, and, for $z \ge 0$,*

$$w(v) \le e^{L-\alpha\|v\|_4^4/2}, \quad Z \ge e^{-J_z}, \quad \frac{d\Pi}{d\,\mathrm{Law}(S_j,h)} \le e^H,$$
$$\Pi(\|v\|_4^4 > z) \le \min(1, e^{H-\alpha z/2}), \qquad \|\|v\|_4\|_{L^q(\Pi)} \le X_{\rm rel}, \quad \|\|h\|_4\|_{L^q(\Pi)} \le Y_{\rm rel} \quad (1 \le q \le 16). \tag{130}$$

Before tilt $E\exp(\|\gamma\|_4^2/(4\sqrt{O})) < 4$. Each Z and $\log Z$ extends as C^1 past $[0,1]$ with

$$(\log Z)' = -N^{-1/2} E_\Pi DV(v)[h], \qquad |(\log Z)'| \le K_1/\sqrt{N},$$
$$\frac{Z_{j,g}(\theta)}{Z_{j,g}(1)} \le e^{K_1/\sqrt{N}}, \qquad \frac{Z_{j,\gamma}(\theta)}{Z_{j,g}(1)} \le e^{2K_1/\sqrt{N}} < 2 \quad (N \ge N_0). \tag{131}$$

The zero-amplitude partitions agree. All derivatives of w through four are globally bounded, with $\|D^k V(v)\| \le v_k(\|v\|_4)$ for $1 \le k \le 4$ and $\|D^k w(v)\| \le w(v) e_k(\|v\|_4)$ for $0 \le k \le 4$.

For $G \in \mathcal{U}_4(r)$, let m_j be its actual normalized Gaussian endpoint expectation and put $\Phi_j = (G \circ O_f - m_j)w$. Then $|m_j| \le 1$, $\Phi_j \in C_b^4(E)$ and

$$\|D^4\Phi_j(v)\|_{\rm op} \le w(v) P_F(\|v\|_4), \qquad E_\Pi[P_F(\|v\|_4)\|h\|_4^4] \le P_F(X_{\rm rel}) Y_{\rm rel}^4. \tag{132}$$

For every integer $N \ge N_0$, each adjacent normalized replacement has error at most $K_4(F)/N^2$ and the actual laws satisfy

$$|P_N^b(G \circ O_f) - Q_\varepsilon(G \circ O_f)| \le \min(2, K_4(F)/N),$$
$$d_{\rm BL}(\mathcal{L}_f(P_N^b), \mathcal{L}_f(Q_\varepsilon)) \le \min\left(2, \frac{\sqrt{6}K_4(F)}{N\zeta^3} + 2\sqrt{r}\zeta\right) \quad (0 < \zeta \le 1),$$
$$d_{\rm BL}(\mathcal{L}_f(P_N^b), \mathcal{L}_f(Q_\varepsilon)) \le \min(2, D_r N^{-1/4}), \qquad D_r^4 = \frac{2048\sqrt{6}}{27} r^{3/2} K_4(F). \tag{133}$$

All supported repetitions, singularities and support geometries retain order, multiplicity and ambient dimension.

Proof. We carry the argument of Section I.4 to these stages using the explicitly assumed full interface, not its fixed-stage theorem. At a.e. point the N independent centered summands have variance $1/4$ and fourth moments at most $3/16$. Their squared coefficient sum is $(N - 1 + \theta^2)/N \le 1$. Expansion of the fourth power yields $E(\sum \beta_i \xi_i)^4 \le (3/16)(\sum \beta_i^2)^2$; only single-index and two-pair terms survive. Tonelli gives integrability on finite D. Hölder and completion of the square give $V(v) \ge \alpha x^4 - B\sqrt{O}x^2 \ge \alpha x^4/2 - L$, $x = \|v\|_4$. Dropping negative quadratic terms only in an upper bound gives $EV \le J_z$; Jensen proves $Z \ge e^{-J_z}$. The resulting actual joint density is bounded by $e^{H-\alpha x^4/2}$, proving its tail.

For a nonnegative T with tail at most $\min(1, e^{a-z})$, $a \ge 0$, tail integration bounds its positive integer moments by those of $a + E_1$ for a unit exponential scalar; Minkowski gives $\|T\|_k \le a + (k!)^{1/k}$. At $T = \alpha x^4/2$, $k = 4$ this gives $\|x\|_{16} \le X_{\rm rel}$. For Gaussian directions, the spatial/probability Minkowski inequality gives $E\|\gamma\|_4^{2k} \le O^{k/2}(k/2)^k$ for $k \ge 2$; at $k = 1$, $E\|\gamma\|_4^2 \le \sqrt{3O}/4 \le \sqrt{O}/2$. The positive series and $k! \ge (k/e)^k$ give $E\exp(\|\gamma\|_4^2/(4\sqrt{O})) \le \sum_{k\ge 0}(e/8)^k < 4$. The actual joint density cap therefore bounds the tilted tail of $T_\gamma = \|\gamma\|_4^2/(4\sqrt{O})$ by $\min(1, e^{H+2-z})$. At $k = 8$, $(8!)^{1/8} < 8$ gives $\|T_\gamma\|_8 < H + 10$, hence the $Y_{\rm rel}$ bound. For a neuron, $\|g^b\|_4 = O^{1/4}/2$ deterministically, even after tilt. Lyapunov gives every real order through sixteen. No post-tilt independence is used.

Quartic Hölder gives the full Fréchet v_k bounds. The exponential partition rule gives the Bell polynomials in (129), including coefficients $1, 6, 3, 4, 1$ at order four. A finite polynomial times $e^{L-\alpha x^4/2}$ is bounded

globally; for $k > 0$ its monomial maximum is $(k/(2\alpha e))^{k/4}$. Thus w has bounded derivatives through four. Differentiating Z is dominated by a constant times $\|h\|_4/\sqrt{N}$, integrable before tilt, on a real neighborhood of the amplitude interval. Hölder under the same actual Π gives $E_\Pi|DV(v)[h]| \le (4UX_{\rm rel}^3 + 2B\sqrt{O}X_{\rm rel})Y_{\rm rel} = K_1$. Integrating the logarithmic derivative along one branch, or along both through their common zero-amplitude partition, proves the two ratios. For $N \ge N_0$, $2K_1/\sqrt{N} \le 1/2$, so the second ratio is at most $\sqrt{e} < 2$.

The fourth product rule uses $|G - m_j| \le 2$ and $\|D^q(G \circ O_f)\| \le F^q$, giving exactly P_F with its second pure e_4 copy and all observation binomials. Its nonnegative monomial x^k has $k \le 12$. Hölder at total order $k + 4 \le 16$ gives $E_\Pi x^k \|h\|_4^4 \le X_{\rm rel}^k Y_{\rm rel}^4$; at $k = 0$ use the fourth direction moment directly. This proves the mixed-moment bound without multiplying dependent marginal expectations.

Taylor through order three has the remainder (105). Conditioning before tilt on S_j, zero Bochner means cancel first terms, global sign symmetry cancels third terms, and the full bounded-bilinear premise cancels second terms for $D^2\Phi_j(S_j)$. Strong measurability, separability and bounded derivatives justify evaluations, expectations and interchanges. Centering at m_j makes the difference divided by $Z_{j,g}(1)$ exactly the normalized endpoint difference. Each of its two remainders has partition ratio at most two, and the integral weight is $1/24$. Their sum is at most $4P_F(X_{\rm rel})Y_{\rm rel}^4/(24N^2) = K_4(F)/N^2$. The N actual normalized mixtures telescope. The final normalized sum of independent Gaussian E-elements has exactly the original Gaussian law, since continuous-linear-functional laws determine Borel laws on separable E. The endpoints are the actual P_N^b, Q_ε.

Ambient L0 contributes two errors $\sqrt{r}\zeta$ and rescales its smooth test by $\sqrt{6}\zeta^{-3}$, proving (133). For $A_0 = \sqrt{6}K_4$, $B_0 = 2\sqrt{r}$, the interior minimizer is $(3A_0/(B_0N))^{1/4}$. Its value has fourth-power coefficient $(2048\sqrt{6}/27)r^{3/2}K_4$. When this minimizer exceeds one, that value exceeds $(4/3)B_0 \ge 8/3 > 2$, and the probability cap suffices. Both cases retain $N \ge N_0$; no covariance inverse or rank reduction occurs. □

J.3 Complete logarithmic enclosure and same-stage width

Theorem J.2 (SVA.d–f: all supported requests)**.** *At each unchanged selected stage, set $c_* = p$ only in the majorants and*

$$\begin{aligned}
&T = \lceil t/2 \rceil, && x_e = T + p + 21, && y_e = T + 2p + 61,\\
&k_1 = 4T + 5p + 127, && n_0 = 8T + 10p + 258, && k_4 = 16T + 20p + 532,\\
&\zeta = 2^{-(d+3)}, && e_{\rm new} = k_4 + 4d + 12 = 16T + 20p + 4d + 544, && N_{\rm new} = 2^{e_{\rm new}}.
\end{aligned} \tag{134}$$

Here T is a scalar half-exponent, distinct from any source transfer interface or field. Then

$$\begin{aligned}
&X_{\rm rel} < 2^{x_e}, \quad Y_{\rm rel} < 2^{y_e}, && K_1 < 2^{k_1 - 2}, \quad 16K_1^2 < 2^{2k_1},\\
&N_0 \le 2^{2k_1} < 2^{n_0}, && K_4(F) \le K_4(\bar{F}) < 2^{k_4}, \quad N_{\rm new} > N_0,\\
&d_{\rm BL}(\mathcal{L}_f(P^b_{N_{\rm new}}), \mathcal{L}_f(Q_\varepsilon)) < 7\eta/8 < \eta.
\end{aligned} \tag{135}$$

The four other errors remain $E_n \le \bar{E}(n) \le \eta$ and $R_{\rm phys}, A_{\rm aux}, S \le \eta$ at the same stages. Thus

$$d_{\rm BL}(\mathcal{L}_f(P^b_{N_{\rm new}}), \mathcal{L}_f(\nu^X)) \le \bar{E}(n) + 4\eta \le 5\eta \le \tau. \tag{136}$$

For the identical old comparison (113)*, $w_{\rm old} = l + 3$, and*

$$e_{\rm new} < 2^{40} < 2^{w_{\rm old}}, \qquad N_{\rm new} < N_{\rm old} = 2^{2^{w_{\rm old}}}. \tag{137}$$

These compare sufficient mathematical whole-plane process-neuron integers at identical model, tests, architectures, target, outer stages and error allocation.

Proof. The ACC predicates give (125) analytically throughout the domain. To display the outer dependencies, $q = \exp(-125/32) < 1/32$ follows from the positive degree-eight exponential partial sum $432528188408407789/8866461766385664 > 32$. The shell bracket increases in q; $\exp(R/256) \le 256/(256 - R) < 2$ gives $E_n \le \bar{E}(n)$. At $n = 80$ the bracket is $159776/961 < 256$ and $\bar{E}(80) < 2^{-38} \le \eta$. The

actual Nelson bounds give $K_\Lambda^2 < 14u^2A < 1$, $K_\sharp \le \max(12U_A, 4)$, $b_{\mathrm{Nelson}} > 3/(16U_A) > 1/16$ and $\log_+(64/(eb_{\mathrm{Nelson}})) < k \le 7$, so $H_2 < H_N \le 3137$. Hence $\sqrt{\bar{\Xi}_2} < 2^{2H_N+1}$ and

$$R_{\mathrm{phys}} < 2^{2H_N+3-p/2} + 2^{2H_N+2-p} \le \eta. \tag{138}$$

The even candidate $4H_N + 2d + 8$ passes both predicates, giving the bound on p. The variance estimate follows from $\operatorname{asinh}(2^p) < \log(3\,2^p) < p + 2$ and $\pi > 3$.

For completeness the original physical Wick exponential and full vector bracket satisfy $\exp(9U_A\bar{c}^2) < 2^{j_e}$ and $2U_AL_{\bar{c}} + F_1 < 64U_Ap^2 + 4c_r \le 2^{h_A}$. The strict margin $R_{\mathrm{aux}} - \max(1000n, R) - s = 1$ includes the open s-neighborhoods of both required compacts. The covariance-defect bound gives $\sqrt{\rho_a(s)} < 2^{p-s/2}$. Thus

$$A_{\mathrm{aux}} < 2^{j_e+h_A+p-s/2} \le \eta, \qquad S < 2^{j_e+h_{\mathrm{stab}}-t} \le \eta. \tag{139}$$

The existence bounds $j_e \le 21600000000 < 2^{35}$, $64U_Ap^2 + 4c_r < 2^{37}$ and $j_e + h_A + p + d < 2^{35}$ imply $h_s \le 36$. They yield $O < 2^{76}$, $h_{\mathrm{stab}} \le 109$, $t < 2^{36}$. The condition $t \ge 34$ ensures $\varepsilon < u$. Every area, variance, geometry and stabilizer dependence used next is retained.

Relax $O < 2^{76}$ to $O < 2^{78}$ and $\sqrt{O} < 2^{39}$. Since $\kappa_a^2 = 2^{2p} - 1$ and $6up < 1$, $B < 2^{2p-1} + 1 \le 2^{2p}$. Put $l_c = t + 4p + 78 \ge 128$. Directly from (129),

$$\begin{aligned} L &< 2^{t+4p+77} = 2^{l_c-1}, \\ J_z &< 2^{78} + 9p^2 < 2^{78} + 2^{32} < 2^{79} \le 2^{l_c}, \\ H &< 2^{l_c+1}, \qquad H+4, H+10 < 2^{l_c+2}, \\ X_{\mathrm{rel}}^4 &< 2^{2t+4p+81}, \qquad Y_{\mathrm{rel}}^4 < 2^{2t+8p+242}. \end{aligned} \tag{140}$$

For both parities of t, $4T \ge 2t$, and the two fourth-power slacks are $4T - 2t + 3 \ge 3$ and $4T - 2t + 2 \ge 2$. Therefore $X_{\mathrm{rel}} < 2^{x_e}$ and $Y_{\mathrm{rel}} < 2^{y_e}$. Odd t rounds upwards; no $\lfloor t/2 \rfloor$ is used.

As $4u < 1$ and $2p + 40 \le 2x_e$,

$$\begin{aligned} v_1(X_{\mathrm{rel}}) &< 2^{3x_e} + 2^{2p+40+x_e} < 2^{3x_e+1}, \\ K_1 &< 2^{3x_e+y_e+1} = 2^{4T+5p+125} = 2^{k_1-2}, \\ 16K_1^2 &< 2^{2k_1}, \qquad N_0 \le 2^{2k_1} < 2^{2k_1+4} = 2^{n_0}. \end{aligned} \tag{141}$$

The right side 2^{2k_1} is an integer. A real number strictly below an integer has ceiling at most that integer; the same-exponent ceiling comparison is weak, followed by a strict four-bit slack. This handles $N_0 = \max(1, \lceil 16K_1^2 \rceil)$ exactly.

Let $W_0 = 2^{3x_e+5} \ge 1$. All four $v_j(X_{\mathrm{rel}}) < W_0$. For v_2 its two terms have upper powers $2x_e + 4$ and $2p + 40 \le 2x_e$; for v_3, v_4 use $x_e + 5$ and 5. The Bell bounds are $1, W_0, 2W_0^2, 5W_0^3, 15W_0^4$, each at most $16W_0^4$. Retaining the centered second e_4, every binomial term and $\bar{F} \le 6$ gives

$$\begin{aligned} P_F(X_{\mathrm{rel}}) \le P_{\bar{F}}(X_{\mathrm{rel}}) &\le 16W_0^4 \left[2 + \sum_{q=1}^{4} \binom{4}{q} 6^q \right] \\ &= 38432W_0^4 < 2^{12x_e+36}, \\ K_4(F) \le K_4(\bar{F}) &< 2^{12x_e+4y_e+36} = 2^{16T+20p+532} = 2^{k_4}. \end{aligned} \tag{142}$$

Here $38432 = 16(1 + 7^4) < 2^{16}$; the scalar coefficient 4112 is not used. We only enlarge the upper by dropping the divisor six in K_4. These whole-domain analytic enclosures retain the actual tilt, volume and variance dependence; finite stage lists do not prove them.

Now $e_{\mathrm{new}} - n_0 = 8T + 10p + 4d + 286 > 0$, so $N_{\mathrm{new}} > N_0$. With $\sqrt{6} < 3$ and $\sqrt{r} \le 2$, the two terms in (133) obey

$$\frac{\sqrt{6}K_4(F)}{N_{\mathrm{new}}\zeta^3} < 3\eta/8, \qquad 2\sqrt{r}\zeta \le \eta/2. \tag{143}$$

Their sum is $< 7\eta/8 < \eta$. The four unchanged later actual-law arrows and the triangle inequality give (136). Strict AP6 identifies the same ν^X; an unknown g_* inequality is not inserted into that step.

For the old guard, $h_R \le 38$, $p < 20000$, $t < 2^{36}$, $o \le 76$ give $v_e < 40040$ and $0 < H_{\mathrm{small}} < 3\,2^{36} + 160335 < 2^{38}$. As $l = t + 4p + o \ge 50$, its bit-length guard holds. The first old candidate $l + 2$ fails its explicit strict predicate; the next succeeds, hence $w_{\mathrm{old}} = l + 3$. This retains the original old full-normalizer bound as its inherited accepted premise. Since $T \le 2^{35}$, $e_{\mathrm{new}} < 2^{39} + 400604 < 2^{40}$, while $w_{\mathrm{old}} \ge 53 > 40$. This proves (137) at identical outer stages and allocations. It proves no necessary or minimal cost, measured speedup or effective common-sequence least-index rule. □

J.4 Exact examples and the separately versioned implementation

For four *distinct* corner centers $(-8,-8),(-8,8),(8,-8),(8,8)$ and tolerance 2^{-12} the accepted example is

$$\begin{aligned} &d = 15, \quad n = 75, \quad p = 6954, \quad s = 4294967296, \\ &t = 1958503808, \quad T = 979251904, \quad e_{\mathrm{new}} = 15668170148, \quad w_{\mathrm{old}} = 1958531694, \\ \bar{E}(n) + 4\eta = \frac{919201379}{6076595200000} < \frac{1}{4096}, &\qquad N_{\mathrm{new}} = 2^{15668170148} < 2^{2^{1958531694}} = N_{\mathrm{old}}. \end{aligned} \tag{144}$$

The historical tuple $(0,0),(8,8),(0,0),(-8,-8)$ has three distinct centers and four ordered slots, including a repeated observation. Its same radius and ambient dimension give the same selected numerical values, while the ordered tuple, request bytes and semantic identity remain distinct. For the single unshifted bump at $1/5$, the accepted odd-t example is

$$n = 73, \quad p = 6574, \quad t = 1658202793, \quad T = 829101397, \quad e_{\mathrm{new}} = 13265754396, \quad w_{\mathrm{old}} = 1658229159. \tag{145}$$

Here $d = 5$ and the upward rounding in (134) matters. These examples use their own selected stages; they are distinct from the older fixed-1/5 NORM/VEC rows in Table 7.

Proposition J.3 (Accepted SVC-relative-accuracy-v1 software contract)**.** *The exact accepted SVC version implements Theorems J.1 and J.2 under their conditional premises and its bounded input/source/resource contract. A successful request retains the exact ordered tuple and unchanged outer minima, returns the derived geometry, source/Nelson/regulator/auxiliary/stabilizer and rational five-error witnesses, and encodes N as the inert node* $\mathrm{pow2}(\mathrm{integer}(e_{\mathrm{new}}))$*. The units of its width cap are $e_{\mathrm{new}} = \log_2 N$. The old comparison is separately typed, with units $w_{\mathrm{old}} = \log_2 \log_2 N_{\mathrm{old}}$, and never controls new selection.*

A separately implemented arithmetic verifier binds exact request bytes, source closure and eight current core/version roots, reconstructs every stage and nested witness with its own arithmetic, and checks exact member sets, types, canonical spelling and scope flags. Success verifies bytes and arithmetic under the accepted mathematical premises. Matching changes to code/proof roots define a different, unaccepted version.

Public availability: the following proof describes the original private SVC package and proof map, which are not exported here.

Proof. The strict request schema, parser and source contract are supplied with the frozen code; the exact locations and hashes are in the proof map. The parser rejects duplicate members, floats/nonfinite tokens, invalid UTF-8/BOM, boolean integer fields, unknown members and unsupported coordinate/tolerance syntax. Canonical integer/rational arithmetic preserves order and multiplicity. It admits the domain in Section J.1, not user coefficients or expressions.

The generator searches the first admissible d, n, k, p, h_s in the stated predicates. The verifier independently uses $d = 5$ at $1/5$ and $d = q + 3$ at 2^{-q}, a rearranged rational shell bound, a Nelson term recurrence, $p = 4H_N + 2d + 8$ and $h_s = 1 + \mathrm{clog}_2(j_e + h_A + p + d)$. The preceding even $p - 2$ fails the first regulator predicate; the preceding dyadic auxiliary exponent fails its predicate. Independent minimal-power calculations give $h_A, o, h_{\mathrm{stab}}, h_R$. No generator selection functions are used by the verifier.

With $T = t \,//\, 2 + (t \bmod 2)$ the verifier independently calculates $k_1 = 3x_e + y_e + 3$, $n_0 = 2k_1 + 4$, $k_4 = 12x_e + 4y_e + 36$ and $e_{\mathrm{new}} = k_4 + 4d + 12$. It checks both parity slacks, the integer-ceiling threshold,

coefficient 38432, joint moment order sixteen, the separate physical variance flag, smoothing fractions and exact rational total. It verifies the typed old guard and strict order without materializing N, 2^p, 2^t or $2^{w_{\mathrm{old}}}$.

The manifest has exactly 208 direct source rows: 195 inherited runtime bodies and thirteen new mathematical/old-contract provenance bodies. Every fixed safe path is bounded-read below the explicit source root, checked for size/hash and required material aliases; symlink escape refuses. Accepted SELECT/OPT/EB1/EC1/ADMIT and SVA decisions/gates retain their exact material relations. The three runtime files, source manifest, schema, software statement/proof and input contract are freshly hashed as eight core bindings. External review and freeze supply a separate version trust root. These checks establish identity and availability, not signatures, classical truth or hostile-filesystem atomicity.

Request and result byte caps are respectively 32768 and 131072. Parser nesting is sixteen; integer tokens have at most twenty characters including sign, strings 4096, member names 256, and value/container nodes 5000. Each source read is capped at one MiB, aggregate source bytes at sixteen MiB. All eight resource members are mandatory. Their exact defaults and allowed intervals are shown in Table 8. A stricter valid cap can refuse before returning a certificate.

The clock begins after request parsing/validation; verifier certificate parsing is inside its measured interval. Source reads, selected ticks and named integer/Fraction watches poll time. Equality at the millisecond ceiling passes; excess, invalid/nonfinite/boolean clock values or backward clocks refuse. Values must be finite monotone integers or floats, and an integer clock has absolute value at most 10^{12}. Named watches cover guarded source-coefficient, spatial/allocation/auxiliary powers; selected rational geometry, Nelson, outer and width exponents; old comparison integers; and the exact total. Logical steps count the documented read/predicate/search ticks. Ordinary Python intermediates and fixed small shifts are not universally watched. Serialization/output, whole-process time, memory and filesystem atomicity are outside the contract. Typed historical measurement metadata is unauthenticated; the verifier measures its own interval.

The exact outcomes are supported-domain success, certificate verified, unsupported input, numerical source gap, source binding refusal, resource exhaustion and refused certificate. Refusals contain no numerical certificate; CLI exit is zero only on success and two for defined refusals. Interpreter/hardware/output-device failures remain outside this exception contract. Executed finite tests and monkeypatch traps support the current implementation correspondence, separately from the analytic proof above and the upstream statement-only proof. The accepted decision governs retained earlier candidate headers in those frozen files. □

Resource member (prefix `max_`)	Default	Allowed interval
`watched_bits`	1024	16–4096
`milliseconds`	30000	1–30000
`steps`	20000	1–50000
`n`	80	1–80
`regulator_exponent`	20000	4–20000
`auxiliary_radius_exponent`	40	1–64
`stabilizer_exponent`	2^{40}	34–2^{50}
`width_exponent`	2^{40}	1–2^{50}

Table 8: Exact SVC selected-resource contract. The last member bounds $e = \log_2 N$. The older `max_width_outer_exponent` member is incompatible and refuses. Named watches and cooperative time have the specific scope stated in the proof.

J.5 Source, history and construction boundaries

The SVA mathematical decision, accepted at its exact conditional scope, retains its original statement that SVC was then unlaunched. The later accepted SVC decision supersedes only that implementation status for the separately reviewed version. Its author/hostile/numerical/code-matching records remain distinct from

SVA's hostile, complete statement-only, arithmetic and actual-proof records. All are internal fresh-session model review, not an outside-expert or statistical-independence credential. Earlier adverse attempts, source gaps, adapted versus unchanged replays and findings/dispositions remain recorded under their original identities. The original private companion index recorded which exact bodies were local.

In the historical private integration, the curated inherited companion tree resolved the 23-packet index and its ninety indexed bindings locally, alongside the seven later baseline packets and the SVA/SVC source trees. That historical availability closed a dated assignment gap. The historical tree did not recursively bundle every quoted path or supply unavailable primary classical proofs. Those private source trees and their original availability map are not attachments to this public edition. The public proof map gives this edition's source locations and hashes. Hashes record exact bytes only.

The deterministic common sequence, its four noneffective least inner indices, quantifiers over all later tests, correlations and reflection lists, and both original architectures remain as in Theorem I.2. The quantitative schedule fixes observations first. MOM proves local H^{-2} moment convergence/UI; the separately derived SEQ-HS statement gives physical/target H^{-s} moment finiteness for real $s > 1$ at its literal scope. The inclusive qualitative cap and strict AP6 numerical cap remain separate. The present single-exponent widths are sufficient whole-plane process-neuron counts; practical/minimal/necessary complexity, full-field rates, global neural Schwartz control, broader domains and priority are separate questions. Public v1 is unchanged.

The standalone source contains the full present argument and every preserved baseline mathematical block. The original private companion proof map binds each new block and reading-guide headline to accepted statement and full-proof clauses. The original reversible editorial delta, exact preservation audit, build, reference/numerical receipts and all-page inspection record are retained as private integration evidence. Fresh source/proof/visual review and coordinator acceptance remain separate.

K A conditional schedule for every dyadic accuracy

This appendix gives all six clauses of the internally accepted conditional UDA-STAGE-RELATIVE-v2 and their complete proof. Its source statement and proof are the exact bytes bound by the current decision. The manuscript presentation awaits its own fresh source and actual-proof reviews. Every integer below is a finite mathematical integer; software resource refusal does not make the mathematical definition partial. Section K.8 supplies the separately accepted UDC implementation contract.

K.1 Model, observations and the literal conditional interface

Fix $m = 1$, $u = 10^{-10}$, $\ell = 1000$ and $K_{\mathrm{cov}} = 1$. The model is the original whole-plane, coefficient-one, same-mass, own-variance Wick interaction $u{:}\phi^4{:}$, with no factorial. The original complete state space is $X = H^{-2}_{\mathrm{loc}}$, and ν^X is the original pressure-identified common Fröhlich–Simon state. The literal P1–P6, Gaussian/Wick, DSS/BD/BBD/Griffiths–Simon/SK-q/P^*, H1/MIX/REG/smoothed-CT/FLOW/TP/T and classical/source premises retain their original finite scopes. The quantitative bridge uses strict AP6, $0 < u/m^2 < 1/8000$. The original fixed g_* and the inequality $10^{-10} < g_*$ remain unknown and unused. NR1's sharper author constants and larger statement-only blind core retain their separate original scopes; excluded NR2 E4/r0 material is absent. The inherited source interface is spelled out in Section K.7; its source bodies are not newly proved.

Let

$$b(t) = \begin{cases} e^{-1/(1-t^2)}, & |t| < 1, \\ 0, & |t| \geq 1, \end{cases} \qquad f_h(z) = b(z_1 - h_1)b(z_2 - h_2).$$

A request is an ordered tuple $(h_1, \dots, h_r)$ with $r \in \{1, 2, 3, 4\}$. Each coordinate is a canonical dyadic $P/2^k \in [-8, 8]$, $0 \leq k \leq 16$: integers and zero have $k = 0$, and P is odd if $k > 0$. Decimal numerators have no plus sign, leading zeros or negative zero. Order and repetitions remain; amplitudes, signs, dilations and rotations are unchanged. The primary accuracy is $\tau = 2^{-q}$ for every integer $q \geq 3$; $\tau = 1/5$ is a separate compatibility case. Set

$$K = \bigcup_{i=1}^{r} [h_{i1} - 1, h_{i1} + 1] \times [h_{i2} - 1, h_{i2} + 1], \qquad R = 1 + \max_{i,j} |h_{ij}|,$$

$$c_r = \lceil \sqrt{r} \rceil, \qquad F_j = \left(\sum_{i=1}^{r} \|f_{h_i}\|_j^2 \right)^{1/2}, \qquad F = F_{4/3}, \quad \overline{F} = 3c_r.$$

For field laws ρ, let $\mathcal{L}_f(\rho) = (\mathcal{O}_f)_*\rho$, where $\mathcal{O}_f(v) = (v(f_{h_i}))_{i=1}^r$ takes values in the full Euclidean $\mathbb{R}^r$. The bounded-Lipschitz distance is

$$d_{\mathrm{BL}}(\rho, \sigma) = \sup_{\|G\|_\infty \leq 1,\ \mathrm{Lip}_{\mathbb{R}^r}(G) \leq 1} |\rho(G) - \sigma(G)|.$$

The two constraints are separate. Write $\mathcal{U}_4(r)$ for real C^4 functions on $\mathbb{R}^r$ satisfying

$$\max_{0 \leq j \leq 4} \sup \|D^j G\|_{\mathrm{op}} \leq 1.$$

order zero means absolute value, and higher orders use full multilinear Euclidean operator norms.

The literal CL1 coefficient uses the same chosen $C_H < 2^{16}$ and $C_{H2} = 2^{384}$ in every expression. With $\kappa = 1/256$, $\epsilon_* = 1/4$ and $c_{\mathrm{CT}} = 81/5$, define

$$\beta_{\mathrm{CT}} = 1 + c_{\mathrm{CT}}\kappa, \qquad \Gamma_G = 4\sqrt{2\pi}\,\kappa,$$

$$
\begin{aligned}
C_* &= 6c_{\mathrm{CT}}\kappa + c_{\mathrm{CT}}^2\kappa^2 + \beta_{\mathrm{CT}}^2(\Gamma_G + 3300u),\\
g_{\mathrm{CL1}}(v) &= 1 + \frac{2}{10^6}\log\frac{1}{1-e^{-4000}} + \frac{\pi}{16}\log(1+1/v), \quad 0 < v \le 1,\\
M(v) &= 32\ell^{-2}\left[\frac{C_H^2 g_{\mathrm{CL1}}(v)^3}{12} + \frac{u^2 C_{H2}^2 g_{\mathrm{CL1}}(v)^7}{9}\right],\\
J_{\mathrm{CL1}} &= e^{8\sqrt{2}\kappa}\beta_{\mathrm{CT}}^2\left[e^{C_*}\int_0^1 \sqrt{M(v)}\,dv + \frac{e^{2C_*}\sqrt{M(1)}}{2}\right],\\
C_f(\mathbf{h};R) &= (2R+1/2)\sqrt{r}\,e^{-2}J_{\mathrm{CL1}}, \qquad\qquad C_f(\mathrm{base};1) = \tfrac{5}{2}e^{-2}J_{\mathrm{CL1}}.
\end{aligned}
$$

Assume the accepted literal base enclosure $C_f(\mathrm{base};1) \le 2^{360}$ with its exact finite-source premises, and put $\overline{C} = 2^{360}c_r(4R+1)/5$. The separate factor $e^{R/256}$ remains. No independently optimized coefficient replaces the source integral.

For every fixed $0 < a < 1$ and finite $D = \Omega = [-R_{\mathrm{aux}}, R_{\mathrm{aux}}]^2$ containing the physical interaction-square closure and every test support, assume the original finite-law interface below. Its finite-positive-stage provenance is EB1/NR1/P6. Bounded SVA or fixed-$n75$ VEC acceptance is not an all-stage premise, and no all-q conclusion is assumed. The physical covariance and variance are

$$
\begin{aligned}
C_a(z) &= (2\pi)^{-2}\int_{\mathbb{R}^2}\frac{e^{i(\omega,\xi)\cdot z}}{1+\omega^2+\xi^2+a^2\omega^2\xi^2}\,d\omega\,d\xi,\\
c = c_a &= \frac{1}{2\pi}\int_0^\infty \frac{d\xi}{\sqrt{(1+\xi^2)(1+a^2\xi^2)}}, \qquad\qquad \kappa_a^2 = a^{-2}-1,\\
c &\le \overline{c}(a) = \frac{\operatorname{asinh}(1/a)+1}{2\pi}.
\end{aligned}
$$

On the separable real Banach space $E = L^4(D)$, γ is the original centered continuous Gaussian endpoint, strongly measurable, with zero Bochner mean, global sign symmetry and covariance $C_{\mathrm{end}}(z,z') = e^{-|z-z'|_1/a}/4$. Its point variance is $1/4$. The Fourier integration variable ξ above is local and does not denote an observation or the standard Gaussian vector used for smoothing.

For $b \in \{\mathrm{product}, \mathrm{Mondrian}\}$, g^b is the restriction of the original whole-plane process neuron. The product neuron is $g(z) = \sigma(z_1)\tau(z_2)/2$ for independent stationary symmetric telegraph processes of flip rate $1/(2a)$. The Mondrian neuron uses the projectively consistent whole-plane axis-parallel partition at budget $1/a$, independently uniformly signed cells, and point values half the cell sign. A rectangle of sides w, h splits at rate $w+h$, vertically with probability $w/(w+h)$ and otherwise horizontally, at a uniform coordinate; daughters evolve independently for the remaining budget. Use the original lower-left half-open representative. Retain the original existence, projective-consistency and measurability premises. Each g^b is strongly measurable and E-valued, half-valued almost everywhere almost surely, has zero Bochner mean, global sign symmetry and covariance C_{end}. Boundary representatives agree in E. For *every* bounded bilinear form $B_0 : E \times E \to \mathbb{R}$, assume

$$\mathbb{E}B_0(g^b, g^b) = \mathbb{E}B_0(\gamma,\gamma).$$

Independent copies may be combined on product probability spaces. These are the original full laws and their restrictions; point-pair covariance does not define this full interface.

For $\Lambda = (-1000n, 1000n)^2$ with $\overline{\Lambda} \subset \Omega$, the actual potential, weight and normalized laws are

$$V(v) = u\int_\Lambda (v^4 - 6cv^2 + 3c^2) + \epsilon\int_{\Omega\setminus\Lambda} v^4 - \frac{\kappa_a^2}{2}\int_\Omega v^2, \quad w(v) = e^{-V(v)},$$

$$F_N^b = N^{-1/2}\sum_{i=1}^{N} g_i^b, \qquad P_N^b = \frac{w\ \mathrm{Law}(F_N^b)}{\mathbb{E}w(F_N^b)}, \qquad Q_\epsilon = \frac{w\ \mathrm{Law}(\gamma)}{\mathbb{E}w(\gamma)}.$$

Their observation laws are exactly the original whole-plane neural and full-Gaussian-tilt laws. The physical c remains in V; a positive majorant c_* never changes it. Width counts whole-plane process neurons, each with countably many internal variables. The ambient L0 premise says that for unit BL G on $\mathbb{R}^r$ and standard Gaussian $\xi \in \mathbb{R}^r$,

$$\begin{aligned} G_\zeta(y) &= \mathbb{E}G(y-\zeta\xi), & \|G - G_\zeta\|_\infty &\le \sqrt{r}\,\zeta, \\ \|G_\zeta\|_\infty &\le 1, & \|DG_\zeta\|_{\mathrm{op}} &\le 1, \\ \|D^j G_\zeta\|_{\mathrm{op}} &\le \sqrt{(j-1)!}\,\zeta^{1-j}, & j &\ge 2. \end{aligned}$$

For $0 < \zeta \le 1$, every norm through order four is at most $\sqrt{6}\zeta^{-3}$, even for singular observation covariance.

The full Gaussian tilt Q_ϵ has the original residual/mass-tilt representation. Its zero-stabilizer residual endpoint is Q_0, the positive-regulator physical law is Q_a, and its own-variance zero-regulator fourth-chaos limit is Q_{phys}. On the actual common Gaussian space, Q_a has density $e^{-U_{a,\Lambda}}/Z_a$, $\mathbb{E}U_{a,\Lambda} = 0$ and $Z_a \ge 1$. The compact-test laws of Q_{phys} are the original classical free finite-volume laws. AP5/AP6 identify the same original target ν^X. These are inherited conditional identifications, with the literal conventions of Section K.7. For $A = |\Lambda| > 0$, $U_A = uA$, $O = |\Omega|$ and $B_{\mathrm{ext}} = O - A$, put

$$\begin{aligned} K_\Lambda^2 &= \frac{75\sqrt{85}}{2\pi^3}u^2 A, & \kappa_0 &= e(1+\sqrt{2}), \\ K_{\mathrm{sharp}} &= \max\{12U_A, (2/e)K_\Lambda\sqrt{\kappa_0}\}, & b_{\mathrm{Nelson}} &= \frac{\pi^2}{48U_A}, \\ T_\Lambda(H) &= \exp[-\exp(\sqrt{b_{\mathrm{Nelson}}H} - 1)], & & \\ \Xi_2 &= e^{2K_{\mathrm{sharp}}} + \int_{K_{\mathrm{sharp}}}^{\infty} 2e^{2H}T_\Lambda(H)\,dH, & & \\ H_2 &= \max\left\{K_{\mathrm{sharp}}, \frac{4}{b_{\mathrm{Nelson}}}\left[\log_+\frac{64}{eb_{\mathrm{Nelson}}}\right]^2\right\}, & \overline{\Xi}_2 &= 2e^{2H_2}, \quad \Xi_2 \le \overline{\Xi}_2, \\ \rho_a(s) &= \frac{\kappa_a^2}{4\pi}\min\{1, 2e^{-\sqrt{2}s}\}, & q_{\mathrm{sp}} &= e^{-1000/256}, \\ T_n &= q_{\mathrm{sp}}^n\left[\frac{2n+1}{1-q_{\mathrm{sp}}} + \frac{2q_{\mathrm{sp}}}{(1-q_{\mathrm{sp}})^2}\right]. & & \end{aligned}$$

Here e is Euler's number and $\log_+ x = \max(0, \log x)$. EC1.a–c apply at every fixed finite positive physical area; Ξ_2 is not volume-uniform. With $\Delta = C_a - C'_{\Omega,a}$ positive semidefinite, $\delta_\Omega(H) = \sup_{z\in H}\Delta(z,z)$ is at most $\rho_a(s)$ whenever the strict Euclidean s-neighborhood of H lies in Ω.

The actual successive errors $Q_\epsilon \to Q_0 \to Q_a \to Q_{\mathrm{phys}} \to \nu^X$, in the stated observation metric (variation for the first arrow), are bounded by

$$\begin{aligned} S &= \min\{2, 6\epsilon B_{\mathrm{ext}}c^2 e^{9U_A c^2}\}, \\ A_{\mathrm{aux}} &= \min\left\{2, e^{9U_A c^2}\left[2U_A(4\sqrt{15}+12)c^{3/2}\sqrt{\delta_\Omega(\overline{\Lambda})} + F_1\sqrt{\delta_\Omega(K)}\right]\right\}, \\ R_{\mathrm{phys}} &= \min\{2, \sqrt{\Xi_2}[4K_\Lambda\sqrt{a} + (a/2)F_2]\}, \end{aligned}$$

$$E_n = 4C_f(\mathbf{h}; R)u\ell^2 e^{R/256}T_n, \qquad n \geq 1, \quad 1000n \geq R.$$

The strict source geometry remains mandatory. On the physical common space, $\mathbb{E}|w_a - w_0| \leq 2K_\Lambda\sqrt{\Xi_2}\sqrt{a}$, $\|w_0\|_2 \leq \sqrt{\Xi_2}$ and $\mathbb{E}|(\Phi_a - \Phi_0)(f)|^2 \leq (a^2/4)\|f\|_2^2$. Full neural/Gaussian normalizers, physical $Z_a \geq 1$ and the residual Jensen normalizer are distinct. Strict AP6 supplies this original-target route without using or establishing $10^{-10} < g_*$.

K.2 Relative constants and the complete old coefficient

At finite positive stages with $0 < \epsilon < u$ and $c_* \geq c$ used only in positive majorants, define

$$\begin{aligned} \alpha &= \epsilon, & U &= u, \\ B &= \kappa_a^2/2 + 6uc_*, & L &= B^2O/(2\alpha), \\ J_z &= 3UO/16 + 3uc_*^2A, & H &= L + J_z. \end{aligned}$$

$$\begin{aligned} v_1(x) &= 4Ux^3 + 2B\sqrt{O}\,x, & v_2(x) &= 12Ux^2 + 2B\sqrt{O}, \\ v_3(x) &= 24Ux, & v_4(x) &= 24U. \end{aligned}$$

$$\begin{aligned} e_0 &= 1, & e_1 &= v_1, & e_2 &= v_1^2 + v_2, \\ e_3 &= v_1^3 + 3v_1v_2 + v_3, \\ e_4 &= v_1^4 + 6v_1^2v_2 + 3v_2^2 + 4v_1v_3 + v_4. \end{aligned}$$

$$\begin{aligned} P_F(x) &= 2e_4(x) + \sum_{j=1}^{4} \binom{4}{j} F^j e_{4-j}(x), \\ X_{\mathrm{rel}} &= [2(H+4)/\alpha]^{1/4}, & Y_{\mathrm{rel}} &= 2O^{1/4}(H+10)^{1/2}, \\ K_1 &= v_1(X_{\mathrm{rel}})Y_{\mathrm{rel}}, & N_0 &= \max\{1, \lceil 16K_1^2 \rceil\}, \\ K_4(F) &= P_F(X_{\mathrm{rel}})Y_{\mathrm{rel}}^4/6. \end{aligned}$$

Derivative norms use $E = L^4(D)$. The scalar moment bounds $X_{\mathrm{rel}}, Y_{\mathrm{rel}}$ are distinct from the field space X. For $P(x) = \sum_k a_k x^k$ with nonnegative coefficients, put

$$\begin{aligned} \beta_0 &= 1, & \beta_k &= \left(\frac{k}{2\alpha e}\right)^{k/4} \quad (k > 0), \\ \mathcal{T}(P) &= e^L \sum_k a_k\beta_k, \\ M_Z &= \mathcal{T}(e_4), & M_A &= \mathcal{T}\left(\sum_{j=0}^{4} \binom{4}{j} F^j e_{4-j}\right), \\ z &= e^{-J_z}, & C_{\mathrm{old}} &= \frac{O(M_A + M_Z)}{96z}, \\ D_{\mathrm{old}}^4 &= \frac{2048\sqrt{6}}{27} r^{3/2} C_{\mathrm{old}}. \end{aligned}$$

One may replace F by $\overline{F}$ and c by p only in positive majorants. The complete old coefficient retains coercivity and both Jensen terms.

K.3 Universal rational stage prescription

For positive rational v, let $\operatorname{clog}_2(v)$ be the least integer $h \ge 0$ with $v \le 2^h$ (the source's `ceil_log2`). For $\tau = 2^{-q}$ set $d = q + 3$; for $\tau = 1/5$ set $d = 5$. Let $\eta = 2^{-d}$ and, in the displayed order, define

$$\begin{aligned}
\overline{\exp R} &= \frac{256}{256 - R},\\
\overline{E}(i) &= \frac{4}{10000}\overline{C}\,\overline{\exp R}\,2^{-5i}\left[(2i+1)\frac{32}{31} + \frac{64}{961}\right],\\
n &= \min\{i \ge 1 : \overline{E}(i) \le \eta\},\\
A &= 4000000n^2, \quad U_A = uA, \quad b_0 = 3/(16U_A),\\
h_K &= \min\{h \ge 0 : 14u^2A \le 2^{2h}\},\\
\overline{K} &= 2^{h_K},\\
L_{10}(k) &= \sum_{j=0}^{10} \frac{k^j}{j!},\\
k &= \min\{i \ge 1 : L_{10}(i) > 32/b_0\},\\
H_N &= 1 + \left\lceil \max\{12U_A, 4\overline{K}, 4k^2/b_0\} \right\rceil,\\
p &= 4H_N + 2h_K + 2d + 8,\\
a &= 2^{-p},\\
j &= \lceil 18U_A p^2 \rceil,\\
B_{\mathrm{aux}} &= 64U_A p^2 + 4c_r,\\
h_{\mathrm{aux}} &= \operatorname{clog}_2(B_{\mathrm{aux}}),\\
h_s &= 1 + \operatorname{clog}_2(j + h_{\mathrm{aux}} + p + d),\\
s &= 2^{h_s},\\
R_{\mathrm{aux}} &= 1000n + s + 1,\\
O &= 4R_{\mathrm{aux}}^2,\\
o &= \operatorname{clog}_2(O),\\
h_{\mathrm{stab}} &= \operatorname{clog}_2(6Op^2),\\
t &= \max\{34, j + h_{\mathrm{stab}} + d\}, \quad \epsilon = 2^{-t},\\
\Lambda &= (-1000n, 1000n)^2,\\
\Omega = D &= [-R_{\mathrm{aux}}, R_{\mathrm{aux}}]^2.
\end{aligned}$$

Use $c_* = p$ in relative majorants and put

$$\begin{aligned}
o_+ &= \max\{o, 78\},\\
T_{1/2} &= \lceil t/2 \rceil,\\
x &= T_{1/2} + p + \left\lceil \frac{o_+ + 3}{4} \right\rceil,\\
y &= T_{1/2} + 2p + \left\lceil \frac{3o_+ + 8}{4} \right\rceil,\\
k_1 &= 3x + y + 3,\\
n_0 &= 2k_1 + 4, \quad k_4 = 12x + 4y + 36,
\end{aligned}$$

$$\zeta = 2^{-d-3},$$
$$e_{\mathrm{new}} = k_4 + 4d + 12, \qquad N_{\mathrm{new}} = 2^{e_{\mathrm{new}}}.$$

The notation $o_+, T_{1/2}$ translates the source's `oplus,T_half`.

At these same outer stages, define

$$h_{\mathrm{root}} = \mathrm{clog}_2(2R_{\mathrm{aux}}), \qquad v_{\mathrm{old}} = 2p + h_{\mathrm{root}} + 2,$$
$$q_{\mathrm{old}} = 4v_{\mathrm{old}} + 3t + 28, \qquad l_{\mathrm{old}} = t + 4p + o,$$
$$H_{\mathrm{small}} = q_{\mathrm{old}} + o + 11 + 4d.$$

Start at $w = l_{\mathrm{old}} + 2$ and increment until both $w > l_{\mathrm{old}} + 2$ and $\mathrm{bitlen}(H_{\mathrm{small}}) \le l_{\mathrm{old}} + 2$ hold. Call the result w_{old} and put $N_{\mathrm{old}} = 2^{2^{w_{\mathrm{old}}}}$. For positive integer M, $\mathrm{bitlen}(M)$ is the least positive h with $M < 2^h$. These are sufficient-width rules, rather than necessary costs or canonical inner least-index definitions.

K.4 The explicitly defined legacy comparator

For $\tau \in \{1/5\} \cup \{2^{-q} : 3 \le q \le 12\}$, use the same ordered tuple, $r, R, c_r, \overline{C}, \overline{E}$ and L_{10}. The suffix L denotes a separately defined legacy rule. Its termination, sufficiency and equality with the universal rule will be proved, rather than assumed. Define in order

$$d_L = \min\{i \ge 0 : 5 \cdot 2^{-i} \le \tau\},$$
$$\eta_L = 2^{-d_L},$$
$$n_L = \min\{1 \le i \le 80 : \overline{E}(i) \le \eta_L\},$$
$$A_L = 4000000 n_L^2,$$
$$U_{AL} = uA_L, \quad b_{0L} = 3/(16U_{AL}),$$
$$k_L = \min\{1 \le i \le 7 : L_{10}(i) > 32/b_{0L}\},$$
$$H_L = 1 + \lceil \max\{12U_{AL}, 4, 4k_L^2/b_{0L}\} \rceil,$$
$$p_L = \min \left\{ \begin{array}{l} i \ge 4 \text{ even} : \\ 2H_L + 3 - i/2 \le -d_L - 1, \\ 2H_L + 2 - i \le -d_L - 1 \end{array} \right\},$$
$$a_L = 2^{-p_L},$$
$$j_L = \lceil 18U_{AL}p_L^2 \rceil,$$
$$B_{\mathrm{aux},L} = 64U_{AL}p_L^2 + 4c_r,$$
$$h_{\mathrm{aux},L} = \mathrm{clog}_2(B_{\mathrm{aux},L}),$$
$$h_{sL} = \min\{i \ge 1 : j_L + h_{\mathrm{aux},L} + p_L - 2^{i-1} \le -d_L\},$$
$$s_L = 2^{h_{sL}},$$
$$R_{\mathrm{aux},L} = 1000 n_L + s_L + 1,$$
$$O_L = 4R_{\mathrm{aux},L}^2,$$
$$o_L = \mathrm{clog}_2(O_L),$$
$$h_{\mathrm{stab},L} = \mathrm{clog}_2(6O_L p_L^2),$$
$$t_L = \max\{34, j_L + h_{\mathrm{stab},L} + d_L\},$$

$$\begin{aligned}\epsilon_L &= 2^{-t_L},\\ \Lambda_L &= (-1000n_L, 1000n_L)^2,\\ D_L = \Omega_L &= [-R_{\mathrm{aux},L}, R_{\mathrm{aux},L}]^2.\end{aligned}$$

For the legacy old-width rule, put

$$\begin{aligned}h_{\mathrm{root},L} &= \mathrm{clog}_2(2R_{\mathrm{aux},L}),\\ v_{\mathrm{old},L} &= 2p_L + h_{\mathrm{root},L} + 2,\\ q_{\mathrm{old},L} &= 4v_{\mathrm{old},L} + 3t_L + 28,\\ l_{\mathrm{old},L} &= t_L + 4p_L + o_L,\\ H_{\mathrm{small},L} &= q_{\mathrm{old},L} + o_L + 11 + 4d_L.\end{aligned}$$

Start at $w = l_{\mathrm{old},L}+2$ and increment by one until both $w > l_{\mathrm{old},L}+2$ and $\mathrm{bitlen}(H_{\mathrm{small},L}) \le l_{\mathrm{old},L}+2$ hold. The selected value is $w_{\mathrm{old},L}$, with $N_{\mathrm{old},L} = 2^{2^{w_{\mathrm{old},L}}}$. The legacy SVC relative exponents are

$$\begin{aligned}T_L &= \lceil t_L/2 \rceil,\\ x_L &= T_L + p_L + 21,\\ y_L &= T_L + 2p_L + 61,\\ k_{1L} &= 4T_L + 5p_L + 127,\\ n_{0L} &= 8T_L + 10p_L + 258,\\ k_{4L} &= 16T_L + 20p_L + 532,\\ \zeta_L &= 2^{-d_L-3},\\ e_L &= k_{4L} + 4d_L + 12,\\ N_L &= 2^{e_L}.\end{aligned}$$

These mathematical rational/integer definitions remain meaningful if a software run refuses for a resource limit. They identify no request or certificate with another and assert no software execution.

K.5 Six exact candidate clauses

Theorem K.1 (UDA.a: finite selection and four downstream errors)**.** *Every allowed request has finite selected stages, with termination witnesses*

$$n \le \left\lceil \frac{d+356}{4} \right\rceil, \quad k \le \lceil 32/b_0 \rceil, \quad h_K \le \lceil 14u^2 A \rceil, \quad \mathrm{clog}_2(v) \le \lceil v \rceil.$$

The last bound holds for every positive argument used. All subsequent formulas are finite compositions. The integer p is the first even integer at least four satisfying both

$$2H_N + h_K + 3 - p/2 \le -d - 1, \qquad 2H_N + 2 - p \le -d - 1,$$

and h_s is the first integer at least one satisfying $j + h_{\mathrm{aux}} + p - 2^{h_s-1} \le -d$. Moreover

$$1 \le R \le 9, \quad K \subset \Lambda_1 \text{ (open)}, \quad F_1 \le 4c_r \le 8, \quad F_2 \le 2c_r \le 4, \quad F \le \overline{F} \le 6,$$
$$C_f(\mathbf{h}; R) \le \overline{C} < 2^{364}, \qquad R_{\mathrm{aux}} - \max(1000n, R) - s = 1,$$
$$0 < a < 1, \quad 0 < \epsilon < u, \quad c \le \overline{c}(a) < p,$$
$$K_\Lambda < \overline{K}, \quad H_2 < H_N, \quad \sqrt{\overline{\Xi}_2} < 2^{2H_N+1},$$
$$E_n \le \overline{E}(n) \le \eta, \qquad R_{\mathrm{phys}} \le \eta, \quad A_{\mathrm{aux}} \le \eta, \quad S \le \eta.$$

Theorem K.2 (UDA.b: arbitrary finite-stage joint laws and derivatives)**.** *At any fixed finite stage satisfying the premises above, for either architecture, $N \ge 1$, $0 \le j \le N-1$, and an independent direction h with neuron or Gaussian law, let*

$$\begin{aligned} S_j &= N^{-1/2}\left[\sum_{i=1}^{j}\gamma_i + \sum_{i=j+1}^{N-1} g_i^b\right], \\ v &= S_j + \theta h/\sqrt{N}, & Z_{j,h}(\theta) &= \mathbb{E}w(v), \\ \Pi_{j,h,\theta} &= w(v)\,\mathrm{Law}(S_j,h)/Z_{j,h}(\theta), & 0 &\le \theta \le 1. \end{aligned}$$

The untilted v has almost-everywhere point variance at most $1/4$ and fourth moment at most $3/16$. Every Z is positive finite. Writing $x(v) = \|v\|_4$, for $z \ge 0$ and every real $1 \le q_0 \le 16$,

$$w(v) \le e^{L-\alpha x(v)^4/2}, \quad Z \ge e^{-J_z}, \quad 0 < \frac{d\Pi}{d\,\mathrm{Law}(S_j,h)} \le e^{H-\alpha x(v)^4/2} \le e^H,$$

$$\Pi(x(v)^4 > z) \le \min\{1, e^{H-\alpha z/2}\},$$

$$\|x(v)\|_{L^{q_0}(\Pi)} \le X_{\mathrm{rel}}, \quad \|\|h\|_4\|_{L^{q_0}(\Pi)} \le Y_{\mathrm{rel}}, \quad \mathbb{E}e^{\|\gamma\|_4^2/(4\sqrt{O})} < 4.$$

Each Z and $\log Z$ extends C^1 to a neighborhood of $[0,1]$, with

$$(\log Z)' = -N^{-1/2}\mathbb{E}_\Pi DV(v)[h], \qquad |(\log Z)'| \le K_1/\sqrt{N}.$$

The zero-amplitude partitions agree. For $N \ge N_0$,

$$\frac{Z_{j,g}(\theta)}{Z_{j,g}(1)} \le e^{K_1/\sqrt{N}}, \qquad \frac{Z_{j,\gamma}(\theta)}{Z_{j,g}(1)} \le e^{2K_1/\sqrt{N}} < 2.$$

The potential is $C^4(E)$, with $\|D^k V(v)\|_{\mathrm{op}} \le v_k(x(v))$ for $1 \le k \le 4$ and $\|D^k w(v)\|_{\mathrm{op}} \le w(v)e_k(x(v))$ for $0 \le k \le 4$. All derivatives of w through four are globally bounded. For $G \in \mathcal{U}_4(r)$, define

$$\begin{aligned} m_j &= \frac{\mathbb{E}[G(\mathcal{O}_f(S_j + \gamma/\sqrt{N}))w(S_j + \gamma/\sqrt{N})]}{Z_{j,\gamma}(1)}, \\ \Phi_j(v) &= [G(\mathcal{O}_f(v)) - m_j]w(v). \end{aligned}$$

Then $|m_j| \le 1$, $\Phi_j \in C_b^4(E)$, $\deg P_F \le 12$, and under each actual joint Π,

$$\|D^4\Phi_j(v)\|_{\mathrm{op}} \le w(v)P_F(x(v)), \qquad \mathbb{E}_\Pi[P_F(x(v))\|h\|_4^4] \le P_F(X_{\mathrm{rel}})Y_{\mathrm{rel}}^4.$$

This is a joint-law estimate, without a product of tilted marginal expectations.

Theorem K.3 (UDA.c: normalized laws, ambient BL and selected width)**.** *At the arbitrary finite stages of UDA.b and for every $N \ge N_0$, each actual adjacent normalized replacement has $\mathcal{U}_4$ error at most $K_4(F)/N^2$, and*

$$\begin{aligned} |P_N^b(G\circ\mathcal{O}_f) - Q_\epsilon(G\circ\mathcal{O}_f)| &\le \min\{2, K_4(F)/N\}, \quad G \in \mathcal{U}_4(r), \\ d_{\mathrm{BL}}(\mathcal{L}_f(P_N^b), \mathcal{L}_f(Q_\epsilon)) &\le \min\{2, \sqrt{6}K_4(F)/(N\zeta^3) + 2\sqrt{r}\zeta\}, \quad 0 < \zeta \le 1, \\ d_{\mathrm{BL}}(\mathcal{L}_f(P_N^b), \mathcal{L}_f(Q_\epsilon)) &\le \min\{2, D_r N^{-1/4}\}, \qquad D_r^4 = \frac{2048\sqrt{6}}{27}r^{3/2}K_4(F). \end{aligned}$$

At every selected stage, including both parities of t,

$$\begin{gathered} X_{\mathrm{rel}} < 2^x, \quad Y_{\mathrm{rel}} < 2^y, \quad K_1 < 2^{k_1-2}, \quad 16K_1^2 < 2^{2k_1}, \\ N_0 \le 2^{2k_1} < 2^{n_0} < N_{\mathrm{new}}, \quad K_4(F) \le K_4(\overline{F}) < 2^{k_4}, \\ d_{\mathrm{BL}}(\mathcal{L}_f(P_{N_{\mathrm{new}}}^b), \mathcal{L}_f(Q_\epsilon)) < 7\eta/8 < \eta. \end{gathered}$$

Theorem K.4 (UDA.d: actual target at every dyadic accuracy)**.** *For both original architectures, every integer $q \geq 3$ and every allowed ordered request,*

$$d_{\mathrm{BL}}(\mathcal{L}_f(P^b_{N_{\mathrm{new}}}), \mathcal{L}_f(\nu^X)) \leq \overline{E}(n) + 4\eta \leq 5\eta \leq 2^{-q}.$$

The separate $1/5$ case has the same conclusion with right side τ. The five actual arrows are

$$P^b_{N_{\mathrm{new}}} \longrightarrow Q_\epsilon \longrightarrow Q_0 \longrightarrow Q_a \longrightarrow Q_{\mathrm{phys}} \longrightarrow \nu^X.$$

Theorem K.5 (UDA.e: complete old coefficient and same-stage order)**.** *At all selected stages, the complete C_{old} gives, for every $N \geq 1$ and $G \in \mathcal{U}_4(r)$,*

$$|P^b_N(G \circ \mathcal{O}_f) - Q_\epsilon(G \circ \mathcal{O}_f)| \leq \min\{2, C_{\mathrm{old}}/N\},$$
$$d_{\mathrm{BL}}(\mathcal{L}_f(P^b_N), \mathcal{L}_f(Q_\epsilon)) \leq \min\{2, D_{\mathrm{old}} N^{-1/4}\}.$$

With $c_* = p$ and $\overline{F}$ in positive majorants,

$$C_{\mathrm{old}} < 2^{\,2^{l_{\mathrm{old}}+2}+q_{\mathrm{old}}+o}, \qquad H_{\mathrm{small}} \leq 7l_{\mathrm{old}} + 47 < 2^{l_{\mathrm{old}}+2},$$
$$w_{\mathrm{old}} = l_{\mathrm{old}} + 3, \qquad d_{\mathrm{BL}}(\mathcal{L}_f(P^b_{N_{\mathrm{old}}}), \mathcal{L}_f(Q_\epsilon)) \leq \eta,$$
$$e_{\mathrm{new}} \leq 12l_{\mathrm{old}} + 553 < 2^{l_{\mathrm{old}}+3} = 2^{w_{\mathrm{old}}}, \qquad N_{\mathrm{new}} < N_{\mathrm{old}}.$$

The sufficient mathematical integers are compared at identical stages, allocations, model, source premises, architectures, tests and target.

Theorem K.6 (UDA.f: exact old-domain compatibility)**.** *For $\tau \in \{1/5\} \cup \{2^{-q} : 3 \leq q \leq 12\}$ the explicit legacy rules terminate. Every common outer variable*

$$d, \eta, n, A, U_A, b_0, k, H_N, p, a, j, B_{\mathrm{aux}}, h_{\mathrm{aux}}, h_s, s, R_{\mathrm{aux}}, O, o, h_{\mathrm{stab}}, t, \epsilon,$$
$$\Lambda, D, \Omega, h_{\mathrm{root}}, v_{\mathrm{old}}, q_{\mathrm{old}}, l_{\mathrm{old}}, H_{\mathrm{small}}, w_{\mathrm{old}}$$

equals its displayed legacy counterpart, with H_N corresponding to H_L and U_A to U_{AL}. Also

$$(T_{1/2}, x, y, k_1, n_0, k_4, \zeta, e_{\mathrm{new}}, N_{\mathrm{new}}) = (T_L, x_L, y_L, k_{1L}, n_{0L}, k_{4L}, \zeta_L, e_L, N_L),$$

and $N_{\mathrm{old}} = N_{\mathrm{old},L}$. In particular

$$h_K = 0, \quad o_+ = 78, \quad n \leq 80, \quad k \leq 7, \quad H_N \leq 3137,$$
$$p \leq 12586, \quad h_s \leq 36, \quad O < 2^{76}, \quad o \leq 76, \quad t < 2^{36},$$
$$x = T_{1/2} + p + 21, \qquad y = T_{1/2} + 2p + 61,$$
$$k_1 = 4T_{1/2} + 5p + 127, \qquad n_0 = 8T_{1/2} + 10p + 258,$$
$$k_4 = 16T_{1/2} + 20p + 532, \qquad e_{\mathrm{new}} = 16\lceil t/2 \rceil + 20p + 4d + 544.$$

This mathematical compatibility identifies no old/new request or certificate and supplies no software extension.

All clauses retain ordered repeats, overlap, touching, separation and singular laws in the full ambient $\mathbb{R}^r$. They assert no independence of interacting observations. Widths are sufficient, possibly astronomical, whole-plane process-neuron counts, each retaining countably many internal variables. Practical, minimal or necessary complexity, finite global scalar weights, measured speedup, canonical least inner indices, an observable-independent effective common sequence, global neural Schwartz control, whole-field rates, larger models/dimensions/test families, novelty priority and public release remain outside these clauses.

K.6 Complete conditional proof

The following is the author derivation of the exact frozen conditional statement. Its premise is the explicit finite interface above and the literal inherited qualifications below. The original source truths and NR1 author-versus-blind-core scopes are retained. In particular, the full laws are supplied at every finite positive stage by EB1/NR1/P6; bounded SVA and fixed-$n75$ VEC acceptances are not extrapolated. Strict AP6 applies because $0 < 10^{-10} < 1/8000$ and identifies the original FS ν^X without the old g_* inequality. Finite diagnostics are evidence, rather than premises.

K.6.1 Observation geometry and the literal coefficient

For $|t| < 1$, each derivative of $b(t)$ is $e^{-1/(1-t^2)}$ times a rational function whose endpoint poles have finite order. For every integer $k \geq 0$, $x^k e^{-x} \to 0$ as $x \to \infty$, for example by the positive exponential series. All derivatives therefore extend by zero at both endpoints. Thus b and its translates are real, nonzero and compact smooth. Since $\max b = e^{-1}$, $f_h \leq e^{-2} < 1$, and its support is the stated closed square of area four. Hence $1 \leq R \leq 9$, all supports lie in $[-9, 9]^2$, and $K \subset \Lambda_1 = (-1000, 1000)^2$, with the latter square open.

Translation invariance and the support/height bounds give $\|f_h\|_1 \leq 4$, $\|f_h\|_2 \leq 2$ and $\|f_h\|_{4/3} \leq 4^{3/4} < 3$, because $64 < 81$. Taking the square sum over the actual ordered r coordinates gives $F_1 \leq 4\sqrt{r} \leq 4c_r$, $F_2 \leq 2c_r$ and $F \leq 3c_r = \overline{F} \leq 6$. Hölder's inequality coordinate by coordinate gives

$$|\mathcal{O}_f(v)|_2^2 \leq \|v\|_4^2 \sum_i \|f_{h_i}\|_{4/3}^2 = F^2 \|v\|_4^2. \tag{1.1}$$

Repeated tests remain in this sum. No support separation, covariance rank or inverse matrix enters the argument.

The same J_{CL1} and chosen H1/H2 coefficients occur in both literal source coefficients, so their ratio is exactly

$$\frac{C_f(\mathbf{h}; R)}{C_f(\mathrm{base}; 1)} = \frac{\sqrt{r}(4R+1)}{5}.$$

The base enclosure gives $C_f \leq \overline{C}$. Since $c_r(4R+1)/5 \leq 74/5 < 16$, $\overline{C} < 2^{364}$. The separate factor $e^{R/256}$ is retained. This proves the observation and coefficient portion of UDA.a.

K.6.2 Universal outer selection and the four actual errors

For $q \geq 3$, $d = q + 3$ is the least integer with $5 \cdot 2^{-d} \leq 2^{-q}$, since $4 < 5 \leq 8$. For $\tau = 1/5$ the least integer is $d = 5$. Thus $d \geq 5$, $\eta = 2^{-d} > 0$ and $5\eta \leq \tau$. There is no upper bound on d in proof steps 2–11.

The positive degree-eight exponential partial at $125/32$ is

$$\sum_{j=0}^{8} \frac{(125/32)^j}{j!} = \frac{432528188408407789}{8866461766385664} > 32.$$

Therefore $0 < q_{\mathrm{sp}} = e^{-125/32} < 1/32$. For $0 \leq v < 1$, the exponential series is termwise at most the geometric series, so $e^v \leq (1-v)^{-1}$. In particular $e^{R/256} \leq 256/(256-R) < 2$. Both positive shell bracket terms increase with $q_{\mathrm{sp}} \in (0, 1)$; substituting $1/32$ gives exactly $(2n+1)32/31 + 64/961$. Since $4u\ell^2 = 4/10000$ and $C_f \leq \overline{C}$,

$$E_n \leq \overline{E}(n). \tag{2.1}$$

The rational bracket is $(1984n+1056)/961 < 4n$ for $n \geq 1$. Also $(4/10000)\overline{\exp R} < 8/10000 < 2^{-10}$ and $\overline{C} < 2^{364}$. Thus

$$\overline{E}(n) < 2^{356} n 2^{-5n} \leq 2^{356-4n}, \qquad n \geq 1, \tag{2.2}$$

where $n \leq 2^n$ follows by induction. The positive integer $n_w = \lceil (d+356)/4 \rceil$ is a termination witness. The defining set for n is nonempty and its minimum is obtained by finite exact rational comparisons. Furthermore $\overline{E}$ decreases strictly: the ratio of successive positive linear brackets is below two, while 2^{-5n} contributes $1/32$. At the selected n, $1000n \geq R$, so the actual AP6 geometry and (2.1) give $E_n \leq \overline{E}(n) \leq \eta$. The physical area $A = 4000000n^2$ is positive finite; $n \leq 80$ is not used.

For positive rational v, $2^h \geq h+1$ for integers $h \geq 0$. Hence $\operatorname{clog}_2(v)$ terminates no later than $h = \lceil v \rceil$. Likewise $4^h \geq h+1$ makes $\lceil 14u^2 A \rceil$ a termination witness for h_K. These rational bounds may be loose. Using $\sqrt{85} < 10$ and $\pi > 3$,

$$K_\Lambda^2 = \frac{75\sqrt{85}}{2\pi^3} u^2 A < 14u^2 A \leq \overline{K}^2.$$

Thus $K_\Lambda < \overline{K}$. Since $e > 2$ and $\sqrt{2} < 2$, the second K_{sharp} branch is $2K_\Lambda \sqrt{(1+\sqrt{2})/e} < 4\overline{K}$. Consequently $K_{\text{sharp}} \leq \max\{12U_A, 4\overline{K}\}$. Also

$$b_{\text{Nelson}} = \frac{\pi^2}{48U_A} > b_0 = \frac{3}{16U_A} > 0, \qquad \frac{64}{e b_{\text{Nelson}}} < \frac{32}{b_0}.$$

The integer $k_w = \lceil 32/b_0 \rceil \geq 1$ satisfies $L_{10}(k_w) \geq 1 + k_w > 32/b_0$, so $k \leq k_w$ is a finite minimum. The positive exponential remainder gives $e^k > L_{10}(k) > 32/b_0 > 64/(e b_{\text{Nelson}})$, hence $\log_+(64/(e b_{\text{Nelson}})) < k$. Therefore the original, volume-dependent Nelson bound obeys

$$\begin{aligned} H_2 &\leq \max\{12U_A, 4\overline{K}, 4k^2/b_0\} < H_N, \\ \sqrt{\Xi_2} &\leq \sqrt{\overline{\Xi}_2} = \sqrt{2} e^{H_2} < 2^{2H_N+1}. \end{aligned} \tag{2.3}$$

Here $\log 2 > 1/2$, because $e^{1/2} = \sqrt{e} < \sqrt{3} < 2$, so $e^z \leq 2^{2z}$ for $z \geq 0$. The strict extra unit in the integer H_N preserves strictness even when the first maximum branch is attained. This uses the actual Ξ_2 at this physical area and its original common-space fourth-chaos/L^2 interface. Finite-regulator stability and Jensen alone would not establish the zero-regulator bound, and no volume-uniform Ξ_2 is claimed.

The selected even integer $p = 4H_N + 2h_K + 2d + 8$ is at least four. The actual regulator bound with $F_2 \leq 4$ gives

$$R_{\text{phys}} < 2^{2H_N + h_K + 3 - p/2} + 2^{2H_N + 2 - p}. \tag{2.4}$$

The first exponent is exactly $-d-1$; the second is $-2H_N - 2h_K - 2d - 6 \leq -d-1$. Their sum is at most η. The first two-summand predicate is equivalent to $p \geq 4H_N + 2h_K + 2d + 8$. This even value satisfies the second predicate, so it is exactly the first even sufficient $p \geq 4$. Thus $0 < a = 2^{-p} < 1$ and $R_{\text{phys}} \leq \eta$. The actual K_Λ contribution has not been replaced by one.

For $p \geq 4$, $\sqrt{1 + 2^{2p}} < 2 \cdot 2^p$, whence $\operatorname{asinh}(2^p) < \log(3 \cdot 2^p) < p+2$, using $\log 2 < 1$ and $\log 3 < 2$. Since $2\pi > 6$,

$$c \leq \bar{c}(a) < (p+3)/6 \leq p. \tag{2.5}$$

Only positive majorants use p; the physical c remains in V and the Wick ordering. Finite-regulator Wick stability is the pointwise identity $z^4 - 6cz^2 + 3c^2 = (z^2 - 3c)^2 - 6c^2 \geq -6c^2$. The physical/residual weights are therefore bounded by $e^{6U_A c^2}$, the physical Jensen normalizer is at least one, and the

residual Jensen normalizer is at least $e^{-3U_A c^2}$, exactly as in the original conditional interface. None requires $U_A < 3$.

Let $j = \lceil 18U_A p^2 \rceil$. By (2.5) and $\log 2 > 1/2$, $e^{9U_A c^2} < 2^j$. Since $\sqrt{15} < 4$ and $p \geq 1$, $(4\sqrt{15}+12)c^{3/2} < 32p^2$. The complete auxiliary bracket satisfies

$$2U_A(4\sqrt{15}+12)c^{3/2} + F_1 < 64U_A p^2 + 4c_r = B_{\text{aux}} \leq 2^{h_{\text{aux}}}. \tag{2.6}$$

All arguments before their clog_2 operations are finite positive rationals. In particular $Z_s = j + h_{\text{aux}} + p + d > 1$, and $h_s = 1 + \operatorname{clog}_2(Z_s)$ is exactly the first integer at least one with $2^{h_s - 1} \geq Z_s$. Thus $s = 2^{h_s} \geq 2$ is finite, and

$$R_{\text{aux}} - \max(1000n, R) - s = 1,$$

because $R \leq 9 < 1000n$. A Euclidean displacement of length less than s changes each coordinate by less than s. The strict s-neighborhoods of both $\overline{\Lambda}$ and K lie in Ω, so the original covariance-defect bound applies to both. Discarding the minimum in $\rho_a(s)$ gives

$$\sqrt{\rho_a(s)} \leq \sqrt{\frac{\kappa_a^2}{2\pi}} e^{-s/\sqrt{2}} < 2^p e^{-s/2} < 2^{p - s/2}. \tag{2.7}$$

The final strict inequality uses $e > 2$, $s > 0$ and no upper bound on p. Multiply by the same physical exponential and both vector terms in (2.6):

$$A_{\text{aux}} < 2^{j + h_{\text{aux}} + p - s/2} \leq 2^{-d} = \eta. \tag{2.8}$$

Since $O = 4R_{\text{aux}}^2 > A$, $B_{\text{ext}} = O - A > 0$. With $h_{\text{stab}} = \operatorname{clog}_2(6Op^2)$ and $t = \max\{34, j + h_{\text{stab}} + d\}$, the actual stabilizer error obeys

$$S < 2^{j + h_{\text{stab}} - t} \leq \eta. \tag{2.9}$$

Also $\epsilon = 2^{-t} \leq 2^{-34} < u$, because $2^{34} > 10^{10}$. Thus $\alpha = \epsilon$ and $U = u$ in every width formula. All later stages are finite compositions of these integers and rational searches, and $\operatorname{clog}_2(v) \leq \lceil v \rceil$ covers each remaining search. No equality oracle, canonical inner index or expansion of an infinite sequence is used. This proves the rest of UDA.a, including the four actual downstream errors.

K.6.3 Arbitrary bridges, coercivity and full normalizers

Fix any finite positive stage satisfying the premises, with $0 < \epsilon < u$; proof steps 3–8 use no selected-stage or bounded-q consequence. Fix an architecture, $N \geq 1$, j, a branch h and $\theta \in [0, 1]$. Before tilt, v is a sum of N independent centered process random elements, $N - 1$ with coefficient $1/\sqrt{N}$ and one with coefficient $\theta/\sqrt{N}$. At almost every point z each unscaled variance is $1/4$; neuron and Gaussian fourth moments are $1/16$ and $3/16$, respectively. Independence of summands and centering give, for real β_i,

$$\begin{aligned} \mathbb{E}\left(\sum_i \beta_i \xi_i(z)\right)^2 &= \frac{1}{4}\sum_i \beta_i^2, \\ \mathbb{E}\left(\sum_i \beta_i \xi_i(z)\right)^4 &= \sum_i \beta_i^4 \mathbb{E}\xi_i(z)^4 + \frac{6}{16}\sum_{i<k} \beta_i^2 \beta_k^2 \\ &\leq \frac{3}{16}\left(\sum_i \beta_i^2\right)^2. \end{aligned} \tag{3.1}$$

Here the squared coefficient sum is $(N-1+\theta^2)/N \le 1$. This includes $N=1$, both endpoint amplitudes and every extreme j. No spatial independence or Gaussian domination of neuronal joint moments is used. Tonelli on finite D gives a finite fourth E-norm moment.

Write $x = \|v\|_4$ and take $c_* \ge c$. The quartic coefficient in V lies in $[\alpha, U]$, the negative quadratic coefficient is pointwise at most B, and the constant is nonnegative. Hölder and completion of the square in x^2 give

$$V(v) \ge \alpha x^4 - B\sqrt{O}\, x^2 \ge \alpha x^4/2 - \frac{B^2 O}{2\alpha} = \alpha x^4/2 - L. \tag{3.2}$$

Therefore $0 < w(v) \le e^{L-\alpha x^4/2} \le e^L$. Every Z is finite and strictly positive, as the expectation of a strictly positive finite random variable. Every term of V is integrable by (3.1). Discarding negative quadratic terms only in the upper bound yields $\mathbb{E}V(v) \le 3UO/16 + 3uc^2A \le J_z$. Jensen gives $Z \ge e^{-J_z}$. For the actual joint law Π of (S_j, h) it follows that

$$\frac{d\Pi}{d\,\mathrm{Law}(S_j,h)} \le e^{H-\alpha x(v)^4/2} \le e^H, \qquad \Pi(x(v)^4 > z) \le \min\{1, e^{H-\alpha z/2}\}. \tag{3.3}$$

These are full neural/Gaussian-tilt partitions. The physical Wick normalizer $Z_a \ge 1$ is neither used here nor asserted for Z.

K.6.4 Joint tilted field and direction moments through sixteen

If $T \ge 0$ and $\mathbb{P}(T > z) \le \min\{1, e^{a-z}\}$ with $a \ge 0$, then T is stochastically dominated by $a + E_1$, where E_1 has unit exponential law. This is a distribution-function comparison, rather than a joint independence assumption. Integrating $qz^{q-1}\mathbb{P}(T > z)$, and integrating by parts for exponential moments, gives for every positive integer q

$$\|T\|_q \le a + (q!)^{1/q}. \tag{4.1}$$

For $T = \alpha x(v)^4/2$, (3.3) gives $a = H$. At order four, $\|T\|_4 \le H + 24^{1/4} < H + 4$, so

$$\|x(v)\|_{L^{16}(\Pi)} \le [2(H+4)/\alpha]^{1/4} = X_{\mathrm{rel}}. \tag{4.2}$$

Lyapunov's inequality gives every real order between one and sixteen.

For the Gaussian direction, Minkowski exchanges L^{2k}(probability) and $L^4(D)$ for $k \ge 2$. Its point variance is $1/4$, hence

$$\mathbb{E}\|\gamma\|_4^{2k} \le O^{k/2}\mathbb{E}|\gamma(z)|^{2k} = O^{k/2}\frac{(2k-1)!!}{4^k} \le O^{k/2}(k/2)^k. \tag{4.3}$$

The equality is the constant point-moment value; the first comparison is an inequality and needs no spatial independence. At $k=1$, use $\mathbb{E}\|\gamma\|_4^2 \le \sqrt{\mathbb{E}\int_D \gamma^4} = \sqrt{3O}/4 \le \sqrt{O}/2$. The positive exponential series for $T_\gamma = \|\gamma\|_4^2/(4\sqrt{O})$ therefore gives

$$\mathbb{E}e^{T_\gamma} \le 1 + \sum_{k\ge1}\frac{k^k}{8^k k!} \le \sum_{k\ge0}(e/8)^k < 8/5 < 4. \tag{4.4}$$

The factorial bound $k! \ge (k/e)^k$ follows by integrating log on $[1,k]$. Tonelli and positivity justify the series interchange. Under the actual joint Π, density domination (3.3) gives $\mathbb{E}_\Pi e^{T_\gamma} < 4e^H$.

Markov's inequality and $\log 4 < 2$ imply $\Pi(T_\gamma > z) \le \min\{1, e^{H+2-z}\}$. Apply (4.1) at order eight and use $(8!)^{1/8} < 8$: $\|T_\gamma\|_{L^8(\Pi)} < H + 10$. Consequently

$$\|\|\gamma\|_4\|_{L^{16}(\Pi)} \le 2O^{1/4}(H+10)^{1/2} = Y_{\mathrm{rel}}. \tag{4.5}$$

For a neuron, $\|g^b\|_4 = O^{1/4}/2$ deterministically, including after tilt, and the same Y_{rel} bounds it. Lyapunov gives every real order through sixteen. Conditional independence of v, h after tilt is never used.

K.6.5 Full Fréchet derivatives and differentiable partitions

Quartic and quadratic integrals on $L^4(D)$ are continuous Fréchet polynomials. For unit L^4 directions h_i, the first derivative is the integral of $4k(z)v^3h_1 - 2b(z)vh_1$, with quartic coefficient $k(z) \le U$ and quadratic coefficient $b(z) \le B$. The constant $3uc^2A$ differentiates to zero. Higher derivatives have coefficients $12k(z), 24k(z), 24k(z)$, with a quadratic contribution only at order two. Hölder for four L^4 factors and the factor $\sqrt{O}$ for two give

$$\begin{aligned}
\|DV(v)\| &\le 4Ux^3 + 2B\sqrt{O}\,x = v_1(x),\\
\|D^2V(v)\| &\le 12Ux^2 + 2B\sqrt{O} = v_2(x),\\
\|D^3V(v)\| &\le 24Ux = v_3(x), \qquad \|D^4V(v)\| \le 24U = v_4(x).
\end{aligned} \tag{5.1}$$

These are full multilinear norms.

The exponential chain rule, written as a sum over partitions of $\{1, \dots, k\}$, expresses $D^k w$ as w times products of derivatives of $-V$. Norms give exactly $e_0, \dots, e_4$. At order four the multiplicities are $1, 6, 3, 4, 1$ for $v_1^4, v_1^2v_2, v_2^2, v_1v_3, v_4$. Thus

$$\|D^k w(v)\|_{\mathrm{op}} \le w(v)e_k(x), \qquad 0 \le k \le 4. \tag{5.2}$$

Each e_k is a nonnegative-coefficient polynomial of degree at most $3k$. By (3.2), its bound is a polynomial times $e^{L-\alpha x^4/2}$. Each monomial is globally bounded: for $k > 0$, $\sup_{x\ge 0} x^k e^{-\alpha x^4/2} = (k/(2\alpha e))^{k/4}$. Hence w and all derivatives through four are globally bounded.

Differentiation of $Z(\theta)$ is dominated by $M\|h\|_4/\sqrt{N}$ for a finite deterministic global derivative bound M. The untilted direction has finite first E-norm moment. Dominated convergence therefore gives a continuous derivative on every compact real θ interval. In fact Z extends C^1 to all real θ and stays positive; therefore $\log Z$ also extends C^1. On $[0, 1]$,

$$(\log Z)' = -N^{-1/2}\mathbb{E}_\Pi DV(v)[h]. \tag{5.3}$$

Under the same joint Π, Hölder at order four for $x^3\|h\|_4$ and at order two for $x\|h\|_4$, together with (4.2),(4.5), gives

$$\mathbb{E}_\Pi|DV(v)[h]| \le 4UX_{\mathrm{rel}}^3Y_{\mathrm{rel}} + 2B\sqrt{O}X_{\mathrm{rel}}Y_{\mathrm{rel}} = K_1. \tag{5.4}$$

Integrate (5.3) between θ and one on the neuron branch to obtain the stated neuron/neuron ratio. At zero both branches have the same S_j, so their partitions agree. Integrating from zero on both branches gives

$$\frac{Z_{j,\gamma}(\theta)}{Z_{j,g}(1)} \le e^{(\theta+1)K_1/\sqrt{N}} \le e^{2K_1/\sqrt{N}}. \tag{5.5}$$

If $N \ge N_0$, then $N \ge 16K_1^2$, so the last expression is at most $e^{1/2} = \sqrt{e} < 2$. This also bounds the neuron/neuron ratio by two. The denominator is the actual neuron endpoint partition, rather than a physical normalizer.

K.6.6 The complete centered product derivative and mixed moments

For $G \in \mathcal{U}_4(r)$, (1.1) bounds $D^q(G \circ \mathcal{O}_f)$ by F^q for $1 \le q \le 4$, and its absolute value by one. The deterministic m_j lies in $[-1, 1]$. The full fourth product rule for $\Phi_j = (G \circ \mathcal{O}_f - m_j)w$ gives

$$\|D^4\Phi_j(v)\|_{\mathrm{op}} \le w(v)\left[2e_4(x) + \sum_{q=1}^{4}\binom{4}{q}F^q e_{4-q}(x)\right] = w(v)P_F(x). \tag{6.1}$$

The leading two copies of e_4 account for $|G - m_j| \le 2$. Every observation binomial term and every Bell multiplicity remains. The lower-order product calculations followed by (3.2) likewise show $\Phi_j \in C_b^4(E)$.

Write $P_F(x) = \sum_{k=0}^{12} a_k x^k$ with $a_k \ge 0$ and $r_h = \|h\|_4$. For $k > 0$, Hölder with conjugate exponents $(k+4)/k$ and $(k+4)/4$ gives

$$\begin{aligned}\mathbb{E}_\Pi x^k r_h^4 &\le (\mathbb{E}_\Pi x^{k+4})^{k/(k+4)}(\mathbb{E}_\Pi r_h^{k+4})^{4/(k+4)} \\ &\le X_{\mathrm{rel}}^k Y_{\mathrm{rel}}^4.\end{aligned} \tag{6.2}$$

At $k = 0$ this is the fourth direction-moment bound. Since $k + 4 \le 16$, all moments used are established. Summing nonnegative coefficients yields $\mathbb{E}_\Pi[P_F(x)r_h^4] \le P_F(X_{\mathrm{rel}})Y_{\mathrm{rel}}^4$. The actual joint law is used throughout; dependent marginal expectations are never multiplied.

K.6.7 Centered adjacent replacement and the normalized telescope

Taylor's theorem for Φ_j at S_j in direction $h/\sqrt{N}$, through order three, has remainder

$$R_h = N^{-2}\int_0^1 \frac{(1-\theta)^3}{6} D^4\Phi_j(S_j + \theta h/\sqrt{N})[h, h, h, h], d\theta. \tag{7.1}$$

Global derivative bounds and the finite untilted fourth E-norm moment justify exchanging expectations, conditioning on S_j and integration. Independence is used before tilt. Conditional on S_j, the first terms vanish by zero Bochner means. The second terms agree by the premise for every bounded bilinear form, applied to the continuous form $D^2\Phi_j(S_j)$. The third terms vanish by global sign symmetry against each continuous homogeneous cubic; the third E-moment guarantees integrability. Constant terms agree. The forms are measurably evaluable because E is separable, the random elements are strongly measurable, and derivatives are continuous. The full bilinear match is an explicit premise, rather than an inference from point-pair covariance.

By definition of m_j, $\mathbb{E}\Phi_j(S_j + \gamma/\sqrt{N}) = 0$. Consequently, for the actual normalized endpoint expectations,

$$L_g(G) - L_\gamma(G) = \frac{\mathbb{E}R_g - \mathbb{E}R_\gamma}{Z_{j,g}(1)}. \tag{7.2}$$

For branch h and amplitude θ, the expected absolute integrand in (7.1), divided by $Z_{j,g}(1)$, is at most

$$\frac{Z_{j,h}(\theta)}{Z_{j,g}(1)}P_F(X_{\mathrm{rel}})Y_{\mathrm{rel}}^4.$$

For $N \ge N_0$ each ratio is at most two by (5.5), and $\int_0^1(1-\theta)^3/6\,d\theta = 1/24$. The two branch remainders sum to

$$\frac{4P_F(X_{\mathrm{rel}})Y_{\mathrm{rel}}^4}{24N^2} = \frac{K_4(F)}{N^2}. \tag{7.3}$$

Let H_j be the law of the normalized N-summand mixture with j Gaussian and $N-j$ neuronal summands, then tilted by the same w and its own normalizer. The neuron endpoint of S_j is H_j and the Gaussian endpoint is H_{j+1}. This follows from the original independent product copies and equality in law, hence also equality after common tilt and normalization. Thus $H_0 = P_N^b$ is the actual neural law. The normalized sum of N independent copies of the centered Gaussian E-valued element γ has exactly the γ law: all finite continuous-linear-functional laws agree, and these determine Borel probabilities on separable E. Hence $H_N = Q_\epsilon$ is the actual full Gaussian tilt. Summing (7.3) over $j = 0, \dots, N-1$ gives $K_4(F)/N$. The bound $|G| \le 1$ supplies the cap two. This proves the normalized part of UDA.c and completes UDA.b.

K.6.8 Ambient smoothing, including singular observation laws

Apply the ambient $\mathbb{R}^r$ L0 premise to a unit BL G. For $0 < \zeta \le 1$, $G_\zeta/(\sqrt{6}\zeta^{-3}) \in \mathcal{U}_4(r)$. The two smoothing errors are each at most $\sqrt{r}\zeta$, so

$$d_{\mathrm{BL}}(\mathcal{L}_f(P_N^b), \mathcal{L}_f(Q_\epsilon)) \le \min\{2, \sqrt{6}K_4(F)/(N\zeta^3) + 2\sqrt{r}\zeta\}. \tag{8.1}$$

Put $A_0 = \sqrt{6}K_4(F)$, $B_0 = 2\sqrt{r}$ and $N_* = 3A_0/B_0$. For $N \ge N_*$, choose $\zeta = (N_*/N)^{1/4} \le 1$. This gives $(4/3)(3A_0)^{1/4}B_0^{3/4}N^{-1/4}$, whose coefficient has fourth power $(2048\sqrt{6}/27)r^{3/2}K_4(F)$. For $N < N_*$ that expression exceeds $(4/3)B_0 \ge 8/3 > 2$, so the cap two gives the same optimized bound. Both cases retain $N \ge N_0$.

Smoothing stays on the full original $\mathbb{R}^r$. If two tests coincide, their coordinates coincide under every law, but (1.1), (6.1) and (8.1) retain both coordinates. Touching, overlap and separation change neither these arguments nor the moment estimates. No covariance inverse, density, rank reduction or independence of interacting observations was assumed.

K.6.9 Stage-dependent logarithmic enclosures for every finite accuracy

Specialize the arbitrary finite-stage result to the universal selection, with $c_* = p$. We have $p \ge 4$, $t \ge 34$, $A < O$, $O \le 2^o$ and $d \le t$. No old-domain bound on n, U_A, p, O, t or e_{new} is used. Since $u < 1/6$ and $p \le 2^{2p-1}$ for $p \ge 4$,

$$B = (2^{2p}-1)/2 + 6up < 2^{2p-1} + p \le 2^{2p}. \tag{9.1}$$

The polynomial-exponential comparison holds at $p = 4$ and persists because the right side multiplies by four and $(p+1)/p \le 5/4$. Similarly $3p^2 < 2^{4p}$ holds at four and persists since $((p+1)/p)^2 \le 25/16 < 16$. These are universal integer inequalities; $6up < 1$ is unnecessary and eventually fails.

Write $l = l_{\mathrm{old}} = t + 4p + o$ and $l_+ = t + 4p + o_+$. The full coercivity and Jensen terms satisfy

$$\begin{aligned} L = B^2O/(2\alpha) &< 2^{t+4p+o-1} = 2^{l-1}, \\ 3uO/16 < O \le 2^o &\le 2^{l-1}, \\ 3up^2A < 3up^2O &< 2^{t+4p+o-1} = 2^{l-1}. \end{aligned}$$

The last bound uses $A < O$, $3up^2 < 3p^2 < 2^{4p}$ and $t \ge 34$. Thus $J_z < 2^l$ and $H = L + J_z < 2^{l+1} \le 2^{l_+ + 1}$. Since $l \ge 50$, both $H + 4, H + 10 < 2^{l_+ + 2}$. Every term of the physical-area contribution $3up^2A$ remains, with $A \le O$; no bounded-U_A shortcut is hidden. Hence

$$X_{\mathrm{rel}}^4 < 2^{2t+4p+o_+ + 3},$$

$$Y_{\mathrm{rel}}^4 = 16O(H+10)^2 < 2^{2t+8p+3o_+ +8}. \tag{9.2}$$

The actual o is first bounded by o_+ in the second line. Since $4T_{1/2} \geq 2t$ at both parities of t, the definitions of x, y give $X_{\mathrm{rel}} < 2^x$, $Y_{\mathrm{rel}} < 2^y$. Strictness survives even when a ceiling has an integer argument.

The quadratic derivative coefficient obeys

$$2B\sqrt{O} < 2^{2p+\lceil o/2\rceil+1}, \qquad 2p+\lceil o/2\rceil+1 \leq 2x. \tag{9.3}$$

Indeed $2\lceil (o_+ +3)/4\rceil \geq (o_+ +3)/2 \geq \lceil o/2\rceil + 1$, and the extra $2T_{1/2}$ is nonnegative. Since $4u < 1$,

$$\begin{aligned} v_1(X_{\mathrm{rel}}) &< 2^{3x} + 2^{2p+\lceil o/2\rceil+1+x} < 2^{3x+1}, \\ K_1 &< 2^{3x+y+1} = 2^{k_1-2}. \end{aligned} \tag{9.4}$$

In particular $16K_1^2 < 2^{2k_1}$. The right side is an integer; the ceiling of a real number strictly below integer M is at most M. Consequently

$$N_0 = \max\{1, \lceil 16K_1^2\rceil\} \leq 2^{2k_1} < 2^{2k_1+4} = 2^{n_0}. \tag{9.5}$$

This handles the ceiling without asserting strictness at that same exponent.

Let $W = 2^{3x+5} \geq 1$. At X_{rel}, (9.3)–(9.4) give $v_1 < W$. For v_2 the two terms are bounded by 2^{2x+4} and 2^{2x}; their sum is below $2^{2x+5} \leq W$. Similarly $v_3 < 2^{x+5} \leq W$ and $v_4 < 2^5 \leq W$. The complete Bell polynomials therefore satisfy

$$e_0 \leq 1, \quad e_1 \leq W, \quad e_2 \leq 2W^2, \quad e_3 \leq 5W^3, \quad e_4 \leq 15W^4.$$

Each is at most $16W^4$. Keeping both centered copies of e_4 and all four observation derivative terms, with $F \leq \overline{F} \leq 6$, gives

$$\begin{aligned} P_F(X_{\mathrm{rel}}) &\leq P_{\overline{F}}(X_{\mathrm{rel}}) \\ &\leq 16W^4 \left[2 + \sum_{j=1}^{4} \binom{4}{j} 6^j\right] = 16(1+7^4)W^4 \\ &= 38432W^4 < 2^{12x+36}. \end{aligned} \tag{9.6}$$

Here $38432 < 2^{16}$ is the full vector coefficient; the smaller scalar coefficient is not substituted. Combining $Y_{\mathrm{rel}}^4 < 2^{4y}$ and enlarging only upward by dropping the divisor six yields

$$K_4(F) \leq K_4(\overline{F}) < 2^{12x+4y+36} = 2^{k_4}. \tag{9.7}$$

All volume, variance, stabilizer and physical-area dependence remains. There is no use of $q \leq 12$, $n \leq 80$, $U_A < 3$, $p < 20000$, $O < 2^{76}$, $t < 2^{36}$ or $e_{\mathrm{new}} < 2^{40}$. This proves UDA.c's universal exponent part from the arbitrary finite-stage joint-law proof.

K.6.10 Sufficient width and the actual five-arrow target triangle

Take $\zeta = 2^{-d-3}$, $e_{\mathrm{new}} = k_4 + 4d + 12$ and $N_{\mathrm{new}} = 2^{e_{\mathrm{new}}}$. These are finite positive mathematical numbers and $0 < \zeta \leq 1$. Exact integer algebra gives

$$e_{\mathrm{new}} - n_0 = 6x + 2y + 4d + 38 > 0. \tag{10.1}$$

Thus $N_{\mathrm{new}} > 2^{n_0} > N_0$. Apply (8.1) with $\sqrt{6} < 3$, $\sqrt{r} \le 2$ and (9.7):

$$\frac{\sqrt{6}K_4(F)}{N_{\mathrm{new}}\zeta^3} < 3 \cdot 2^{k_4 - e_{\mathrm{new}} + 3d + 9} = 3\eta/8,$$
$$2\sqrt{r}\zeta \le 4 \cdot 2^{-d-3} = \eta/2. \tag{10.2}$$

The actual neural-to-full-Gaussian-tilt distance is below $7\eta/8 < \eta$. This completes the selected-width part of UDA.c; steps 3–8 already prove all its arbitrary finite-stage clauses.

The four quantities bounded in step 2 belong to the same actual $Q_\epsilon, Q_0, Q_a, Q_{\mathrm{phys}}, \nu^X$ chain in the source premises, with the same physical and auxiliary squares. They do not depend on N. Physical and full endpoint-reference normalizers have stayed separate; Q_ϵ is the full Gaussian tilt, distinct from a physical zero-regulator law. Strict AP6 identifies the last endpoint with the original pressure target. On the full original $\mathbb{R}^r$, the triangle gives

$$\begin{aligned} d_{\mathrm{BL}}(\mathcal{L}_f(P^b_{N_{\mathrm{new}}}), \mathcal{L}_f(\nu^X)) &\le E_{\mathrm{width}} + S + A_{\mathrm{aux}} + R_{\mathrm{phys}} + E_n \\ &\le \overline{E}(n) + 4\eta \le 5\eta \le \tau. \end{aligned} \tag{10.3}$$

The width error has the strict margin (10.2). This holds for every $q \ge 3$ with $d = q + 3$ and for the separate 1/5 case with $d = 5$. It proves UDA.d for the unchanged target, architectures and observations. The optional probability cap two does not repair an allocation or a failed stage.

K.6.11 The complete old coefficient and same-stage comparison

First derive the old law coefficient at arbitrary finite stages from the same full reference laws; bounded SVA/ACC acceptance is not extrapolated. For $\Psi \in C^4_b(E)$ with fourth derivative norm bounded by M, expand around the common S_j through order three. The centering, every-bilinear match and global sign-symmetry cancellations of step 7 apply. Each fourth remainder is at most $M\|h\|_4^4/(24N^2)$. Tonelli and the point moments give $\mathbb{E}\|g\|_4^4 = O/16$, $\mathbb{E}\|\gamma\|_4^4 = 3O/16$. After N replacements,

$$|\operatorname{Law}(F^b_N)(\Psi) - \operatorname{Law}(\gamma)(\Psi)| \le \frac{OM}{96N}. \tag{11.1}$$

The normalized Gaussian iid sum has exactly the γ law on separable E, as above. No tilt, independence after tilt, normalizer premise or threshold N_0 is required in this raw estimate.

The polynomial-exponential bound (3.2) and the complete derivatives (5.1)–(5.2) give

$$\begin{aligned} \sup \|D^4 w\| &\le \mathcal{T}(e_4) = M_Z, \\ \sup \|D^4[(G \circ \mathcal{O}_f)w]\| &\le \mathcal{T}\left(\sum_{j=0}^{4} \binom{4}{j} F^j e_{4-j}\right) = M_A. \end{aligned} \tag{11.2}$$

For $k > 0$, differentiation shows $\sup_{x \ge 0} x^k e^{-\alpha x^4/2} = \beta_k$, and $\beta_0 = 1$. Summing nonnegative coefficients bounds the polynomial supremum from above. Lower derivatives are also bounded, so (11.1) applies legitimately. Let A_N, A_γ be weighted G expectations and Z_N, Z_γ their actual full partitions. Then $\delta_A \le OM_A/(96N)$ and $\delta_Z \le OM_Z/(96N)$. Both partitions are at least $z = e^{-J_z}$ by the separate endpoint-reference Jensen estimate, while $|A_\gamma/Z_\gamma| \le 1$. The exact identity

$$\frac{A_N}{Z_N} - \frac{A_\gamma}{Z_\gamma} = \frac{A_N - A_\gamma}{Z_N} + \frac{A_\gamma}{Z_\gamma}\frac{Z_\gamma - Z_N}{Z_N}$$

gives normalized error at most $(\delta_A + \delta_Z)/z = C_{\mathrm{old}}/N$, using one power of $1/z$. It is also at most two. The same ambient L0 smoothing and optimizer as in step 8, with the cap two handling an optimizer above the allowed smoothing scale, give

$$d_{\mathrm{BL}}(\mathcal{L}_f(P_N^b), \mathcal{L}_f(Q_\epsilon)) \le \min\{2, D_{\mathrm{old}} N^{-1/4}\}, \quad D_{\mathrm{old}}^4 = \frac{2048\sqrt{6}}{27} r^{3/2} C_{\mathrm{old}}, \quad N \ge 1. \tag{11.3}$$

This proves the first part of UDA.e at all finite stages. Nonnegative coefficients ensure that replacing F by $\overline{F}$ and c by p only enlarges the old coefficient; V itself is unchanged.

At universal selected stages let $h_{\mathrm{root}} = \mathrm{clog}_2(2R_{\mathrm{aux}})$, $v = v_{\mathrm{old}} = 2p + h_{\mathrm{root}} + 2$ and $V_0 = 2^v$. Then $\sqrt{O} = 2R_{\mathrm{aux}} \le 2^{h_{\mathrm{root}}}$. By (9.1), the coefficient sums of v_1, v_2 are strictly below $4 + 2^{2p+h_{\mathrm{root}}+1}$ and $12 + 2^{2p+h_{\mathrm{root}}+1}$, respectively. Since $p \ge 4$, $h_{\mathrm{root}} \ge 0$, each is at most V_0. Those of v_3, v_4 are below $24 < V_0$. The Bell coefficient sums are consequently bounded by $1, V_0, 2V_0^2, 5V_0^3, 15V_0^4$, each at most $16V_0^4$. The polynomial for $M_A + M_Z$ has coefficient sum

$$P_{\overline{F}}(1) \le 16V_0^4 \left[1 + \sum_{j=0}^{4} \binom{4}{j} 6^j\right] = 38432 V_0^4 < 2^{4v+16}. \tag{11.4}$$

This is the whole vector polynomial, including both e_4 copies, of degree at most twelve. For $1 \le k \le 12$, $\alpha = 2^{-t}$ and $e > 2$ give

$$\beta_k = \left(\frac{k}{2\alpha e}\right)^{k/4} < [2^{t+4}]^{k/4} \le 2^{3t+12}.$$

This also covers $\beta_0 = 1$. Thus the complete polynomial part is below $2^{q_{\mathrm{old}}}$, where $q_{\mathrm{old}} = 4v + 3t + 28$. Step 9 gave the full $H = L + J_z < 2^{l_{\mathrm{old}}+1}$ without a bounded-area shortcut. Since $e^H \le 2^{2H}$,

$$C_{\mathrm{old}} = (O/96) e^H \sum_k a_k \beta_k < 2^{\,2^{l_{\mathrm{old}}+2} + q_{\mathrm{old}} + o}. \tag{11.5}$$

Neither Jensen remainder, the physical-area term nor the $O/96$ prefactor has been discarded downward. The same-stage old remainder guards also hold: $o \le l_{\mathrm{old}} - 1$ and $\mathrm{clog}_2(3U_A p^2) \le l_{\mathrm{old}} - 1$, by step 9's strict bound.

The vector coefficient in (11.3) is below 2^{11}: $(2048\sqrt{6}/27) < 256$ follows from $\sqrt{6} < 5/2$, and $r^{3/2} \le 8$. Hence $N \ge 2^{11+4d} C_{\mathrm{old}}$ suffices for old first-arrow error at most η. The old symbolic guard must retain both $2^{l_{\mathrm{old}}+2}$ and $H_{\mathrm{small}} = q_{\mathrm{old}} + o + 11 + 4d$. Since $O \ge 1$ gives $h_{\mathrm{root}} \le o$, and $t \ge d$, exact expansion yields

$$\begin{aligned} H_{\mathrm{small}} &= 8p + 4h_{\mathrm{root}} + 3t + o + 47 + 4d \\ &\le 8p + 5o + 7t + 47 \le 7l_{\mathrm{old}} + 47. \end{aligned} \tag{11.6}$$

For every integer $l \ge 5$, $7l + 47 < 2^{l+2}$: it holds at five and its strict gap grows because $2^{l+2} > 7$. Here $l_{\mathrm{old}} \ge 50$. Thus $\mathrm{bitlen}(H_{\mathrm{small}}) \le l_{\mathrm{old}} + 2$. The first candidate $w = l_{\mathrm{old}} + 2$ fails the explicit strict predicate; the next, $w_{\mathrm{old}} = l_{\mathrm{old}} + 3$, satisfies it and the unchanged guard. Equations (11.5)–(11.6) give

$$\log_2[2^{11+4d} C_{\mathrm{old}}] < 2^{l_{\mathrm{old}}+2} + H_{\mathrm{small}} < 2^{l_{\mathrm{old}}+3} = \log_2 N_{\mathrm{old}}. \tag{11.7}$$

Thus N_{old} is sufficient at these same stages, with first-arrow error at most η by (11.3). The coefficient is unchanged, and no necessary-width lower bound is proved.

For integer a, $\lceil a/4 \rceil \le (a+3)/4$. Expand

$$\begin{aligned} e_{\text{new}} &= 16T_{1/2} + 20p + 12\left\lceil \frac{o_+ + 3}{4} \right\rceil + 4\left\lceil \frac{3o_+ + 8}{4} \right\rceil + 4d + 48 \\ &\le 16T_{1/2} + 20p + 6o_+ + 4d + 77. \end{aligned}$$

Since $T_{1/2} \le (t+1)/2$, $o_+ \le o + 78$ and $d \le t$,

$$e_{\text{new}} \le 12t + 20p + 6o + 553 \le 12l_{\text{old}} + 553. \tag{11.8}$$

For every integer $l \ge 8$, $12l + 553 < 2^{l+3}$: it holds at eight and the gap grows because $2^{l+3} > 12$. Hence

$$e_{\text{new}} < 2^{l_{\text{old}}+3} = 2^{w_{\text{old}}}, \qquad N_{\text{new}} = 2^{e_{\text{new}}} < 2^{2^{w_{\text{old}}}} = N_{\text{old}}. \tag{11.9}$$

This proves UDA.e universally. It compares sufficient mathematical integers; there is no fixed bound $e_{\text{new}} < 2^{40}$.

K.6.12 Exact recovery of each formerly supported stage and formula

Version 2 gives the complete legacy comparator explicitly. The comparisons here concern those definitions, so their equality needs no external ACC/SVC statement. Version 1's original statement and proof remain preserved; its standalone UDA.f omitted the comparator definitions. The repair specifies the theorem conclusion fully without adding an analytic premise.

On the old tolerance set, $d \in \{5, \dots, 15\}$. At $n = 80$ the spatial bracket is $159776/961 < 256$, so $\overline{E}(80) < 2^{-38} \le \eta$. The unbounded minimum n therefore equals the old first witness in $1, \dots, 80$. Then $U_A = n^2/2500 \le 64/25 < 3$ and

$$14u^2 A \le 358400000000/10^{20} < 1.$$

Thus $h_K = 0$ and $\overline{K} = 1$ exactly. Also $b_0 > 1/16$ and $32/b_0 < 512$. The degree-ten rational partial at seven exceeds 512, so the unbounded minimum k is the same old first witness in $1, \dots, 7$. Since $12U_A < 36$ and $4k^2/b_0 < 4 \cdot 49 \cdot 16 = 3136$, the identical integer formula gives $H_N \le 3137$. The regulator formula reduces to $p = 4H_N + 2d + 8 \le 12586$, exactly the legacy first even two-predicate witness. This proves equality of n, k, H_N, p, rather than merely another sufficient choice.

At these bounds $j = \lceil 18U_A p^2 \rceil \le 21600000000 < 2^{35}$ and $B_{\text{aux}} < 76800000008 < 2^{37}$. Hence $h_{\text{aux}} \le 37$ and $j + h_{\text{aux}} + p + d < 2^{35}$. The original first auxiliary predicate gives $h_s \le 36$. Step 2 showed that the new formula is exactly this same minimum, so $h_s, s, R_{\text{aux}}, O, o, h_{\text{stab}}, t$ recover the legacy stages exactly. Moreover $R_{\text{aux}} = 1000n + s + 1 < 2^{37}$, $O < 2^{76}$, $o \le 76$, and thus $o_+ = 78$. From $6Op^2 < 2^{109}$ we get $h_{\text{stab}} \le 109$ and $t = \max\{34, j + h_{\text{stab}} + d\} < 2^{36}$. Every common value uses the same literal predicate and tie rule; no finite diagnostic-grid inference is required.

With $o_+ = 78$, the ceilings are $\lceil 81/4 \rceil = 21$ and $\lceil 242/4 \rceil = 61$. Therefore

$$\begin{aligned} x &= T_{1/2} + p + 21, & y &= T_{1/2} + 2p + 61, \\ k_1 &= 3x + y + 3 = 4T_{1/2} + 5p + 127, \\ n_0 &= 2k_1 + 4 = 8T_{1/2} + 10p + 258, \\ k_4 &= 12x + 4y + 36 = 16T_{1/2} + 20p + 532, \\ e_{\text{new}} &= k_4 + 4d + 12 = 16\lceil t/2 \rceil + 20p + 4d + 544. \end{aligned}$$

These are exactly the legacy SVA/SVC exponents and width at those identical stages, including odd-t rounding. Step 11 also gives the identical old index $w_{\mathrm{old}} = l_{\mathrm{old}} + 3$, with the same l_{old}.

For completeness, $d_L = d$, $\eta_L = \eta$ by their definition. The same spatial predicates and $n \le 80$ give $n_L = n$, hence $A_L = A$, $U_{AL} = U_A$, $b_{0L} = b_0$. The identical finite Nelson predicates and $k \le 7$ give $k_L = k$; $h_K = 0$ gives $H_L = H_N$. The first-even predicates give $p_L = p$, $a_L = a$. Successive substitution gives

$$j_L = j,\ B_{\mathrm{aux},L} = B_{\mathrm{aux}},\ h_{\mathrm{aux},L} = h_{\mathrm{aux}},\ h_{sL} = h_s,\ s_L = s,$$
$$R_{\mathrm{aux},L} = R_{\mathrm{aux}},\ O_L = O,\ o_L = o,\ h_{\mathrm{stab},L} = h_{\mathrm{stab}},\ t_L = t,\ \epsilon_L = \epsilon,$$
$$\Lambda_L = \Lambda, \quad D_L = D, \quad \Omega_L = \Omega.$$

Substitution in the old-index definitions gives $h_{\mathrm{root},L} = h_{\mathrm{root}}$, $v_{\mathrm{old},L} = v_{\mathrm{old}}$, $q_{\mathrm{old},L} = q_{\mathrm{old}}$, $l_{\mathrm{old},L} = l_{\mathrm{old}}$ and $H_{\mathrm{small},L} = H_{\mathrm{small}}$. Their guard succeeds by step 11 and the first strict predicate selects $w_{\mathrm{old},L} = w_{\mathrm{old}}$, so $N_{\mathrm{old},L} = N_{\mathrm{old}}$. Finally, the displayed legacy relative formulas and the preceding algebra give

$$(T_L, x_L, y_L, k_{1L}, n_{0L}, k_{4L}, \zeta_L, e_L, N_L) = (T_{1/2}, x, y, k_1, n_0, k_4, \zeta, e_{\mathrm{new}}, N_{\mathrm{new}}).$$

Every variable in UDA.f is defined and its equality proved. The separate 1/5 case uses $d = 5$ without conflating its request with a dyadic one. This completes UDA.f.

K.6.13 Evidence, retained adverse facts and scope

Steps 1–12 are symbolic proofs over all declared integers q; a finite case list supplies no universal quantifier. The source author records a portable audit reconstructing rational stages and checking arithmetic and enclosures beyond $q = 12$, in noncompact volume/variance regimes and at both t parities. Those recorded runs are author evidence, distinct from fresh independent review or field-law simulation. Those historical runs are attributed to their recorded roles. Any current manuscript replay is separately identified in the companion execution receipt.

The product neuron shows why Gaussian joint-moment domination is not a permitted shortcut. At rectangle corners $(0,0),(a,0),(0,a),(a,a)$ its fourfold product is exactly $1/16$, whereas the Gaussian pairing sum is $(2e^{-2} + e^{-4})/16 < 1/16$. The proof uses one-point moments before tilt and coercivity/density domination under the actual joint tilt. The given full bilinear interface remains explicit.

All original source/classical/strict-AP6/NR1/unknown-g_* qualifications remain, as do the physical-versus-endpoint covariances and normalizers. Old-domain bounded quantities and numerical guards are not theorem premises. Existing ACC/SVC software and certificates keep their original domains and identities. The mathematical derivation supplies the every-q implication. The separate UDC contract below gives its finite partial software realization. Canonical least inner indices, practical/minimal/necessary costs, full-field rates, global neural Schwartz control, broader models/test families and novelty/priority remain separate. Both widths count whole-plane process neurons with countably many internal variables. The accepted source decisions and the fresh manuscript reviews have distinct scopes. □

K.7 Literal inherited source interface and attribution

The accepted statement includes a statement-only transcription of the inherited SVA source appendix. The direct finite interfaces stated earlier are its usable premises. The original private

source-availability map identifies which exact historical bodies were supplied in that integration packet. Historical paths retain their attribution and source-access limits; supplying a body does not newly prove it, authenticate its authorship or remove an original qualification. Source mesh ε and flow time s are local to this subsection, distinct from the outer stabilizer ϵ and selected auxiliary radius. Gaussian coupling fields are distinct from the field state space X. The historical global EB1 hypothesis involving g_* is not asserted; strict AP6 supplies the target route. The old EB1 coefficient is retained for scope and rederived at every finite stage in proof step 11. Its bounded SVA acceptance does not supply a universal premise. The source records the inherited ADMIT statement digest

`3e361ceb474a571783015c802c75a2314a58005052f07eb46e09bfb18034c997`

For $m > 0$, let P_m be the real whole-plane Gaussian free field with covariance $(-\Delta + m^2)^{-1}$ and Fourier measure $dp/(2\pi)^2$. For a centered axis-parallel square $B = (-b, b)^2$, $b > 0$, let J_B be its coefficient-one fourth Wick integral on this same Gaussian space, with the same mass and no factorial. Put

$$U_B = uJ_B, \quad Z_B(h) = \mathbb{E}_{P_m} e^{-U_B + h\phi(\mathbf{1}_B)}, \quad \nu_B = e^{-U_B} P_m / Z_B(0).$$

The Gaussian L^2 extension defines $\phi(\mathbf{1}_B)$, and every subsequent tilt uses that same random variable. The source in $Z_B(h)$ is confined to B; its boundary is Lebesgue null. The following remain named premises with their original authorship, source qualifications and accepted scopes.

P1: finite ferromagnetic DSS and SK-q. On the shifted torus $T = \varepsilon(\mathbb{Z} + 1/2)^2/L\mathbb{Z}^2$, with L/ε even and at least four, use physical inner product $\varepsilon^2 \sum$, operator $A = -\Delta^\varepsilon + m^2$, and the plane lattice variance c_ε. For every profile $q : T \to [0, 1]$, the law

$$d\nu \propto \exp\left[-u\varepsilon^2 \sum_x q_x {:}\psi_x^4{:}_{c_\varepsilon}\right] dP_{A^{-1}}$$

satisfies the accepted SK-q inequality. At every finite real physical field H, its tilt by $\exp(\varepsilon^2 \sum_x H_x \psi_x)$ satisfies

$$0 \le \mathrm{Cov}_{\nu^H}(\psi_x, \psi_y) \le \mathbb{E}_\nu(\psi_x \psi_y).$$

The analogous clauses hold with added physical mass $1/s$ for every finite $s > 0$. These finite DSS-q/SKQ clauses include Gaussian sites and retain the frozen BD/BBD/Griffiths–Simon, Ising-FKG and Robbins–Stirling qualifications. They have no g_* premise.

P2: exact finite-positive-scale P^*. If

$$0 < \varepsilon \le \min(1/4, 18/(5m)), \quad L \ge \max(1, \sqrt{2}/m), \quad L/\varepsilon \in 2\mathbb{N}, \quad L/\varepsilon \ge 4, \quad 0 < u \le m^2/8000,$$

then core P^* using P1, $S_{\max} = \infty$, $\Phi = 1$ and $\kappa = 0$ gives, for finite $s > 0$ and every profile $q : T \to [0, 1]$,

$$\chi_{s,q} = \max_x \varepsilon^2 \sum_y S_{s,q}(x, y), \qquad S_{s,q}(x, y) = \mathbb{E}_{\mu_{s,q}}(\psi_x \psi_y),$$

$$d\mu_{s,q} \propto \exp\left[-u\varepsilon^2 \sum_x q_x {:}\psi_x^4{:}_{c_\varepsilon}\right] dP_{(A+1/s)^{-1}},$$

$$\chi_{s,q} \le \frac{s}{1 + m^2 s} + \frac{2s}{2 + m^2 s}.$$

This combines the accepted $\|S - C_s\|_{1,0} \le \tau_s$ with $\tau_s = 2s/(2 + m^2 s)$ and the exact Gaussian row sum. P2 includes no original-law endpoint conclusion.

P3: fixed-square Gaussian/Wick transfer. At fixed $B = (-b, b)^2$, take commensurate $\varepsilon = b/N \to 0$ on the shifted plane lattice. Accepted T1–T4/GQ/W(b)/W-laws place the plane Gaussian field, the variables

$$J_B^\varepsilon = \varepsilon^2 \sum_{x \in B} :\psi_x^4:_{c_\varepsilon}, \qquad Y_f^\varepsilon = \varepsilon^2 \sum_x f(x)\psi_x,$$

and their continuum counterparts on one Gaussian space. For every real compact smooth f, $J_B^\varepsilon \to J_B$ and $Y_f^\varepsilon \to \phi(f)$ in Gaussian L^2. For each fixed finite $p \geq 1$, $\sup_{\varepsilon \text{ sufficiently small}} \mathbb{E} e^{-puJ_B^\varepsilon} < \infty$. The continuum density has positive finite normalizer and finite Gaussian L^p density at every finite p. These finite-volume statements have no weak-coupling or g_* premise. Coefficient-one Wick and finite-law identifications are precisely those of the original construction.

P4: fixed-mesh Gaussian periodization. At fixed ε, torus Gaussian covariances converge entrywise on every finite set of fixed sites to the plane lattice covariance as $L \to \infty$ through admissible even multiples. The plane variance used for Wick ordering is the same on both sides. This is the accepted H1-PREQ Gaussian periodization/envelope input.

P5: original Fröhlich–Simon source facts. Use the original same-mass whole-plane free-boundary convention with $P_h(X) = uX^4 - hX$, $u > 0$. The free pressure $\alpha_m(P_h)$ is the normalized log-partition limit. Its dyadic-square realization is attributed to Fröhlich and Simon [8, Proposition 2.1 and the remark after (2.3), pp. 502–503]. The equality of the two original maximal states $\nu_{m,u,+} = \nu_{m,u,-}$ is equivalent to differentiability of $h \mapsto \alpha_m(P_h)$ at zero by the original Corollary 4.3(ii), p. 510. Theorem 5.2, p. 512, then gives ordinary exhaustion convergence of the classical free finite-volume states to their common state, read via the compact-test generating functional at every real frequency. These are attributed published facts; the source reports a source match, and their original proofs are not reproved here. The historical `SOURCE-MAP.md/json` record paths, hashes, locations and primary-PDF access in the upstream packet; those files are not supplied to this role. P5 uses the converse in the actual Corollary 4.3(ii); the manuscript's narrower C1(iii) formulation by itself would not suffice.

P6: additional explicit bridge inputs. For AP5–AP6 retain the following frozen accepted inputs at their actual parameter scopes. First, retain the all-$m, u > 0$ physical/classical identity at historical `CT.tex:2135-2166`, Gaussian common-space/Wick laws, and the original complete local H^{-2} field-space, cutoff/Fourier coordinate, Borel-determining and compactness definitions. Second, the all-semibounded-polynomial FS C1(ii) local density and positive boundary factorization at historical `CT.tex:701-711` give

$$dv_{m,u}|_{\mathcal{C}_B} = e^{-uJ_B - \alpha_m(uX^4)|B|} g_{\partial B}\, dP_m,$$

with positive boundary-measurable g and finite local Gaussian L^p densities. Either recorded boundary-algebra reading is retained. Third, retain the finite DSS, SKQ, P^*, H1-PREQ, H1, MIX, REG, smoothed-CT, FLOW, TP and T interfaces reproduced at historical `CL1-ST.md:138-1828`, qualified by their separate decisions and later source-normalization correction. That file's new CL1 conclusion is excluded as a premise. FLOW's own cap is $0 < u \leq m^2/8000$; TP/T1–T4 are fixed-square transfers.

Finally, retain the raw fixed-stage NR1/L0 and Gaussian/common-space/Wick inequalities of the historical EB-ST accepted-input appendix, at $m, u > 0$ and fixed positive regulator/stabilizer/finite

geometry. In particular NR1 gives $\delta_Z \le d_D M_Z/(96N)$, $\delta_A \le d_D M_A/(96N)$, with $Z_Q \ge z$. L0 smooths unit BL functions with error at most $\sqrt{r}\delta$ and order-through-four operator bounds at most $\sqrt{6}\delta^{-3}$ for $0 < \delta \le 1$. Keep the sharper-author/larger-blind-core scope, Gaussian mass tilt, independent covariance-defect coupling and actual physical common-space Wick bounds/normalizers. These are finite raw statements. Old EB1/FV/CL1 conclusions with global g_* hypotheses are excluded from this bridge's premises. Prior argument ports give author provenance, rather than a proof a blind solver must read. The source records the literal statements, separate decision hashes and ranges in historical `support/UPSTREAM-STATEMENTS.md`, `support/` decisions and `SOURCE-MAP.json`; their bodies are unavailable to this role. P6 excludes AHKZ C2(ii) and C2(iii).

AP1: original finite-lattice endpoint and source bound. Under P1–P5 and $0 < u \le m^2/8000$, for all lattices, tori and profiles of P2 the original zero-field law obeys

$$\max_x \varepsilon^2 \sum_y \mathbb{E}_\nu(\psi_x \psi_y) \le 3/m^2.$$

For real vectors f on T and real t,

$$0 \le \log \mathbb{E}_\nu \exp\left[t\varepsilon^2 \sum_x f_x \psi_x\right] \le \frac{3t^2}{2m^2}\varepsilon^2 \sum_x f_x^2.$$

AP2: actual free-boundary continuum source bound. For centered squares B, real compact smooth f and real t,

$$0 \le \log \mathbb{E}_{\nu_B} e^{t\phi(f)} \le \frac{3t^2\|f\|_2^2}{2m^2}.$$

The same holds for real $f \in L^2(\mathbb{R}^2)$ via the Gaussian extension. In particular $0 \le \log[Z_B(h)/Z_B(0)] \le 3h^2|B|/(2m^2)$ for real h.

AP3: pressure and the exact original FS target. With P5's pressure,

$$0 \le \alpha_m(uX^4 - hX) - \alpha_m(uX^4) \le 3h^2/(2m^2), \quad h \in \mathbb{R}.$$

It is differentiable at zero with derivative zero. The actual two original FS maximal states coincide; write their common value as $v_{m,u}$. For every $\ell > 0$, finite real compact-smooth tuple f, and centered $\Lambda_n = (-n\ell, n\ell)^2$, their joint test laws under the original classical free ν_{m,u,Λ_n} converge to those under $v_{m,u}$.

AP4: reference-pair identity and the fixed instance. For $M > 0$, set $s = m/M$ and $\lambda = u/s^2$. Then $\lambda/M^2 = u/m^2 \le 1/8000$, and under the manuscript's exact dilation $(T_s\psi)(f) = \psi(s^{-2}f(\cdot/s))$,

$$v_{m,u} = (T_s)_* v_{M,\lambda}.$$

The source-defined target is therefore the original common FS maximal state and the same reference/dilation expression as in the construction, with identical physical finite-volume approximants, rather than an arbitrary subsequential limit. The fixed $m = 1, u = 10^{-10}$ has this original-target existence/equality/free-limit witness under P1–P5 since $10^{-10} < 1/8000$.

AP5: physical source identity, field version and interactingness. Also assuming P6, the physical zero-regulator finite-volume laws have exactly the classical ν_B compact-joint laws. They converge weakly on the original $X = H^{-2}_{\mathrm{loc}}$ to a unique probability ν^X representing $v_{m,u}$; all compact-test polynomial moments converge and are uniformly integrable at every fixed positive order. For every $u > 0$ in this ratio range, $v_{m,u}$ is non-Gaussian by the retained positive local quartic boundary-density structure. This interactingness check is separate from equality of the two maximal states.

AP6: strict numerical-budget applicability. Also assume P6, $0 < u/m^2 < 1/8000$ and a fixed finite compact-smooth tuple. Finite-stage definitions/constants are precisely the inherited `EB-ST.md:7-55`. For $K \subset [-R_{\mathrm{obs}}, R_{\mathrm{obs}}]^2$ use the exact `CL1-ST.md:47-68` coefficient with $R = R_{\mathrm{obs}}$, its H1/MIX constants furnished by the separately accepted finite interfaces after their actual premises are discharged. The formula has $\epsilon_* = \min(1/4, 1/m)$, $K_{\mathrm{cov}} = 1$, $c_{\mathrm{CT}} = 16.2$ and $\kappa = m/256$; AP6 itself supplies no numerical C_f. If $K \subset \Lambda_{n_0}$ open, the actual continuum finite-box interpolation obeys, for $n \geq n_0$, $t \in [0,1]$ and $G \in C^1_b$ with separately bounded value and Euclidean gradient,

$$\begin{aligned}&\left|\mathrm{Cov}_{\nu_{n,t}}\left(G(Y_f), u[J_{\Lambda_{n+1}} - J_{\Lambda_n}]\right)\right| \\ &\qquad \leq C_f u\, 4\ell^2(2n+1)e^{-\kappa \operatorname{dist}(K, S_n)}.\end{aligned}$$

For $n\ell \geq R_{\mathrm{obs}}$, define

$$q = e^{-\kappa\ell}, \quad T_n = q^n\left[\frac{2n+1}{1-q} + \frac{2q}{(1-q)^2}\right], \qquad E_n = 4C_f u \ell^2 e^{\kappa R_{\mathrm{obs}}} T_n.$$

The actual physical observation law then has BL_1 distance at most E_n from the same source target $v_{m,u}$, equivalently ν^X. For both original process-neuron architectures and every positive finite stage in EB-ST, the same five-term expression holds:

$$\begin{aligned}&d_{\mathrm{BL}}\left(\mathcal{L}_f(P^b_{N,\epsilon,\Omega,a;\Lambda_n}), \mathcal{L}_f(\nu^X)\right) \\ &\qquad \leq \min\{2, \min(2, D_{\mathrm{law}} N^{-1/4}) + S + A_{\mathrm{aux}} + R_{\mathrm{phys}} + E_n\}.\end{aligned}$$

Here $D_{\mathrm{law}}, S, A_{\mathrm{aux}}, R_{\mathrm{phys}}$ are the exact finite-stage formulas of historical `EB-ST.md:32-44,72-98`, with the required geometry and physical covariance. BL_1 retains its two separate constraints. With certified coefficient/test/support envelopes, the EC1.e reverse-stage argument applies using the distinct `FS-pressure-target-v1` certificate in place of an old g_* inequality. It supplies no unsupplied envelope and neither amends prior acceptance nor certifies the original observable-independent sequence.

K.7.1 Inherited EB1 finite normalized-law and removal interfaces

The recorded EB1 statement digest is

`d6b400d4d47c6f8e28f7344b678b2ed2114504336c1635585a192bcf25319c83`

Definitions and EB1.a–d,f retain their finite-stage scopes, with AP6 supplying the explicit-cap final arrow. EB1.e,g are unused and omitted; excluded E4/r0 material is absent. The historical global header fixed $m, u, \ell > 0$ with $0 < u/m^2 < \min(g_*, 1/8000)$ and an admissible reference pair/target. Its g_* condition is preserved as historical scope, rather than assumed for the present strict-AP6 route. The physical squares $\Lambda_n = (-n\ell, n\ell)^2$ restrict the interaction; all field laws occupy

the whole plane. The observations, support and BL_1 convention are those already defined. The physical multiplier is $(m^2+\omega^2+\xi^2+a^2\omega^2\xi^2)^{-1}$, $0<a<1/m$, $\kappa_a^2=a^{-2}-m^2$, $c=C_a(0)$. The endpoint covariance is $e^{-|z-z'|_1/a}/4$, a distinct covariance. Stabilizer $\epsilon>0$ and Ω are as above. A width is a count of iid product/Mondrian whole-plane process neurons, each with countably many variables. Lattice mesh, spectral cutoff and covariance-flow time are not approximation parameters in this chain.

For the fixed-stage width clause allow nested centered physical squares $\Lambda_0\subset\Lambda_1$, including equality, enclosed by Ω, and a bounded box D containing Ω and all supports. Put $d_D=|D|$, $O=|\Omega|$, $E_D=L^4(D)$, $Y=\mathcal{O}_f$, and

$$V_i(v)=u\int_{\Lambda_i}(v^4-6cv^2+3c^2)+\epsilon\int_{\Omega\setminus\Lambda_i}v^4-\frac{\kappa_a^2}{2}\int_\Omega v^2,$$
$$V_t=(1-t)V_0+tV_1,\qquad 0\le t\le 1.$$

Let Q_N^b be the law of the normalized neuronal sum restricted to D, and Q_γ the endpoint Gaussian law. Write $w_t=e^{-V_t}$, $Z_Q=Q(w_t)$, $A_Q(t,G)=Q[G(Y)w_t]$, $\rho_{Q,t}=Y_*(w_tQ/Z_Q)$ and $L_Q(t,G)=A_Q/Z_Q$. Their $t=i$ endpoints are the actual associated neural/full-Gaussian-tilt observation laws. The inherited constants are

$$\alpha=\min(u,\epsilon),\qquad U=\max(u,\epsilon),$$
$$B_N=\kappa_a^2/2+6uc,\qquad L_N=B_N^2O/(2\alpha),$$
$$F=\left(\sum_i\|f_i\|_{4/3,D}^2\right)^{1/2}.$$

Use $v_1=4Ux^3+2B_N\sqrt{O}\,x$, $v_2=12Ux^2+2B_N\sqrt{O}$, $v_3=24Ux$, $v_4=24U$, and precisely the $e_0,\dots,e_4$ Bell polynomials displayed earlier. With $\beta_0=1$, $\beta_k=(k/(2\alpha e))^{k/4}$ for $k>0$, put

$$\mathcal{T}_N\left(\sum p_kx^k\right)=e^{L_N}\sum p_k\beta_k\quad(p_k\ge0),$$
$$M_Z=\mathcal{T}_N(e_4),\qquad M_A=\mathcal{T}_N\left(\sum_{j=0}^4\binom{4}{j}F^je_{4-j}\right),$$
$$z=e^{-3UO/16-3uc^2|\Lambda_1|},\qquad C_{\mathrm{law}}=d_D(M_A+M_Z)/(96z),$$
$$D_{\mathrm{law}}=\tfrac{4}{3}(3\sqrt{6}C_{\mathrm{law}})^{1/4}(2\sqrt{r})^{3/4}$$
$$=2^{23/8}3^{-5/8}r^{3/8}C_{\mathrm{law}}^{1/4},\qquad D'_{\mathrm{law}}=\sqrt{6}C_{\mathrm{law}}+2\sqrt{r}.$$

Euler's number e is distinct from e_j.

For one physical Λ, set $A=|\Lambda|$, $B_{\mathrm{ext}}=|\Omega\setminus\Lambda|$, $F_p=(\sum_i\|f_i\|_p^2)^{1/2}$ and $L_c=(4\sqrt{15}+12)c^{3/2}$. The literal laws are

$$P=P^b_{N,\epsilon,\Omega,a;\Lambda},\quad Q_\epsilon=\nu^v_{\epsilon,\Omega,a;\Lambda}=\iota_*\operatorname{Res}_\epsilon,\quad Q_0=\nu^{(2)}_{\Omega,a;\Lambda}=\iota_*\operatorname{Res}_0,$$
$$Q_a=\nu_{a,\Lambda},\quad Q_{\mathrm{phys}}=\nu_{0,\Lambda},\quad Q_{\mathrm{target}}=\nu=\nu^X_\infty.$$

The residual law has normalized density $e^{-W_{a,\Lambda}-\epsilon J}$ against $G'_{\Omega,a}$, where $W=u\int_\Lambda(x^4-6c_ax^2+3c_a^2)$ and $J=\int_{\Omega\setminus\Lambda}x^4$. Q_ϵ is the positive-stabilizer full tilt and Q_0 its residual endpoint. Q_a is the normalized $e^{-U_{a,\Lambda}}$ tilt on the actual common realization Φ_a; $U_{0,\Lambda}$ is the fourth-chaos limit. Classical

finite-volume compact-test laws equal those of Q_{phys}, and ν is the existing target. Mass, factorial and normalization stay fixed. Define

$$\Delta_{\Omega,a} = C_a - C'_{\Omega,a}, \qquad \delta_\Omega(H) = \sup_{x\in H} \Delta_{\Omega,a}(x,x), \quad \delta_\Omega(\varnothing) = 0,$$

$$\xi_a(s) = \kappa_a^2 \int_{|x|>s} C_a(x)^2\, dx,$$

$$\kappa_0 = e(1+\sqrt{2}), \qquad \kappa_2 = (2\pi)^{-1}\sqrt{17\pi/16},$$

$$M_6 = (2\pi)^{-5/3}(5\pi)^{5/6} m^{-1/3}, \qquad K_\Lambda = u\sqrt{384 A \kappa_2 M_6^3},$$

$$K_{\mathrm{sharp}} = \max\{12uA, (2/e)K_\Lambda\sqrt{\kappa_0/m}\},$$

$$T_\Lambda(H) = \exp[-\exp((\pi/2)\sqrt{H/(12uA)} - 1)],$$

$$\Xi_p(\Lambda) = e^{pK_{\mathrm{sharp}}} + \int_{K_{\mathrm{sharp}}}^{\infty} p e^{pH} T_\Lambda(H)\, dH.$$

EB1.a: actual fixed-stage neural law. For either architecture and every $N \geq 1$, uniformly in $t \in [0,1]$,

$$\sup_{G\in\mathcal{U}_4(r)} |L_{Q_N^b}(t,G) - L_{Q_\gamma}(t,G)| \leq \min\{2, C_{\mathrm{law}}/N\},$$

$$d_{\mathrm{BL}}(\rho_{Q_N^b,t}, \rho_{Q_\gamma,t}) \leq \sqrt{6} C_{\mathrm{law}}/(N\delta^3) + 2\sqrt{r}\delta, \quad 0 < \delta \leq 1,$$

$$\sup_t d_{\mathrm{BL}}(\rho_{Q_N^b,t}, \rho_{Q_\gamma,t}) \leq \min\{2, D_{\mathrm{law}} N^{-1/4}\}, \qquad D_{\mathrm{law}} \leq D'_{\mathrm{law}}.$$

All derivative norms are full multilinear operator norms and endpoint identification is included. The inherited finite clause covers all widths, singular/repeated/sign-reversed tests, tests outside Ω, constant G, $u = \epsilon$ and empty shells. Zero shell response does not identify neural and full-tilt laws. Empty/all-zero tuples have zero error directly; $r = 0$ does not need smoothing. Uniformity is only for fixed positive parameters and bounded geometry. The current theorem retains its narrower original ordered bump family.

EB1.b: stabilizer. With variation $\sup_{\mathrm{Borel}\ |H|\leq 1}$, whose maximum is two,

$$\|Q_\epsilon - Q_0\|_{\mathrm{var}} \leq \min\{2, 6\epsilon B_{\mathrm{ext}} c_a^2 e^{9uAc_a^2}\} = S.$$

This includes $\epsilon = 0$ through the residual laws and gives zero when $B_{\mathrm{ext}} = 0$.

EB1.c: auxiliary region. In the same tuple metric,

$$d_{\mathrm{BL}}(\mathcal{L}_f(Q_0), \mathcal{L}_f(Q_a)) \leq \min\left\{2, e^{9uAc_a^2}[2uAL_c\sqrt{\delta_\Omega(\overline{\Lambda})} + F_1\sqrt{\delta_\Omega(K)}]\right\}.$$

If Ω contains all points at Euclidean distance less than $s > 0$ from compact H, then $\delta_\Omega(H) \leq \xi_a(s)$, which tends to zero at fixed a as $s \to \infty$. Thus the error vanishes for fixed a, Λ, f as Ω grows. It suffices that $\overline{\Lambda} \cup K \subset [-R_H, R_H]^2$ and $0 < s < R_{\mathrm{aux}} - R_H$. This inherited clause asserts no uniform-in-a or exponential auxiliary rate by itself.

EB1.d: physical regulator. At fixed Λ and $0 < a < 1/m$,

$$d_{\mathrm{BL}}(\mathcal{L}_f(Q_a), \mathcal{L}_f(Q_{\mathrm{phys}})) \le \min\{2, \Xi_2(\Lambda)^{1/2}[4K_\Lambda\sqrt{a} + (a/2)F_2]\}.$$

It tends to zero at fixed Λ, f; neither Ξ_2 nor K_Λ is asserted uniform in Λ.

EB1.f: compatible budget and literal limits. For the fixed tuple, let accepted CL1-FV1 furnish n_0 with $K \subset \Lambda_{n_0}$. For $n \ge n_0$, $n\ell \ge R_{\mathrm{obs}}$, use $\kappa = m/256$, $q = e^{-\kappa\ell}$ and the same T_n, E_n as AP6. Choose the EB1.a constants with $\Lambda_0 = \Lambda_1 = \Lambda_n$. For every admissible positive a, ϵ, bounded Ω enclosing $\overline{\Lambda}_n$, a box D containing Ω and supports, and $N \ge 1$, the five-term budget is the AP6-displayed expression with its historical ν target. It passes exactly through $Q_\epsilon, Q_0, Q_a, Q_{\mathrm{phys}}, \nu$ and implies the literal iterated limit: N first, then $\epsilon \to 0$, $R_{\mathrm{aux}} \to \infty$, $a \to 0$, $n \to \infty$. A tolerance schedule reverses these dependencies: $n, a, R_{\mathrm{aux}}, \epsilon, N$. Historical C_f, g_* retain their possibly noneffective source meaning; there is no arbitrary simultaneous schedule, practical numerical width, new dimension, stronger coupling or priority assertion in that clause.

K.7.2 Inherited EC1.a–c coefficient bounds

The corrected EC1 statement digest recorded in the source is

`1406f33ad2370b8f5c59751ec73d06ada0c3a9c159fd75e3ec6b4abbb748b4fe`

These are rules for the physical variance, covariance tail and fixed-square Nelson constant; the accepted positive-accuracy selector is not an oracle for the present tuple. Use the exact massive covariance/Fourier convention at historical `CONSTRUCTION.tex:303-330`,

$$C_a(z) = (2\pi)^{-2} \int_{\mathbb{R}^2} \frac{e^{ip\cdot z}}{m^2 + \omega^2 + \xi^2 + a^2\omega^2\xi^2}\, dp,$$

$$m > 0, \quad 0 < a < 1/m, \quad \kappa_a^2 = a^{-2} - m^2 > 0, \quad c_a = C_a(0).$$

Tail norms are Euclidean. The physical covariance differs from the variance-$1/4$ endpoint covariance. Retain the exact EB1 $K_\Lambda, K_{\mathrm{sharp}}, T_\Lambda, \Xi_p$ for $u > 0$ and positive finite area A, with $\log_+ x = \max(0, \log x)$. For width clauses use the same possibly equal nested squares, $O > 0$, $d_D > 0$, $\alpha = \min(u, \epsilon) > 0$, $U = \max(u, \epsilon)$, $B_N = \kappa_a^2/2 + 6uc_a$ and tuple norm F. No convention changes.

EC1.a: point variance and certified enclosures. For $h = ma \in (0, 1)$,

$$\max\left\{\frac{1}{4}, \frac{\log(1/h) - (1-h^2)/2}{2\pi}\right\} \le c_a \le \overline{c}(a) = \frac{\operatorname{asinh}(1/h) + 1}{2\pi},$$

$$\operatorname{asinh}(x) = \log(x + \sqrt{1+x^2}).$$

Hence $c_a = (2\pi)^{-1}\log(1/(ma)) + O(1)$ as $a \downarrow 0$ at fixed m. Upper-bound uses, including positive-coefficient EB1 widths, may replace c_a by $\overline{c}(a)$. For $Q > 0$ and a partition $0 = q_0 < \cdots < q_M = Q$, define

$$H(q) = [(m^2 + q^2)(1 + a^2q^2)]^{-1/2},$$
$$c_{\mathrm{lo}} = (2\pi)^{-1} \sum_j (q_{j+1} - q_j) H(q_{j+1}),$$

$$c_{\mathrm{hi}} = (2\pi)^{-1}\left[\sum_j (q_{j+1}-q_j)H(q_j) + 1/(aQ)\right].$$

Then $c_{\mathrm{lo}} \le c_a \le c_{\mathrm{hi}}$ and

$$c_{\mathrm{hi}} - c_{\mathrm{lo}} \le \frac{\max_j (q_{j+1}-q_j)/m + 1/(aQ)}{2\pi}.$$

Thus positive computable m, a satisfying the strict regulator condition give computability of c_a by certified elementary operations.

EC1.b: exponential squared-covariance tail. For $s \ge 0$,

$$\int_{|z|>s} C_a(z)^2\,dz \le \min\{1/(4\pi m^2), e^{-\sqrt{2}ms}/(2\pi m^2)\},$$

$$\xi_a(s) \le \rho_a(s) = \frac{\kappa_a^2}{4\pi m^2}\min\{1, 2e^{-\sqrt{2}ms}\}.$$

At fixed a this vanishes explicitly. The rate depends on m and the prefactor retains its regulator dependence. The same strict source geometry is mandatory: if the Euclidean s-neighborhood of compact H lies in Ω, $\delta_\Omega(H) \le \rho_a(s)$. Thus if $\overline{\Lambda} \cup K \subset [-R_H, R_H]^2$ and $0 < s < R_{\mathrm{aux}} - R_H$, the auxiliary error is at most

$$\min\{2, H_{\mathrm{aux}}\sqrt{\rho_a(s)}\}, \qquad H_{\mathrm{aux}} = e^{9uA\overline{c}(a)^2}[2uA(4\sqrt{15}+12)\overline{c}(a)^{3/2} + F_1].$$

No regulator-uniform defect modulus or interchange of limits is asserted.

EC1.c: elementary bounds and enclosures for Ξ_p. For every fixed $p \ge 1$ set

$$b = \frac{\pi^2}{48uA}, \qquad H_p = \max\left\{K_{\mathrm{sharp}}, \frac{4}{b}\left[\log_+ \frac{32p}{eb}\right]^2\right\}.$$

Then

$$e^{pK_{\mathrm{sharp}}} \le \Xi_p(\Lambda) \le e^{pH_p} + e^{-pH_p} \le 2e^{pH_p}.$$

These finite elementary values give explicit certified $\Xi_2^{1/2}$ and fixed-volume regulator bounds. The exact equivalent formula $K_\Lambda^2 = [75\sqrt{85}/(2\pi^3 m)]u^2 A$ avoids fractional cube-root powers in numerical evaluation.

For $T \ge H_p$ and a partition $K_{\mathrm{sharp}} = x_0 < \cdots < x_M = T$, the lower and upper enclosures for Ξ_p are respectively

$$e^{pK_{\mathrm{sharp}}} + \sum_j (x_{j+1}-x_j)pe^{px_j}T_\Lambda(x_{j+1}),$$

$$e^{pK_{\mathrm{sharp}}} + \sum_j (x_{j+1}-x_j)pe^{px_{j+1}}T_\Lambda(x_j) + e^{-pT}.$$

The degenerate interval $T = K_{\mathrm{sharp}}$ has empty sum. Increasing T and refining the partition makes the enclosure arbitrarily tight, proving computability for computable m, u, A, p without evaluating any CL1 source coefficient. At fixed m, u, p, as A grows,

$$\log \Xi_p \ge 12puA,$$

$$\log \Xi_p \le p \max\left\{K_{\mathrm{sharp}}, \frac{192uA}{\pi^2}\left[\log_+ \frac{1536puA}{e\pi^2}\right]^2\right\} + \log 2.$$

This upper bound grows as $O_{m,u,p}(A \log^2 A)$ and concerns the source Nelson bound, without a volume-uniform moment assertion.

K.7.3 Inherited OPT coefficients and physical Gaussian facts

The accepted OPT statement digest recorded in the frozen source is

`1c178d728e6ffbf744b19eff30433fff75961b67365dcbfec0ef17f196e61e5b`

Use the same chosen explicit H1 coefficient $C_H < 2^{16}$ in every C_f formula. OPT.a–b supply the usable H2 value $C_{H2} = 2^{384}$ in the literal J expression and its scalar $R = 1$ coefficient $C_f \le 2^{360}$. Their finite-source assumptions and attribution remain. The source appendix describes these as inherited inputs for SELECT; that historical label does not change the present UDA interface. The finite-lattice variables below are local to that inherited theorem, and its original scalar tuple is unused. Reproving the earlier coefficient theorem is outside the present translation.

OPT.a: explicit finite H2 coefficient. At the fixed $m, u, \ell, K_{\mathrm{cov}} = 1$, on the entire remaining finite-lattice domain, with actual posterior, actual outer law and literal A1–A6 from the finite-interface appendix, the physical Wick covariance obeys

$$\left\| \varepsilon^2 \sum_{z \in S_n^\varepsilon} e^{d_L(x,z)/128} \left| \mathrm{Cov}(:\psi_z^4:_c, :\psi_x^3:_c \mid Y_s) \right| \right\|_{L^2(\nu_s)} \le 2^{384} g(s)^{7/2}.$$

This includes every permitted ε, n, t, L, all $0 < s \le 1$, both t endpoints, zero-profile sites and coincident x, z. The squared envelope is $2^{768} g(s)^7$, with finite required square-root integral. No unquantified spatial score constant or translated-moment premise is added.

OPT.b: literal bump coefficient. With the earlier explicit H1 formula, OPT.a's H2 coefficient, $R = 1$, $n_0 = 1$, $\epsilon_* = 1/4$, $\kappa = 1/256$ and $c_{\mathrm{CT}} = 81/5$, the literal CL1 expression obeys $C_f \le 2^{360}$. The formula includes its source integral, endpoint, every prefactor and $[m^2(1+m^2)]^{-1}$. AP6 gives its applicability to the same original pressure-identified FS target under P1–P6. Earlier acceptances $C_{H2} \le 2^{1800}$ and $C_f \le 2^{1810}$ do not by themselves accept the changed coefficients.

The inherited Gaussian/common-space facts are premises already used in the EB1 proof, rather than conclusions of SELECT's new source estimate. They hold for every fixed finite real compact-smooth tuple, including singular test covariance. At $m = 1$, $0 < a < 1$, the physical Φ_a has the multiplier stated earlier, with measure $dp/(2\pi)^2$. On the accepted common realization, $\mathbb{E}|(\Phi_a - \Phi_0)(f)|^2 \le (a^2/4)\|f\|_2^2$ for every such real f. The continuous Gaussian endpoint has covariance $e^{-|z-w|_1/a}/4$, distinct from the physical point variance c_a.

For each Ω, $C'_{\Omega,a}$ is an actual continuous covariance kernel with $0 \le C'_{\Omega,a} \le C_a$ in positive-semidefinite kernel order. Thus $\Delta = C_a - C'_{\Omega,a}$ is positive semidefinite. There exist independent centered continuous Gaussian fields γ', ψ with covariances C', Δ. Then $X_c = \gamma' + \psi$ has physical regulated reference law, and $Y_c = \gamma'$ has the mass-tilt reference law. The weights $w_X = e^{-W_{a,\Lambda}(X_c)}$ and $w_Y = e^{-W_{a,\Lambda}(Y_c)}$ yield the actual normalized Q_a, Q_0. Their common upper bound is $e^{6uAc_a^2}$; the Y normalizer is at least $e^{-3uAc_a^2}$. If the strict Euclidean s-neighborhood of compact H lies in Ω, $\delta_\Omega(H) \le \kappa_a^2 \int_{|z|>s} C_a(z)^2\, dz$.

On the physical common Gaussian space $U_{a,\Lambda}$ is the coefficient-one own-variance Wick interaction, and $U_{0,\Lambda}$ its fourth-chaos limit. Then $\mathbb{E}U_{a,\Lambda} = 0$, $Z_a = \mathbb{E}e^{-U_{a,\Lambda}} \geq 1$. For $w_a = e^{-U_{a,\Lambda}}$, $w_0 = e^{-U_{0,\Lambda}}$,

$$\mathbb{E}|w_a - w_0| \leq 2K_\Lambda \Xi_2(\Lambda)^{1/2}\sqrt{a}, \qquad \|w_0\|_2 \leq \Xi_2(\Lambda)^{1/2}.$$

All stated physical moments are finite at each fixed physical square. Q_{phys} has the original classical ν_B compact-test laws, as AP5 also supplies.

Finally, accepted L0 is an ambient Euclidean premise. For G with separate unit sup and Lipschitz bounds on $\mathbb{R}^r$, standard Gaussian $\xi \in \mathbb{R}^r$ and $G_\delta(y) = \mathbb{E}G(y - \delta\xi)$, G_δ is smooth, $\|G - G_\delta\|_\infty \leq \sqrt{r}\delta$, $\|G_\delta\|_\infty \leq 1$, $\|DG_\delta\|_{\mathrm{op}} \leq 1$ and $\|D^j G_\delta\|_{\mathrm{op}} \leq \sqrt{(j-1)!}\delta^{1-j}$ for $j \geq 2$. For $0 < \delta \leq 1$, all norms through four are at most $\sqrt{6}\delta^{-3}$. It holds in the original ambient dimension irrespective of covariance rank.

K.8 The separately versioned finite-resource implementation

UDA's every-q statement concerns finite mathematical integers. The separately accepted UDC-dyadic-accuracy-v1 implements a finite partial realization: success gives an exact symbolic certificate, and a valid mathematical request may resource-refuse. Its frozen source files retain their author-candidate status flags; the external exact-current UDC decision supplies their internal acceptance. Neither a success flag nor a newly self-hashed core supplies acceptance. The unchanged older ACC and SVC interfaces retain their own names, domains and bytes.

The version identities are:

Request
`UDC-dyadic-accuracy-request/1`

Certificate
`UDC-dyadic-certificate/1`

Backend
`accepted-UDA-STAGE-RELATIVE-v2-bounded-v1`

Source bundle
`UDC-UDA-STAGE-RELATIVE-v2-sources-20261028-v1`

The following entry-point and source-root descriptions refer to the original private UDC packet, which is not included in this public document export. Its original private source map supplies the paths and digests described below. The command-line entry points are `dyadic_accuracy_search.py` and the separate `verify_dyadic_accuracy.py`. Both use the adjacent `sources/` tree by default; `-source-root` relocates an identical complete tree. Every declared source body is local and hash-bound, with no live checkout fallback. The companion source map binds these names to their actual supplied relative paths and full digests. In this packet the core root is `inputs/accepted/UDC/`, with its direct tree at `inputs/accepted/UDC/sources/`. The eight bound core bodies are the producer, verifier, `dyadic_common.py`, `SOURCE-PINS.json`, `STATEMENT-ONLY.md`, `PROOF.md`, `INPUT-CONTRACT.md` and `INPUT-SCHEMA.json`. The reviewed frozen core supplies the external version trust root for those exact hashes.

Every fixed request member and member set is exact. The request retains the original model, metric, architectures and ordered translated-bump domain. A dyadic tolerance has `kind=dyadic` and a canonical decimal-string `q`, such as `"13"`; the separate compatibility form is `kind=one-fifth`.

Decimal q has no sign, leading zero, decimal point, exponent notation or whitespace. Values below three are unsupported. Canonical positive text exceeding the chosen representation ceiling resource-refuses before conversion to an integer or a q-dependent allocation. There is no mathematical upper q. Coordinate numerators are signed canonical decimal strings, with exponent $0, \ldots, 16$, odd numerator for a positive denominator exponent, and exponent zero for integers and zero. The exact coordinate bound $|P| \le 8\,2^k$ uses only fixed small shifts.

All ten limits are required. The first four are ordinary bounded integer fields; each stage limit is a typed integer node with `kind=integer` and canonical decimal text. A node's text must also fit `max_integer_digits`. The defaults and inclusive hard ranges are:

Limit	Default; hard range
`max_integer_digits`	1024; $1, \ldots, 4096$ decimal magnitude digits
`max_watched_bits`	8192; $16, \ldots, 8192$ bits of selected magnitudes
`max_milliseconds`	30000; $1, \ldots, 30000$ cooperative milliseconds
`max_steps`	20000; $1, \ldots, 50000$ named logical ticks
`max_n`	10^{100}; $1, \ldots, 10^{2048}$
`max_k`	10^{100}; $1, \ldots, 10^{2048}$
`max_regulator_exponent`	10^{400}; $4, \ldots, 10^{2048}$
`max_auxiliary_radius_exponent`	8191; $1, \ldots, 8191$
`max_stabilizer_exponent`	10^{1000}; $34, \ldots, 10^{2048}$
`max_width_exponent`	10^{1000}; $1, \ldots, 10^{2048}$

Equality at stage, representation, bit, transport, work and timing caps passes. A conservative small-power preflight can refuse when its preconstruction bound $\mathrm{bitlen}(b)j$ exceeds the watched-bit cap even though a particular eventual value could fit. These are selected software capacities, rather than necessary mathematical resources. The width cap bounds $e_{\mathrm{new}} = \log_2 N_{\mathrm{new}}$; w_{old} has the different unit $\log_2 \log_2 N_{\mathrm{old}}$.

Strict parsing and refusal precedence. API requests and certificate results are bytes. Request transport is capped at 32768 bytes and certificate-result transport at 131072 bytes; CLI reads stop after cap plus one. Before numerical or source work the parser checks strict UTF-8, no BOM, nesting at most sixteen, at most 5000 value/container nodes, strings at most 4096 characters, member names at most 256 characters, and integer tokens at most twenty characters including a sign. Duplicate members, floats, nonfinite tokens, malformed syntax, trailing data and unsupported types refuse; booleans are distinguished from integers. Byte capacity is reached before decoding, nesting before recursive JSON parsing, integer-token capacity during parsing, and the remaining structure capacities before semantic validation. A reached finite capacity returns `RESOURCE_EXHAUSTED`; it can prevent a later syntax condition from being reached. Malformed syntax and unsupported mathematical values return `UNSUPPORTED_INPUT` when reached. Certificate parsing preserves capacity refusal and translates malformed syntax to `REFUSED_CERTIFICATE`. This retains the corrected parser category without declaring a large valid q outside the theorem.

Every derived mathematical integer is a typed bounded decimal node; rationals have typed numerator and positive denominator. Powers are inert `pow2` nodes with typed exponents. The tolerance, a, ϵ, η, ζ and both widths remain symbolic. No 2^q, 2^p, 2^t, N_{new} or $2^{w_{\mathrm{old}}}$ is materialized. Every success node is watched and checked against its decimal digit ceiling. The hard 8192-bit watched bound implies fewer than 2468 decimal magnitude digits, below the 4096-character string cap and the interpreter conversion guard. The producer reparses its own encoded result before

reporting success. Thus successful output contains neither an oversized bare integer token nor an unreadable decimal node. A representation failure is an explicit resource refusal, even when the corresponding mathematical integer exists.

Exact factored arithmetic. Let $a, b > 0$ be integers and z a finite signed integer. To decide $a2^z \le b$ when $z \ge 0$, compare $\operatorname{bitlen}(a) + z$ with $\operatorname{bitlen}(b)$. Unequal lengths decide the comparison; equal lengths require the guarded exact integer shift. When $z < 0$, compare $\operatorname{bitlen}(a)$ with $\operatorname{bitlen}(b) - z$, with the same equal-length rule. The verifier separately computes the displacement $z + \operatorname{bitlen}(a) - \operatorname{bitlen}(b)$ and uses its sign before a guarded equal-length comparison. Both rules follow directly from $2^{h-1} \le M < 2^h$ for a positive integer of binary length h. Equality returns true. The guard checks the nonnegative shift exponent plus one before constructing that power. These exact predicates work even when the displayed exponent is far beyond a materializable power; no floating logarithm or astronomical shift is needed.

For a positive rational a/b, the producer starts at $h = \max(0, \operatorname{bitlen}(a) - \operatorname{bitlen}(b))$ and increases once if the factored test $a/b \le 2^h$ fails. Binary-length bounds put the least answer in precisely these two positions; it checks the answer and predecessor. The verifier independently uses zero for $a \le b$, and otherwise

$$\operatorname{clog}_2(a/b) = \operatorname{bitlen}\left(\left\lfloor \frac{a-1}{b} \right\rfloor\right). \tag{146}$$

Indeed $\lfloor (a-1)/b \rfloor = \lceil a/b \rceil - 1$, whose binary length is the least h with $a/b \le 2^h$. Its predicate and predecessor are also checked. This proves every non-strict dyadic tie rule in both paths. In particular $h_K = \lceil \operatorname{clog}_2(14u^2A)/2 \rceil$ is the least integer with $14u^2A \le 2^{2h_K}$, including exact powers of four and arguments at most one.

Least spatial and Nelson stages. Both paths use the exact geometry and $\overline{C}$ above. The producer writes $\overline{E}(i) = C(i)2^{-5i}$ with

$$C(i) = \frac{\overline{C}\,\overline{\exp R}}{2500}\left[\frac{32(2i+1)}{31} + \frac{64}{961}\right].$$

The verifier independently reduces the bracket to $(1984i + 1056)/961$. The spatial predicate is exactly $C(i)2^{d-5i} \le 1$. The preceding mathematical proof shows strict monotonicity and the positive upper witness $\lceil (d + 356)/4 \rceil$. The producer bisects from zero to the lesser of this witness and `max_n`; failure of the high predicate is a stage refusal. The verifier separately doubles from one, clamps at the same finite cap, and refines its false/true bracket. Each refinement strictly reduces the positive endpoint gap. The final predicate and predecessor prove the same least n. The selected ratio $\overline{E}(n)/\eta$ is materialized only with a separately guarded finite power and watched reduced rational. This is a capacity check on that ratio, rather than construction of η.

At this n, $A = 4000000n^2$, $U_A = A/10^{10}$ and $b_0 = 3/(16U_A)$ are exact rational values. The producer forms $L_{10}(k)$ using direct guarded nonnegative polynomial powers and factorials. The verifier uses the distinct recurrence $T_0 = 1$, $T_j = T_{j-1}k/j$, summing through ten. Both give the same degree-ten polynomial and strict predicate $L_{10}(k) > 32/b_0$. The finite witness $\lceil 32/b_0 \rceil$, producer bisection, independent verifier doubling/refinement, finite `max_k` and checked predecessor prove the same least k. Monotonicity follows from the nonnegative coefficients and positive linear term. Also $1 + \lceil \max(12U_A, 4\overline{K}, 4k^2/b_0) \rceil$ equals one plus the maximum of the individual ceilings. Thus the two implementations use exactly H_N, without substituting the older compact bounds. Both derive $p = 4H_N + 2h_K + 2d + 8$. Its first physical exponent is exactly $-d - 1$, its second is at most $-d - 1$,

and the preceding even integer fails the first predicate. The actual K_Λ and volume-dependent Ξ_2 remain in the conditional error estimate.

Auxiliary radius and stabilizer. At that same p, both compute $j = \lceil 18U_A p^2 \rceil$ and the complete $B_{\mathrm{aux}} = 64U_A p^2 + 4c_r$. By the proved logarithmic rule, $h_s = 1 + \mathrm{clog}_2(j + h_{\mathrm{aux}} + p + d)$ is the first integer at least one satisfying $j + h_{\mathrm{aux}} + p - 2^{h_s - 1} \le -d$. Predicate and predecessor are checked before accepting the selected stage. The guarded $s = 2^{h_s}$ gives $R_{\mathrm{aux}} = 1000n + s + 1$ and $O = 4R_{\mathrm{aux}}^2$. Since $R \le 9 < 1000n$, the strict source margin is one and $O > A$. Finally $t = \max(34, j + \mathrm{clog}_2(6Op^2) + d)$ gives the stabilizer exponent at most $-d$ and $0 < \epsilon \le 2^{-34} < u$. At every stage, equality with a cap passes and strict exceed refuses. The accepted mathematical argument then supplies the actual three $R_{\mathrm{phys}}, A_{\mathrm{aux}}, S$ bounds, with their full physical and common-space premises.

Relative threshold, parity and complete old comparison. Both implementations independently reconstruct $o_+ = \max(o, 78)$, $T_{1/2} = \lceil t/2 \rceil$, x, y, k_1, n_0, k_4 and e_{new}. They verify $4x \ge 2t + 4p + o_+ + 3$ and $4y \ge 2t + 8p + 3o_+ + 8$. The proof above keeps both t parities, the actual joint tilted moments through sixteen, every centered Bell/product term and the vector coefficient 38432. Consequently $K_1 < 2^{k_1 - 2}$ and $16K_1^2 < 2^{2k_1}$. Since the right side is an integer, $N_0 \le 2^{2k_1} < 2^{n_0}$. The exact positive difference $e_{\mathrm{new}} - n_0 = 6x + 2y + 4d + 38$ proves threshold safety. $K_4 < 2^{k_4}$ and $\zeta = 2^{-d-3}$ give normalized-law error below $3\eta/8$ and smoothing at most $\eta/2$. The actual first arrow is below $7\eta/8$. Software reconstruction uses these proved inequalities without replacing the interacting tilt by independent marginal laws.

At exactly those stages each path reconstructs the complete old coefficient majorants $h_{\mathrm{root}}, v_{\mathrm{old}}, q_{\mathrm{old}}, l_{\mathrm{old}}$ and H_{small}. It checks both positive remainder guards, $o \le l_{\mathrm{old}} - 1$ and $\mathrm{clog}_2(3U_A p^2) \le l_{\mathrm{old}} - 1$, and checks $H_{\mathrm{small}} \le 7l_{\mathrm{old}} + 47$ with $\mathrm{bitlen}(H_{\mathrm{small}}) \le l_{\mathrm{old}} + 2$. Thus the first old candidate fails its explicit strict inequality and $w_{\mathrm{old}} = l_{\mathrm{old}} + 3$ passes both original guards. The old sufficiency proof above retains coercivity and both Jensen terms. The new bound $e_{\mathrm{new}} \le 12l_{\mathrm{old}} + 553$ and $\mathrm{bitlen}(e_{\mathrm{new}}) \le w_{\mathrm{old}}$ prove $e_{\mathrm{new}} < 2^{w_{\mathrm{old}}}$, hence $2^{e_{\mathrm{new}}} < 2^{2^{w_{\mathrm{old}}}}$, without constructing either count. The obsolete compact $e < 2^{40}$ and $6up < 1$ guards do not appear.

The actual target budget and legacy equality. The five certificate arrows are the same $P_N \to Q_\epsilon \to Q_0 \to Q_a \to Q_{\mathrm{phys}} \to \nu^X$. Their upper factors in units of symbolic η are $(1, 1, 1, 1, \overline{E}(n)/\eta)$, whose sum is at most five. The allocated total divided by tolerance is $5/8$ for $d = q + 3$, and $25/32$ for the separate $1/5$ request with $d = 5$. The stronger first-arrow bound is separately retained. UDA.f proves exact equality of every listed legacy outer variable, both width formulas and all relative exponents on the eleven old settings. Both software paths check the bounded consequences and $e = 16\lceil t/2 \rceil + 20p + 4d + 544$ when that branch applies. This is mathematical stage equality; the new request and certificate identities remain distinct. Finite replay cases check executed paths, while the complete proof supplies the universal quantifier.

Source and core integrity; independent arithmetic. The embedded manifest digest binds all 541 direct source rows, including each exact relative path, size and hash. Per-source reads are capped at 1 MiB and aggregate source reads at 16 MiB. Resolved paths must remain inside the supplied root; absolute paths, parent escapes and symlink escapes refuse. Required original SELECT/OPT/EB1/EC1/ADMIT/SVA gates and material-role aliases, strict AP6 and unknown-g_* flags, and the exact-current UDA statement, proof, freeze, decision and open gate are checked among explicit digest aliases. An unavailable required body is a source gap; changed

size/hash/manifest/identity is a binding refusal; an over-cap read can resource-refuse first. Every declared body is bundled. These are byte identity checks, with the externally reviewed frozen core as the version trust root; a self-rewritten core or manifest is a different, unaccepted version. Source presence supplies no new analytic truth or authorship proof.

The verifier imports only the common syntax, binding, constants and resource utilities. It imports no producer stage/search/width routine and no older arithmetic module. Its independent bracket/refinement, Nelson recurrence, rational-power formula and integer algebra reconstruct the entire expected typed object. Exact canonical encoded comparison checks every nested witness, symbolic unit, source/core binding, scope flag, extra or missing member and Boolean/integer distinction. It freshly binds current code and source bytes. Historical logical steps, selected peak bits and elapsed time are checked as typed provenance, with `historical_execution_authenticated=false`. A fabricated but well-typed historical time may verify; the verifier's own cooperative measurement is separate. Source/core hashes authenticate neither historical elapsed time nor external scientific acceptance.

Selected watches, work and timing scope. Every search loop has a finite bracket/refinement variant or fixed degree, with named logical-step polling. Ticks count selected source/core/manifest reads, predicates, comparisons, logarithmic minimality checks, search iterations and polynomial/recurrence terms. They are software actions, rather than CPU instructions or a complexity bound. Output conversion, watches and time polls do not add ticks. Strict step exceed refuses. Selected integer and reduced-rational numerator/denominator watches cover the parsed stage caps and q, all selected stage values, factored comparison arguments, direct polynomial powers or recurrence terms, radii, area, relative/old exponents, ratio and typed output. A power-of-two allocation is guarded by exponent plus one; a degree at most twelve small power uses its conservative preconstruction bit bound. Fixed small constants and all Python arithmetic intermediates are not universally instrumented. These watches supply no all-intermediate or process-memory theorem.

Cooperative time begins after request parsing and validation in both APIs; the verifier parses the certificate result after creating its clock. Source reads, ticks, watches and final polls check elapsed time. Clock reads must be finite non-Boolean integers/floats, integer magnitude at most 10^{12}, callable and monotonically nondecreasing. A bad or backward clock resource-refuses. Equality at the millisecond limit passes; strict exceed refuses. An operation is not interrupted between polls. Serialization, output, interpreter startup and whole-process duration are outside that interval. Arbitrary interruption, interpreter/hardware/output-device failure and process-memory guarantees are outside the defined certificate contract. CLI output uses exclusive creation, preserving prior result bytes on retries.

The producer's `SUPPORTED_DOMAIN_SUCCESS` supplies a conditional numerical certificate. The verifier's `CERTIFICATE_VERIFIED` reports fresh binding and separate arithmetic. The defined refusals are:

- `UNSUPPORTED_INPUT`: malformed or unsupported request.
- `NUMERICAL_SOURCE_GAP`: required frozen source unavailable.
- `SOURCE_BINDING_REFUSAL`: changed binding or identity.
- `RESOURCE_EXHAUSTED`: a reached declared finite capacity.
- `REFUSED_CERTIFICATE`: malformed, incompatible or false result.

Every defined refusal has `numeric_certificate_available=false` and omits a numerical certificate. Success and verification keep self-acceptance false. The pre-correction parser core, original harness failures and the minor corrected forty-to-thirty-nine case count remain historical evidence;

the current parser categories and exact current source decision govern this version.

A recorded request beyond the old eleven settings. The accepted portable example uses the four distinct ordered corners $(-8,-8),(-8,8),(8,-8),(8,8)$, both architectures, and $q=13$. Its certificate records

$$\begin{aligned} d &= 16, & n &= 76, & H_N &= 2417, & p &= 9708,\\ h_s &= 33, & R_{\mathrm{aux}} &= 8590010593, & t &= 3919396758, & & \\ e_{\mathrm{new}} &= 31355368832, & w_{\mathrm{old}} &= 3919435662. & & & & \end{aligned}$$

The exact spatial ratio is $\overline{E}(n)/\eta = 691456/11411875$. Thus the symbolic counts are $2^{31355368832}$ and $2^{2^{3919435662}}$ in their distinct units. This is recorded exact conditional arithmetic; it is neither an actual field simulation nor a measurement of the cost of constructing those process neurons. Any replay performed for this manuscript has its own hashed receipt. The universal theorem comes from the proof above, independently of this finite witness.

All physical covariance/Wick/common-space, strong-L^4 random-element, zero-Bochner-mean, sign, every bounded-bilinear and actual endpoint interfaces retain their original conditional scopes at each selected finite positive stage. NR1's sharper author versus larger blind core, classical/source truth, strict AP6 and unknown original fixed g_* remain explicit. Width counts whole-plane process neurons with countably many internal variables. The software computes no canonical least inner indices, practical/minimal/necessary cost, finite global-weight field simulator, full-field rate, global neural Schwartz control, broader model/test family or priority. Its internal exact-version acceptance and this manuscript's fresh review are distinct; the immutable public v1 is unchanged.

L Renormalized local squares and finite-neural approximation

Guide to the local-square update

The preceding all-dyadic theorem controls its stated family of linear measurements. This addendum integrates the separately accepted local-square results CC1-SMOOTH-BYPASS-v1 and CA1-v1: a quantitative approximation of the renormalized square and a comparison of the actual normalized finite-neural joint law with the original target pair. All named premises remain below. Internal acceptance of the source results was recorded on 31 October 2026. Their complete manuscript integration received source and actual-proof matching review on 6 November; the public REVIEW document states the review limits.

For the same target, physical subtraction and Borel prescription, CC1 gives

$$\mathbb{E}_{\nu_\infty}|Q_s - Q_0| \le 12 L_f \sqrt{s} < 36\sqrt{s}.$$

Coupling the unchanged first coordinate supplies precisely CA1.5b's joint bounded-Lipschitz input. Under the *union* of S1–S4, H1–H7 and Appendix S, take

$$r(s) = 36\sqrt{s}, \qquad S_r(\tau) = 2^{-2\,\mathrm{L2plus}(36/\tau)}.$$

The target allocation is $\eta/8$, hence $j = \mathrm{L2plus}(288/\eta)$. CA1 then selects finite positive stages of joint error at most every rational $0 < \eta \le 1$. This is conditional mathematical termination. Useful sampling, expectation-evaluation runtime, finite-bit global simulation, interaction-resolving gap, matched advantage, priority and larger-model claims remain separate. G6 remains **STOP / INCOMPLETE**; no full G1–G9 capability is promoted. Neither the stronger numerical L^2 density certificate nor a later tail improvement enters this update.

L.1 Conventions and complete retained premises

Use $z = (t, y)$, $p = (w, q)$ in two Euclidean dimensions,

$$\hat{g}(p) = \int e^{-ip\cdot z} g(z)\, dz, \quad d\lambda(p) = dp/(2\pi)^2, \quad m = 1, \quad u = 10^{-10}, \quad \ell = 1000.$$

There is no factorial in u. Write $\Lambda = B_n = (-1000n, 1000n)^2$, $n \in \mathbb{N}$, $n \ge 1$, and $A_n = 4{,}000{,}000n^2$. Wick ordering is $:x^2:_c = x^2 - c$ and $:x^4:_c = x^4 - 6cx^2 + 3c^2$. All finite-dimensional norms, derivative operator norms and Lipschitz constants are Euclidean. BL_1 imposes separately $\sup|G| \le 1$ and $\mathrm{Lip}(G) \le 1$, so its probability distance is at most two. Source lattice mesh, flow time and profile are distinct from a, β, s, η and the shell parameter q.

H1: original target, common Gaussian space and physical laws. The original real complete separable space is $X = H^{-2}_{\mathrm{loc}}(\mathbb{R}^2; \mathbb{R})$. Smooth $0 \le \chi_j \le 1$ equal one on $[-j, j]^2$, are supported in $(-j - 1/2, j + 1/2)^2$ and satisfy $\chi_{j+1} = 1$ on $\operatorname{supp}\chi_j$. On $\mathbb{T}_j = \mathbb{R}^2/[2(j+1)\mathbb{Z}]^2$, put

$$V_j = 4(j+1)^2, \quad e_{j,k}(z) = V_j^{-1/2} e^{i\pi k\cdot z/(j+1)}, \quad \lambda_{j,k} = 1 + \pi^2|k|^2/(j+1)^2,$$
$$p_j(v)^2 = \sum_{k\in\mathbb{Z}^2} \lambda_{j,k}^{-2} |v(\chi_j \overline{e_{j,k}})|^2, \qquad d_X(v, w) = \sum_{j\ge 1} 2^{-j} \min(1, p_j(v - w)).$$

Compact smooth evaluations are continuous and generate the stated Borel sigma-algebra. The original strict AP3/AP5/AP6 route identifies ν_∞ as the X representation of the unique common original Fröhlich–Simon maximal state $v_{(1,10^{-10})}$, preserving every such evaluation, and gives $\nu_n \Rightarrow \nu_\infty$ in X. The strict cap $0 < 10^{-10} < 1/8000$ retains P1–P6 below; the older unknown g_* route is distinct.

On the real Hermitian isonormal space $(\Gamma, \mathcal{F}, P)$, $H = L^2(\hat{C}_0\, d\lambda)$, use accepted Borel X versions and continuous positive-a point versions

$$\hat{C}_0 = (1 + w^2 + q^2)^{-1}, \quad \hat{C}_a = (1 + w^2 + q^2 + a^2 w^2 q^2)^{-1}, \quad 0 < a < 1,$$

$$\tau_a = (\hat{C}_a/\hat{C}_0)^{1/2}, \quad \tau_0 = 1, \quad \Phi(g) = W_H(\hat{g}), \quad \Phi_a(g) = W_H(\hat{g}\tau_a),$$
$$c_a = \int \hat{C}_a \, d\lambda = \frac{1}{2\pi}\int_0^\infty \frac{dq}{\sqrt{(1+q^2)(1+a^2q^2)}}, \qquad U_{a,n} = u\int_\Lambda (\Phi_a^4 - 6c_a\Phi_a^2 + 3c_a^2).$$

$U_{0,n}$ is the accepted Gaussian L^2 limit; these are centered fourth chaos. At every permitted $a \geq 0$ define individually

$$Z_{a,n} = \mathbb{E}_P e^{-U_{a,n}}, \quad r_{a,n} = e^{-U_{a,n}}/Z_{a,n}, \quad \mu_{a,n} = r_{a,n}P, \quad \nu_{a,n} = (\Phi_a)_\# \mu_{a,n}, \quad \nu_n = \nu_{0,n}.$$

The actual same-mass, own-variance, whole-plane free-boundary Wick/classical identification, Gaussian versions, isometry $\mathbb{E}I_2(K)I_2(L) = 2\langle K, L\rangle$, Gaussian moments and homogeneous-chaos hypercontractivity remain accepted/classical inputs. Version equalities are fixed-parameter or specified countable-family equalities.

H2: fixed-box quantitative physical/common-space input HC. Define

$$F_2 = \|f\|_2, \quad H_f = F_2\sqrt{2/\pi}, \quad L_f = F_2\sqrt{2\sqrt{5}/\pi}, \quad \kappa_2 = \frac{\sqrt{17}}{8\sqrt{\pi}}, \quad M_6 = (2\pi)^{-5/3}(5\pi)^{5/6},$$
$$K_n = u\sqrt{384A_n\kappa_2 M_6^3} < n/10^6, \quad h_2(n) = \max(n^2/100, n^4/125), \quad m_2(n) = 1 + \lceil 4h_2(n)\rceil,$$
$$\ell_n = \lceil m_2(n)/2\rceil, \quad M_n = 2^{\ell_n}, \quad C_n = 3 + 4n/10^6, \qquad R_n(a) = M_n\{aF_2/2 + (H_f + 4K_n)\sqrt{a}\}.$$

Retain CMP-FV-v1 P1–P4 and literal C2/C4/C5/C6 at every fixed n:

$$1 \leq Z_{a,n} \leq M_n, \quad \|r_{a,n}\|_2 \leq M_n, \quad \|r_{a,n} - r_{0,n}\|_1 \leq 4K_nM_n\sqrt{a}.$$

For the H4 own-smoothed pairs,

$$d_{\mathrm{BL}}\big(\mathrm{Law}_{\mu_{a,n}}(\Phi_a(f), Y_{a,s}), \mathrm{Law}_{\mu_{0,n}}(\Phi(f), Y_s)\big) \leq \min(2, R_n(a)). \tag{HC}$$

The common-space density and actual field pushforward are distinct. Fixed-box raw-square and Borel-prescription qualifications remain; $Q_0(\Phi_a) = \int f(\Phi_a^2 - c_a)$ is not hypothesized.

H3: finite-linear response, moments and local target density. For every real compact smooth h and $t \in \mathbb{R}$,

$$\mathbb{E}_\zeta e^{tv(h)} \leq e^{(3/2)t^2\|h\|_2^2}, \qquad \zeta \in \{\nu_n : n \geq 1\} \cup \{\nu_\infty\}.$$

Accepted MOM all-positive local p_j moments, uniform integrability and polynomial-local-growth expectation transfer remain at their exact scopes. The unrestricted finite-linear-vector AP6 interface is literal as follows. For $g = (g_i)$ supported in $K \subset [-R, R]^2$ inside open B_{n_0}, set

$$\overline{F}_g = (2R + 1/2)(\sum_i \|g_i\|_\infty^2)^{1/2}, \quad C_g = P_*\overline{F}_g, \quad q = e^{-125/32},$$
$$\Theta_n = q^n\{(2n+1)/(1-q) + 2q/(1-q)^2\},$$
$$d_{\mathrm{BL}}(g_\#\nu_n, g_\#\nu_\infty) \leq 4C_g u\ell^2 e^{R/256}\Theta_n \quad (n \geq n_0,\ n\ell \geq R).$$

It includes singular/repeated tuples and the norm-preserving BL extension. The chosen OPT coefficient version has the same $C_H < 2^{16}$, $C_{H2} \leq 2^{384}$ and $C_{\mathrm{base}} = P_*(5/2)\|f\|_\infty \leq 2^{360}$. For literal identification of P_* put

$$\bar{\delta} = 2\ell^{-2}\log[1/(1 - e^{-4\ell})], \quad g_{\mathrm{flow}}(t) = 1 + \bar{\delta} + (\pi/16)\log(1 + 1/t),$$
$$c_{\mathrm{CT}} = 81/5, \quad b_{\mathrm{CT}} = 1 + c_{\mathrm{CT}}/256, \quad \Gamma_G = 4\sqrt{2\pi}/256,$$
$$C_* = 6c_{\mathrm{CT}}/256 + c_{\mathrm{CT}}^2/256^2 + b_{\mathrm{CT}}^2(\Gamma_G + 3300u),$$

$$M_{\mathrm{flow}}(t) = 32\ell^{-2}\{C_H^2 g_{\mathrm{flow}}(t)^3/12 + u^2 C_{H2}^2 g_{\mathrm{flow}}(t)^7/9\},$$
$$P_* = e^{8\sqrt{2}/256} b_{\mathrm{CT}}^2 \{e^{C_*}\int_0^1 \sqrt{M_{\mathrm{flow}}(t)}\,dt + e^{2C_*}\sqrt{M_{\mathrm{flow}}(1)}/2\}.$$

Source time t is separate from s. This general AP6 interface is not an extension of UDA's restricted ordered one-through-four bump theorem.

Let $D_{\mathrm{loc}} = (-3,3)^2$ and let $\mathcal{C}_D$ be generated by real compact smooth evaluations supported inside it. The same target has a normalized density relative to the *restricted whole-plane mass-one* Gaussian law:

$$k_D \ge 0, \quad \mathbb{E}_{P_1} k_D = 1, \quad K_D = \|k_D\|_{L^2(P_1)} < \infty,$$
$$k_D = e^{-uJ_D - \alpha_1(uX^4)|D|} g_{\partial D}, \qquad g_{\partial D} > 0 \text{ a.e.}, \quad D = D_{\mathrm{loc}}.$$

The boundary-algebra alternatives and all-semibounded-polynomial scope remain: the factor is measurable either for the intersection of neighborhood algebras or for the completed algebra generated by boundary-supported H^{-1} evaluations. Pressure and boundary factor belong to this density; no finite-box partition function or Dirichlet Gaussian is substituted. A numerical upper certificate for K_D is additional only in CA1.5.

H4: fixed observables and Gaussian composite estimates. Define

$$b(t) = \begin{cases} e^{-1/(1-t^2)}, & |t| < 1, \\ 0, & |t| \ge 1, \end{cases} \quad f(t,y) = b(t)b(y), \quad I = \int f, \quad \rho = f/I,$$
$$R_2 = \|\rho\|_2, \quad D_\rho = \|\nabla\rho\|_2, \quad \rho_s = s^{-2}\rho(\cdot/s), \quad h_{s,z}(x) = \rho_s(z-x), \quad 0 < s \le 1,$$
$$c_s = \int |\hat\rho(sp)|^2 \hat C_0(p)\,d\lambda, \quad c_{a,s} = \int |\hat\rho(sp)|^2 \hat C_a(p)\,d\lambda,$$
$$X_{\mathrm{obs}}(v) = v(f), \quad Q_s(v) = \int f(z)\{v(h_{s,z})^2 - c_s\}\,dz,$$
$$Y_{a,s} = \int f\{(\rho_s * \Phi_a)^2 - c_{a,s}\}, \qquad Y_s = Q_s(\Phi).$$

Q_s is the actual continuous/Borel local X functional. Set $Q_0(v) = \lim_{j\to\infty} Q_{2^{-2j}}(v)$ on its Borel convergence set, zero elsewhere, $j \ge 1$. This is the same physical local Borel Wick prescription under ν_n, ν_∞. On the Gaussian space,

$$Q_0(\Phi) = I_2(\hat f(p+q)) \text{ a.s.}, \qquad \|Y_s - Q_0(\Phi)\|_2 \le L_f\sqrt{s}.$$

The Gaussian null set and target-local absolute-continuity qualifications remain. Write $\mathcal{L}_{\mathrm{target}} = \mathrm{Law}_{\nu_\infty}(X_{\mathrm{obs}}, Q_0)$, with second coordinate $O_2(f)$. Literal envelopes:

$$1/27 < I < 1, \quad F_2 < 1/2, \quad \|f\|_{4/3} < 1, \quad R_2 \le 54, \quad D_\rho \le 81, \quad H_f < 2, \quad L_f < 3.$$

The weaker $I \le 4, F_2 \le 2$ are allowed. All $h_{s,z}$ on $\mathrm{supp}\, f$ are supported in $[-2,2]^2$, strictly inside D and every open B_n. The fixed query is $F(x,y) = \cos((x+y)/2)$.

H5: original actual source laws RL2/SP1–SP4. The product neuron is $g^{\mathrm{product}}(t,y) = \sigma(t)\tau(y)/2$ for independent stationary symmetric two-sided telegraphs with flip rate $1/(2a)$. The Mondrian neuron is $g^{\mathrm{Mondrian}}(z) = \eta_{\mathrm{cell}(z)}/2$ from the original stationary projective axis-parallel whole-plane partition at budget $1/a$, with conditionally independent fair cell signs. A rectangle of sides w, h splits at rate $w+h$, vertically with probability $w/(w+h)$ and uniform cut; daughters evolve independently for the remaining budget. Use the lower-left half-open representative modulo area-null boundaries. Whole-plane existence, stationarity/projectivity, path measurability, copies and actual architecture/potential identification remain SP1–SP4.

The original centered continuous Gaussian endpoint γ has

$$C_{\mathrm{end}}(z-z') = e^{-|z-z'|_1/a}/4, \qquad \hat{C}_{\mathrm{end}} = \frac{a^2}{(1+a^2w^2)(1+a^2q^2)}.$$

For finite $R_{\mathrm{aux}} \geq 1000n$, set $D_{\mathrm{aux}} = [-R_{\mathrm{aux}}, R_{\mathrm{aux}}]^2$, $d = 4R_{\mathrm{aux}}^2$, $E = L^4(D_{\mathrm{aux}}; \mathbb{R})$, $K = a^{-2} - 1$, $F_N^{\mathfrak{b}} = N^{-1/2}\sum_i g_i^{\mathfrak{b}}$, $\rho_N^{\mathfrak{b}} = \mathrm{Law}_E(F_N^{\mathfrak{b}})$, $\rho_\gamma = \mathrm{Law}_E(\gamma)$, $\mathfrak{b} \in \{\mathrm{product}, \mathrm{Mondrian}\}$. These actual strongly measurable E random elements have zero Bochner means, global sign symmetry, fourth-norm moments $d/16$ and $3d/16$, respectively, and finite second-norm moments. For *every* bounded real bilinear form B_0, including nonsymmetric ones,

$$\mathbb{E}B_0(g^{\mathfrak{b}}, g^{\mathfrak{b}}) = \mathbb{E}B_0(\gamma, \gamma).$$

Independent full-law copies exist; $N^{-1/2}\sum \gamma_i$ has exactly ρ_γ. The finite-rank/every-bilinear route retains accepted NR1 Astra/max attribution and exact-author/larger-blind-core qualification; RL2 consolidation and normalized-law work retain recorded Sol/max attribution.

For $\beta > 0$ define on E

$$W(v) = u\int_\Lambda (v^4 - 6c_a v^2 + 3c_a^2), \quad J_{\mathrm{out}}(v) = \int_{D_{\mathrm{aux}}\setminus\Lambda} v^4,$$
$$V(v) = W(v) + \beta J_{\mathrm{out}}(v) - (K/2)\int_{D_{\mathrm{aux}}} v^2, \quad w = e^{-V},$$
$$Z_N^{\mathfrak{b}} = \rho_N^{\mathfrak{b}}(w), \quad Z_\gamma = \rho_\gamma(w), \quad P_N^{\mathfrak{b}} = w\rho_N^{\mathfrak{b}}/Z_N^{\mathfrak{b}}, \quad Q_\beta = w\rho_\gamma/Z_\gamma.$$

T_s denotes convolution after zero extension from D_{aux}, agreeing with the whole-plane convolution on $\mathrm{supp}\, f$. Put

$$B_T(v,h) = \int f(Tv)(Th), \quad J_{T,k}(v) = (\int_{D_{\mathrm{aux}}} fv, B_T(v,v) - Ik),$$
$$J_s = J_{T_s,c_s}, \qquad \mathcal{L}_{\mathrm{neural}}^{\mathfrak{b}} = (J_s)_{\#}P_N^{\mathfrak{b}}.$$

These are full tilted actual source laws with distinct normalizers. No covariance-to-nonlinear-law identification is assumed.

H6: literal HP mass tilt, residual laws and defect coupling. For the whole-plane endpoint law G_{end} put

$$M_{D_{\mathrm{aux}},a} = \mathbb{E}_{G_{\mathrm{end}}} e^{(K/2)\int_{D_{\mathrm{aux}}} v^2} \in (0,\infty), \quad G' = e^{(K/2)\int_{D_{\mathrm{aux}}} v^2} G_{\mathrm{end}}/M_{D_{\mathrm{aux}},a},$$
$$z_\beta = \mathbb{E}_{G'} e^{-W-\beta J_{\mathrm{out}}}, \quad z_0 = \mathbb{E}_{G'} e^{-W}, \quad Q_{\mathrm{res}}^\beta = e^{-W-\beta J_{\mathrm{out}}} G'/z_\beta, \quad Q_{\mathrm{res}}^0 = e^{-W} G'/z_0.$$

The genuine centered continuous whole-plane Gaussian $G' = G'_{D_{\mathrm{aux}},a}$ has the uniformly convergent covariance series, $x_0 = z, x_{j+1} = z'$,

$$C'(z,z') = C_{\mathrm{end}}(z-z') + \sum_{j\geq 1} K^j \int_{D_{\mathrm{aux}}^j} \prod_{i=0}^{j} C_{\mathrm{end}}(x_i - x_{i+1})\, dx_1 \cdots dx_j.$$

The all-Borel whole-path/restriction mass-tilt identity and $Z_\gamma = M_{D_{\mathrm{aux}},a} z_\beta$ remain; Q_β is the E restriction of Q_{res}^β. The physical positive-a law is $e^{-W}G_a/Z_{a,n}$ with its own normalizer.

$\Delta_{D_{\mathrm{aux}},a} = C_a - C'$ is a continuous positive semidefinite kernel; C' point variances lie in $[0, c_a]$. Independent continuous centered Gaussian $Y \sim G'$ and $\psi \sim G_\Delta$ exist with $X_a = Y + \psi \sim G_a$. For compact H put $\delta_D(H) = \sup_{z\in H} \Delta(z,z)$. If D_{aux} contains its Euclidean r neighborhood, $r > 0$,

$$\delta_D(H) \leq \xi_a(r) = K\int_{|x|>r} C_a(x)^2\, dx.$$

Separate fixed-D couplings suffice. All physical/common-space, CT G-tilt/G-order/G-diff, classical and source qualifications remain.

H7: literal normalized NC1 stage estimate. Here let $\alpha = \min(u,\beta)$, $A = \max(u,\beta)$, $B = K/2 + 6uc_a$, $L = B^2 d/(2\alpha)$. These α, L are distinct from pressure and the CC1 smoothing constant. For $x \ge 0$ put

$$\begin{aligned}
&v_1 = 4Ax^3 + 2B\sqrt{d}\,x, && v_2 = 12Ax^2 + 2B\sqrt{d}, && v_3 = 24Ax, && v_4 = 24A,\\
&e_1 = v_1, && e_2 = v_1^2 + v_2, && e_3 = v_1^3 + 3v_1v_2 + v_3, &&\\
&e_4 = v_1^4 + 6v_1^2v_2 + 3v_2^2 + 4v_1v_3 + v_4, &&&&&&\\
&q_1 = 1 + 2x, && q_2 = 2, && p_1 = q_1, &&\\
&p_2 = q_1^2 + q_2, && p_3 = q_1^3 + 3q_1q_2, && p_4 = q_1^4 + 6q_1^2q_2 + 3q_2^2. &&
\end{aligned}$$

For a nonnegative-coefficient $P = \sum_j P_j x^j$, let $a_0 = 1$, $a_j = [j/(2\alpha e)]^{j/4}$ for $j > 0$, $\mathcal{T}(P) = e^L \sum_j P_j a_j$, and

$$P_Z = e_4, \quad P_J = e_4 + 4p_1e_3 + 6p_2e_2 + 4p_3e_1 + p_4, \quad M_Z = \mathcal{T}(P_Z), \quad M_J = \mathcal{T}(P_J),$$
$$z = e^{-3Ad/16 - 3uc_a^2 A_n}, \quad C = \frac{d(M_J + M_Z)}{96z}, \quad D_J = \frac{4}{3}(3\sqrt{6}C)^{1/4}(2\sqrt{2})^{3/4}.$$

The exact imported conclusions are positive bounded Borel w, $Z_N^{\mathfrak{b}}, Z_\gamma \ge z > 0$, all-Borel actual source-law identities of restrictions and full tilts, and, uniformly in $\mathfrak{b}$, $N \ge 1$, $0 < s \le 1$ and every finite common k,

$$d_{\mathrm{BL}}((J_{T_s,k})_\# P_N^{\mathfrak{b}}, (J_{T_s,k})_\# Q_\beta) \le \min(2, D_J N^{-1/4}),$$
$$d_{\mathcal{U}_4}((J_{T_s,k})_\# P_N^{\mathfrak{b}}, (J_{T_s,k})_\# Q_\beta) \le \min(2, C/N).$$

$\mathcal{U}_4$ bounds value and derivative operator norms through order four by one. For $0 < \eta_W \le 1$, $N \ge \lceil 512\sqrt{3}C\eta_W^{-4} \rceil$ suffices for error $\le \eta_W$. All $a, \beta, n, R_{\mathrm{aux}}$ dependence remains. No $a = 0, \beta = 0, s = 0$ stage is included.

S1: CC1 model and source qualification. CC1 uses exactly H1's model, boxes, Fourier convention, original X Borel space and common FS state, the actual weak-X limit of the original same-mass free-boundary Wick-quartic laws. The strict route retains literal P1–P6 and accepted H0–H5 source hypotheses; the original numerical g_* inequality is uncomputed and unused. P1–P6 include finite ferromagnetic DSS/SK-q for all finite real physical fields, finite-positive-scale P_* susceptibility, fixed-square Gaussian/Wick exponential transfer, fixed-mesh periodization, original FS pressure/maximal-state/free-limit facts, and accepted physical/common-space/local-density/finite-response interfaces. Individual source, classical, H1/MIX/REG/FLOW/TP/T and NR1 qualifications remain. RL2/SP1–SP4 and NC1 HP/HC condition the later assembly; CC1 alone makes no new assertion about those arrows.

S2: Gaussian reference, local interaction and Nelson input. P_1 is the centered whole-plane Gaussian with multiplier $\hat{C}_0$; only its sigma-algebra is restricted. The common realization and Φ_a, τ_a, c_a are H1's. Let $D = (-3,3)^2$, $A_D = 36$ and $U_D = uJ_D$, the literal coefficient-one mass-one Gaussian fourth Wick integral, equivalently the Gaussian L^2 limit as $a \downarrow 0$ of

$$U_{a,D} = u \int_D [\Phi_a(z)^4 - 6c_a\Phi_a(z)^2 + 3c_a^2]\,dz.$$

$\mathbb{E}_{P_1} U_D = 0$. Retain Gaussian construction, general-rectangle Nelson estimate, second-chaos isometry, finite-chaos hypercontractivity, Plancherel and elementary measure/real analysis at their named classical scopes. Put

$$\kappa_0 = e(1 + \sqrt{2}), \quad K_{A_D} = u\sqrt{384 A_D \kappa_2 M_6^3},$$
$$K_{A_D}^\sharp = \max(12uA_D, (2/e)K_{A_D}\sqrt{\kappa_0}), \qquad \beta_{A_D} = \pi/(2\sqrt{12uA_D}).$$

The retained zero-regulator tail is $P_1(U_D < -v) \le \exp[-\exp(\beta_{A_D}\sqrt{v} - 1)]$ for $v \ge K_{A_D}^\sharp$. All parameters refer to this same interaction and whole-plane mass-one reference.

S3: actual normalized factorization and numerical source. Write $\alpha_{\rm pr} = \alpha_1(uX^4)$ for the original normalized log-partition pressure. The density is

$$k_D = e^{-U_D - 36\alpha_{\rm pr}} g_{\partial D}, \quad k_D \geq 0, \quad \mathbb{E}_{P_1} k_D = 1.$$

The same positive boundary factor and BC1/BC2 readings in H3 remain; no isolated finite-box normalizer enters. Inherited H5 supplies $K_D < \infty$ without its value. The accepted primary supplement FS1977-v1 matches Fröhlich–Simon Lemma 7.1 and Theorem 7.2 proof, printed p. 517, equation (7.1), for the original Section 4 maximal states including their common value. For an l-square E, mass m_0, and $q \geq 4(1-e^{-m_0 l})^{-2}$, every nonnegative bounded $\mathcal{C}_E$ measurable H satisfies

$$\nu_\infty(H) \leq e^{-|E|\alpha_1(uX^4)} \|He^{-uJ_E}\|_{L^q(P_1)}.$$

Here $m_0 = 1, E = D, l = 6$. This attributed fact retains literal same-mass, pressure, polynomial and state-class scope [8, Lemma 7.1 and proof of Theorem 7.2, p. 517]. The accepted investigation inspected primary text and page image; this integration transcribes that frozen premise. Its GRS checkerboard dependency is retained [10]. The numerical clause is a matched primary input, not a consequence of the old qualitative H5 alone.

S4: CC1 observables. Use H4's $b, f, I, F_2, \rho, \rho_s, h_{s,z}, c_s, Q_s$ and countable Borel Q_0. Set

$$r_s(p) = \hat{\rho}(sp), \quad K_f(p,q) = \hat{f}(p+q), \quad O_2^P = I_2(K_f), \quad \sigma = F_2/\sqrt{2\pi}, \quad L_f = F_2\sqrt{2\sqrt{5}/\pi}.$$

The isometry has factor two. Under the actual target $Q_0 = O_2(f)$. All smoothed tests stay strictly inside D. Subtraction is exactly c_s under target and physical laws. The joint query remains $X_{\rm obs}(v) = v(f)$, $F(x,y) = \cos((x+y)/2)$.

Appendix S: full retained source scope

These are attributed accepted-source hypotheses with original source and review qualifications, rather than fresh proofs.

P1. On $\mathbb{T} = \epsilon(\mathbb{Z} + 1/2)^2/L\mathbb{Z}^2$, L/ϵ even and at least four, use physical inner product $\epsilon^2 \sum$, $A = -\Delta^\epsilon + m^2$ and the *plane* variance c_ϵ. For every profile $q : \mathbb{T} \to [0,1]$, the law proportional to $e^{-u\epsilon^2 \sum q_x : \psi_x^4 :_{c_\epsilon}} dP_{A^{-1}}$ under every finite real physical field H satisfies $0 \leq \mathrm{Cov}_{\nu^H}(\psi_x, \psi_y) \leq \mathbb{E}_\nu \psi_x \psi_y$. The DSS/SK-$q$ clauses with added mass $1/t$ hold at all finite $t > 0$. Retain BD/BBD/Griffiths–Simon, Ising-FKG and Robbins–Stirling qualifications; no g_* premise is added.

P2. For $0 < \epsilon \leq \min(1/4, 18/(5m))$, $L \geq \max(1, \sqrt{2}/m)$, admissible even $L/\epsilon \geq 4$ and $0 < u \leq m^2/8000$, the original core P_* at $S_{\max} = \infty, \Phi = 1, \kappa = 0$ gives at every finite $t > 0$ and profile q

$$\chi_{t,q} = \max_x \epsilon^2 \sum_y \mathbb{E}_{\mu_{t,q}} \psi_x \psi_y \leq t/(1+m^2 t) + 2t/(2+m^2 t),$$

$$d\mu_{t,q} \propto e^{-u\epsilon^2 \sum q_x : \psi_x^4 :_{c_\epsilon}} dP_{(A+1/t)^{-1}}.$$

It retains $\|S - C_t\|_{1,0} \leq 2t/(2+m^2 t)$; P2 alone supplies no original-law endpoint conclusion.

P3. At fixed $B = (-b, b)^2$, commensurate plane meshes $\epsilon = b/N$ have the accepted T1–T4/GQ/W(b)/W-laws common Gaussian realization: $J_B^\epsilon = \epsilon^2 \sum_{x \in B} : \psi_x^4 :_{c_\epsilon}$ and $Y_f^\epsilon = \epsilon^2 \sum f(x)\psi_x$ converge in Gaussian L^2 to $J_B, \Phi(f)$. For each fixed finite $p \geq 1$, $\sup_{\epsilon \text{ sufficiently small}} \mathbb{E} e^{-puJ_B^\epsilon} < \infty$. Positive finite continuum normalizers, every finite Gaussian L^p density, coefficient-one Wick conventions and exact finite-law identifications remain at all $m, u > 0$.

P4. At fixed mesh torus covariances converge entrywise on finite fixed site sets to plane covariance through admissible even periods, with the same plane Wick variance. This is the accepted H1-PREQ periodization/envelope premise.

P5. Use the original FS same-mass whole-plane free-boundary convention and $P_h(X) = uX^4 - hX$. Its normalized pressure and dyadic-square realization are original Proposition 2.1 and the remark after (2.3). Corollary 4.3(ii)'s converse equates differentiability at $h = 0$ with equality of the maximal states; Theorem 5.2 supplies ordinary-exhaustion convergence to the common state, in the compact generating-functional reading at every real frequency [8]. These are attributed published facts; full original FS proofs were not supplied/read in this integration.

P6. Retain (i) the all-$m, u > 0$ physical/classical identity and full original local H^{-2}/Borel/compactness definitions; (ii) all-semibounded-polynomial original FS local L^p density and positive boundary factorization under either boundary-algebra reading; (iii) exact finite DSS/SKQ/P_*/H1-PREQ/ H1/MIX/REG/smoothed-CT/FLOW/TP/T interfaces, individual accepted decisions and source-normalization correction, including FLOW's $0 < u \le m^2/8000$ cap and fixed-square TP/T1–T4 scope; and (iv) raw fixed-stage NR1/L0 and Gaussian/common-space/Wick normalizers, original author-constant/larger-blind-core qualification, actual Gaussian mass-tilt/independent-defect coupling and physical own-variance identification. Old EB1/FV/CL1 with global unknown g_* are excluded outside their literal domains; AHKZ C2(ii)/(iii) are not added.

Named FS, AHKZ, GRS and SG, DSS, BD, BBD and Griffiths–Simon, Gaussian/Wick/Nelson, H1/MIX/FLOW/TP/T, NR1 attribution/review and source-access qualifications remain. Original FS/OPT full bodies are not a fresh audit here. Frozen CV1/NC1/CMP/RL2 decisions preserve all inherited adverse findings at original object, severity and disposition. The original private proof map distinguishes actual reading in that integration from these transcribed premises.

L.2 Complete CC1 conclusions

Theorem L.1 (CC1-SMOOTH-BYPASS-v1, under S1–S4)**.** *The following conclusions hold for the fixed original target.*

CC1.1 *The local Gaussian exponential certificate is*

$$K_{A_D} < 128u, \quad K^{\sharp}_{A_D} = 432u, \quad \beta_{A_D} > 1500, \quad \mathbb{E}_{P_1} e^{-10U_D} \le 19/9 < 4, \quad \|e^{-U_D}\|_{10} \le 4/3.$$

CC1.2 *For the literal pressure and boundary factor,*

$$\alpha_{\mathrm{pr}} \ge 0, \qquad \|g_{\partial D}\|_{5/4} \le 1, \qquad \|k_D\|_{10/9} \le 4/3.$$

This is a different exponent from $K_D = \|k_D\|_2$ and does not certify the latter numerically.

CC1.3 *The same Borel Q_0 agrees with O_2^P under P_1 and its convergence set has full target probability. For $0 < s \le 1$ and real $r \ge 1$,*

$$\|Q_s - Q_0\|_{L^r(\nu_\infty)} \le (4/3)^{1/r}(10r-1)L_f\sqrt{s},$$
$$\|Q_0\|_{L^r(\nu_\infty)} \le (4/3)^{1/r}(10r-1)\sigma.$$

In particular $\mathbb{E}_{\nu_\infty}|Q_s - O_2(f)| \le 12L_f\sqrt{s} < 36\sqrt{s}$. Every fixed positive moment order of $\{Q_s : 0 < s \le 1\}$ is uniformly integrable under the target.

CC1.4 *On positive rationals define* $\mathrm{L2plus}(x) = \max(0, \lceil \log_2 x \rceil)$ *by exact integer comparisons. For positive rational τ, choose $j = \mathrm{L2plus}(36/\tau)$ and $s = 2^{-2j}$. Then $0 < s \le 1$ and the target L^1 smoothing error is $\le \tau$. For an allocation $\epsilon/4$, $j = \mathrm{L2plus}(144/\epsilon)$ suffices. Symbolic dyadic output is permitted.*

CC1.5 *The stronger numerical L^2 density norm remains uncertified by CC1. The Gaussian reference, target, positive boundary factor, pressure sign and subtraction are unchanged. CC1 alone supplies target-local smoothing, without raw fixed-a square limit, finite-neural assembly, sampler, normalized estimator, expectation runtime, matched comparator, positive interaction gap, full-work benefit or novelty-priority claim. No full G1–G9 flag is promoted. Its historical acceptance is governed by the frozen separate review/audit decisions, rather than the original author candidate header.*

L.3 Complete CA1 conclusions

Theorem L.2 (CA1.1: actual-law eight-charge budget)**.** *Assume H1–H7 and Appendix S. Fix $n \geq 1$, $0 < a < 1$, $\beta > 0$, finite $R_{\mathrm{aux}} \geq 1000n$, $N \geq 1$, $0 < s \leq 1$, $T > 0$ finite. Let $K_f = [-1,1]^2$, $K_* = [-2,2]^2$, $L_c = (4\sqrt{15}+12)c_a^{3/2}$, and put*

$$\begin{aligned}
E_{\mathrm{width}} &= D_J N^{-1/4},\\
E_{\mathrm{stab}} &= \min(2, 6\beta |D_{\mathrm{aux}} \setminus \Lambda| c_a^2 e^{9uA_n c_a^2}),\\
E_{\mathrm{aux}} &= \min\big(2, e^{9uA_n c_a^2}[2uA_n L_c \sqrt{\delta_D(\overline{\Lambda})} + I\sqrt{\delta_D(K_f)} + 2I\sqrt{c_a}\sqrt{\delta_D(K_*)}]\big),\\
E_{\mathrm{reg}} &= R_n(a) \leq M_n C_n \sqrt{a}, \qquad E_{\mathrm{var}} = d_{a,s} = I(c_s - c_{a,s}), \quad 0 \leq E_{\mathrm{var}} \leq 27(a/s)^2,\\
C_s^* &= (9/5)2^{360}\sqrt{1 + 1/(Is^4)}, \qquad E_n^*(s) = (1/2500)e^{1/128} C_s^* \Theta_n,\\
B_s &= 3IR_2^2/s^2 \leq 324/s^2, \qquad \Lambda_T = \max(1, 2T),\\
E_{\mathrm{vol}} &= \Lambda_T E_n^*(s), \qquad E_{\mathrm{clip}} = 4\sqrt{2}e^{-T^2/(4B_s)}, \qquad E_{\mathrm{target}} = K_D L_f \sqrt{s}.
\end{aligned}$$

For both architectures with the same parameters,

$$d_{\mathrm{BL}}(\mathcal{L}^{\mathfrak{b}}_{\mathrm{neural}}, \mathcal{L}_{\mathrm{target}}) \leq \min(2, E_{\mathrm{width}} + E_{\mathrm{stab}} + E_{\mathrm{aux}} + E_{\mathrm{reg}} + E_{\mathrm{var}} + E_{\mathrm{vol}} + E_{\mathrm{clip}} + E_{\mathrm{target}}). \tag{CA1-BL}$$

The seven successive normalized pair laws, with r_D denoting restriction to D_{aux}, are

$$\begin{aligned}
&(J_s)_\# P_N^{\mathfrak{b}},\\
&(J_s)_\# Q_\beta = (J_s r_D)_\# Q_{\mathrm{res}}^\beta,\\
&(J_s r_D)_\# Q_{\mathrm{res}}^0,\\
&(X_{\mathrm{obs}}, Q_s)_\# \nu_{a,n},\\
&(X_{\mathrm{obs}}, Q_s)_\# \nu_n,\\
&(X_{\mathrm{obs}}, Q_s)_\# \nu_\infty,\\
&(X_{\mathrm{obs}}, Q_0)_\# \nu_\infty.
\end{aligned}$$

The regulator arrow has charges $E_{\mathrm{reg}}, E_{\mathrm{var}}$ and the volume arrow has $E_{\mathrm{vol}}, E_{\mathrm{clip}}$, each once. Every law retains its normalizer in H1,H3,H5,H6. Alternatively E_{target} may be replaced by any supplied certified joint-law bound $r(s)$ for the last arrow; the $K_D L_f \sqrt{s}$ and $r(s)$ options are alternatives, never summed. For the fixed cosine one may replace E_{width} by C/N, leaving other charges unchanged. These bound expectation differences; no Gaussian-comparator separation follows.

Proposition L.3 (CA1.2: subtractions and order)**.** *For each path/restriction,*

$$\begin{aligned}
J_{T,k_2} = J_{T,k_1} - (0, I(k_2 - k_1)), \qquad Q_s(\Phi_a) = Y_{a,s} - d_{a,s},\\
J_{T_s,c_s} = J_{T_s,c_{a,s}} - (0, d_{a,s}),\\
[B_{\mathrm{id}}(v,v) - Ic_a] - [B_{T_s}(v,v) - Ic_s] = \int f[v^2 - (T_s v)^2] + I(c_s - c_a).
\end{aligned}$$

Common translation preserves both distances exactly; translating one law costs at most $I|k_2 - k_1|$. The raw action uses exact c_a. The untilted raw endpoint/neural variance is $1/4$, and its smoothed value is

$c_{\mathrm{end},s} = \int |\hat{\rho}(sp)|^2 \hat{C}_{\mathrm{end}}\, d\lambda$*; interacting expectations are separate. Endpoint-own to physical subtraction is the shift* $+(0, I(c_{\mathrm{end},s} - c_s))$*, absent from* (CA1-BL) *because adjacent laws use common* c_s*. No ordering of* c_s *and raw* c_a *is asserted. Successive limits are allowed in the order* $N \to \infty$*,* $\beta \to 0$*,* $R_{\mathrm{aux}} \to \infty$*,* $a \to 0$ *at fixed* n, s*,* $n \to \infty$ *at fixed* s *with clipping controlled, then* $s \to 0$ *at the target; reverse nested tolerance choices are allowed. The diagonals below are separately controlled. At fixed* a, N *the neural physical-*C_0*-subtracted coordinate tends to* $-\infty$ *as* $s \to 0$*; fixed-stage tightness does not follow.*

Proposition L.4 (CA1.3: explicit tail and finite rational extension)**.** *For every* $0 < a < 1$ *and* $r > 0$*,*

$$\int |x|^2 C_a(x)^2\, dx \le 1/16, \qquad \xi_a(r) \le K/(16r^2). \tag{CA1-TAIL}$$

For integers $n, k \ge 1$*,* $a = 2^{-2k}$ *and rational* $0 < \eta_A, \eta_S, \eta_W \le 1$*, the following* EXT *returns finite* $R_{\mathrm{aux}}, \beta, N$ *with auxiliary, stabilizer and width charges at most* η_A, η_S, η_W *for both architectures:*

$$\bar{c} = 2k + 3, \quad \bar{E} = 4^{\lceil 9uA_n\bar{c}^2 \rceil}, \quad \bar{H} = 56uA_n\bar{c}^2 + 4 + 8\bar{c},$$
$$r = \max(1, \lceil \bar{E}\bar{H}/(4a\eta_A) \rceil), \quad R_{\mathrm{aux}} = 1000n + r + 1, \quad d = 4R_{\mathrm{aux}}^2,$$
$$\beta = \min(u, \eta_S/(6d\bar{c}^2\bar{E})), \quad \alpha = \beta, \quad A = u,$$
$$\bar{B} = K/2 + 6u\bar{c}, \quad \bar{L} = \bar{B}^2 d/(2\alpha), \quad \overline{Z}_{\exp} = 3ud/16 + 3u\bar{c}^2 A_n.$$

Form $\overline{P}_Z, \overline{P}_J$ *from H7 by replacing* B *by* $\bar{B}$ *and* $\sqrt{d}$ *by the exact rational* $2R_{\mathrm{aux}}$*. Put* $t_0 = 1$*,* $t_j = \max(1, j/(2\alpha))^j$ *for* $1 \le j \le 12$ *and*

$$\bar{C} = \frac{d}{96} 4^{\lceil \bar{L} \rceil + \lceil \overline{Z}_{\exp} \rceil} \sum_{j=0}^{12} (\overline{P}_Z + \overline{P}_J)_j t_j,$$
$$N = \max(1, \lceil 1024 \bar{C} \eta_W^{-4} \rceil). \tag{EXT}$$

Subscripts are polynomial coefficients, absent coefficients zero. Large powers may be symbolic. Exact c_a, c_s *and original laws remain in the model;* $\bar{c}, \bar{C}$ *only choose sufficient bounds.*

Theorem L.5 (CA1.4: analytic convergence using finiteness)**.** *For* $n \ge 1$ *take*

$$s_n = 2^{-n}, \quad T_n = 36\sqrt{n}/s_n, \quad k_n = \max(1, \ell_n + n + \lceil \log_2 C_n \rceil, \lceil (3n+5)/4 \rceil),$$
$$a_n = 2^{-2k_n},$$

then use $\mathrm{EXT}(n, k_n, 2^{-n}, 2^{-n}, 2^{-n})$ *for* $R_{\mathrm{aux},n}, \beta_n, N_n$*. Both actual neural laws satisfy*

$$d_{\mathrm{BL}}(\mathcal{L}^{\mathfrak{b}}_{\mathrm{neural},n}, \mathcal{L}_{\mathrm{target}}) \le \min\big(2, 5\,2^{-n} + 2^{362} n^{3/2} 2^{-2n} + 4\sqrt{2} e^{-n} + 3K_D 2^{-n/2}\big) \longrightarrow 0. \tag{CA1-DIAG}$$

The sequence needs no numerical K_D*; its unknown finite value remains in the bound.*

Theorem L.6 (CA1.5: conditional numerical sufficient selector)**.** *Supply a certified finite rational* $\bar{K}_D \ge K_D$ *for H3's actual normalized target-local density, with boundary and pressure, and rational* $0 < \eta \le 1$*. Define* L2plus *as in CC1.4, put* $e = \eta/2$*, and in order take*

$$j = \mathrm{L2plus}(12\bar{K}_D/e), \quad s = 2^{-2j}, \quad b_{\mathrm{clip}} = \mathrm{L2plus}(32/e), \quad T = 36 b_{\mathrm{clip}}/s,$$
$$n = \max(1, \lceil (364 + 6j + \mathrm{L2plus}(b_{\mathrm{clip}}/e))/4 \rceil),$$
$$k = \max(1, \ell_n + \mathrm{L2plus}(8C_n/e), j + \lceil \mathrm{L2plus}(216/e)/4 \rceil), \quad a = 2^{-2k}.$$

Invoke $\mathrm{EXT}(n, k, \eta/8, \eta/8, \eta/4)$*. The finite tuple* $(s, T, n, a, R_{\mathrm{aux}}, \beta, N)$ *gives* (CA1-BL) *at most* η *for both architectures. Its eight allocations are*

charge	*width*	*stab.*	*aux.*	*reg.*	*variance*	*volume*	*clip*	*target*
allowance	$\eta/4$	$\eta/8$	$\eta/8$	$\eta/16$	$\eta/16$	$\eta/8$	$\eta/8$	$\eta/8$.

Every operation terminates for finite rational inputs. This is a mathematical sufficient procedure, without a bounded-resource executable claim. Without $\bar{K}_D$*, CA1.4 remains analytic and this numerical route stays conditional. No additional density certificate is extracted from an evolving sibling or qualitative finiteness.*

Theorem L.7 (CA1.5b: alternate certified smoothing bypass)**.** *Instead of $\bar{K}_D$, supply an explicit certified $r(s) \geq 0$, on positive rational $s \leq 1$, such that*

$$d_{\mathrm{BL}}((X_{\mathrm{obs}}, Q_s)_{\#}\nu_\infty, \mathcal{L}_{\mathrm{target}}) \leq r(s), \qquad r(s) \to 0,$$

and a specified terminating algorithm S_r returning on every rational $0 < \tau \leq 1$ a rational $0 < s \leq 1$ with certified $r(s) \leq \tau$. It concerns exactly the original normalized target and identical physical c_s/Q_0 prescription, retaining all its source premises as additional hypotheses. Qualitative existence or an unverified program is insufficient; a certified L^1 smoothing modulus suffices.

For rational $0 < \eta \leq 1$ take $e = \eta/2$, $s = S_r(\eta/8)$, $h = \mathrm{L2plus}(1/s)$, and

$$b_{\mathrm{clip}} = \mathrm{L2plus}(32/e), \quad T = 36b_{\mathrm{clip}}/s,$$
$$n = \max(1, \lceil (364 + 3h + \mathrm{L2plus}(b_{\mathrm{clip}}/e))/4 \rceil),$$
$$k = \max(1, \ell_n + \mathrm{L2plus}(8C_n/e), \lceil (2h + \mathrm{L2plus}(216/e))/4 \rceil), \quad a = 2^{-2k}.$$

Then $\mathrm{EXT}(n, k, \eta/8, \eta/8, \eta/4)$ gives actual joint distance $\leq \eta$, with the same eight allocations and E_{target} replaced by $r(s)$. This terminating conditional selector needs no certified numerical L^2 density norm. H3's conditioning and finiteness remain in its analytic source basis; the supplied smoothing guarantee is exactly the additional bypass premise.

Remark L.8 (CA1.6: literal scope). CA1 concerns the fixed smoothed physical-C_0-subtracted joint observation of the original two process-neuron architectures in the original two-dimensional massive coefficient-one quartic model. Every finite stage has positive a, β, s and finite geometry and width. Whole-plane process-neuron counts do not specify a finite global random parameter vector. The raw growing-volume route, volume-uniform raw Wick Cauchy/UI, finite-bit generation, useful estimator/sampler, total work, positive cosine gap, matched interacting-covariance Gaussian comparator, comparative advantage, priority, broader-QFT and publication claims remain separate. Full G1–G9 capability flags remain false. No historical access violation, missing proof, parser/software defect, unreviewed implication or raw severity is repaired by this integration.

L.4 Complete proof of CC1

Source identity and local normalization. S1 fixes the actual common FS maximal state, represented on original X. The strict P1–P6/AP3/AP5 bridge identifies it with the same physical free-boundary limit. Both original Section 4 maximal states are among those covered by FS Theorem 7.2, so S3 applies to this identified target. The older unknown g_* route is unused. The reference is whole-plane P_1 restricted to $\mathcal{C}_D$, rather than a Dirichlet reference. Gaussian/common-space fourth-Wick uniqueness identifies source J_D with the anisotropic own-c_a limit on D. The general fixed-square Gaussian/Wick transfer and general-rectangle construction apply at this local region independently of the exhaustion boxes B_n.

A $\mathcal{C}_D$-measurable version of U_D follows by compactly contained interior squares and compactly supported mollifiers. Fourth-chaos pairing bounds the squared L^2 norm of an omitted boundary layer E_0 by $24u^2|E_0|\|C_0\|_4^4 \to 0$; the accepted covariance bounds make $\|C_0\|_4$ finite. Interior mollifier limits are local, and Gaussian L^2 uniqueness identifies their limit with U_D. This locality does not require a positive-a common-space representative to be a local functional of Φ.

Set $V_D = e^{-U_D}$ and $g = g_{\partial D}$. The exact density is $k_D = e^{-36\alpha_{\mathrm{pr}}}V_D g$, $\mathbb{E}k_D = 1$, with both minus signs and no division by $\mathbb{E}V_D$. For every centered finite square E, $\mathbb{E}U_E = 0$ and finite negative-exponential moments give $Z_E = \mathbb{E}e^{-U_E} \geq e^{-\mathbb{E}U_E} = 1$. P5/AP3 identifies pressure as the same-mass normalized log-partition limit, hence

$$\alpha_{\mathrm{pr}} = \lim |E|^{-1} \log Z_E \geq 0, \qquad 0 < e^{-36\alpha_{\mathrm{pr}}} \leq 1.$$

Existence and source identification of pressure remain premises.

Numerical Gaussian exponential certificate. The general-rectangle Nelson input has semiboundedness coefficient $6u$ and degree parameter two. Since $\pi > 3$,

$$\kappa_2 < 1, \quad M_6 = (5/(4\pi))^{5/6} < 1, \quad 384A_D = 13824 < 128^2, \quad K_{A_D} < 128u.$$

Also $2 < e < 3$, $\sqrt{2} < 3/2$ imply $\kappa_0 < 15/2 < 8$ and $(2/e)\sqrt{\kappa_0} < 3$. Thus the second candidate for $K^{\sharp}_{A_D}$ is $< 384u$, whereas $12uA_D = 432u$, proving $K^{\sharp}_{A_D} = 432u$. As $432/10^{10} < 1/1000^2$, $\beta_{A_D} > 1500$.

Take $h_0 = 1/100 > K^{\sharp}_{A_D}$. For $v \geq h_0$, let $R(v) = e^{\beta_{A_D}\sqrt{v}-1}/v$. Then

$$(\log R)'(v) = (\beta_{A_D}\sqrt{v} - 2)/(2v) > 0, \qquad R(h_0) > 100 > 20.$$

The tail therefore gives $P_1(U_D < -v) \leq e^{-20v}$ for $v \geq h_0$. Layer cake cut at e^{10h_0} yields

$$\begin{aligned}\mathbb{E}e^{-10U_D} &\leq e^{10h_0} + \int_{h_0}^{\infty} 10e^{10v} P_1(U_D < -v)\, dv \\ &\leq e^{1/10} + e^{-1/10} \leq 10/9 + 1 = 19/9 < 4.\end{aligned}$$

Here $e^x \leq (1-x)^{-1}$, $0 \leq x < 1$, follows by termwise comparison of positive series. Since $(4/3)^{10} > 4$, $\|V_D\|_{10} \leq 4^{1/10} < 4/3$. This proves CC1.1 without numerical quadrature or a global finite-box density envelope. Its rational arithmetic and the attributed analytic tail have distinct evidentiary roles.

The exact boundary factor and lower exponent. For $l = 6, m_0 = 1$, $e > 2$ gives

$$4(1 - e^{-6})^{-2} < 4(1 - 1/64)^{-2} = 16384/3969 < 5.$$

Thus S3 permits exactly $q = 5$: $\nu_\infty(H) \leq e^{-36\alpha_{\mathrm{pr}}}\|HV_D\|_5$. This is the published chessboard conclusion for the actual Section 4 FS states, with its retained GRS dependency.

For bounded nonnegative local G, let $A_M = \{1/M \leq V_D \leq M\}$ and $H_M = GV_D^{-1}\mathbf{1}_{A_M}$. The test is bounded and local. Substituting the exact density into the source inequality and cancelling the positive pressure factor gives

$$\mathbb{E}[Gg\mathbf{1}_{A_M}] \leq \|G\mathbf{1}_{A_M}\|_5 \leq \|G\|_5.$$

Since U_D is finite a.s., A_M increases to full measure. Monotone convergence and truncation extend $\mathbb{E}Gg \leq \|G\|_5$ to all nonnegative local $G \in L^5$. Duality gives $\|g\|_{5/4} \leq 1$. The factor is a local equivalence class through $k_D = e^{-36\alpha_{\mathrm{pr}}}V_D g$ and positive V_D; its original boundary-measurable representative in either reading is unchanged.

Since $1/(10/9) = 1/10 + 1/(5/4)$, Hölder gives

$$\|k_D\|_{10/9} \leq e^{-36\alpha_{\mathrm{pr}}}\|V_D\|_{10}\|g\|_{5/4} \leq 4/3.$$

This proves CC1.2, with no inference of an L^2 bound from a smaller exponent. Equivalently the source inequality and Hölder give

$$\nu_\infty(H) \leq \|H\|_{10}\|V_D\|_{10} \leq (4/3)\|H\|_{10}$$

first for bounded nonnegative local H, then by monotone truncation whenever its Gaussian L^{10} norm is finite. This justifies unbounded observables below.

Gaussian square, smoothing and the one Borel version. Write $C(p) = (1 + |p|^2)^{-1}$. Parameter integration $1/(AB) = \int_0^1 [tA + (1-t)B]^{-2}\, dt$ and translation give

$$\int_{\mathbb{R}^2} C(p)C(k-p)\, dp = \pi \int_0^1 [1 + t(1-t)|k|^2]^{-1}\, dt \leq \pi.$$

With exactly $d\lambda = dp/(2\pi)^2$, Tonelli and Plancherel show

$$\|K_f\|^2_{H\otimes H} = \iint |\hat{f}(p+q)|^2 C(p)C(q)\, d\lambda(p)d\lambda(q) \leq F_2^2/(4\pi).$$

Thus $O_2^P = I_2(K_f)$ exists and $\|O_2^P\|_2 \leq \sigma$, including the isometry's factor two. For $s > 0$, $Q_s(\Phi)$ has kernel $(r_s \otimes r_s)K_f$. First use finite spectral cutoffs, Wick pairing and Bochner integration; dominated tensor

convergence and Gaussian Fubini identify the limit with the measurable spatial integral. Its centering is the physical mass-one c_s at every s.

Evenness, nonnegativity, unit integral and support of ρ give $|r_s| \le 1$ and $0 \le 1 - r_s(p) \le \min(2, s^2|p|^2)$, by $1 - \cos\theta \le \theta^2/2$ and $|z|^2 \le 2$ on its support. Denote inverse transforms of $C, r_s C, r_s^2 C$ by C_0, C_{s0}, C_{ss}. Their L^2 norms are bounded by $M_2 = 1/(2\sqrt{\pi})$. The radial Plancherel split at $|p| = 1/s$ gives, with $D_s = \|C_{s0} - C_0\|_2$,

$$D_s^2 \le s^4 \int_{|p|\le 1/s} \frac{|p|^4}{(1+|p|^2)^2}\, d\lambda + 4 \int_{|p|>1/s} |p|^{-4}\, d\lambda \le 5s^2/(4\pi),$$
$$\|C_{ss} - C_{s0}\|_2 \le D_s.$$

For integrable G_0, translated-product Cauchy–Schwarz gives $\iint |f(x)f(y)G_0(x-y)|\, dxdy \le F_2^2 \|G_0\|_1$. Wick pairing and $\|A^2 - B^2\|_1 \le \|A - B\|_2(\|A\|_2 + \|B\|_2)$ yield

$$\|Q_s(\Phi) - O_2^P\|_2^2 = 2 \iint f(x)f(y)(C_{ss}^2 + C_0^2 - 2C_{s0}^2)(x-y)\, dxdy$$
$$\le 8F_2^2 M_2 D_s \le (2\sqrt{5}/\pi)F_2^2 s = L_f^2 s.$$

For distributional rigor, finite momentum cutoffs converge in tensor norm by the preceding bound, covariances converge in L^2 and their squares in L^1, so the translated-product estimate passes both formula and bound to the limit. No spatial positivity of r_s or covariance difference is assumed. Second-chaos hypercontractivity gives for every real $R \ge 2$

$$\|Q_s(\Phi) - O_2^P\|_R \le (R-1)L_f\sqrt{s}, \qquad \|O_2^P\|_R \le (R-1)\sigma.$$

For positive s, distributional smoothing makes $z \mapsto v(h_{s,z})$ continuous. Its compact integral is a limit of Borel Riemann sums, all using tests inside $[-2,2]^2$, so Q_s is Borel and $\mathcal{C}_D$ local. At $s_j = 2^{-2j}$, the sum of squared Gaussian errors is finite. Tonelli gives almost-sure convergence to O_2^P. The convergence event and limit are local, hence $Q_0(\Phi) = O_2^P$ a.s. Target-local absolute continuity transfers this same full-probability event. Equality of source and X representations on compact evaluations transfers the same countable construction, giving target $O_2(f) = Q_0$ without another subsequence or observable. Finally $F_2 \le 2, \pi > 3, \sqrt{5} < 3$ imply $\sigma < 1, L_f < 3$.

All orders and numerical target smoothing. For real $r \ge 1$, put $E_s = Q_s - Q_0$. Its rth absolute power is local. Hypercontractivity at $R = 10r$ and the preceding local transfer inequality give

$$\mathbb{E}_{\nu_\infty}|E_s|^r \le (4/3)\|E_s\|_{10r,P_1}^r \le (4/3)[(10r-1)L_f\sqrt{s}]^r,$$

and likewise $\mathbb{E}_{\nu_\infty}|Q_0|^r \le (4/3)[(10r-1)\sigma]^r$. Taking roots proves CC1.3; $r = 1$ yields $12L_f\sqrt{s} < 36\sqrt{s}$. No unknown K_D, boundary norm or finite-box M_n occurs.

For $R \ge 1$, Minkowski gives

$$\sup_{0<s\le 1} \|Q_s\|_{L^R(\nu_\infty)} \le (4/3)^{1/R}(10R-1)(\sigma + L_f).$$

For any positive moment order a_0, choose $R > \max(a_0, 1)$. Markov/Hölder bounds the a_0-moment tail uniformly by a constant times the threshold to power $-(R - a_0)$. This proves uniform integrability of $\{|Q_s|^{a_0} : 0 < s \le 1\}$, at the target-local scope.

Terminating smoothing selector. For positive rational x, L2plus(x) is obtained by finite integer numerator/denominator comparisons with powers of two. At $x = 36/\tau$, $2^j \ge x$, including $j = 0$ if $x \le 1$. Thus $s = 2^{-2j}$ satisfies $0 < s \le 1$ and $36\sqrt{s} = 36\,2^{-j} \le \tau$. Replacing τ by $\epsilon/4$ gives the stated $144/\epsilon$ formula. This is exact rational selection with symbolic dyadic output, not interacting expectation evaluation. Historical finite selector examples are finite checks separate from this proof. CC1.4 follows.

Retained unsuccessful routes and limits. The original direct L^2 Hölder route gives

$$\mathbb{E}_{\nu_\infty}|Q_s - Q_0| \le K_D L_f \sqrt{s}, \qquad K_D = e^{-36\alpha_{\rm pr}}\|V_D g\|_2 \le e^{-36\alpha_{\rm pr}}\|V_D\|_4\|g\|_4.$$

Old qualitative finiteness of $\|g\|_4$ gave no numerical certificate. Normalization alone only fixes $\mathbb{E}V_D g = e^{36\alpha_{\rm pr}}$ and does not bound $\|g\|_4$. Setting $g = 1$ or dividing by $\mathbb{E}V_D$ would change the target. The smaller-exponent proof retains both factors.

Finite-box marginalization gives $\|\mathbb{E}[r_{0,n} \mid \mathcal{C}_D]\|_2 \le \|r_{0,n}\|_2 \le M_n$, which grows with n. The unused raw triangle has sufficient charges $M_n L_f\sqrt{s}$ and $2^{362}n^{3/2}s^{-3}2^{-5n}$. Since $\ell_n \ge 2n^4/125$, forcing the former small makes that crude second majorant grow. This is failure of sufficient majorants, with no nonconvergence or all-method impossibility implication.

The linear source bound only gives

$$\mathbb{E}_{\nu_\infty}|Q_s| \le 3I\|\rho_s\|_2^2 + Ic_s \le 4IR_2^2/s^2 \le 432/s^2, \quad \mathbb{E}_{\nu_\infty}|Q_s - Q_t| \le 432(s^{-2} + t^{-2}).$$

These divergent upper bounds, and the explicitly scale-dependent H^{-2} local moment constant, supply no small-scale Wick-square Cauchy modulus. An unused conditional-boundary proposal would require a correctly normalized $\mathbb{E}_{\mathrm{cond},\xi}|Q_s - Q_0| \le C(\xi)\sqrt{s}$ with explicit finite $\mathbb{E}_{\nu_\infty}C(\xi)$, shifted-Dirichlet Wick conversion, conditional normalizer and quantitative target-boundary trace control. The linear L^2 source interface supplies none of this boundary control. Finite-tuple AP6 reconstruction likewise cannot certify the missing infinite-dimensional density-norm tail by observed stabilization.

The FS source matching retains the GRS checkerboard dependency and the original classical/P1–P6 scope. Historical source investigators read the primary FS line/page, without claiming a fresh secondary-proof audit. Frozen original author candidate headers and operational/access findings remain historical; their subsequent accepted decisions govern CC1. There is no new outside-expert, statistical-independence or OS-isolation claim. The stronger L^2 certificate, raw fixed-a limit, assembly from CC1 alone, useful sampling/work/comparison/gap and priority remain outside CC1.5. This completes the full positive proof and its retained limits.

L.5 Complete CA1 proof

P0: target, local prescription and normalization ledger. H1 fixes the original pressure-identified target, rather than a new state or subsequence. Same-mass whole-plane reference, free-boundary interaction and physical/classical identification are literal source conditions. $8000 < 10^{10}$ proves the strict cap, without any old g_* inference.

All $h_{s,z}$ are supported in $[-2,2]^2$, strictly inside $D_{\rm loc}$, every B_n and $D_{\rm aux}$. The accepted CV1 continuity estimate is

$$|Q_s(v) - Q_s(w)| \le IA_s^2 p_3(v-w)[p_3(v) + p_3(w)], \quad A_s^2 = s^{-2}R_2^2 + 2s^{-4}D_\rho^2 + s^{-6}\|\Delta\rho\|_2^2.$$

Alternatively countable Riemann sums show local Borel measurability. Zero extension of an E class gives exactly the whole-plane convolution on $\operatorname{supp} f$, so the linear coordinate and the actual Borel pair agree.

Each probability has its own normalization before observation:

law	reference and weight	normalizer
$P_N^{\mathfrak{b}}$	$\rho_N^{\mathfrak{b}},\ e^{-W-\beta J_{\rm out}+(K/2)\int_{D_{\rm aux}} v^2}$	$Z_N^{\mathfrak{b}}$
Q_β	ρ_γ, same full weight	Z_γ
$Q_{\rm res}^\beta$	$G',\ e^{-W-\beta J_{\rm out}}$	z_β
$Q_{\rm res}^0$	$G',\ e^{-W}$	z_0
$\nu_{a,n}$	$G_a,\ e^{-W}$	$Z_{a,n}$
ν_n	$\Phi_\#(e^{-U_{0,n}}P/Z_{0,n})$	$Z_{0,n}$
ν_∞	$P_1\vert_{\mathcal{C}_D},\ k_D$	$\mathbb{E}_{P_1}k_D = 1$

The pure Gaussian mass normalizer $M_{D_{\rm aux},a}$ is separate; HP gives $Z_\gamma = M_{D_{\rm aux},a}z_\beta$ and the all-Borel restriction identity. This does not identify Q_β with $\nu_{a,n}$ at finite geometry and β. Endpoint covariance $C_{\rm end}$, full tilt Q_β and physical reference C_a remain distinct. No extra local Z_D is inserted into k_D.

P1: fixed-stage width and its normalization. Use the literal NC1 theorem H7. Nonnegative unit-mass convolution after zero extension is an L^4 contraction. Hölder gives $\|v \mapsto \int fv\| \le \|f\|_{4/3} < 1$ and $\|B_T\| \le F_2 < 1/2$ for T_s and T_{id}. For $x = \|v\|_4$, $J_{T,k}$ therefore has first derivative at most $q_1 = 1 + 2x$, second at most $q_2 = 2$, higher derivatives zero. The finite subtraction changes none. For $H(v) = G(J_{T,k}(v))$ with $G \in \mathcal{U}_4$, the scalar chain rule gives $p_1, \dots, p_4$; fourth-order multiplicities $1, 6, 3$ correspond to four first derivatives, one second and two first, and two second derivatives.

Differentiation and Hölder give $v_1, \dots, v_4$ for V; in particular $\|v\|_2^2 \le \sqrt{d}\, x^2$, with quadratic coefficient at most B. The exponential chain rule bounds $D^4 w$ by we_4, while Leibniz multiplicities $1, 4, 6, 4, 1$ give $D^4(Hw)$ at most wP_J. Moreover

$$V(v) \ge \alpha x^4 - B\sqrt{d}\, x^2 \ge (\alpha/2)x^4 - L, \qquad \max_{x \ge 0} x^j e^{-\alpha x^4/2} = [j/(2\alpha e)]^{j/4} \quad (j > 0).$$

The dropped constant $3uc_a^2 A_n$ is positive; the $j = 0$ maximum is one. Summing nonnegative coefficients gives M_Z, M_J and bounded lower derivatives and functions, uniformly in positive s and finite common k.

Actual iid neuron signs of magnitude $1/2$ give $\mathbb{E}F_N(z)^4 = (3 - 2/N)/16 \le 3/16$, while $\mathbb{E}\gamma(z)^4 = 3/16$. Tonelli, dropping nonpositive quadratic terms, gives $\mathbb{E}V \le 3Ad/16 + 3uc_a^2 A_n$ under either reference. Jensen proves the two own normalizers at least z, while quartic damping makes them finite positive. Their equality is never assumed.

For an E-C^4 function Ψ with $\sup \|D^4\Psi\| \le M$, successively replace the N independent normalized neurons by independent normalized endpoint copies. Integral Taylor remainder is $M\|h\|_4^4/24$. Zero Bochner mean cancels linear terms, the EVERY-bounded-bilinear identity at the independent base cancels all quadratic terms, and global sign symmetry cancels cubics; lower terms are integrable. Each replacement costs at most $M(d/16 + 3d/16)/(24N^2) = Md/(96N^2)$. The telescope ends at the exact Gaussian-sum law ρ_γ. Numerator and denominator errors are $dM_J/(96N)$ and $dM_Z/(96N)$. For $A_N = \rho_N^{\mathfrak{b}}(Hw)$ and $A_\gamma = \rho_\gamma(Hw)$,

$$\frac{A_N}{Z_N^{\mathfrak{b}}} - \frac{A_\gamma}{Z_\gamma} = \frac{A_N - A_\gamma}{Z_N^{\mathfrak{b}}} + \frac{A_\gamma}{Z_\gamma}\frac{Z_\gamma - Z_N^{\mathfrak{b}}}{Z_N^{\mathfrak{b}}}.$$

Since $|A_\gamma/Z_\gamma| \le 1$ and $Z_N^{\mathfrak{b}} \ge z$, the bound is C/N, with one z^{-1} and both errors.

For joint $G \in \mathrm{BL}_1(\mathbb{R}^2)$, Gaussian convolution at scale $0 < \delta \le 1$ approximates it uniformly within $\sqrt{2}\delta$. Its first derivative is at most one and its jth derivative operator norm for $2 \le j \le 4$ is at most $\sqrt{(j-1)!}\,\delta^{1-j}$: directional integration by parts and the degree-$(j-1)$ Wick polynomial L^2 bound establish this for arbitrary unit directions. Thus

$$\text{width error} \le \sqrt{6}C/(N\delta^3) + 2\sqrt{2}\delta.$$

Optimization gives $D_J N^{-1/4}$, capped separately by two. Taking $\delta = \eta_W/(4\sqrt{2})$ proves the sufficient $512\sqrt{3}C\eta_W^{-4}$ width. These smoothing costs are internal to E_{width}, distinct from physical s and counted once. The fixed cosine has derivative norms $\le 2^{-j/2}$ and lies in $\mathcal{U}_4$ and BL_1, hence its C/N alternative. This recalls the first actual-law arrow with all NC1/RL2/NR1 qualifications.

P2: stabilizer between residual normalized probabilities. Put $c = c_a$. Since $x^4 - 6cx^2 + 3c^2 = (x^2 - 3c)^2 - 6c^2$, $W \ge -6uA_n c^2$ and write $M = e^{6uA_n c^2}$. Under G', point variance $r(z) \in [0, c]$ gives

$$\mathbb{E}_{G'} W = 3u \int_\Lambda (r(z) - c)^2 \le 3uA_n c^2, \quad \mathbb{E}_{G'} J_{\mathrm{out}} \le 3|D_{\mathrm{aux}} \setminus \Lambda|c^2.$$

Jensen gives $z_0 \ge z_{0,\mathrm{low}} = e^{-3uA_n c^2}$, distinct from P1's z. Let $h_\beta = e^{-W - \beta J_{\mathrm{out}}}$ and $h_0 = e^{-W}$. The nonnegative $\Delta_h = \mathbb{E}_{G'}(h_0 - h_\beta)$ satisfies $\Delta_h \le 3\beta M|D_{\mathrm{aux}} \setminus \Lambda|c^2$ by $1 - e^{-t} \le t$. The exact subtraction

$$h_\beta/z_\beta - h_0/z_0 = (h_\beta - h_0)/z_0 + h_\beta(1/z_\beta - 1/z_0)$$

has L^1 norm at most $2\Delta_h/z_0$. Thus every bounded Borel whole-path test changes by at most E_{stab}. HP's all-Borel restriction identity applies on the Q_β side and pushforward cannot increase the bound. This is the single stabilizer charge.

P3: auxiliary coupling of action and observation. Use independent HP paths $X_a = Y + \psi$, $X_a \sim G_a, Y \sim G'$. Their e^{-W} tilts are exactly $\nu_{a,n}$ and Q^0_{res}, with normalizers $Z_{a,n}$ and z_0. For $p_c(x) = x^4 - 6cx^2 + 3c^2$, factoring and $(|x| + |y|)(x^2 + y^2) \le 2(|x|^3 + |y|^3)$ give

$$|p_c(x) - p_c(y)| \le |x - y|\{2(|x|^3 + |y|^3) + 6c(|x| + |y|)\}.$$

Centered Gaussian marginals of variance at most c have sixth moment at most $15c^3$ and second at most c. The bracket has L^2 norm $\le L_c = (4\sqrt{15}+12)c^{3/2}$. Cauchy–Schwarz, integration and the defect diagonal yield

$$\mathbb{E}|W(X_a) - W(Y)| \le uA_nL_c\sqrt{\delta_D(\overline{\Lambda})}, \quad \mathbb{E}|e^{-W(X_a)} - e^{-W(Y)}| \le MuA_nL_c\sqrt{\delta_D(\overline{\Lambda})}.$$

The latter uses the M-Lipschitz bound for e^{-t} at $t \ge -6uA_nc^2$.

Positivity of f, PSD covariance and Minkowski give $\mathbb{E}|X_a(f) - Y(f)| \le I\sqrt{\delta_D(K_f)}$. For $z \in \operatorname{supp} f$, let $A_z = T_s(X_a - Y)(z)$ and $B_z = T_s(X_a + Y)(z)$. Unit nonnegative convolution gives $\|A_z\|_2 \le \sqrt{\delta_D(K_s)}$, $\|B_z\|_2 \le 2\sqrt{c}$, where $K_s = [-1 - s, 1 + s]^2 \subset K_*$. Cauchy–Schwarz, without independence of these variables, yields

$$\mathbb{E}|B_{T_s}(X_a, X_a) - B_{T_s}(Y, Y)| \le 2I\sqrt{c}\sqrt{\delta_D(K_*)}.$$

The common subtraction cancels, so the expected Euclidean pair difference is at most the sum of its two coordinate bounds.

For $G \in \mathrm{BL}_1$, write $H_X = G(J_s(X_a))$, $H_Y = G(J_s(Y))$, $w_X = e^{-W(X_a)}, w_Y = e^{-W(Y)}$, $z_X = \mathbb{E}w_X, z_Y = \mathbb{E}w_Y = z_0$. Direct normalization subtraction gives

$$\left|\frac{\mathbb{E}w_YH_Y}{z_Y} - \frac{\mathbb{E}w_XH_X}{z_X}\right| \le \frac{2\mathbb{E}|w_X - w_Y| + \mathbb{E}(w_X|H_Y - H_X|)}{z_Y} \le \frac{2\mathbb{E}|w_X - w_Y| + M\mathbb{E}\|J_s(Y) - J_s(X_a)\|}{z_{0,\mathrm{low}}}.$$

Since $M/z_{0,\mathrm{low}} = e^{9uA_nc^2}$, this is exactly E_{aux} with probability cap two. Densities, own normalizers and nonlinear path observations change together.

P4: regulator, physical variance and translations. HC uses the common-space comparison

$$|\mathbb{E}_{\mu_{a,n}}G(A) - \mathbb{E}_{\mu_{0,n}}G(B)| \le \|r_{a,n} - r_{0,n}\|_1 + M_n\|A - B\|_{L^2(P;\mathbb{R}^2)}$$

for $A = (\Phi_a(f), Y_{a,s})$, $B = (\Phi(f), Y_s)$. The density term is $4K_nM_n\sqrt{a}$, the Gaussian linear difference at most $aF_2/2$, and the own-smoothed second-chaos difference at most $H_f\sqrt{a}$, uniformly in positive s. These are literal accepted CMP/HC bounds for actual anisotropic τ_a. They give $R_n(a)$, without a finite-box raw smoothing term. Since $a < 1, F_2/2 \le 1, H_f < 2$, $K_n < n/10^6$, $R_n(a) \le M_nC_n\sqrt{a}$.

With $D_0 = 1 + w^2 + q^2$, $T_0 = w^2q^2 \le D_0^2/4$, algebra gives

$$0 \le \hat{C}_0 - \hat{C}_a = \frac{a^2T_0}{D_0(D_0 + a^2T_0)} \le a^2/4,$$

so Plancherel and $\|\rho\|_2 = F_2/I$ yield

$$0 \le d_{a,s} = I\int |\hat{\rho}(sp)|^2(\hat{C}_0 - \hat{C}_a)\,d\lambda \le F_2^2a^2/(4Is^2) \le 27(a/s)^2.$$

At every positive-a physical path, $Q_s(\Phi_a) = Y_{a,s} - d_{a,s}$. Translating one normalized law costs at most $d_{a,s}$, without an M_n factor. The regulator arrow is $R_n(a) + d_{a,s}$.

Changing common subtraction k_1 to k_2 gives $J_{T,k_2} = J_{T,k_1} - (0, I(k_2 - k_1))$. Translation preserves both test classes, hence common translation preserves distances exactly; one-law translation costs at most $I|k_2 - k_1|$. Using common c_s throughout P1–P3 adds no endpoint-natural-centering charge. The four quantities $c_{\mathrm{end},s}, c_{a,s}, c_s, c_a$ differ. Untilting gives $\mathbb{E}B_{T_s} = Ic_{\mathrm{end},s}$ and $\mathbb{E}B_{\mathrm{id}} = I/4$, with no interacting-centering consequence. Endpoint-own to physical translation is $+(0, I(c_{\mathrm{end},s} - c_s))$. Finally exact subtraction gives the raw-to-smoothed identity in CA1.2, whose term $I(c_s - c_a)$ has no asserted sign and cannot be replaced by $d_{a,s}$. The action always uses exact $:v^4:_{c_a}$, including when upper envelopes later choose stages.

P5: direct nonlinear volume, clipping and target smoothing. Fix $s > 0$. H3 implies $\mathbb{E}_\zeta v(h)^2 \le 3\|h\|_2^2$ for $\zeta = \nu_n, \nu_\infty$. For any scalar source Y with the corresponding subgaussian proxy σ_h^2, an independent standard Gaussian and Tonelli give $\mathbb{E}e^{\lambda Y^2} \le (1 - 2\lambda\sigma_h^2)^{-1/2}$ for $0 \le \lambda < (2\sigma_h^2)^{-1}$; this does not make Y Gaussian. Here all $h_{s,z}$ have proxy $\sigma_s^2 = 3R_2^2/s^2$.

Take a finite rectangular partition (A_i) of $[-1,1]^2$ with $z_i \in A_i$, exact weights $w_i = \int_{A_i} f$, discarding zero weights. Put

$$Z_i(v) = \sqrt{w_i} v(h_{s,z_i}), \quad S_{\mathcal{P}} = \sum_i Z_i^2, \quad Q_{\mathcal{P}} = S_{\mathcal{P}} - Ic_s, \quad S_s = \int f(z) v(h_{s,z})^2 \, dz.$$

Weights sum exactly to I. Jensen with weights w_i/I or $f(z)\,dz/I$ gives $\mathbb{E}_\zeta e^{S_{\mathcal{P}}/(4B_s)} \le \sqrt{2}$ and $\mathbb{E}_\zeta e^{S_s/(4B_s)} \le \sqrt{2}$, $B_s = I\sigma_s^2$. Both tails above T^2 are bounded by $\sqrt{2}e^{-T^2/(4B_s)}$. No field-evaluation independence is used.

If the largest cell diameter is $\delta_{\mathcal{P}}$, translated mollifier L^2 control and source moments give

$$\begin{gathered}
\|h_{s,z} - h_{s,z_i}\|_2 \le D_\rho |z - z_i| s^{-2}, \\
\mathbb{E}_\zeta |v(h_{s,z})^2 - v(h_{s,z_i})^2| \le 6R_2 D_\rho |z - z_i| s^{-3}, \\
\mathbb{E}_\zeta |Q_s - Q_{\mathcal{P}}| \le 6IR_2 D_\rho \delta_{\mathcal{P}} s^{-3}.
\end{gathered}$$

Exact total weight cancels subtraction; quadrature tends to zero at each fixed n, s, T, uniformly for these two endpoint laws.

Apply H3 to $g = (f, \sqrt{w_i} h_{s,z_i})$, $R = 2, n_0 = 1$. The square-sum coefficient is exactly $e^{-4}[1 + 1/(Is^4)]$, independently of the number of cells. The same P_* gives $C_g \le C_s^*$ and AP6 tuple distance $\le E_n^*(s)$. For joint $G \in \mathrm{BL}_1$ define $H_T(x,z) = G(x, \min(\|z\|_2^2, T^2) - Ic_s)$. Its value is at most one and Lipschitz constant at most $\Lambda_T = \max(1, 2T)$, so H_T/Λ_T is an admissible AP6 test, including singular/repeated tuples. Clipping costs at most twice the tail at each endpoint. Thus

$$|\nu_n G(X_{\mathrm{obs}}, Q_{\mathcal{P}}) - \nu_\infty G(X_{\mathrm{obs}}, Q_{\mathcal{P}})| \le \Lambda_T E_n^*(s) + 4\sqrt{2} e^{-T^2/(4B_s)}.$$

Removing quadrature costs at most $12IR_2 D_\rho \delta_{\mathcal{P}} s^{-3} \to 0$. The right side is independent of partition and G, so the supremum proves the actual nonlinear $E_{\mathrm{vol}} + E_{\mathrm{clip}}$ arrow. This is a mathematical observation limit, rather than a numerical work procedure.

For the last arrow the normalized local density and H4 give

$$\mathbb{E}_{\nu_\infty} |Q_s - Q_0| = \mathbb{E}_{P_1}[k_D |Y_s - Q_0(\Phi)|] \le K_D \|Y_s - Q_0(\Phi)\|_2 \le K_D L_f \sqrt{s}.$$

Only local variables enter. Summability of squared errors at 2^{-2j} gives the one Gaussian full-probability convergence set; locality transfers it by target absolute continuity, and each ν_n uses the same prescription by its common-space density. The endpoint is the intended physical Borel pair, rather than a new pair-law-only variable. A BL test changes by at most this L^1 coordinate difference. Alternatively a certified bound $r(s)$ for this identical joint-law arrow costs $r(s)$ once, with no extra density factor.

Taking the triangle over the seven normalized laws proves CA1.1 with exactly eight charges and cap two. Width smoothing, variance translation, two-endpoint clipping and target smoothing are counted once. The cosine belongs to the class and inherits the budget and its sharper width option.

P6: actual anisotropic tail and rational EXT. Let $A_q = 1 + q^2$, $B_q = 1 + a^2 q^2$. Then $\hat{C}_a(w,q) = (A_q + B_q w^2)^{-1}$ and $\partial_w \hat{C}_a = -2B_q w/(A_q + B_q w^2)^2$. Substitute $w = \sqrt{A_q/B_q}\, t$, then $t = \tan\theta$, to obtain

$$\int_{\mathbb{R}} \frac{w^2\, dw}{(A_q + B_q w^2)^4} = \frac{\pi}{16 A_q^{5/2} B_q^{3/2}}.$$

Including exactly the Fourier measure,

$$\int |\partial_w \hat{C}_a|^2 \, d\lambda = \frac{1}{16\pi} \int_{\mathbb{R}} \frac{\sqrt{1 + a^2 q^2}}{(1 + q^2)^{5/2}} \, dq.$$

The q derivative has the same integral. For $0 < a < 1$,

$$\int |\nabla \hat{C}_a|^2 \, d\lambda \le \frac{1}{8\pi} \int_{\mathbb{R}} (1+q^2)^{-2} \, dq = 1/16.$$

These integrals establish derivative L^2 integrability, while $\hat{C}_a \le \hat{C}_0$ establishes $\hat{C}_a \in L^2$. Distributional Fourier differentiation and Plancherel give $\int |x|^2 C_a(x)^2 \, dx \le 1/16$. Its L^2 representative is the accepted continuous positive-a covariance. The deterministic integral tail is $\int_{|x|>r} C_a^2 \le 1/(16r^2)$, proving (CA1-TAIL) by HP, without spatial positivity or altered regulator.

For $a = 2^{-2k}$, $k \ge 1$, the accepted exact variance bound gives

$$c_a \le [\operatorname{arsinh}(1/a) + 1]/(2\pi) \le [\log(3/a) + 1]/(2\pi) < 2k + 3 = \bar{c},$$

using $\log 3 < 2, \log 2 < 1, 2\pi > 1$. Thus $\bar{c} \ge 1$ and $\bar{E} \ge e^{9uA_n c_a^2}$ since $e^x \le 4^{\lceil x \rceil}$ for $x \ge 0$. Also $L_c \le 28\bar{c}^2$ and, using the weaker $I \le 4$, $2uA_nL_c + I + 2I\sqrt{c_a} \le 56uA_n\bar{c}^2 + 4 + 8\bar{c} = \bar{H}$. The selected $R_{\text{aux}} = 1000n + r + 1$ contains the closed Euclidean r neighborhoods of $\overline{\Lambda}, K_f, K_*$ with surplus margin. The tail and $\sqrt{K} \le 1/a$ imply $\sqrt{\delta_D(H)} \le 1/(4ar)$ at each compact, hence $E_{\text{aux}} \le \bar{E}\bar{H}/(4ar) \le \eta_A$. All these choices precede β. Because $|D_{\text{aux}} \setminus \Lambda| \le d$, the chosen positive β gives $E_{\text{stab}} \le 6\beta d\bar{c}^2\bar{E} \le \eta_S$. No β limit is interchanged with expanding geometry.

For width, $\beta \le u$ gives $\alpha = \beta, A = u$. Increasing c_a to $\bar{c}$ increases B, L and all nonnegative polynomial coefficients; $\sqrt{d} = 2R_{\text{aux}}$ is exact and $z^{-1} \le e^{\overline{Z}_{\exp}}$. For $j > 0$,

$$[j/(2\alpha e)]^{j/4} \le \max(1, j/(2\alpha))^j = t_j :$$

if $j/(2\alpha) \le 1$ the left side is at most one; otherwise use $e \ge 1$ and $j/4 \le j$. The maximal polynomial degree is twelve. Coefficientwise comparison and $e^{\bar{L}+\overline{Z}_{\exp}} \le 4^{\lceil \bar{L} \rceil + \lceil \overline{Z}_{\exp} \rceil}$ prove $C \le \bar{C}$. Thus $N \ge 1024\bar{C}\eta_W^{-4} \ge 512\sqrt{3}C\eta_W^{-4}$. Moreover $D_J^4 = (8192\sqrt{3}/27)C < 1024C$ directly bounds the displayed E_{width} by η_W. Every coefficient, ceiling and power is finite rational/integer. The action still uses exact c_a and observation exact c_s; envelopes only choose resources. This proves all CA1.3 claims without a tail or stage-normalizer oracle.

P7: analytic diagonal. The chosen k_n gives

$$2^{\ell_n} C_n 2^{-k_n} \le 2^{-n}, \qquad 27\, 2^{-4k_n+2n} \le 27\, 2^{-n-5} < 2^{-n},$$

so regulator and variance each cost at most 2^{-n} and $0 < a_n \le 1/4$. The first six nonnegative Taylor terms of $e^{125/32}$ exceed 32 by exact rational arithmetic. Thus $q < 1/32$, and

$$\Theta_n \le 4n2^{-5n}, \quad C_s^* < 2^{364}s^{-2}, \quad E_n^*(s) \le 2^{355}ns^{-2}2^{-5n}.$$

The last uses $e^{1/128} \le 128/127$ and $4(128/127)/2500 < 1/512$. At $T_n = 36\sqrt{n}/s_n$, $\Lambda_T = 72\sqrt{n}/s_n < 2^7\sqrt{n}/s_n$, and $T_n^2/(4B_{s_n}) \ge n$. Consequently

$$E_{\text{vol}} \le 2^{362}n^{3/2}2^{-2n}, \quad E_{\text{clip}} \le 4\sqrt{2}e^{-n}, \quad E_{\text{target}} \le 3K_D 2^{-n/2}.$$

EXT bounds each of width, stabilizer and auxiliary by 2^{-n}. Adding the two regulator/variance terms gives $5\,2^{-n}$ and proves (CA1-DIAG). Every term tends to zero for fixed finite K_D. Each stage is finite positive and works for both architectures. No value of K_D defines the sequence, although finiteness alone yields no accuracy-dependent stopping index from this bound.

P8: numerical selector under the density certificate. A normalized density has $K_D \ge \|k_D\|_1 = 1$, so $\bar{K}_D \ge 1$. For rational $x > 0$, $x \le 2^{\text{L2plus}(x)}$ by exact finite comparisons. With $e = \eta/2$ the selector has $j \ge 4$, $b_{\text{clip}} \ge 5$, $0 < s \le 1$, $T > 1/2$ and $\Lambda_T = 72b_{\text{clip}}/s$. The order is s, T, n, a, auxiliary box, stabilizer, width.

First $L_f < 3$ and the choice of j give $E_{\text{target}} \le 3\bar{K}_D 2^{-j} \le e/4 = \eta/8$. Next $B_s \le 324/s^2$ gives $T^2/(4B_s) \ge b_{\text{clip}}^2$. Using $\sqrt{2} < 2, \exp(1) > 2$, $E_{\text{clip}} \le 8\,2^{-b_{\text{clip}}^2} \le 8\,2^{-b_{\text{clip}}} \le e/4$. For volume use $n \le 2^n$:

$$E_{\text{vol}} \le 2^{362} b_{\text{clip}} s^{-3} n 2^{-5n} \le 2^{362} b_{\text{clip}} 2^{6j} 2^{-4n} \le e/4$$

by $4n \ge 364 + 6j + \text{L2plus}(b_{\text{clip}}/e)$. Open-box support and $n\ell \ge 2$ already hold at $n = 1$. Writing $L_A = \text{L2plus}(8C_n/e)$ and $L_S = \text{L2plus}(216/e)$, the selected k satisfies $k \ge \ell_n + L_A$, $4(k-j) \ge L_S$, hence

$$E_{\text{reg}} \le 2^{\ell_n} C_n 2^{-k} \le e/8 = \eta/16, \qquad E_{\text{var}} \le 27\,2^{-4(k-j)} \le e/8.$$

The variance charge is outside M_n. These five physical-to-target allowances sum to e. EXT at the already fixed n, k gives $\eta/8, \eta/8, \eta/4$ for auxiliary, stabilizer and width, totaling the other $\eta/2$. All eight allocations sum exactly to η and the cosine inherits the same guarantee.

Termination consists of finite power-of-two comparisons, rational h_2, m_2, ℓ_n, C_n, dyadics s, a, finite integer $\bar{E}, r$, positive rational β, finitely many polynomial coefficients of degree at most twelve, the finite integer exponent $\lceil \bar{L} \rceil + \lceil \overline{Z}_{\exp} \rceil$, rational $\bar{C}$ and rounded integer N. Unlimited exact arithmetic terminates; a finite symbolic expression also describes the enormous chosen integer. No finite resource or performance theorem is asserted. The density certificate is an extra mathematical input, not verified by this algorithm. Without it CA1.4 and conditional CA1.5 remain; M_n, a Dirichlet density, bare boundary factor or pressure-free expression cannot replace K_D.

P8b: alternate certified target modulus. Under exactly CA1.5b's contract, $S_r(\eta/8)$ terminates with rational $s \in (0, 1]$ and certified last-arrow cost at most $\eta/8$. Let $h = \text{L2plus}(1/s)$, so $s^{-1} \le 2^h$. The P7 majorant and choice of n give

$$E_{\text{vol}} \le 2^{362} b_{\text{clip}} s^{-3} n 2^{-5n} \le 2^{362} b_{\text{clip}} 2^{3h} 2^{-4n} \le e/4.$$

The same clipping choice gives $E_{\text{clip}} \le e/4$, and $k \ge \ell_n + \text{L2plus}(8C_n/e)$ gives $E_{\text{reg}} \le e/8$. The condition $4k \ge 2h + \text{L2plus}(216/e)$ yields $E_{\text{var}} \le 27\,2^{-4k} s^{-2} \le 27\,2^{-4k+2h} \le e/8$. The four non-target physical charges have exactly P8's allowances, and $r(s)$ occupies the single target allowance. EXT gives the same other three. The sum is η. Composition of the terminating S_r with finite rational operations terminates; no monotonicity or rounding of r, s is used. Dyadic s is unnecessary on this route. This proves CA1.5b with neither a numerical L^2 certificate nor an unproved integration step.

P9: computability, order and retained adverse paths. The selector's explicit envelopes are rational/integer once its extra certificate and accuracy are supplied: 2^{360}, box/support constants, $M_n, C_n, \bar{c}, \bar{E}, \bar{H}, \beta$, coefficients, $\bar{C}, N$. The accepted OPT 2^{360} version remains a source premise, not a new OPT proof. HP/HC are source-qualified Gaussian/physical theorems rather than covariance-matching inferences.

Exact positive-stage $c_a, c_s, c_{a,s}$ are specified computable reals. Frozen accepted CMP C8 procedures bound the c_a integral tail beyond $q = R$ by $1/(2\pi a R)$, and the $c_s, c_{a,s}$ tails outside $[-R, R]^2$ by $R_2^2/[s^2(1+R^2)]$. Compact elementary quadrature and explicit bump derivative bounds enclose the rest. These are mathematical procedures, not executed quadratures at astronomical selected parameters. They are unnecessary for the rational sufficient selector and compute no normalized interacting expectation. Nonlinear normalizers remain positive finite source-law integrals with the stated lower/enclosure bounds; no numerical values or sampling access follow. CA1.5's extra input is $\bar{K}_D$; CA1.5b's is exactly the certified smoothing/selector contract.

Order is essential. Every finite neural path satisfies $|F_N| \le \sqrt{N}/2$ area-a.e.; nonnegative unit convolution gives $B_{T_s} \le IN/4$. But $c_s \to \infty$: on each fixed momentum ball $|\hat{\rho}(sp)|^2 \to 1$, while expanding-ball integrals of $\hat{C}_0$ diverge logarithmically. Thus its physical-C_0-subtracted coordinate is at most $IN/4 - Ic_s \to -\infty$ at fixed a, N. Under the continuous endpoint full tilt the squared convolution has a finite pathwise limit, with the same divergence of subtraction. Uniform NC1 comparison with a moving common subtraction supplies no tightness. The stated successive limits, reverse nested choices and separately estimated diagonals preserve the regulator/smoothing order, proving the remaining CA1.2 assertions.

The unused triangle through finite-box raw Q_0 adds $M_n L_f \sqrt{s}$. Its majorant grows as $2^{\text{constant } n^4}$, conflicting with the selected crude smoothed-volume majorant when both are forced to vanish. This is failure of these sufficient estimates, without nonconvergence or an all-method lower bound. The actual direct smoothed chain has no such arrow, so the charge is absent by law choice rather than bookkeeping omission. Growing-volume raw transfer stays separately open.

Original adverse records remain: CMP GAUSS/AUTHOR MAJOR startup-access findings and software-v1 NUM DISAGREEMENTS/MAJOR parser-resource defect; RL2's MAJOR missing-body issue; and all CV1/NC1 raw findings, severities, assessments, separate confirmations and remaining source/positive obligations. This integration does not reread unprovided historical raw logs or declare their original compliance repaired. Historical author pre-execution hook failures remain distinct from the present author's own retained command failures. Frozen acceptance decisions supersede the historical candidate-status headers for accepted mathematics; they do not erase failures or establish expert/model-error independence.

CA1 establishes conditional actual normalized joint-law assembly and finite sufficient scales. It establishes no useful estimator, free exact-real draw, finite-bit source generation, global-action cost, overlap/mixing, complete runtime, matched Gaussian comparator, positive selected-cosine gap, comparative benefit, new priority or other model. Full G1–G9 flags remain false. The original source review and the present derived-manuscript review are distinct. This completes CA1.1–CA1.6 and all retained limits.

L.6 The same-target CC1–CA1 integration

Corollary L.9 (Certified composite selector under the union of premises)**.** *Assume all S1–S4 and H1–H7/Appendix S for their identical original target, Gaussian reference, bump, c_s subtraction and one Borel Q_0. CC1 supplies CA1.5b with*

$$r(s) = 36\sqrt{s}, \qquad S_r(\tau) = 2^{-2\,\mathrm{L2plus}(36/\tau)}.$$

For every rational $0 < \eta \le 1$, CA1.5b therefore selects finite positive s, a, β and finite n, R_{aux}, N for both architectures with $d_{\mathrm{BL}}(\mathcal{L}^{\mathfrak{b}}_{\mathrm{neural}}, \mathcal{L}_{\mathrm{target}}) \le \eta$. At its target allocation,

$$j = \mathrm{L2plus}(288/\eta), \qquad s = 2^{-2j}, \qquad h = 2j.$$

The alternative CA1.5 density-certificate theorem and the analytic CA1.4 sequence retain their own distinct premises and bounds.

Proof. S1/H1 identify exactly the same original FS state and physical free-boundary limit, and S2–S4/H3–H4 identify the same restricted whole-plane Gaussian, pressure/boundary density, bump, physical c_s and countable local Q_0. For any joint $G \in \mathrm{BL}_1$, coupling both observations on one target field keeps X_{obs} identical and gives

$$|\mathbb{E}_{\nu_\infty} G(X_{\mathrm{obs}}, Q_s) - \mathbb{E}_{\nu_\infty} G(X_{\mathrm{obs}}, Q_0)| \le \mathbb{E}_{\nu_\infty} |Q_s - Q_0| \le 36\sqrt{s}.$$

CC1.4 terminates with $r(S_r(\tau)) \le \tau$. Thus all literal CA1.5b additional premises hold under the union. Take $\tau = \eta/8$ to get the $288/\eta$ index, and apply its proved selector without adding a second target charge or replacing the target density norm. Since $s = 2^{-2j}$ and $j \ge 0$, $\mathrm{L2plus}(1/s) = 2j$ exactly. Other stages, own normalizers, eight allocations and positive-stage/order conditions are those already proved. No later-tail theorem or stronger L^2 certificate is invoked. □

This update integrates accepted conditional mathematics into the preserved manuscript. Its finite symbolic choices are not executed giant field stages. The separate G6 STOP / INCOMPLETE decision remains unchanged; earlier public editions remain separately archived. Historical internal acceptance of source mathematics, fresh manuscript review, finite algebra checks, local LaTeX build and visual document QA are separate records in the companion report.

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